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Abhishek Mishra

# Special Integrals

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Abhishek Mishra
Department of Computer Science
and Information Systems
Birla Institute of Technology and Science
Pilani, Rajasthan, India

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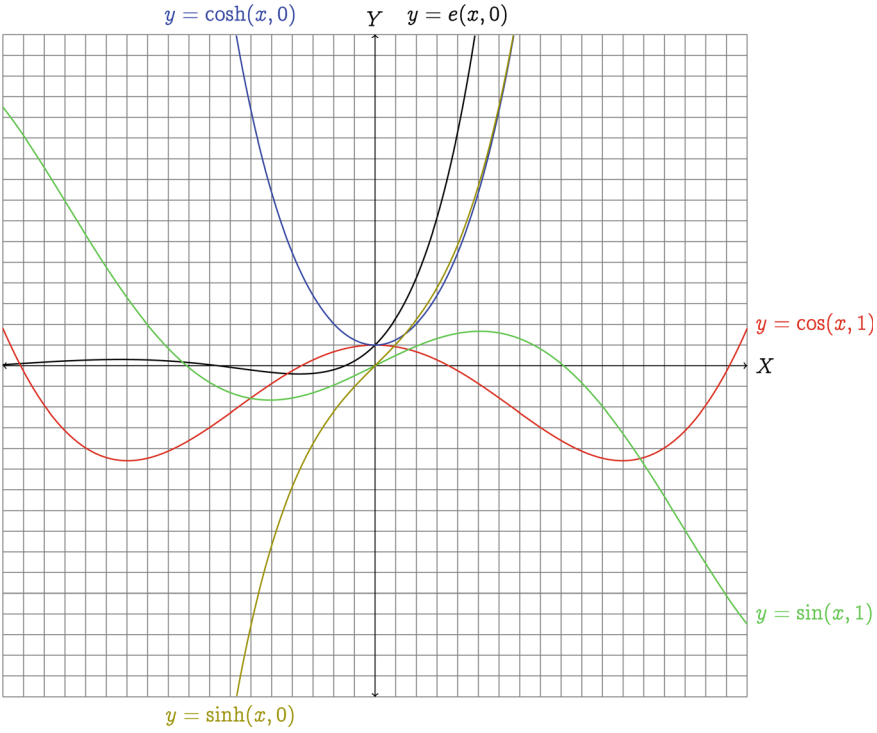
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PSEUDO-EXPONENTIAL FUNCTIONS IN $[-18; 18] \times [-16; 16]$

*Dedicated to my philosopher father, from whom I Inherited the habit of reading books.*

# Preface

Having extensive experience in teaching theoretical computer science courses, I have written many lecture notes and designed many original problems for my students to solve. I was thinking of writing a book using these lecture notes and problems. I had to decide on the topic for my first book. I developed my interest in theoretical computer science during my Ph.D. However, even before that, calculus was my first love during my schooldays. Having already done some original work in calculus during my schooldays, I decided to write this book on it.

My motivation for writing this book dates back to the summer of 1997. After taking my class 10 board exam, I was trying to find the sum of the following infinite series.

$$\sum_{j=0}^{\infty} \frac{1}{(j!)^2} = \frac{1}{(0!)^2} + \frac{1}{(1!)^2} + \frac{1}{(2!)^2} + \frac{1}{(3!)^2} + \ldots = ?$$

I succeeded in finding an integral representation for the series as [4.22]

$$\frac{1}{\pi} \int_0^{\pi} e^{2\cos\theta}\, d\theta = \sum_{j=0}^{\infty} \frac{1}{(j!)^2} = \frac{1}{(0!)^2} + \frac{1}{(1!)^2} + \frac{1}{(2!)^2} + \frac{1}{(3!)^2} + \ldots$$

This motivated me to explore further integral representations of series. This resulted in about a 150 page-long manuscript listing about 380 identities (Mishra, 1997).

The present book is based on the above manuscript. The manuscript contains several theorems without proof that are used in the derivation of the identities. I have proved all the theorems in the chapters. I have given the identities as problems to solve. I found very few of the identities in the manuscript to be incorrect, which I have not included in the book. Some of the identities had minor corrections. I have included the corrected versions of these identities in the book.

Later, I was able to find 32 of the identities in the existing literature. [1.1–1.6], [1.15], and [2.9–2.10] are Lehmer's identities (Lehmer, 1985). [1.50] is a special case of Eq. (2.2.2) on page 37 in Sect. 2.2 of the book by Borwein and Borwein

(1987). Seventy-two similar identities, mainly involving the binomial coefficients, are given in Chap. 1.

[1.73] and [1.83] are Ramanujan's identities. They are given as entry 24 on p. 396 and entry 29 on p. 399, respectively, in Chap. 31 of the book by Berndt (1994). Twenty similar identities involving powers of 2 are given in Chap. 1.

[2.5] and [2.20] are given as Examples XXI-8 and XXI-7, p. 158, respectively; and [2.21] is given as Eq. 3 in Sect. 129 on p. 156 in the book by Loney (1928). Fifty-two similar identities involving the circular and hyperbolic trigonometrical functions are given in Chap. 2.

[2.8–2.9] are given as Theorems 6.6.1–6.6.2 on p. 119, [1.37] is given as Eq. (6.6.10) on p. 120, and [2.10] is given as Eq. (6.6.21) on p. 122 in Sect. 6.6; [4.143] is given as Theorem 6.4.1 on p. 113 in Sect. 6.4; and [4.147] is given as Proposition 12.5.1 on p. 246 in Sect. 12.5 in the book by Boros and Moll (2006).

[4.2] is given as Example 3 on p. 113 in Sect. 6.21 of the book by Whittaker and Watson (2016). [4.83] and [5.18] are Eq. (6) on p. 20, [4.91] and [5.19] are Eq. (7) on p. 21, [4.77] and [5.12] are Eq. (8) on p. 21, [4.85] and [5.13] are Eq. (9) on p. 21 in Sect. 2.2; and [4.22] and [4.24] are special cases of Eq. (9) on p. 79 in Sect. 3.71 of the book by Watson (2011). Three hundred eighteen similar identities involving special integrals and series sums are given in the three chapters (Chaps. 4–6). This is the unique contribution of the present book. To the best of my knowledge, this is the first book with similar topics covered.

The three prerequisites for reading this book are knowledge of *Calculus* (Thomas et al., 2018; Apostol, 2017; Hardy, 1921; Piskunov, 1996a,b; Maron, 1973; Edwards, 1896; Bromwich, 2018), *Complex Analysis* (Ahlfors, 2017; Brown and Churchill, 2017a; Whittaker and Watson, 2016; Lang, 2013; Convay, 1995), and *Fourier Series* (Brown and Churchill, 2017b; Tolstov, 1977; Carslaw, 2018). In *Calculus*, you should be able to differentiate and integrate functions of a single variable. Knowledge of double integrals and differential equations is desirable but not essential in reading major portions of the book. In *Complex Analysis*, you should be familiar with the complex elementary functions and Taylor's and Maclaurin's Series. In *Fourier Series*, you should be able to compute the coefficients of the Fourier series.

Part I of the book has six chapters. In Chap. 1 ("Binomial Series"), you will learn the differentiation and integration methods for summing a series. You will learn how to apply these techniques to find the sum of some well-known series. You will also learn the technique of summing some interesting series involving integral and fractional powers of 2. The problems listed in this chapter mainly involve the binomial coefficients.

In Chap. 2 ("Trigonometrical Series"), you will learn the use of complex variables and the techniques of Chap. 1 to derive some interesting trigonometrical series. The problems listed in this chapter are series sum identities involving the circular and hyperbolic trigonometrical functions.

In Chap. 3 ("Bessel Functions"), first you will learn about the Bessel coefficients $J_n(z)$ and $I_n(z)$. After that, you will learn about their generalizations: the Bessel functions $J_v(z)$ and $I_v(z)$. The problems listed in this chapter are some of the properties of the Bessel functions.

In Chap. 4 ("Special Integrals"), you will learn the use of complex variables in the integrals and how to use them to get some integral representations of some series sums. The problems listed in this chapter are some special integrals derived using the theorems listed in the chapter.

In Chap. 5 ("Generalized Bessel Functions"), first you learn about the Mittag-Leffler functions $E_\alpha(z)$ (Mittag-Leffler, 1903; Erdélyi, 1955) and their various generalizations: $E_{\alpha,\beta}(z)$ by Wiman (1905), $E^{\gamma}_{\alpha,\beta}(z)$ by Prabhakar (1971), and $E^{\gamma,q}_{\alpha,\beta}(z)$ by Shukla and Prajapati (2007). Then you will learn about the generalized Bessel functions $\phi(\rho, \beta; z)$ as defined by Wright (1933). Using the ideas of Chap. 3, as a special case of $\phi(\rho, \beta; z)$, the pseudo-exponential functions are defined. You will learn how to get their integral representations in the problems listed in this chapter. You will learn that these functions exhibit similar differentiation and integration properties as compared to the exponential function and the circular and hyperbolic trigonometrical functions. The problems listed in this chapter involve the pseudo-exponential functions and the properties of the above-listed functions.

Chapter 6 ("Series Sums Using Special Integrals"), is a continuation of Chap. 4, in which you will learn some more interesting properties of the pseudo-exponential functions. You will also learn a generalized theorem for series sums. As a special case of this theorem, you can derive some of the theorems of Chap. 4. The problems listed in this chapter are some series sums obtained by making use of special integrals.

Part II of the book has solutions to the problems in the six chapters in Part I. You should attempt the problems before referring to the solutions. If you can find a better solution, I will be very grateful to you for knowing the solution. The Appendix ("Mapping from the Manuscript to the Book") tells where to find the corresponding entries of the manuscript in the book.

I am highly grateful to my cousin, Vishal Mishra. He came from Delhi to our place at Banaras Hindu University in Varanasi to complete his Bachelor of Technology in Computer Science and Engineering at the Institute of Technology during the period 1992–1996. During his four years of stay with us, I studied in the school: classes 6–9. I got inspired by him and started asking him mathematics questions. Class 6 was not fruitful for me because I was frightened of algebra. But in class 7, I got deeply interested in mathematics. Under his expert guidance, within six months, I was studying class 12 mathematics. His method of explaining the mathematical concepts was excellent. Taking full advantage of his mentorship, in class 9, I started studying his *Calculus* (Thomas and Finney, 1995). As a result, by class 10, I became proficient in calculus, complex analysis, and Fourier series, and was able to write the manuscript (Mishra, 1997).

I am highly grateful to the anonymous referee for giving me more insights and references that helped to improve the quality of this book. I am also highly grateful to Shamim Ahmad from Springer Nature for the book review process.

I am thankful to my colleagues Sundaresan Raman, Kamlesh Tiwari, Ashutosh Bhatia, and Sudeept Mohan. They helped me in my application for sabbatical leave for writing this and other books. Unfortunately, it was unsuccessful, and I had to drop the idea of writing other books.

Last but not least, I would like to thank my family for their support and encouragement. My mother was happy to know that I was writing a book. My sons Ishitva and Ivaan took away my file for keeping the drafts of the book for playing! My wife Gitika constantly enquired about the status of my book. She was eagerly waiting for my book to be published. This motivated me to make extra efforts to complete the book.

I have reviewed the book several times and found some errors in each revision. If you can find some more errors in the book and inform me, I will be very grateful to you.

Pilani, India Abhishek Mishra

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# Notations

| | |
|---|---|
| $[m.n]$ | Problem $n$ in Chapter $m$ |
| $(m.n)$ | Equation $n$ in Chapter $m$ |
| $\mathbb{N}$ | The set of all natural numbers: $\{1, 2, 3, 4, \dots\}$ |
| $\mathbb{N} - 1$ | The set of all non-negative integers: $\{n - 1 \mid n \in \mathbb{N}\}$ |
| $\mathbb{N} + 1$ | The set of all natural numbers greater than 1: $\{n + 1 \mid n \in \mathbb{N}\}$ |
| $-\mathbb{N}$ | The set of all negative integers: $\{-n \mid n \in \mathbb{N}\}$ |
| $2\mathbb{N}$ | The set of all even natural numbers: $\{2n \mid n \in \mathbb{N}\}$ |
| $2\mathbb{N} - 2$ | The set of all non-negative even integers: $\{2n - 2 \mid n \in \mathbb{N}\}$ |
| $2\mathbb{N} - 1$ | The set of all odd natural numbers: $\{2n - 1 \mid n \in \mathbb{N}\}$ |
| $2\mathbb{N} + 1$ | The set of all odd natural numbers greater than 1: $\{2n + 1 \mid n \in \mathbb{N}\}$ |
| $-(2\mathbb{N} - 1)$ | The set of all negative odd integers: $\{-(2n - 1) \mid n \in \mathbb{N}\}$ |
| $-2\mathbb{N}$ | The set of all negative even integers: $\{-2n \mid n \in \mathbb{N}\}$ |
| $\mathbb{Z}$ | The set of all integers: $\{sn \mid n \in \mathbb{N}, s \in \{0, 1, -1\}\}$ |
| $2\mathbb{Z} + 1$ | The set of all odd integers: $\{2z + 1 \mid z \in \mathbb{Z}\}$ |
| $2\mathbb{Z}$ | The set of all even integers: $\{2z \mid z \in \mathbb{Z}\}$ |
| $\mathbb{R}$ | The set of all real numbers |
| $\mathbb{R}^+$ | The set of all positive real numbers: $\{x \mid x \in \mathbb{R}, x > 0\}$ |
| $\mathbb{R}^-$ | The set of all negative real numbers: $\{x \mid x \in \mathbb{R}, x < 0\}$ |
| $\mathbb{C}$ | The set of all complex numbers |
| LHS | Left-Hand Side |
| RHS | Right-Hand Side |
| $(f(j))_{j=1}^{n}$ | The ordered $n$-tuple $(f(1), f(2), \dots, f(n-1), f(n))$ |
| $S^2$ | The 2-dimensional Cartesian product $S \times S$ |
| $xS$ or $Sx$ | $x$ multiplied by each element of the set $S$: $\{xs \mid s \in S\}$ |
| $[m..n]$ | The closed integer interval $\{p \mid m \leq p \leq n, p \in \mathbb{Z}\}$ |
| $[x, y]$ | The closed real interval $\{z \mid x \leq z \leq y, z \in \mathbb{R}\}$ |
| $(x, y)$ | The open real interval $\{z \mid x < z < y, z \in \mathbb{R}\}$ |

| | |
|---|---|
| $[x, y)$ | The semi-closed real interval $\{z \vert x \leq z < y, z \in \mathbb{R}\}$ |
| $\forall$ | For all |
| $\exists$ | There exists |
| $\forall(n, r) \in$ | |
| $(\mathbb{N} - 1) \times [0..n]$ | $= \{(n, r) \vert n \in \mathbb{N} - 1, r \in [0..n]\}$ |
| $\forall(z, \theta) \in \mathbb{Z}\times$ | |
| $\left((4z - 1)\frac{\pi}{n}, (4z + 1)\frac{\pi}{n}\right)$ | $= \left\{(z, \theta) \vert z \in \mathbb{Z}, \theta \in \left((4z - 1)\frac{\pi}{n}, (4z + 1)\frac{\pi}{n}\right)\right\}$ |
| $\lfloor x \rfloor$ | The greatest integer $\leq x$ |
| $\lceil x \rceil$ | The smallest integer $\geq x$ |
| $0^0$ | $= 1$ |
| $[n \geq 0]$ | $= 1$ if $n \geq 0$; $= 0$, otherwise |
| $\sum_{j=m}^{n} a_j \sum_{k=p}^{q} b_k$ | $= \left(\sum_{j=m}^{n} a_j\right)\left(\sum_{k=p}^{q} b_k\right)$ |
| $\Re(z)$ | Real part of the complex number $z$ |
| $\Re(\mathbb{C})^+$ | The set of all complex numbers with positive real part: $\{w \vert w \in \mathbb{C}, \Re(w) > 0\}$ |
| $\zeta(x)$ | The Riemann's zeta function defined by $\zeta(x) = \sum_{j=1}^{\infty} \frac{1}{j^x}$ for $x > 1$ |

# Part I
# Special Integrals

# Chapter 1
# Binomial Series




In this chapter, we will learn about differentiation and integration methods for series sums. We will apply these techniques to get some well-known series sums. The problems listed in this chapter mainly involve the binomial coefficients. Many of the problems are special cases of problems in Chap. 2. The remaining problems can be solved by the techniques developed in this chapter.

We will also learn about summing series involving powers of 2. All these series have one thing in common. They involve two types of terms: several similar terms and one dissimilar term. The dissimilar term consumes the similar terms successively, finally yielding a dissimilar term, which is the sum of the series.

## 1.1 The Differentiation Method for Series Sum

The idea of using derivatives for finding series sums is straightforward. Given an identity involving a variable $x$, simply differentiate both sides of the identity with respect to $x$ to get a new identity.

**Definition 1.1** (*The Differentiation Method for Series Sum*)
Let

$$f(x) = \sum_{j=0}^{n} a_j x^j, \tag{1.1}$$

be a function. In the differentiation method for series sum, we differentiate (1.1) with respect to $x$ to get

$$f'(x) = \sum_{j=1}^{n} j a_j x^{j-1}.$$

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The above method can also be applied when the series in (1.1) is infinite. We can get an alternative method for the sum of *Arithmetic Progressions (AP)* using this method.

**Example 1.2** (*Finding the Sum of First n Natural Numbers*)
Prove that $\forall n \in \mathbb{N}$,

$$\sum_{j=1}^{n} j = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

***Proof*** The sum of the *Geometric Progression (GP)* $\left(x^j\right)_{j=0}^{n}$ is given by

$$\sum_{j=0}^{n} x^j = \frac{x^{n+1} - 1}{x - 1}. \tag{1.2}$$

Differentiating (1.2) with respect to $x$, we get

$$\begin{aligned} \sum_{j=1}^{n} j x^{j-1} &= \frac{(n+1)x^n(x-1) - \left(x^{n+1} - 1\right)}{(x-1)^2} \\ &= \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}. \end{aligned} \tag{1.3}$$

Taking the limit as $x \to 1$ in (1.3), we get

$$\begin{aligned} &\lim_{x\to 1} \sum_{j=1}^{n} j x^{j-1} = \lim_{x\to 1} \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2} \\ \iff &\sum_{j=1}^{n} j = \lim_{y\to 0} \frac{n(1+y)^{n+1} - (n+1)(1+y)^n + 1}{y^2} \\ &= \lim_{y\to 0} \left( n \left( \frac{1}{y^2} + (n+1)\frac{1}{y} + \frac{(n+1)n}{2} + y \sum_{j=3}^{n+1} \binom{n+1}{j} y^{j-3} \right) \right. \\ &\quad \left. -(n+1) \left( \frac{1}{y^2} + \frac{n}{y} + \frac{n(n-1)}{2} + y \sum_{j=3}^{n} \binom{n}{j} y^{j-3} \right) + \frac{1}{y^2} \right) \\ &= \lim_{y\to 0} \frac{n - (n+1) + 1}{y^2} + \lim_{y\to 0} \frac{n(n+1) - (n+1)n}{y} \\ &\quad + \lim_{y\to 0} \frac{n^2(n+1) - n(n+1)(n-1)}{2} \\ &\quad + \lim_{y\to 0} y \left( \sum_{j=3}^{n+1} n \binom{n+1}{j} y^{j-3} - \sum_{j=3}^{n} (n+1) \binom{n}{j} y^{j-3} \right) \end{aligned}$$

$$= \frac{n(n+1)(n-(n-1))}{2}$$
$$= \frac{n(n+1)}{2}.$$

□

**Example 1.3** (*Finding the Sum of First n Odd Natural Numbers*)
Prove that $\forall n \in \mathbb{N}$,

$$\sum_{j=1}^{n}(2j-1) = 1+3+5+\cdots+2n-1 = n^2.$$

***Proof*** The sum of the GP $\left(x^{2j-1}\right)_{j=1}^{n}$ is given by

$$\sum_{j=1}^{n} x^{2j-1} = \frac{x^{2n+1}-x}{x^2-1}. \tag{1.4}$$

Differentiating (1.4) with respect to $x$, we get

$$\begin{aligned}\sum_{j=1}^{n}(2j-1)x^{2j-2} &= \frac{\left((2n+1)x^{2n}-1\right)\left(x^2-1\right)-2x\left(x^{2n+1}-x\right)}{\left(x^2-1\right)^2} \\ &= \frac{(2n-1)x^{2n+2}-(2n+1)x^{2n}+x^2+1}{\left(x^2-1\right)^2}.\end{aligned} \tag{1.5}$$

Taking the limit as $x \to 1$ in (1.5), we get

$$\lim_{x\to 1}\sum_{j=1}^{n}(2j-1)x^{2j-2} = \lim_{x\to 1}\frac{(2n-1)x^{2n+2}-(2n+1)x^{2n}+x^2+1}{\left(x^2-1\right)^2}$$
$$\iff \sum_{j=1}^{n}(2j-1)$$
$$= \lim_{y\to 0}\frac{(2n-1)(1+y)^{2n+2}-(2n+1)(1+y)^{2n}+(1+y)^2+1}{((1+y)^2-1)^2}$$
$$= \lim_{y\to 0}\frac{1}{y^2(2+y)^2}\left((2n-1)\left(1+(2n+2)y+\frac{(2n+2)(2n+1)}{2}y^2\right.\right.$$
$$\left.+\, y^3\sum_{j=3}^{2n+2}\binom{2n+2}{j}y^{j-3}\right)$$
$$-(2n+1)\left(1+2ny+\frac{2n(2n-1)}{2}y^2+y^3\sum_{j=3}^{2n}\binom{2n}{j}y^{j-3}\right)$$
$$\left.+(1+2y+y^2)+1\right)$$

$$
\begin{aligned}
&= \lim_{y\to 0} \frac{(2n-1)-(2n+1)+1+1}{y^2(2+y)^2} + \lim_{y\to 0} \frac{(2n-1)(2n+2)-2n(2n+1)+2}{y(2+y)^2} \\
&\quad + \lim_{y\to 0} \frac{(2n-1)(n+1)(2n+1)-n(2n+1)(2n-1)+1}{(2+y)^2} \\
&\quad + \lim_{y\to 0} \frac{y}{(2+y)^2} \left( \sum_{j=3}^{2n+2} (2n-1)\binom{2n+2}{j} y^{j-3} - \sum_{j=3}^{2n} (2n+1)\binom{2n}{j} y^{j-3} \right) \\
&= \frac{(2n+1)(2n-1)((n+1)-n)+1}{4} \\
&= n^2.
\end{aligned}
$$

□

With more lengthy calculations, we can compute the sum of squares and cubes of consecutive natural numbers.

**Example 1.4** (*Finding the Sum of Squares of First n Natural Numbers*)
Prove that $\forall n \in \mathbb{N}$,

$$\sum_{j=1}^{n} j^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

***Proof*** Multiplying (1.3) by $x$, we get

$$\sum_{j=1}^{n} j x^j = \frac{n x^{n+2} - (n+1)x^{n+1} + x}{(x-1)^2}. \tag{1.6}$$

Differentiating (1.6) with respect to $x$, we get

$$
\begin{aligned}
\sum_{j=1}^{n} j^2 x^{j-1} &= \frac{1}{(x-1)^4} \Big( \big(n(n+2)x^{n+1} - (n+1)^2 x^n + 1\big)(x-1)^2 \\
&\quad -2(x-1)\left(nx^{n+2} - (n+1)x^{n+1} + x\right)\Big) \\
&= \frac{1}{(x-1)^3} \Big( \big(n(n+2)x^{n+1} - (n+1)^2 x^n + 1\big)(x-1) \\
&\quad -2\left(nx^{n+2} - (n+1)x^{n+1} + x\right)\Big) \\
&= \frac{1}{(x-1)^3} \Big( (n(n+2) - 2n)x^{n+2} \\
&\quad + \left(2(n+1) - n(n+2) - (n+1)^2\right)x^{n+1} + (n+1)^2 x^n - x - 1\Big) \\
&= \frac{n^2 x^{n+2} - \left(2n^2 + 2n - 1\right)x^{n+1} + (n+1)^2 x^n - x - 1}{(x-1)^3}.
\end{aligned} \tag{1.7}
$$

Taking the limit as $x \to 1$ in (1.7), we get

$$\lim_{x\to 1}\sum_{j=1}^{n} j^2 x^{j-1} = \lim_{x\to 1}\frac{n^2x^{n+2} - \left(2n^2+2n-1\right)x^{n+1} + (n+1)^2x^n - x - 1}{(x-1)^3}$$

$$\Longleftrightarrow \sum_{j=1}^{n} j^2 = \lim_{y\to 0}\frac{1}{y^3}\left(n^2(1+y)^{n+2} - \left(2n^2+2n-1\right)(1+y)^{n+1}\right.$$
$$\left.+(n+1)^2(1+y)^n - (1+y) - 1\right)$$

$$= \lim_{y\to 0}\frac{1}{y^3}\left(n^2\left(1+(n+2)y + \frac{(n+2)(n+1)}{2}y^2 + \frac{(n+2)(n+1)n}{6}y^3\right.\right.$$
$$\left.+y^4\sum_{j=4}^{n+2}\binom{n+2}{j}y^{j-4}\right)$$
$$-\left(2n^2+2n-1\right)\left(1+(n+1)y + \frac{(n+1)n}{2}y^2 + \frac{(n+1)n(n-1)}{6}y^3\right.$$
$$\left.+y^4\sum_{j=4}^{n+1}\binom{n+1}{j}y^{j-4}\right)$$
$$+(n+1)^2\left(1+ny + \frac{n(n-1)}{2}y^2 + \frac{n(n-1)(n-2)}{6}y^3\right.$$
$$\left.\left.+y^4\sum_{j=4}^{n}\binom{n}{j}y^{j-4}\right) - (1+y) - 1\right)$$

$$= \lim_{y\to 0}\frac{n^2 - \left(2n^2+2n-1\right) + (n+1)^2 - 1 - 1}{y^3}$$
$$+\lim_{y\to 0}\frac{n^2(n+2) - \left(2n^2+2n-1\right)(n+1) + n(n+1)^2 - 1}{y^2}$$
$$+\lim_{y\to 0}\frac{n^2(n+2)(n+1) - n\left(2n^2+2n-1\right)(n+1) + n(n+1)^2(n-1)}{2y}$$
$$+\lim_{y\to 0}\left(\frac{n^3(n+2)(n+1)}{6} - \frac{n(2n^2+2n-1)(n+1)(n-1)}{6}\right.$$
$$\left.+\frac{n(n+1)^2(n-1)(n-2)}{6}\right) + \lim_{y\to 0} y\left(n^2\sum_{j=4}^{n+2}\binom{n+2}{j}y^{j-4}\right.$$
$$\left.-\left(2n^2+2n-1\right)\sum_{j=4}^{n+1}\binom{n+1}{j}y^{j-4} + (n+1)^2\sum_{j=4}^{n}\binom{n}{j}y^{j-4}\right)$$

$$= \frac{n(n+1)\left(n^2(n+2) - \left(2n^2+2n-1\right)(n-1) + (n+1)(n-1)(n-2)\right)}{6}$$
$$= \frac{n(n+1)(2n+1)}{6}.$$

□

**Example 1.5** (*Finding the Sum of Cubes of First n Natural Numbers*)
Prove that $\forall n \in \mathbb{N}$,

$$\sum_{j=1}^{n} j^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

***Proof*** Multiplying (1.7) by $x$, we get

$$\sum_{j=1}^{n} j^2 x^j = \frac{n^2 x^{n+3} - \left(2n^2+2n-1\right)x^{n+2} + (n+1)^2 x^{n+1} - x^2 - x}{(x-1)^3}. \tag{1.8}$$

Differentiating (1.8) with respect to $x$, we get

$$\begin{aligned}
\sum_{j=1}^{n} j^3 x^{j-1} &= \frac{1}{(x-1)^6}\left(\left(n^2(n+3)x^{n+2} - \left(2n^2+2n-1\right)(n+2)x^{n+1}\right.\right.\\
&\quad \left.+(n+1)^2(n+1)x^n - 2x - 1\right)(x-1)^3\\
&\quad \left.-3(x-1)^2\left(n^2x^{n+3} - \left(2n^2+2n-1\right)x^{n+2} + (n+1)^2x^{n+1} - x^2 - x\right)\right)\\
&= \frac{1}{(x-1)^4}\left(\left(n^2(n+3)x^{n+2} - \left(2n^2+2n-1\right)(n+2)x^{n+1}\right.\right.\\
&\quad \left.+(n+1)^3x^n - 2x - 1\right)(x-1)\\
&\quad \left.-3\left(n^2x^{n+3} - \left(2n^2+2n-1\right)x^{n+2} + (n+1)^2x^{n+1} - x^2 - x\right)\right)\\
&= \frac{1}{(x-1)^4}\left(\left(n^2(n+3) - 3n^2\right)x^{n+3} + \left(3\left(2n^2+2n-1\right)\right.\right.\\
&\quad \left.-n^2(n+3) - \left(2n^2+2n-1\right)(n+2)\right)x^{n+2}\\
&\quad + \left((n+1)^3 + \left(2n^2+2n-1\right)(n+2) - 3(n+1)^2\right)x^{n+1}\\
&\quad \left.-(n+1)^3x^n + (3-2)x^2 + (3+2-1)x + 1\right)\\
&= \frac{1}{(x-1)^4}\left(n^3x^{n+3} - \left(3n^3+3n^2-3n+1\right)x^{n+2}\right.\\
&\quad \left.+\left(3n^3+6n^2-4\right)x^{n+1} - \left(n^3+3n^2+3n+1\right)x^n + x^2 + 4x + 1\right).
\end{aligned} \tag{1.9}$$

Taking the limit as $x \to 1$ in (1.9), we get

$$\lim_{x\to 1}\sum_{j=1}^{n} j^3 x^{j-1} = \lim_{x\to 1}\frac{1}{(x-1)^4}\left(n^3x^{n+3} - \left(3n^3+3n^2-3n+1\right)x^{n+2}\right.$$
$$\left. + \left(3n^3+6n^2-4\right)x^{n+1} - \left(n^3+3n^2+3n+1\right)x^n + x^2 + 4x + 1\right)$$
$$\Longleftrightarrow \sum_{j=1}^{n} j^3 = \lim_{y\to 0}\frac{1}{y^4}\left(n^3(1+y)^{n+3} - \left(3n^3+3n^2-3n+1\right)(1+y)^{n+2}\right.$$
$$\left. + \left(3n^3+6n^2-4\right)(1+y)^{n+1}\right.$$
$$\left. - \left(n^3+3n^2+3n+1\right)(1+y)^n + (1+y)^2 + 4(1+y) + 1\right)$$

$$= \lim_{y\to 0}\frac{1}{y^4}\left(n^3\left(1+(n+3)y + \frac{(n+3)(n+2)}{2}y^2 + \frac{(n+3)(n+2)(n+1)}{6}y^3\right.\right.$$
$$\left. + \frac{(n+3)(n+2)(n+1)n}{24}y^4 + y^5\sum_{j=5}^{n+3}\binom{n+3}{j}y^{j-5}\right)$$
$$- \left(3n^3+3n^2-3n+1\right)\left(1+(n+2)y + \frac{(n+2)(n+1)}{2}y^2\right.$$
$$\left. + \frac{(n+2)(n+1)n}{6}y^3 + \frac{(n+2)(n+1)n(n-1)}{24}y^4 + y^5\sum_{j=5}^{n+2}\binom{n+2}{j}y^{j-5}\right)$$
$$+ \left(3n^3+6n^2-4\right)\left(1+(n+1)y + \frac{(n+1)n}{2}y^2 + \frac{(n+1)n(n-1)}{6}y^3\right.$$
$$\left. + \frac{(n+1)n(n-1)(n-2)}{24}y^4 + y^5\sum_{j=5}^{n+1}\binom{n+1}{j}y^{j-5}\right)$$
$$- \left(n^3+3n^2+3n+1\right)\left(1+ny + \frac{n(n-1)}{2}y^2 + \frac{n(n-1)(n-2)}{6}y^3\right.$$
$$\left. + \frac{n(n-1)(n-2)(n-3)}{24}y^4 + y^5\sum_{j=5}^{n}\binom{n}{j}y^{j-5}\right) + (1+y)^2 + 4(1+y) + 1\Big)$$

$$= \lim_{y\to 0}\frac{1}{y^4}\left(n^3 - \left(3n^3+3n^2-3n+1\right) + \left(3n^3+6n^2-4\right)\right.$$
$$\left. - \left(n^3+3n^2+3n+1\right) + 1 + 4 + 1\right)$$
$$+ \lim_{y\to 0}\frac{1}{y^3}\left(n^3(n+3) - \left(3n^3+3n^2-3n+1\right)(n+2)\right.$$
$$\left. + \left(3n^3+6n^2-4\right)(n+1) - n\left(n^3+3n^2+3n+1\right) + 2 + 4\right)$$
$$+ \lim_{y\to 0}\frac{1}{2y^2}\left(n^3(n+3)(n+2) - \left(3n^3+3n^2-3n+1\right)(n+2)(n+1)\right.$$

$$
\begin{aligned}
&+n\left(3n^3+6n^2-4\right)(n+1)-n\left(n^3+3n^2+3n+1\right)(n-1)+2\big)\\
&+\lim_{y\to 0}\frac{n(n+1)}{6y}\left(n^2(n+3)(n+2)-\left(3n^3+3n^2-3n+1\right)(n+2)\right.\\
&+\left(3n^3+6n^2-4\right)(n-1)-(n+1)^2(n-1)(n-2)\big)\\
&+\lim_{y\to 0}\frac{n(n+1)}{24}\left(n^3(n+3)(n+2)-\left(3n^3+3n^2-3n+1\right)(n+2)(n-1)\right.\\
&+\left(3n^3+6n^2-4\right)(n-1)(n-2)-(n+1)^2(n-1)(n-2)(n-3)\big)\\
&+\lim_{y\to 0} y\left(n^3\sum_{j=5}^{n+3}\binom{n+3}{j}y^{j-5}-\left(3n^3+3n^2-3n+1\right)\sum_{j=5}^{n+2}\binom{n+2}{j}y^{j-5}\right.\\
&\left.+\left(3n^3+6n^2-4\right)\sum_{j=5}^{n+1}\binom{n+1}{j}y^{j-5}-\left(n^3+3n^2+3n+1\right)\sum_{j=5}^{n}\binom{n}{j}y^{j-5}\right)\\
&=\frac{n(n+1)}{24}\left(n^5+5n^4+6n^3-\left(3n^3+3n^2-3n+1\right)\left(n^2+n-2\right)\right.\\
&\quad\left.+\left(3n^3+6n^2-4\right)\left(n^2-3n+2\right)-\left(n^2-1\right)\left(n^3-4n^2+n+6\right)\right)\\
&=\left(\frac{n(n+1)}{2}\right)^2.
\end{aligned}
$$

□

## 1.2 The Integration Method for Series Sum

Similar to the differentiation method of series sum, given an identity in variable $x$, we can integrate both sides of the identity with respect to $x$ to get a new identity.

**Definition 1.6** (*The Integration Method for Series Sum*)
Let

$$f(x)=\sum_{j=0}^{n}a_j x^j \tag{1.10}$$

be a function. In the integration method for series sum, we integrate (1.10) with respect to $x$ from 0 to $x$ to get

$$\int_0^x f(x)\,dx=\sum_{j=0}^{n}\frac{a_j}{j+1}x^{j+1}.$$

We can also apply the above method to infinite series sums. A simple application of this method to GP gives an identity involving a *Harmonic Progression (HP)* and a series for $\tan^{-1} x$.

**Example 1.7** (*Finding the Sum of Reciprocals of First n Natural Numbers*) Prove that $\forall n \in \mathbb{N}$,

$$\begin{aligned}\sum_{j=1}^{n} \frac{1}{j} &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \\ &= \sum_{j=1}^{n} \frac{(-1)^{j-1}}{j} \binom{n}{j} \\ &= \frac{\binom{n}{1}}{1} - \frac{\binom{n}{2}}{2} + \frac{\binom{n}{3}}{3} + \cdots + (-1)^{n-1} \frac{\binom{n}{n}}{n}.\end{aligned} \tag{1.11}$$

***Proof*** The sum of the GP $\left(x^j\right)_{j=0}^{n-1}$ is given by

$$\sum_{j=0}^{n-1} x^j = \frac{x^n - 1}{x - 1}. \tag{1.12}$$

Integrating (1.12) with respect to $x$ from 0 to 1, we get

$$\begin{aligned}&\sum_{j=0}^{n-1} \frac{x^{j+1}}{j+1}\bigg|_0^1 = \int_0^1 \frac{x^n - 1}{x - 1}\, dx \\ \iff &\sum_{j=1}^{n} \frac{1}{j} = \int_{-1}^{0} \frac{(1+y)^n - 1}{y}\, dy \\ &= \int_{-1}^{0} \frac{1}{y} \left( \sum_{j=0}^{n} \binom{n}{j} y^j - 1 \right) dy \\ &= \sum_{j=1}^{n} \binom{n}{j} \int_{-1}^{0} y^{j-1}\, dy \\ &= \sum_{j=1}^{n} \binom{n}{j} \frac{y^j}{j}\bigg|_{-1}^{0} \\ &= \sum_{j=1}^{n} \frac{(-1)^{j-1}}{j} \binom{n}{j}.\end{aligned}$$

□

**Example 1.8** (*Infinite Series for* $\tan^{-1} x$)
Prove that $\forall x \in [-1, 1]$,

$$\tan^{-1} x = \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j+1}}{2j+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \tag{1.13}$$

***Proof*** The sum of the GP $\left((-1)^j x^{2j}\right)_{j=0}^{\infty}$ is given by

$$\sum_{j=0}^{\infty} (-1)^j x^{2j} = \frac{1}{1+x^2}. \tag{1.14}$$

Integrating (1.14) with respect to $x$ from 0 to $x$, we get

$$\begin{aligned} &\sum_{j=0}^{\infty} (-1)^j \frac{x^{2j+1}}{2j+1}\bigg|_0^x = \tan^{-1} x\big|_0^x \\ \iff &\sum_{j=0}^{\infty} (-1)^j \frac{x^{2j+1}}{2j+1} = \tan^{-1} x. \end{aligned}$$

□

**Example 1.9** (*Infinite Series for* $\tanh^{-1} x$)
Prove that $\forall x \in (-1, 1)$,

$$\tanh^{-1} x = \sum_{j=0}^{\infty} \frac{x^{2j+1}}{2j+1} = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \cdots \tag{1.15}$$

***Proof*** Replace $x$ with $ix$ in (1.13). □

## 1.3 A Recursive Formula for $\pi$

As an example of terms involving powers of 2, we will derive a recursive formula for $\pi$.

**Example 1.10** (*A Recursive Formula for* $\pi$)
Let the sequence $(y_j)_{j=1}^{\infty}$ be defined as

$$y_j = \begin{cases} 2, & \text{if } j = 1; \\ \sqrt{2^{2j-1} - \sqrt{2^{4j-2} - 2^{2j} y_{j-1}^2}}, & \text{if } j \geq 2. \end{cases} \tag{1.16}$$

Prove that

$$\lim_{n\to\infty} y_n = \pi.$$

***Proof*** We will prove by induction that $\forall n \in \mathbb{N}$,

$$y_n = 2^n \sin\frac{\pi}{2^n}. \tag{1.17}$$

Let the induction hypothesis $P(n)$ be (1.17).

*Basis Step:* $P(1)$ states that

$$y_1 = 2 = 2\sin\frac{\pi}{2},$$

which is true. Assuming $P(n)$ to be true, from (1.16) and (1.17), we have

$$\begin{aligned}
y_{n+1} &= \sqrt{2^{2n+1} - \sqrt{2^{4n+2} - 2^{2n+2} y_n^2}} \\
&= \sqrt{2^{2n+1} - \sqrt{2^{4n+2} - 2^{2n+2} \cdot 2^{2n} \sin^2\frac{\pi}{2^n}}} \\
&= \sqrt{2^{2n+1} - \sqrt{2^{4n+2} - 2^{4n+2} \sin^2\frac{\pi}{2^n}}} \\
&= \sqrt{2^{2n+1} - 2^{2n+1}\sqrt{1 - \sin^2\frac{\pi}{2^n}}} \\
&= \sqrt{2^{2n+1} - 2^{2n+1}\cos\frac{\pi}{2^n}} \\
&= 2^n\sqrt{2\left(1 - \cos\frac{\pi}{2^n}\right)} \\
&= 2^n\sqrt{4\sin^2\frac{\pi}{2^{n+1}}} \\
&= 2^{n+1}\sin\frac{\pi}{2^{n+1}},
\end{aligned}$$

proving $P(n+1)$ to be true. From (1.17), we have

$$\begin{aligned}
\lim_{n\to\infty} y_n &= \lim_{n\to\infty} 2^n \sin\frac{\pi}{2^n} \\
&= \lim_{n\to\infty} \pi \left(\frac{\sin\frac{\pi}{2^n}}{\frac{\pi}{2^n}}\right) \\
&= \pi \cdot 1 \\
&= \pi.
\end{aligned}$$

□

**Definition 1.11** (*Repeated Exponential Square Root Functions*)
$\forall n \in \mathbb{N}$, we define the function $R_n : \mathbb{N}^n \to \mathbb{R}^+$ as follows:

$$R_n\left((m_j)_{j=1}^n\right) = \sqrt{2^{m_1} + \sqrt{2^{m_2} + \sqrt{\cdots + \sqrt{2^{m_{n-1}} + \sqrt{2^{m_n}}}}}}.$$

**Example 1.12** (*Solving the Recursive Formula for* $\pi$)
Prove that the solution of (1.16) is given by

$$y_n = \begin{cases} 2, & \text{if } n = 1; \\ R_1(3), & \text{if } n = 2; \\ \sqrt{2^{2n-1} - R_{n-1}\left(\left(2^j n - 2^j + 1\right)_{j=2}^n\right)}, & \text{if } n \geq 3. \end{cases} \tag{1.18}$$

***Proof*** We will prove (1.18) by induction. Let the induction hypothesis $P(n)$ be (1.18).
*Basis Step:* $P(1)$ states that $y_1 = 2$, which is already given in (1.16). $P(2)$ states that $y_2 = R_1(3) = \sqrt{2^3}$, which can be proved using (1.17) as follows:

$$y_2 = 2^2 \sin\frac{\pi}{2^2} = \frac{2^2}{\sqrt{2}} = \sqrt{2^3}.$$

*Induction Step:* Assuming $P(n)$ to be true, using (1.16) and (1.18), we can prove $P(n+1)$ to be true as follows:

$$\begin{aligned} y_{n+1} &= \sqrt{2^{2n+1} - \sqrt{2^{4n+2} - 2^{2n+2} y_n^2}} \\ &= \sqrt{2^{2n+1} - \sqrt{2^{4n+2} - 2^{2n+2}\left(2^{2n-1} - R_{n-1}\left(\left(2^j n - 2^j + 1\right)_{j=2}^n\right)\right)}} \\ &= \sqrt{2^{2n+1} - \sqrt{2^{4n+1} + 2^{2n+2} R_{n-1}\left(\left(2^j n - 2^j + 1\right)_{j=2}^n\right)}} \\ &= \sqrt{2^{2n+1} - \sqrt{2^{4n+1} + R_{n-1}\left(\left(2^j n + 2^j + 2^j n - 2^j + 1\right)_{j=2}^n\right)}} \\ &= \sqrt{2^{2n+1} - \sqrt{2^{4n+1} + R_{n-1}\left(\left(2^{j+1} n + 1\right)_{j=2}^n\right)}} \\ &= \sqrt{2^{2n+1} - R_n\left(\left(2^{j+1} n + 1\right)_{j=1}^n\right)} \\ &= \sqrt{2^{2n+1} - R_n\left(\left(2^j n + 1\right)_{j=2}^{n+1}\right)}. \end{aligned}$$

□

As an example of a limit involving powers of 2, we will derive an alternative definition of $\log x$.

**Example 1.13** (*An Alternative Definition of* $\log x$)
Prove that $\forall x \in \mathbb{R}^+$,

$$\log x = \lim_{n\to\infty} 2^n \left(x^{1/2^n} - 1\right). \tag{1.19}$$

***Proof*** We have

$$\begin{aligned}
\lim_{n\to\infty} 2^n \left(x^{1/2^n} - 1\right) &= \lim_{n\to\infty} 2^n \left(e^{\log(x)/2^n} - 1\right) \\
&= \lim_{n\to\infty} 2^n \sum_{j=1}^{\infty} \frac{\log^j x}{j! 2^{nj}} \\
&= \log x + \lim_{n\to\infty} \sum_{j=2}^{\infty} \frac{\log^j x}{j! 2^{n(j-1)}} \\
&= \log x.
\end{aligned}$$

□

## Problems

[1.1] (Lehmer, 1985) Prove that

$$\begin{aligned}
\frac{1}{3} + \frac{2\pi\sqrt{3}}{27} &= \sum_{j=1}^{\infty} \binom{2j}{j}^{-1} \\
&= \frac{1}{\binom{2}{1}} + \frac{1}{\binom{4}{2}} + \frac{1}{\binom{6}{3}} + \frac{1}{\binom{8}{4}} + \cdots
\end{aligned}$$

[1.2] (Lehmer, 1985) Prove that

$$\begin{aligned}
\frac{1}{5} + \frac{4\sqrt{5}}{25} \log\left(\frac{1+\sqrt{5}}{2}\right) &= \sum_{j=1}^{\infty} (-1)^{j-1} \binom{2j}{j}^{-1} \\
&= \frac{1}{\binom{2}{1}} - \frac{1}{\binom{4}{2}} + \frac{1}{\binom{6}{3}} - \frac{1}{\binom{8}{4}} + \cdots
\end{aligned}$$

[1.3] (Lehmer, 1985) Prove that

$$\frac{\pi\sqrt{3}}{9} = \sum_{j=1}^{\infty} \frac{1}{j}\binom{2j}{j}^{-1}$$
$$= \frac{1}{1\binom{2}{1}} + \frac{1}{2\binom{4}{2}} + \frac{1}{3\binom{6}{3}} + \frac{1}{4\binom{8}{4}} + \cdots$$

[1.4] (Lehmer, 1985) Prove that

$$\frac{2\sqrt{5}}{5}\log\left(\frac{1+\sqrt{5}}{2}\right) = \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{j}\binom{2j}{j}^{-1}$$
$$= \frac{1}{1\binom{2}{1}} - \frac{1}{2\binom{4}{2}} + \frac{1}{3\binom{6}{3}} - \frac{1}{4\binom{8}{4}} + \cdots$$

[1.5] (Lehmer, 1985) Prove that

$$\frac{\pi^2}{18} = \sum_{j=1}^{\infty} \frac{1}{j^2}\binom{2j}{j}^{-1}$$
$$= \frac{1}{1^2\binom{2}{1}} + \frac{1}{2^2\binom{4}{2}} + \frac{1}{3^2\binom{6}{3}} + \frac{1}{4^2\binom{8}{4}} + \cdots$$

[1.6] (Lehmer, 1985) Prove that

$$2\log^2\left(\frac{1+\sqrt{5}}{2}\right) = \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{j^2}\binom{2j}{j}^{-1}$$
$$= \frac{1}{1^2\binom{2}{1}} - \frac{1}{2^2\binom{4}{2}} + \frac{1}{3^2\binom{6}{3}} - \frac{1}{4^2\binom{8}{4}} + \cdots$$

[1.7] Prove that

$$2-\frac{\pi\sqrt{3}}{3}=\sum_{j=1}^{\infty}\frac{1}{j(2j+1)}\binom{2j}{j}^{-1}$$
$$=\frac{1}{1\cdot 3\cdot\binom{2}{1}}+\frac{1}{2\cdot 5\cdot\binom{4}{2}}+\frac{1}{3\cdot 7\cdot\binom{6}{3}}+\frac{1}{4\cdot 9\cdot\binom{8}{4}}+\cdots$$

[1.8] Prove that

$$2\sqrt{5}\log\left(\frac{1+\sqrt{5}}{2}\right)-2=\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{j(2j+1)}\binom{2j}{j}^{-1}$$
$$=\frac{1}{1\cdot 3\cdot\binom{2}{1}}-\frac{1}{2\cdot 5\cdot\binom{4}{2}}+\frac{1}{3\cdot 7\cdot\binom{6}{3}}-\frac{1}{4\cdot 9\cdot\binom{8}{4}}+\cdots$$

[1.9] Prove that

$$\frac{\pi}{3}-\frac{1}{4}+4\sqrt{3}-4\log\left(\frac{e^2}{2}\left(1+\frac{\sqrt{3}}{2}\right)\right)=\sum_{j=1}^{\infty}\frac{1}{(2j+1)(2j+2)^2 2^{4j}}\binom{2j}{j}$$
$$=\frac{\binom{2}{1}}{3\cdot 4^2\cdot 2^4}+\frac{\binom{4}{2}}{5\cdot 6^2\cdot 2^8}+\frac{\binom{6}{3}}{7\cdot 8^2\cdot 2^{12}}+\frac{\binom{8}{4}}{9\cdot 10^2\cdot 2^{16}}+\cdots$$

[1.10] Prove that

$$4\sqrt{5}+\frac{1}{4}-2\log\left(\frac{1+\sqrt{5}}{2}\right)-4\log\left(\frac{e^2}{2}\left(1+\frac{\sqrt{5}}{2}\right)\right)$$
$$=\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{(2j+1)(2j+2)^2 2^{4j}}\binom{2j}{j}$$
$$=\frac{\binom{2}{1}}{3\cdot 4^2\cdot 2^4}-\frac{\binom{4}{2}}{5\cdot 6^2\cdot 2^8}+\frac{\binom{6}{3}}{7\cdot 8^2\cdot 2^{12}}-\frac{\binom{8}{4}}{9\cdot 10^2\cdot 2^{16}}+\cdots$$

[1.11] Prove that

$$\frac{\pi^2}{18}+\frac{2\pi\sqrt{3}}{3}-4=\sum_{j=1}^{\infty}\frac{1}{j^2(2j+1)}\binom{2j}{j}^{-1}$$
$$=\frac{1}{1^2\cdot 3\cdot\binom{2}{1}}+\frac{1}{2^2\cdot 5\cdot\binom{4}{2}}+\frac{1}{3^2\cdot 7\cdot\binom{6}{3}}+\frac{1}{4^2\cdot 9\cdot\binom{8}{4}}+\cdots$$

[1.12] Prove that

$$4+2\log^2\left(\frac{1+\sqrt{5}}{2}\right)-4\sqrt{5}\log\left(\frac{1+\sqrt{5}}{2}\right)=\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{j^2(2j+1)}\binom{2j}{j}^{-1}$$
$$=\frac{1}{1^2\cdot 3\cdot\binom{2}{1}}-\frac{1}{2^2\cdot 5\cdot\binom{4}{2}}+\frac{1}{3^2\cdot 7\cdot\binom{6}{3}}-\frac{1}{4^2\cdot 9\cdot\binom{8}{4}}+\cdots$$

[1.13] Prove that

$$\frac{3}{2}-\frac{\pi\sqrt{3}}{6}-\frac{\pi^2}{18}=\sum_{j=1}^{\infty}\frac{1}{j(2j+1)(2j+2)}\binom{2j}{j}^{-1}$$
$$=\frac{1}{1\cdot 3\cdot 4\cdot\binom{2}{1}}+\frac{1}{2\cdot 5\cdot 6\cdot\binom{4}{2}}+\frac{1}{3\cdot 7\cdot 8\cdot\binom{6}{3}}+\frac{1}{4\cdot 9\cdot 10\cdot\binom{8}{4}}+\cdots$$

[1.14] Prove that

$$2\log^2\left(\frac{1+\sqrt{5}}{2}\right)+\sqrt{5}\log\left(\frac{1+\sqrt{5}}{2}\right)-\frac{3}{2}$$
$$=\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{j(2j+1)(2j+2)}\binom{2j}{j}^{-1}$$
$$=\frac{1}{1\cdot 3\cdot 4\cdot\binom{2}{1}}-\frac{1}{2\cdot 5\cdot 6\cdot\binom{4}{2}}+\frac{1}{3\cdot 7\cdot 8\cdot\binom{6}{3}}-\frac{1}{4\cdot 9\cdot 10\cdot\binom{8}{4}}+\cdots$$

[1.15] (Lehmer, 1985) Prove that

$$\frac{\pi}{3} = \sum_{j=0}^{\infty} \frac{1}{(2j+1)2^{4j}} \binom{2j}{j}$$
$$= 1 + \frac{\binom{2}{1}}{3 \cdot 2^4} + \frac{\binom{4}{2}}{5 \cdot 2^8} + \frac{\binom{6}{3}}{7 \cdot 2^{12}} + \cdots$$

[1.16] Prove that

$$2 \log\left(\frac{1+\sqrt{5}}{2}\right) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)2^{4j}} \binom{2j}{j}$$
$$= 1 - \frac{\binom{2}{1}}{3 \cdot 2^4} + \frac{\binom{4}{2}}{5 \cdot 2^8} - \frac{\binom{6}{3}}{7 \cdot 2^{12}} + \cdots$$

[1.17] Prove that

$$\frac{\pi}{3} + 2\sqrt{3} - \frac{9}{2} = \sum_{j=1}^{\infty} \frac{1}{(2j+1)(2j+2)2^{4j}} \binom{2j}{j}$$
$$= \frac{\binom{2}{1}}{3 \cdot 4 \cdot 2^4} + \frac{\binom{4}{2}}{5 \cdot 6 \cdot 2^8} + \frac{\binom{6}{3}}{7 \cdot 8 \cdot 2^{12}} + \frac{\binom{8}{4}}{9 \cdot 10 \cdot 2^{16}} + \cdots$$

[1.18] Prove that

$$2\sqrt{5} - \frac{7}{2} - 2 \log\left(\frac{1+\sqrt{5}}{2}\right) = \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{(2j+1)(2j+2)2^{4j}} \binom{2j}{j}$$
$$= \frac{\binom{2}{1}}{3 \cdot 4 \cdot 2^4} - \frac{\binom{4}{2}}{5 \cdot 6 \cdot 2^8} + \frac{\binom{6}{3}}{7 \cdot 8 \cdot 2^{12}} - \frac{\binom{8}{4}}{9 \cdot 10 \cdot 2^{16}} + \cdots$$

[1.19] Prove that

$$\frac{2\sqrt{3}}{3} = \sum_{j=0}^{\infty} \frac{1}{2^{4j}} \binom{2j}{j}$$
$$= 1 + \frac{\binom{2}{1}}{2^4} + \frac{\binom{4}{2}}{2^8} + \frac{\binom{6}{3}}{2^{12}} + \cdots$$

[1.20] Prove that

$$\frac{2\sqrt{5}}{5} = \sum_{j=0}^{\infty} \frac{(-1)^j}{2^{4j}} \binom{2j}{j}$$
$$= 1 - \frac{\binom{2}{1}}{2^4} + \frac{\binom{4}{2}}{2^8} - \frac{\binom{6}{3}}{2^{12}} + \cdots$$

[1.21] Prove that

$$\frac{8\sqrt{3}}{9} = \sum_{j=0}^{\infty} \frac{(2j+1)}{2^{4j}} \binom{2j}{j}$$
$$= 1 + \frac{3\binom{2}{1}}{2^4} + \frac{5\binom{4}{2}}{2^8} + \frac{7\binom{6}{3}}{2^{12}} + \cdots$$

[1.22] Prove that

$$\frac{8\sqrt{5}}{25} = \sum_{j=0}^{\infty} (-1)^j \frac{(2j+1)}{2^{4j}} \binom{2j}{j}$$
$$= 1 - \frac{3\binom{2}{1}}{2^4} + \frac{5\binom{4}{2}}{2^8} - \frac{7\binom{6}{3}}{2^{12}} + \cdots$$

[1.23] Prove that

$$\frac{16\sqrt{3}}{9} = \sum_{j=0}^{\infty} \frac{(2j+1)^2}{2^{4j}} \binom{2j}{j}$$
$$= 1^2 + \frac{3^2\binom{2}{1}}{2^4} + \frac{5^2\binom{4}{2}}{2^8} + \frac{7^2\binom{6}{3}}{2^{12}} + \cdots$$

[1.24] Prove that

$$\frac{16\sqrt{5}}{125} = \sum_{j=0}^{\infty} (-1)^j \frac{(2j+1)^2}{2^{4j}} \binom{2j}{j}$$
$$= 1^2 - \frac{3^2\binom{2}{1}}{2^4} + \frac{5^2\binom{4}{2}}{2^8} - \frac{7^2\binom{6}{3}}{2^{12}} + \cdots$$

[1.25] Prove that

$$\frac{160\sqrt{3}}{27} = \sum_{j=0}^{\infty} \frac{(2j+1)^3}{2^{4j}} \binom{2j}{j}$$
$$= 1^3 + \frac{3^3\binom{2}{1}}{2^4} + \frac{5^3\binom{4}{2}}{2^8} + \frac{7^3\binom{6}{3}}{2^{12}} + \cdots$$

[1.26] Prove that

$$-\frac{32\sqrt{5}}{125} = \sum_{j=0}^{\infty} (-1)^j \frac{(2j+1)^3}{2^{4j}} \binom{2j}{j}$$
$$= 1^3 - \frac{3^3\binom{2}{1}}{2^4} + \frac{5^3\binom{4}{2}}{2^8} - \frac{7^3\binom{6}{3}}{2^{12}} + \cdots$$

[1.27] Prove that

$$\frac{2368\sqrt{3}}{81} = \sum_{j=0}^{\infty} \frac{(2j+1)^4}{2^{4j}} \binom{2j}{j}$$
$$= 1^4 + \frac{3^4\binom{2}{1}}{2^4} + \frac{5^4\binom{4}{2}}{2^8} + \frac{7^4\binom{6}{3}}{2^{12}} + \cdots$$

[1.28] Prove that

$$-\frac{448\sqrt{5}}{625} = \sum_{j=0}^{\infty} (-1)^j \frac{(2j+1)^4}{2^{4j}} \binom{2j}{j}$$
$$= 1^4 - \frac{3^4\binom{2}{1}}{2^4} + \frac{5^4\binom{4}{2}}{2^8} - \frac{7^4\binom{6}{3}}{2^{12}} + \cdots$$

[1.29] Prove that

$$\pi = 3 + 6\sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{(6j)^2 - 1}$$
$$= 3 + 6\left(\frac{1}{6^2-1} - \frac{1}{12^2-1} + \frac{1}{18^2-1} - \cdots\right).$$

[1.30] Prove that

$$\frac{\pi}{e^{\pi} - e^{-\pi}} = \sum_{j=2}^{\infty} \frac{(-1)^j}{j^2+1}$$
$$= \frac{1}{2^2+1} - \frac{1}{3^2+1} + \frac{1}{4^2+1} - \frac{1}{5^2+1} + \cdots$$

[1.31] Prove that

$$\frac{e^2 + 2e - 1}{2(e^2-1)} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j^2\pi^2 + 1}$$
$$= 1 - \frac{1}{\pi^2+1} + \frac{1}{2^2\pi^2+1} - \frac{1}{3^2\pi^2+1} + \cdots$$

[1.32] Prove that

$$\frac{3-\csc 1}{2} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j^2\pi^2 - 1}$$
$$= 1 - \frac{1}{\pi^2 - 1} + \frac{1}{2^2\pi^2 - 1} - \frac{1}{3^2\pi^2 - 1} + \cdots$$

[1.33] Prove that

$$6 = \sum_{j=1}^{\infty} (4j+1)\binom{2j+2}{4}^{-1}$$
$$= \frac{5}{\binom{4}{4}} + \frac{9}{\binom{6}{4}} + \frac{13}{\binom{8}{4}} + \frac{17}{\binom{10}{4}} + \cdots$$

[1.34] Prove that $\forall n \in \mathbb{N}$,

$$\frac{1}{n+1} = \sum_{j=0}^{n} \frac{(-1)^j}{j+1}\binom{n}{j}$$
$$= \frac{\binom{n}{0}}{1} - \frac{\binom{n}{1}}{2} + \frac{\binom{n}{2}}{3} - \cdots + (-1)^n \frac{\binom{n}{n}}{n+1}.$$

[1.35] Prove that $\forall n \in \mathbb{N}$,

$$\frac{1}{n+1}\sum_{j=1}^{n} \frac{1}{j+1} = \frac{1}{n+1}\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots \frac{1}{n+1}\right)$$
$$= \sum_{j=1}^{n} (-1)^{j-1} \frac{j}{(j+1)^2}\binom{n}{j}$$
$$= \frac{1\binom{n}{1}}{2^2} - \frac{2\binom{n}{2}}{3^2} + \frac{3\binom{n}{3}}{4^2} - \cdots + (-1)^{n-1} \frac{n\binom{n}{n}}{(n+1)^2}.$$

[1.36] Prove that $\forall n \in \mathbb{N}+1$,

$$\begin{aligned} 0 &= \sum_{j=1}^{n} (-1)^{j-1} j \binom{n}{j} \\ &= 1\binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - \cdots + (-1)^{n-1} n \binom{n}{n}. \end{aligned}$$

[1.37] (Boros and Moll, 2006) Prove that $\forall n \in \mathbb{N}$,

$$\begin{aligned} \frac{2^{2n}}{2n+1}\binom{2n}{n}^{-1} &= \sum_{j=0}^{n} \frac{(-1)^j}{2j+1}\binom{n}{j} \\ &= \frac{\binom{n}{0}}{1} - \frac{\binom{n}{1}}{3} + \frac{\binom{n}{2}}{5} - \cdots + (-1)^n \frac{\binom{n}{n}}{2n+1}. \end{aligned}$$

[1.38] Prove that $\forall n \in 2\mathbb{N}-2$,

$$\begin{aligned} (-1)^{n/2} &= \sum_{j=0}^{n/2} (-1)^j \frac{\prod_{k=0}^{j-1} \left(n^2 - (2k)^2\right)}{(2j)!} \\ &= 1 - \frac{n^2}{2!} + \frac{n^2\left(n^2-2^2\right)}{4!} - \cdots + (-1)^{n/2} \\ &\quad \frac{n^2\left(n^2-2^2\right)\left(n^2-4^2\right)\cdots\left(n^2-(n-2)^2\right)}{n!}. \end{aligned}$$

[1.39] Prove that $\forall n \in 2\mathbb{N}$,

$$\begin{aligned} (-1)^{n/2+1} &= \sum_{j=0}^{n/2-1} (-1)^j \frac{\prod_{k=1}^{j} \left(n^2 - (2k)^2\right)}{(2j+1)!} \\ &= 1 - \frac{n^2-2^2}{3!} + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)}{5!} - \cdots \\ &\quad + (-1)^{n/2-1} \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)\left(n^2-6^2\right)\cdots\left(n^2-(n-2)^2\right)}{(n-1)!}. \end{aligned}$$

[1.40] Prove that $\forall n \in 2\mathbb{N}$,

$$(-1)^{n/2}\frac{(1-n^2)}{3} = \sum_{j=0}^{n/2-1}(-1)^j \frac{\prod_{k=1}^{j}\left(n^2-(2k)^2\right)}{(2j)!}$$
$$= 1 - \frac{n^2-2^2}{2!} + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)}{4!} - \cdots$$
$$+ (-1)^{n/2-1}\frac{\left(n^2-2^2\right)\left(n^2-4^2\right)\left(n^2-6^2\right)\cdots\left(n^2-(n-2)^2\right)}{(n-2)!}.$$

[1.41] Prove that $\forall n \in 2\mathbb{N}-1$,

$$\frac{(-1)^{(n-1)/2}}{n} = \sum_{j=0}^{(n-1)/2}(-1)^j \frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j+1)!}$$
$$= 1 - \frac{\left(n^2-1^2\right)}{3!} + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{5!} - \cdots$$
$$+ (-1)^{(n-1)/2}\frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\cdots\left(n^2-(n-2)^2\right)}{n!}.$$

[1.42] Prove that $\forall n \in 2\mathbb{N}-1$,

$$(-1)^{(n-1)/2}n = \sum_{j=0}^{(n-1)/2}(-1)^j \frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j)!}$$
$$= 1 - \frac{\left(n^2-1^2\right)}{2!} + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{4!} - \cdots$$
$$+ (-1)^{(n-1)/2}\frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\cdots\left(n^2-(n-2)^2\right)}{(n-1)!}.$$

[1.43] Prove that $\forall n \in 2\mathbb{N}+1$,

$$(-1)^{(n+1)/2}\frac{n}{3} = \sum_{j=0}^{(n-3)/2} (-1)^j \frac{\prod_{k=1}^{j}\left(n^2-(2k+1)^2\right)}{(2j+1)!}$$
$$= 1 - \frac{\left(n^2-3^2\right)}{3!} + \frac{\left(n^2-3^2\right)\left(n^2-5^2\right)}{5!} - \cdots$$
$$+ (-1)^{(n-3)/2}\frac{\left(n^2-3^2\right)\left(n^2-5^2\right)\left(n^2-7^2\right)\cdots\left(n^2-(n-2)^2\right)}{(n-2)!}.$$

[1.44] Prove that $\forall x \in \mathbb{R} - [-1, 1]$,

$$\frac{x(x+1)}{(x-1)^3} = \sum_{j=1}^{\infty}\frac{j^2}{x^j}$$
$$= \frac{1^2}{x} + \frac{2^2}{x^2} + \frac{3^2}{x^3} + \frac{4^2}{x^4} + \cdots$$

[1.45] Prove that $\forall x \in \mathbb{R} - [-1, 1]$,

$$\frac{x(x-1)}{(x+1)^3} = \sum_{j=1}^{\infty}(-1)^{j-1}\frac{i^2}{x^j}$$
$$= \frac{1^2}{x} - \frac{2^2}{x^2} + \frac{3^2}{x^3} - \frac{4^2}{x^4} + \cdots$$

[1.46] Prove that $\forall x \in \mathbb{R} - [-1, 1]$,

$$\frac{x(x^2+4x+1)}{(x-1)^4} = \sum_{j=1}^{\infty}\frac{j^3}{x^j}$$
$$= \frac{1^3}{x} + \frac{2^3}{x^2} + \frac{3^3}{x^3} + \frac{4^3}{x^4} + \cdots$$

[1.47] Prove that $\forall x \in \mathbb{R} - [-1, 1]$,

$$\frac{x(x^2 - 4x + 1)}{(x+1)^4} = \sum_{j=1}^{\infty} (-1)^{j-1} \frac{j^3}{x^j} = \frac{1^3}{x} - \frac{2^3}{x^2} + \frac{3^3}{x^3} - \frac{4^3}{x^4} + \cdots$$

[1.48] Prove that $\forall x \in \mathbb{R} - [-1, 1]$,

$$\frac{x(x+1)(x^2 + 10x + 1)}{(x-1)^5} = \sum_{j=1}^{\infty} \frac{j^4}{x^j} = \frac{1^4}{x} + \frac{2^4}{x^2} + \frac{3^4}{x^3} + \frac{4^4}{x^4} + \cdots$$

[1.49] Prove that $\forall x \in \mathbb{R} - [-1, 1]$,

$$\frac{x(x-1)(x^2 - 10x + 1)}{(x+1)^5} = \sum_{j=1}^{\infty} (-1)^{j-1} \frac{j^4}{x^j} = \frac{1^4}{x} - \frac{2^4}{x^2} + \frac{3^4}{x^3} - \frac{4^4}{x^4} + \cdots$$

[1.50] (Borwein and Borwein, 1987) Prove that

$$\frac{\pi}{2} \coth \pi - \frac{1}{2} = \sum_{j=1}^{\infty} \frac{1}{j^2 + 1} = \frac{1}{1^2 + 1} + \frac{1}{2^2 + 1} + \frac{1}{3^2 + 1} + \frac{1}{4^2 + 1} + \cdots$$

[1.51] Prove that

$$\frac{\pi}{4} \tanh \frac{\pi}{2} = \sum_{j=0}^{\infty} \frac{1}{(2j+1)^2 + 1} = \frac{1}{1^2 + 1} + \frac{1}{3^2 + 1} + \frac{1}{5^2 + 1} + \frac{1}{7^2 + 1} + \cdots$$

[1.52] Prove that

$$\frac{2\pi^2+\pi\sinh 2\pi-4\sinh^2\pi}{8\sinh^2\pi}=\sum_{j=1}^{\infty}\frac{1}{\left(j^2+1\right)^2}$$
$$=\frac{1}{\left(1^2+1\right)^2}+\frac{1}{\left(2^2+1\right)^2}+\frac{1}{\left(3^2+1\right)^2}+\frac{1}{\left(4^2+1\right)^2}+\cdots$$

[1.53] Prove that

$$\frac{\pi(\sinh\pi-\pi)}{16\cosh^2\frac{\pi}{2}}=\sum_{j=0}^{\infty}\frac{1}{\left((2j+1)^2+1\right)^2}$$
$$=\frac{1}{\left(1^2+1\right)^2}+\frac{1}{\left(3^2+1\right)^2}+\frac{1}{\left(5^2+1\right)^2}+\frac{1}{\left(7^2+1\right)^2}+\cdots$$

[1.54] Prove that

$$\frac{1}{2}-\frac{\pi\sqrt{3}}{12}=\sum_{j=1}^{\infty}\frac{1}{(6j)^2-1}$$
$$=\frac{1}{6^2-1}+\frac{1}{12^2-1}+\frac{1}{18^2-1}+\frac{1}{24^2-1}+\cdots$$

[1.55] Prove that

$$\frac{\pi\sqrt{3}}{36}=\sum_{j=0}^{\infty}\frac{1}{(6j+3)^2-1}$$
$$=\frac{1}{3^2-1}+\frac{1}{9^2-1}+\frac{1}{15^2-1}+\frac{1}{21^2-1}+\cdots$$

[1.56] Prove that

$$\frac{\pi^2}{36}+\frac{\pi\sqrt{3}}{24}-\frac{1}{2}=\sum_{j=1}^{\infty}\frac{1}{\left((6j)^2-1\right)^2}$$
$$=\frac{1}{\left(6^2-1\right)^2}+\frac{1}{\left(12^2-1\right)^2}+\frac{1}{\left(18^2-1\right)^2}+\frac{1}{\left(24^2-1\right)^2}+\cdots$$

[1.57] Prove that

$$\frac{\pi(2\pi - 3\sqrt{3})}{216} = \sum_{j=0}^{\infty} \frac{1}{\left((6j+3)^2 - 1\right)^2}$$
$$= \frac{1}{\left(3^2-1\right)^2} + \frac{1}{\left(9^2-1\right)^2} + \frac{1}{\left(15^2-1\right)^2} + \frac{1}{\left(21^2-1\right)^2} + \cdots$$

[1.58] Prove that

$$\frac{1}{e^2 - 1} = \sum_{j=1}^{\infty} \frac{1}{j^2\pi^2 + 1}$$
$$= \frac{1}{\pi^2+1} + \frac{1}{2^2\pi^2+1} + \frac{1}{3^2\pi^2+1} + \frac{1}{4^2\pi^2+1} + \cdots$$

[1.59] Prove that

$$\frac{e-1}{4(e+1)} = \sum_{j=0}^{\infty} \frac{1}{(2j+1)^2\pi^2 + 1}$$
$$= \frac{1}{\pi^2+1} + \frac{1}{3^2\pi^2+1} + \frac{1}{5^2\pi^2+1} + \frac{1}{7^2\pi^2+1} + \cdots$$

[1.60] Prove that

$$\frac{2 + \sinh 2 - 4\sinh^2 1}{8\sinh^2 1} = \sum_{j=1}^{\infty} \frac{1}{\left(j^2\pi^2 + 1\right)^2}$$
$$= \frac{1}{\left(\pi^2+1\right)^2} + \frac{1}{\left(2^2\pi^2+1\right)^2} + \frac{1}{\left(3^2\pi^2+1\right)^2} + \frac{1}{\left(4^2\pi^2+1\right)^2} + \cdots$$

[1.61] Prove that

$$\frac{\sinh 1 - 1}{16\cosh^2 \frac{1}{2}} = \sum_{j=0}^{\infty} \frac{1}{\left((2j+1)^2\pi^2 + 1\right)^2}$$
$$= \frac{1}{\left(\pi^2+1\right)^2} + \frac{1}{\left(3^2\pi^2+1\right)^2} + \frac{1}{\left(5^2\pi^2+1\right)^2} + \frac{1}{\left(7^2\pi^2+1\right)^2} + \cdots$$

[1.62] Prove that

$$\frac{1-\cot 1}{2} = \sum_{j=1}^{\infty} \frac{1}{j^2\pi^2 - 1}$$
$$= \frac{1}{\pi^2-1} + \frac{1}{2^2\pi^2-1} + \frac{1}{3^2\pi^2-1} + \frac{1}{4^2\pi^2-1} + \cdots$$

[1.63] Prove that

$$\frac{1}{4}\tan\frac{1}{2} = \sum_{j=0}^{\infty} \frac{1}{(2j+1)^2\pi^2 - 1}$$
$$= \frac{1}{\pi^2-1} + \frac{1}{3^2\pi^2-1} + \frac{1}{5^2\pi^2-1} + \frac{1}{7^2\pi^2-1} + \cdots$$

[1.64] Prove that

$$\frac{2+\sin 2 - 4\sin^2 1}{8\sin^2 1} = \sum_{j=1}^{\infty} \frac{1}{\left(j^2\pi^2-1\right)^2}$$
$$= \frac{1}{\left(\pi^2-1\right)^2} + \frac{1}{\left(2^2\pi^2-1\right)^2} + \frac{1}{\left(3^2\pi^2-1\right)^2} + \frac{1}{\left(4^2\pi^2-1\right)^2} + \cdots$$

[1.65] Prove that

$$\frac{1-\sin 1}{16\cos^2\frac{1}{2}} = \sum_{j=0}^{\infty} \frac{1}{\left((2j+1)^2\pi^2-1\right)^2}$$
$$= \frac{1}{\left(\pi^2-1\right)^2} + \frac{1}{\left(3^2\pi^2-1\right)^2} + \frac{1}{\left(5^2\pi^2-1\right)^2} + \frac{1}{\left(7^2\pi^2-1\right)^2} + \cdots$$

[1.66] Prove that $\forall n \in \mathbb{N}$,

$$\frac{\lfloor n!e \rfloor}{n!} = \sum_{j=0}^{n} \frac{1}{j!}$$
$$= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}.$$

[1.67] Prove that $\forall n \in 2\mathbb{N} - 1$,

$$\frac{1}{n!}\left\lfloor \frac{n!}{e} \right\rfloor = \frac{1}{(n+1)!}\left\lfloor \frac{(n+1)!}{e} \right\rfloor = \sum_{j=0}^{n} \frac{(-1)^j}{j!}$$
$$= \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \cdots - \frac{1}{n!}.$$

[1.68] Prove that

$$\frac{10100}{970299} = \sum_{j=1}^{\infty} \frac{j^2}{100^j}$$
$$= 0.010409162536496482\ldots$$

[1.69] Prove that

$$\frac{\pi^2 \coth \pi^2 - 1}{2\pi^2} = \sum_{j=1}^{\infty} \frac{1}{j^2 + \pi^2}$$
$$= \frac{1}{1^2 + \pi^2} + \frac{1}{2^2 + \pi^2} + \frac{1}{3^2 + \pi^2} + \frac{1}{4^2 + \pi^2} + \cdots$$

[1.70] Prove that $\forall (n, x) \in (\mathbb{N} - 1) \times \left( \mathbb{R} - \left\{ -\frac{1}{j} \middle| j \in [1..n] \right\} \right)$,

$$\sum_{j=0}^{n} \frac{(-1)^j}{1+jx}\binom{n}{j} = 1 - \frac{\binom{n}{1}}{1+x} + \frac{\binom{n}{2}}{1+2x} - \cdots + (-1)^n \frac{\binom{n}{n}}{1+nx}$$
$$= \frac{n!x^n}{\prod_{j=0}^{n}(1+jx)}$$
$$= \frac{n!x^n}{(1+x)(1+2x)(1+3x)\cdots(1+nx)}.$$

[1.71] Prove that $\forall (m,n) \in (\mathbb{N}-1)^2$, if $m = n$, then

$$\begin{aligned}
&\sum_{j=0}^{m-n} (-1)^j \binom{n+j}{n} \binom{m}{n+j} \\
&= \binom{n}{n}\binom{m}{n} - \binom{n+1}{n}\binom{m}{n+1} + \binom{n+2}{n}\binom{m}{n+2} - \cdots \\
&\qquad + (-1)^{m-n} \binom{m}{n}\binom{m}{m} \\
&= 1.
\end{aligned}$$

[1.72] Prove that $\forall (m,n) \in (\mathbb{N}-1)^2$, if $m \neq n$, then

$$\begin{aligned}
&\sum_{j=0}^{m-n} (-1)^j \binom{n+j}{n} \binom{m}{n+j} \\
&= \binom{n}{n}\binom{m}{n} - \binom{n+1}{n}\binom{m}{n+1} + \binom{n+2}{n}\binom{m}{n+2} - \cdots \\
&\qquad + (-1)^{m-n} \binom{m}{n}\binom{m}{m} \\
&= 0.
\end{aligned}$$

[1.73] (Berndt, 1994) Prove that

$$\forall \theta \in \mathbb{R} - \{0\} - \left\{ \frac{1}{n\pi} \mid n \in \mathbb{Z} - \{0\} \right\} - \left\{ \frac{1}{2^{m-1}(2n+1)\pi} \mid n \in \mathbb{Z}, m \in \mathbb{N} \right\},$$

$$\begin{aligned}
\theta &= \cot \frac{1}{\theta} + \sum_{j=1}^{\infty} \frac{1}{2^j} \tan \frac{1}{2^j \theta} \\
&= \cot \frac{1}{\theta} + \frac{1}{2} \tan \frac{1}{2\theta} + \frac{1}{4} \tan \frac{1}{4\theta} + \frac{1}{8} \tan \frac{1}{8\theta} + \cdots
\end{aligned}$$

[1.74] Prove that $\forall (x,n) \in (\mathbb{R} - \{-1, 1\}) \times \mathbb{N}$,

$$\begin{aligned}
\frac{1}{x-1} &= \sum_{j=0}^{n-1} \frac{2^j}{x^{2^j}+1} + \frac{2^n}{x^{2^n}-1} \\
&= \frac{1}{x+1} + \frac{2}{x^2+1} + \cdots + \frac{2^{n-1}}{x^{2^{n-1}}+1} + \frac{2^n}{x^{2^n}-1}.
\end{aligned}$$

[1.75] Prove that $\forall (x, n) \in (\mathbb{R} - \{-1, 1\}) \times \mathbb{N}$,

$$\begin{aligned}\frac{x}{(x-1)^2} &= \sum_{j=0}^{n-1} \frac{2^{2j} x^{2^j}}{\left(x^{2^j}+1\right)^2} + \frac{2^{2n} x^{2^n}}{\left(x^{2^n}-1\right)^2} \\ &= \frac{x}{(x+1)^2} + \frac{2^2 x^2}{\left(x^2+1\right)^2} + \cdots + \frac{2^{2n-2} x^{2^{n-1}}}{\left(x^{2^{n-1}}+1\right)^2} + \frac{2^{2n} x^{2^n}}{\left(x^{2^n}-1\right)^2}.\end{aligned}$$

[1.76] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times \mathbb{N}, x \neq \pm y, xy \neq \pm 1$,

$$\begin{aligned}&\frac{x\left(y^2-1\right)}{(x-y)(xy-1)} = \sum_{j=0}^{n-1} \frac{2^j x^{2^j}\left(y^{2^{j+1}}-1\right)}{\left(x^{2^j}+y^{2^j}\right)\left(x^{2^j} y^{2^j}+1\right)} + \frac{2^n x^{2^n}\left(y^{2^{n+1}}-1\right)}{\left(x^{2^n}-y^{2^n}\right)\left(x^{2^n} y^{2^n}-1\right)} \\ &= \frac{x\left(y^2-1\right)}{(x+y)(xy+1)} + \frac{2x^2\left(y^4-1\right)}{\left(x^2+y^2\right)\left(x^2 y^2+1\right)} + \cdots \\ &\quad + \frac{2^{n-1} x^{2^{n-1}}\left(y^{2^n}-1\right)}{\left(x^{2^{n-1}}+y^{2^{n-1}}\right)\left(x^{2^{n-1}} y^{2^{n-1}}+1\right)} + \frac{2^n x^{2^n}\left(y^{2^{n+1}}-1\right)}{\left(x^{2^n}-y^{2^n}\right)\left(x^{2^n} y^{2^n}-1\right)}.\end{aligned}$$

[1.77] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times \mathbb{N}, x \neq \pm y$,

$$\begin{aligned}\frac{y}{x-y} &= \sum_{j=0}^{n-1} \frac{2^j y^{2^j}}{x^{2^j}+y^{2^j}} + \frac{2^n y^{2^n}}{x^{2^n}-y^{2^n}} \\ &= \frac{y}{x+y} + \frac{2y^2}{x^2+y^2} + \cdots + \frac{2^{n-1} y^{2^{n-1}}}{x^{2^{n-1}}+y^{2^{n-1}}} + \frac{2^n y^{2^n}}{x^{2^n}-y^{2^n}}.\end{aligned}$$

[1.78] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times \mathbb{N}, x \neq \pm y, xy \neq \pm 1$,

$$\begin{aligned}&\frac{y(xy-1)+x-y}{(x-y)(xy-1)} \\ &= \sum_{j=0}^{n-1} 2^j \frac{y^{2^j}\left(x^{2^j} y^{2^j}+1\right)+x^{2^j}+y^{2^j}}{\left(x^{2^j}+y^{2^j}\right)\left(x^{2^j} y^{2^j}+1\right)} + 2^n \frac{y^{2^n}\left(x^{2^n} y^{2^n}-1\right)+x^{2^n}-y^{2^n}}{\left(x^{2^n}-y^{2^n}\right)\left(x^{2^n} y^{2^n}-1\right)} \\ &= \frac{y(xy+1)+x+y}{(x+y)(xy+1)} + 2\frac{y^2\left(x^2 y^2+1\right)+x^2+y^2}{\left(x^2+y^2\right)\left(x^2 y^2+1\right)} + \cdots \\ &\quad + 2^{n-1} \frac{y^{2^{n-1}}\left(x^{2^{n-1}} y^{2^{n-1}}+1\right)+x^{2^{n-1}}+y^{2^{n-1}}}{\left(x^{2^{n-1}}+y^{2^{n-1}}\right)\left(x^{2^{n-1}} y^{2^{n-1}}+1\right)} + 2^n \frac{y^{2^n}\left(x^{2^n} y^{2^n}-1\right)+x^{2^n}-y^{2^n}}{\left(x^{2^n}-y^{2^n}\right)\left(x^{2^n} y^{2^n}-1\right)}.\end{aligned}$$

[1.79] Prove that $\forall(x,\theta,n)\in\mathbb{R}^2\times\mathbb{N}$,

$$\frac{x\cos\theta-1}{x^2-2x\cos\theta+1}=\sum_{j=0}^{n-1}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^n\left(x^{2^n}\cos 2^n\theta-1\right)}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}$$
$$=\frac{x\cos\theta+1}{x^2+2x\cos\theta+1}+\frac{2\left(x^2\cos 2\theta+1\right)}{x^4+2x^2\cos 2\theta+1}+\cdots$$
$$+\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}+\frac{2^n\left(x^{2^n}\cos 2^n\theta-1\right)}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}.$$

[1.80] Prove that $\forall(x,\theta,n)\in\mathbb{R}^2\times\mathbb{N}$,

$$\frac{x\sin\theta}{x^2-2x\cos\theta+1}=\sum_{j=0}^{n-1}\frac{2^jx^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^nx^{2^n}\sin 2^n\theta}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}$$
$$=\frac{x\sin\theta}{x^2+2x\cos\theta+1}+\frac{2x^2\sin 2\theta}{x^4+2x^2\cos 2\theta+1}+\cdots$$
$$+\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}+\frac{2^nx^{2^n}\sin 2^n\theta}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}.$$

[1.81] Prove that $\forall(x,\theta,n)\in\mathbb{R}^2\times(\mathbb{N}-1)$,

$$\frac{x^{2^{n+2}}-2x^{2^{n+1}}\cos 2^{n+1}\theta+1}{x^2-2x\cos\theta+1}=\prod_{j=0}^{n}\left(x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1\right)$$
$$=\left(x^2+2x\cos\theta+1\right)\left(x^4+2x^2\cos 2\theta+1\right)\left(x^8+2x^4\cos 4\theta+1\right)$$
$$\cdots\left(x^{2^{n+1}}+2x^{2^n}\cos 2^n\theta+1\right).$$

[1.82] Let

$$T(x,\theta,n)=\sum_{j=0}^{n-1}\tan^{-1}\frac{x^{2^j}+\cos 2^j\theta}{\sin 2^j\theta}+\tan^{-1}\frac{x^{2^n}-\cos 2^n\theta}{\sin 2^n\theta}$$
$$=\tan^{-1}\frac{x+\cos\theta}{\sin\theta}+\tan^{-1}\frac{x^2+\cos 2\theta}{\sin 2\theta}+\cdots$$
$$+\tan^{-1}\frac{x^{2^{n-1}}+\cos 2^{n-1}\theta}{\sin 2^{n-1}\theta}+\tan^{-1}\frac{x^{2^n}-\cos 2^n\theta}{\sin 2^n\theta}.$$

Prove that $\forall(x,\theta,n)\in\mathbb{R}\times\left(\mathbb{R}-\left\{\frac{m\pi}{2^j}\mid j\in[0..n],m\in\mathbb{Z}\right\}\right)\times(\mathbb{N}-1)$,

$$T(x,\theta,n)-\tan^{-1}\frac{x-\cos\theta}{\sin\theta}\in\frac{\pi}{2}\mathbb{Z}.$$

[1.83] (Berndt, 1994) Prove that $\forall x\in\mathbb{R}^+-\{1\}$,

$$\frac{1}{\log x}=\frac{1}{x-1}+\sum_{j=1}^{\infty}\frac{1}{2^j\left(1+x^{1/2^j}\right)}$$
$$=\frac{1}{x-1}+\frac{1}{2\left(1+x^{1/2}\right)}+\frac{1}{4\left(1+x^{1/4}\right)}+\frac{1}{8\left(1+x^{1/8}\right)}+\cdots$$

[1.84] Prove that $\forall x\in\mathbb{R}^+-\{1\}$,

$$\frac{1}{\log^2 x}=\frac{x}{(x-1)^2}+\sum_{j=1}^{\infty}\frac{x^{1/2^j}}{2^{2j}\left(1+x^{1/2^j}\right)^2}$$
$$=\frac{x}{(x-1)^2}+\frac{x^{1/2}}{2^2\left(1+x^{1/2}\right)^2}+\frac{x^{1/4}}{4^2\left(1+x^{1/4}\right)^2}+\frac{x^{1/8}}{8^2\left(1+x^{1/8}\right)^2}+\cdots$$

[1.85] Prove that $\forall n\in\mathbb{N}$, $\exists(a_k)_{k=1}^n\in\mathbb{R}^n$ and $\exists(b_k)_{k=1}^n\in\mathbb{R}^n$ such that

$$\frac{1}{\log^n x}=\sum_{k=1}^{n}\frac{a_k}{(x-1)^k}+\sum_{k=1}^{n}\sum_{j=1}^{\infty}\frac{b_k}{2^{jn}\left(1+x^{1/2^j}\right)^k}.$$

[1.86] Prove that $\forall(x,\theta)\in\mathbb{R}^+\times\mathbb{R}$, such that $\theta^2+\log^2 x\neq 0$,

$$\frac{\log x}{\theta^2+\log^2 x}=\frac{x\cos\theta-1}{x^2-2x\cos\theta+1}+\sum_{j=1}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}$$

$$=\frac{x\cos\theta-1}{x^2-2x\cos\theta+1}+\frac{x^{1/2}\cos\dfrac{\theta}{2}+1}{2\left(x+2x^{1/2}\cos\dfrac{\theta}{2}+1\right)}$$

$$+\frac{x^{1/4}\cos\dfrac{\theta}{4}+1}{4\left(x^{1/2}+2x^{1/4}\cos\dfrac{\theta}{4}+1\right)}+\frac{x^{1/8}\cos\dfrac{\theta}{8}+1}{8\left(x^{1/4}+2x^{1/8}\cos\dfrac{\theta}{8}+1\right)}+\cdots$$

[1.87] Prove that $\forall(x,\theta)\in\mathbb{R}^+\times\mathbb{R}$, such that $\theta^2+\log^2 x\neq 0$,

$$\frac{\theta}{\theta^2+\log^2 x}=\frac{x\sin\theta}{x^2-2x\cos\theta+1}+\sum_{j=1}^{\infty}\frac{x^{1/2^j}\sin\dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}$$

$$=\frac{x\sin\theta}{x^2-2x\cos\theta+1}+\frac{x^{1/2}\sin\dfrac{\theta}{2}}{2\left(x+2x^{1/2}\cos\dfrac{\theta}{2}+1\right)}$$

$$+\frac{x^{1/4}\sin\dfrac{\theta}{4}}{4\left(x^{1/2}+2x^{1/4}\cos\dfrac{\theta}{4}+1\right)}+\frac{x^{1/8}\sin\dfrac{\theta}{8}}{8\left(x^{1/4}+2x^{1/8}\cos\dfrac{\theta}{8}+1\right)}+\cdots$$

[1.88] Prove that $\forall(x,\theta)\in\mathbb{R}^+\times\mathbb{R}$, $\theta\neq\pm\log x$,

$$\frac{\log x}{\log^2 x-\theta^2}=\frac{x\cosh\theta-1}{x^2-2x\cosh\theta+1}+\sum_{j=1}^{\infty}\frac{x^{1/2^j}\cosh\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cosh\dfrac{\theta}{2^j}+1\right)}$$

$$=\frac{x\cosh\theta-1}{x^2-2x\cosh\theta+1}+\frac{x^{1/2}\cosh\dfrac{\theta}{2}+1}{2\left(x+2x^{1/2}\cosh\dfrac{\theta}{2}+1\right)}$$

$$+\frac{x^{1/4}\cosh\dfrac{\theta}{4}+1}{4\left(x^{1/2}+2x^{1/4}\cosh\dfrac{\theta}{4}+1\right)}+\frac{x^{1/8}\cosh\dfrac{\theta}{8}+1}{8\left(x^{1/4}+2x^{1/8}\cosh\dfrac{\theta}{8}+1\right)}+\cdots$$

[1.89] Prove that $\forall(x,\theta)\in\mathbb{R}^+\times\mathbb{R}$, $\theta\neq\pm\log x$,

$$\frac{\theta}{\log^2 x-\theta^2}=\frac{x\sinh\theta}{x^2-2x\cosh\theta+1}+\sum_{j=1}^{\infty}\frac{x^{1/2^j}\sinh\dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cosh\dfrac{\theta}{2^j}+1\right)}$$

$$=\frac{x\sinh\theta}{x^2-2x\cosh\theta+1}+\frac{x^{1/2}\sinh\dfrac{\theta}{2}}{2\left(x+2x^{1/2}\cosh\dfrac{\theta}{2}+1\right)}$$

$$+\frac{x^{1/4}\sinh\dfrac{\theta}{4}}{4\left(x^{1/2}+2x^{1/4}\cosh\dfrac{\theta}{4}+1\right)}+\frac{x^{1/8}\sinh\dfrac{\theta}{8}}{8\left(x^{1/4}+2x^{1/8}\cosh\dfrac{\theta}{8}+1\right)}+\cdots$$

[1.90] Prove that $\forall(x,y)\in\left(\mathbb{R}^+\right)^2$, $x\neq y$, $xy\neq 1$,

$$\frac{\log x^2}{\log^2 x-\log^2 y}=\frac{y(xy-1)+x-y}{(x-y)(xy-1)}$$

$$+\sum_{j=1}^{\infty}\frac{y^{1/2^j}\left(x^{1/2^j}y^{1/2^j}+1\right)+x^{1/2^j}+y^{1/2^j}}{2^j\left(x^{1/2^j}+y^{1/2^j}\right)\left(x^{1/2^j}y^{1/2^j}+1\right)}$$

$$= \frac{y(xy-1)+x-y}{(x-y)(xy-1)} + \frac{y^{1/2}\left(x^{1/2}y^{1/2}+1\right)+x^{1/2}+y^{1/2}}{2\left(x^{1/2}+y^{1/2}\right)\left(x^{1/2}y^{1/2}+1\right)}$$
$$+ \frac{y^{1/4}\left(x^{1/4}y^{1/4}+1\right)+x^{1/4}+y^{1/4}}{4\left(x^{1/4}+y^{1/4}\right)\left(x^{1/4}y^{1/4}+1\right)}$$
$$+ \frac{y^{1/8}\left(x^{1/8}y^{1/8}+1\right)+x^{1/8}+y^{1/8}}{8\left(x^{1/8}+y^{1/8}\right)\left(x^{1/8}y^{1/8}+1\right)} + \cdots$$

[1.91] Prove that $\forall (x, y) \in \left(\mathbb{R}^+\right)^2, x \neq y, xy \neq 1$,

$$\frac{\log y^2}{\log^2 x - \log^2 y} = \frac{x(y^2-1)}{(x-y)(xy-1)} + \sum_{j=1}^{\infty} \frac{x^{1/2^j}\left(y^{1/2^{j-1}}-1\right)}{2^j\left(x^{1/2^j}+y^{1/2^j}\right)\left(x^{1/2^j}y^{1/2^j}+1\right)}$$
$$= \frac{x(y^2-1)}{(x-y)(xy-1)} + \frac{x^{1/2}(y-1)}{2\left(x^{1/2}+y^{1/2}\right)\left(x^{1/2}y^{1/2}+1\right)}$$
$$+ \frac{x^{1/4}\left(y^{1/2}-1\right)}{4\left(x^{1/4}+y^{1/4}\right)\left(x^{1/4}y^{1/4}+1\right)} + \frac{x^{1/8}\left(y^{1/4}-1\right)}{8\left(x^{1/8}+y^{1/8}\right)\left(x^{1/8}y^{1/8}+1\right)} + \cdots$$

[1.92] Prove that $\forall (x, n) \in (\mathbb{R} - \{-1, 1\}) \times \mathbb{N}$,

$$\frac{1+x}{1-x} = \sum_{j=0}^{n-1} 2^j \frac{x^{2^j}-1}{x^{2^j}+1} + 2^n \frac{1+x^{2^n}}{1-x^{2^n}}$$
$$= \frac{x-1}{x+1} + 2\frac{x^2-1}{x^2+1} + \cdots + 2^{n-1}\frac{x^{2^{n-1}}-1}{x^{2^{n-1}}+1} + 2^n \frac{1+x^{2^n}}{1-x^{2^n}}.$$

# Chapter 2
# Trigonometrical Series




In this chapter, we will learn about the application of De Moivre's theorem and Euler's formula in the derivation of trigonometrical sums and products. We will also apply the differentiation and integration methods of series sums as described in the previous chapter. Circular and hyperbolic trigonometrical identities are given as problems in this chapter.

## 2.1 Trigonometrical Series

De Moivre's theorem states that $\forall(\theta, m) \in \mathbb{R} \times \mathbb{Z}$,

$$(\cos\theta + i\sin\theta)^m = \cos m\theta + i\sin m\theta = e^{im\theta}.$$

By Euler's formula, $\forall\theta \in \mathbb{R}$, we have

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

We can find the sum of the GP $\left(e^{ij\theta}\right)_{j=0}^{n}$, and equate its real and imaginary parts to get some trigonometrical sums.

**Example 2.1** (*Trigonometrical Series* 1 *and* 2)
Prove that $\forall(\theta, n) \in (\mathbb{R} - 2\pi\mathbb{Z}) \times (\mathbb{N} - 1)$,

$$\sum_{j=0}^{n} \cos j\theta = 1 + \cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{\sin\dfrac{(n+1)\theta}{2}\cos\dfrac{n\theta}{2}}{\sin\dfrac{\theta}{2}}, \quad (2.1)$$

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and

$$\sum_{j=0}^{n} \sin j\theta = \sin\theta + \sin 2\theta + \sin 3\theta + \cdots + \sin n\theta = \frac{\sin\dfrac{(n+1)\theta}{2}\sin\dfrac{n\theta}{2}}{\sin\dfrac{\theta}{2}}. \quad (2.2)$$

***Proof*** Let

$$C = \sum_{j=0}^{n} \cos j\theta, \quad (2.3)$$

and

$$S = \sum_{j=0}^{n} \sin j\theta. \quad (2.4)$$

Multiplying (2.4) by $i$ and adding the result in (2.3), we get

$$\begin{aligned} C + iS &= \sum_{j=0}^{n} (\cos j\theta + i \sin j\theta) \\ &= \sum_{j=0}^{n} e^{ij\theta} \\ &= \frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1} \\ &= \frac{e^{i(n+1)\theta/2}\left(e^{i(n+1)\theta/2} - e^{-i(n+1)\theta/2}\right)}{e^{i\theta/2}\left(e^{i\theta/2} - e^{-i\theta/2}\right)} \\ &= e^{in\theta/2}\frac{\left(e^{i(n+1)\theta/2} - e^{-i(n+1)\theta/2}\right)/2i}{\left(e^{i\theta/2} - e^{-i\theta/2}\right)/2i} \\ &= \left(\cos\frac{n\theta}{2} + i\sin\frac{n\theta}{2}\right)\frac{\sin\dfrac{(n+1)\theta}{2}}{\sin\dfrac{\theta}{2}} \\ &= \frac{\sin\dfrac{(n+1)\theta}{2}\cos\dfrac{n\theta}{2}}{\sin\dfrac{\theta}{2}} + i\frac{\sin\dfrac{(n+1)\theta}{2}\sin\dfrac{n\theta}{2}}{\sin\dfrac{\theta}{2}}. \end{aligned} \quad (2.5)$$

By equating the real and imaginary parts of (2.5), we get (2.1) and (2.2) respectively. □

**Example 2.2** (*Finding the Sum of First n Natural Numbers*)
Prove that $\forall n \in \mathbb{N}$,

$$\sum_{j=1}^{n} j = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}. \tag{2.6}$$

***Proof*** Dividing the LHS of (2.2) by $\sin\theta$ and taking the limit as $\theta \to 0$, we get

$$\begin{aligned} \lim_{\theta\to 0} \sum_{j=0}^{n} \frac{\sin j\theta}{\sin\theta} &= \sum_{j=0}^{n} \lim_{\theta\to 0} \frac{\sin j\theta}{\sin\theta} \\ &= \sum_{j=1}^{n} \lim_{\theta\to 0} j \frac{\frac{\sin j\theta}{j\theta}}{\frac{\sin\theta}{\theta}} \\ &= \sum_{j=1}^{n} j. \end{aligned} \tag{2.7}$$

Dividing the RHS of (2.2) by $\sin\theta$ and taking the limit as $\theta \to 0$, we get

$$\begin{aligned} &\lim_{\theta\to 0} \frac{\sin\frac{(n+1)\theta}{2} \sin\frac{n\theta}{2}}{\sin\theta \sin\frac{\theta}{2}} \\ &= \lim_{\theta\to 0} \left(\frac{n+1}{2}\right)\left(\frac{n}{2}\right)\left(\frac{1}{2}\right)^{-1} \left(\frac{\sin\frac{(n+1)\theta}{2}}{\frac{(n+1)\theta}{2}}\right)\left(\frac{\sin\frac{n\theta}{2}}{\frac{n\theta}{2}}\right)\left(\frac{\sin\theta}{\theta}\right)^{-1}\left(\frac{\sin\frac{\theta}{2}}{\frac{\theta}{2}}\right)^{-1} \\ &= \frac{n(n+1)}{2}. \end{aligned} \tag{2.8}$$

By equating (2.7) and (2.8), we get (2.6). □

**Example 2.3** (*Trigonometrical Series* 3)
Prove that $\forall(\theta, n) \in (\mathbb{R} - 2\pi\mathbb{Z}) \times (\mathbb{N} - 1)$,

$$\begin{aligned} &\sum_{j=1}^{n} j \sin j\theta = \sin\theta + 2\sin 2\theta + 3\sin 3\theta + \cdots + n\sin n\theta \\ &= \frac{n\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2} - (n+1)\cos\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}}{2\sin\frac{\theta}{2}} \\ &\quad + \frac{\sin\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}. \end{aligned}$$

***Proof*** Differentiating (2.1) with respect to $\theta$, we get

$$\begin{aligned}
&\sum_{j=1}^{n} -j\sin j\theta \\
&= \frac{\left(\frac{n+1}{2}\cos\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2} - \frac{n}{2}\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}\right)\sin\frac{\theta}{2}}{\sin^2\frac{\theta}{2}} \\
&\quad - \frac{\frac{1}{2}\sin\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}\cos\frac{\theta}{2}}{\sin^2\frac{\theta}{2}} \\
\Longleftrightarrow\ &\sum_{j=1}^{n} j\sin j\theta \\
&= \frac{n\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2} - (n+1)\cos\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}}{2\sin\frac{\theta}{2}} \\
&\quad + \frac{\sin\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}.
\end{aligned}$$

□

**Example 2.4** (*Trigonometrical Series* 4)
Prove that $\forall(\theta, n) \in (\mathbb{R} - 2\pi\mathbb{Z}) \times \mathbb{N}$,

$$\begin{aligned}
&\sum_{j=1}^{n} j\cos j\theta = \cos\theta + 2\cos 2\theta + 3\cos 3\theta + \cdots + n\cos n\theta \\
&= \frac{(n+1)\cos\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2} + n\sin\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}}{2\sin\frac{\theta}{2}} \\
&\quad - \frac{\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}.
\end{aligned}$$

***Proof*** Differentiating (2.2) with respect to $\theta$, we get

$$\begin{aligned}\sum_{j=1}^{n} j\cos j\theta &= \frac{\left(\frac{n+1}{2}\cos\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}+\frac{n}{2}\sin\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}\right)\sin\frac{\theta}{2}}{\sin^2\frac{\theta}{2}}\\ &\quad-\frac{\frac{1}{2}\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}\cos\frac{\theta}{2}}{\sin^2\frac{\theta}{2}}\\ &= \frac{(n+1)\cos\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}+n\sin\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2}}{2\sin\frac{\theta}{2}}\\ &\quad-\frac{\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}.\end{aligned}$$

□

Now, we will apply De Moivre's theorem and the binomial theorem to expand $\cos n\theta$ and $\sin n\theta$ in a series of powers of $\cos\theta$ and $\sin\theta$.

**Example 2.5** (*Trigonometrical Series* 5 *and* 6)
Prove that $\forall(\theta, n) \in \mathbb{R}\times\mathbb{N}$,

$$\begin{aligned}\cos n\theta &= \sum_{j=0}^{\lfloor n/2\rfloor}(-1)^j\binom{n}{2j}\cos^{n-2j}\theta\sin^{2j}\theta\\ &= \cos^n\theta-\binom{n}{2}\cos^{n-2}\theta\sin^2\theta+\binom{n}{4}\cos^{n-4}\theta\sin^4\theta\\ &\quad+\cdots(-1)^{\lfloor n/2\rfloor}\cos^{n-2\lfloor n/2\rfloor}\theta\sin^{2\lfloor n/2\rfloor}\theta,\end{aligned}\tag{2.9}$$

and

$$\begin{aligned}\sin n\theta &= \sum_{j=1}^{\lceil n/2\rceil}(-1)^{j-1}\binom{n}{2j-1}\cos^{n-2j+1}\theta\sin^{2j-1}\theta\\ &= \binom{n}{1}\cos^{n-1}\theta\sin\theta-\binom{n}{3}\cos^{n-3}\theta\sin^3\theta+\binom{n}{5}\cos^{n-5}\theta\sin^5\theta\\ &\quad+\cdots(-1)^{\lceil n/2\rceil-1}\cos^{n-2\lceil n/2\rceil+1}\theta\sin^{2\lceil n/2\rceil-1}\theta.\end{aligned}\tag{2.10}$$

***Proof*** Using De Moivre's theorem and the binomial theorem, we have

$$\begin{aligned}
&(\cos\theta + i\sin\theta)^n = \sum_{j=0}^{n}\binom{n}{j} i^j \cos^{n-j}\theta \sin^j\theta \\
\Longleftrightarrow\ &\cos n\theta + i\sin n\theta \\
&= \sum_{j=0}^{\lfloor n/2\rfloor}(-1)^j\binom{n}{2j}\cos^{n-2j}\theta\sin^{2j}\theta \\
&\quad + i\sum_{j=1}^{\lceil n/2\rceil}(-1)^{j-1}\binom{n}{2j-1}\cos^{n-2j+1}\theta\sin^{2j-1}\theta.
\end{aligned} \tag{2.11}$$

By equating the real and imaginary parts of (2.11), we get (2.9) and (2.10) respectively. □

We can also evaluate a quadratic fraction using two methods: first, by making use of the complex variables, and second, by making use of the binomial theorem. As a result, we will get expansions of $\sin n\theta/\sin\theta$ and $\cos n\theta$ in a series of powers of $\cos\theta$.

**Example 2.6** (Loney, 1928) (*Trigonometrical Series* 7)
Prove that $\forall(\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times 2\mathbb{N}$,

$$\begin{aligned}
\frac{\sin n\theta}{\sin\theta} &= (-1)^{n/2+1} n \sum_{j=0}^{n/2-1}(-1)^j \frac{\prod_{k=1}^{j}\left(n^2-(2k)^2\right)}{(2j+1)!}\cos^{2j+1}\theta \\
&= (-1)^{n/2+1} n\left(\cos\theta - \frac{\left(n^2-2^2\right)}{3!}\cos^3\theta + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)}{5!}\cos^5\theta\right. \\
&\quad \left. + \cdots + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)\left(n^2-6^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-1)!}\cos^{n-1}\theta\right).
\end{aligned} \tag{2.12}$$

***Proof*** We will compute the value of $1/\left(1-2x\cos\theta+x^2\right)$ using two methods and equate the coefficient of $x^{n-1}$ to get the value of $\sin n\theta/\sin\theta$ as a series in powers of $\cos\theta$.
*First Method:* Using Euler's formula, we have

$$\begin{aligned}
\frac{1}{1-2x\cos\theta+x^2} &= \frac{1}{1-x\left(e^{i\theta}+e^{-i\theta}\right)+x^2} \\
&= \frac{1}{\left(1-xe^{i\theta}\right)\left(1-xe^{-i\theta}\right)} \\
&= \frac{1}{x\left(e^{i\theta}-e^{-i\theta}\right)}\left(\frac{1}{1-xe^{i\theta}} - \frac{1}{1-xe^{-i\theta}}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2ix\sin\theta}\left(\sum_{j=0}^{\infty} x^j e^{ij\theta} - \sum_{j=0}^{\infty} x^j e^{-ij\theta}\right) \\
&= \frac{1}{2ix\sin\theta}\sum_{j=0}^{\infty} x^j \left(e^{ij\theta} - e^{-ij\theta}\right) \\
&= \frac{1}{2ix\sin\theta}\sum_{j=0}^{\infty} 2ix^j \sin j\theta \\
&= \sum_{j=1}^{\infty} \frac{\sin j\theta}{\sin\theta} x^{j-1}.
\end{aligned}
\tag{2.13}
$$

*Second Method:* Using the binomial theorem, we have

$$
\begin{aligned}
\frac{1}{1-2x\cos\theta+x^2} &= \frac{1}{1+x(x-2\cos\theta)} \\
&= \sum_{j=0}^{\infty} (-1)^j x^j (x-2\cos\theta)^j.
\end{aligned}
\tag{2.14}
$$

For $(\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times 2\mathbb{N}$, we equate the coefficients of $x^{n-1}$ in (2.13) and (2.14) to get

$$
\begin{aligned}
\frac{\sin n\theta}{\sin\theta} &= \sum_{j=0}^{n/2-1} (-1)^{n/2+j} \binom{n/2+j}{2j+1} (-2\cos\theta)^{2j+1} \\
&= (-1)^{n/2+1} \sum_{j=0}^{n/2-1} (-1)^j \frac{2^{2j+1} \prod_{k=0}^{2j} \left(\frac{n}{2}+j-k\right)}{(2j+1)!} \cos^{2j+1}\theta \\
&= (-1)^{n/2+1} \sum_{j=0}^{n/2-1} (-1)^j \frac{\prod_{k=0}^{2j} (n+2j-2k)}{(2j+1)!} \cos^{2j+1}\theta \\
&= (-1)^{n/2+1} n \sum_{j=0}^{n/2-1} (-1)^j \frac{\prod_{k=1}^{j} \left(n^2-(2k)^2\right)}{(2j+1)!} \cos^{2j+1}\theta.
\end{aligned}
$$

□

**Example 2.7** (Loney, 1928) (*Trigonometrical Series* 8)
Prove that $\forall(\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times (2\mathbb{N} - 1)$,

$$\begin{aligned}\frac{\sin n\theta}{\sin\theta} &= (-1)^{(n-1)/2}\sum_{j=0}^{(n-1)/2}(-1)^j\frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j)!}\cos^{2j}\theta\\ &= (-1)^{(n-1)/2}\left(1-\frac{\left(n^2-1^2\right)}{2!}\cos^2\theta+\frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{4!}\cos^4\theta\right.\\ &\quad\left.+\cdots+\frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-1)!}\cos^{n-1}\theta\right).\end{aligned}\tag{2.15}$$

***Proof*** For $(\theta, n)\in(\mathbb{R}-\mathbb{Z}\pi)\times(2\mathbb{N}-1)$, we equate the coefficients of $x^{n-1}$ in (2.13) and (2.14) to get

$$\begin{aligned}\frac{\sin n\theta}{\sin\theta} &= \sum_{j=0}^{(n-1)/2}(-1)^{(n-1)/2+j}\binom{(n-1)/2+j}{2j}(-2\cos\theta)^{2j}\\ &= (-1)^{(n-1)/2}\sum_{j=0}^{(n-1)/2}(-1)^j\frac{2^{2j}\prod_{k=0}^{2j-1}\left(\frac{n-1}{2}+j-k\right)}{(2j)!}\cos^{2j}\theta\\ &= (-1)^{(n-1)/2}\sum_{j=0}^{(n-1)/2}(-1)^j\frac{\prod_{k=0}^{2j-1}(n-1+2j-2k)}{(2j)!}\cos^{2j}\theta\\ &= (-1)^{(n-1)/2}\sum_{j=0}^{(n-1)/2}(-1)^j\frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j)!}\cos^{2j}\theta.\end{aligned}$$

□

**Example 2.8** (Loney, 1928) (*Trigonometrical Series* 9)
Prove that $\forall(\theta, n)\in(\mathbb{R}-\mathbb{Z}\pi)\times(2\mathbb{N}-2)$,

$$\begin{aligned}\cos n\theta &= (-1)^{n/2}\sum_{j=0}^{n/2}(-1)^j\frac{\prod_{k=0}^{j-1}\left(n^2-(2k)^2\right)}{(2j)!}\cos^{2j}\theta\\ &= (-1)^{n/2}\left(1-\frac{n^2}{2!}\cos^2\theta+\frac{n^2\left(n^2-2^2\right)}{4!}\cos^4\theta\right.\\ &\quad\left.+\cdots+\frac{n^2\left(n^2-2^2\right)\left(n^2-4^2\right)\ldots\left(n^2-(n-2)^2\right)}{n!}\cos^n\theta\right).\end{aligned}\tag{2.16}$$

***Proof*** We will compute the value of $(1 - x\cos\theta)/\left(1 - 2x\cos\theta + x^2\right)$ using two methods and equate the coefficient of $x^n$ to get the value of $\cos n\theta$ as a series in powers of $\cos\theta$.
*First Method:* Using Euler's formula, we have

$$\begin{aligned}\frac{1 - x\cos\theta}{1 - 2x\cos\theta + x^2} &= \frac{1 - x\dfrac{e^{i\theta} + e^{i\theta}}{2}}{1 - x\left(e^{i\theta} + e^{-i\theta}\right) + x^2} \\ &= \frac{2 - x\left(e^{i\theta} + e^{-i\theta}\right)}{2\left(1 - xe^{i\theta}\right)\left(1 - xe^{-i\theta}\right)} \\ &= \frac{1}{2}\left(\frac{1}{1 - xe^{i\theta}} + \frac{1}{1 - xe^{-i\theta}}\right) \\ &= \frac{1}{2}\left(\sum_{j=0}^{\infty} x^j e^{ij\theta} + \sum_{j=0}^{\infty} x^j e^{-ij\theta}\right) \\ &= \sum_{j=0}^{\infty} x^j \frac{e^{ij\theta} + e^{-ij\theta}}{2} \\ &= \sum_{j=0}^{\infty} x^j \cos j\theta.\end{aligned} \tag{2.17}$$

*Second Method:* Using the binomial theorem, we have

$$\begin{aligned}\frac{1 - x\cos\theta}{1 - 2x\cos\theta + x^2} &= \frac{1 - x\cos\theta}{1 + x(x - 2\cos\theta)} \\ &= (1 - x\cos\theta)\sum_{j=0}^{\infty}(-1)^j x^j (x - 2\cos\theta)^j.\end{aligned} \tag{2.18}$$

For $(\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times 2\mathbb{N}$, we equate the coefficients of $x^n$ in (2.17) and (2.18) to get

$$\begin{aligned}\cos n\theta &= \sum_{j=0}^{n/2}(-1)^{n/2+j}\binom{n/2 + j}{2j}(-2\cos\theta)^{2j} \\ &\quad - \sum_{j=0}^{n/2-1}(-1)^{n/2+j}\binom{n/2 + j}{2j + 1}(-2\cos\theta)^{2j+1}\cos\theta \\ &= \sum_{j=0}^{n/2}(-1)^{n/2+j}\binom{n/2 + j}{2j}(-2\cos\theta)^{2j} \\ &\quad - \sum_{j=1}^{n/2}(-1)^{n/2+j-1}\binom{n/2 + j - 1}{2j - 1}(-2\cos\theta)^{2j-1}\cos\theta\end{aligned}$$

$$= (-1)^{n/2} \left( 1 + \sum_{j=1}^{n/2} (-1)^j 2^{2j-1} \left( \frac{2 \prod_{k=0}^{2j-1} \left( \frac{n}{2} + j - k \right)}{(2j)!} \right. \right.$$

$$\left. \left. - \frac{\prod_{k=0}^{2j-2} \left( \frac{n}{2} + j - 1 - k \right)}{(2j-1)!} \right) \cos^{2j} \theta \right)$$

$$= (-1)^{n/2} \left( 1 + \sum_{j=1}^{n/2} (-1)^j 2^{2j-1} \left( \frac{n+2j}{2j} - 1 \right) \right.$$

$$\left. \frac{\prod_{k=0}^{2j-2} \left( \frac{n}{2} + j - 1 - k \right)}{(2j-1)!} \cos^{2j} \theta \right)$$

$$= (-1)^{n/2} \left( 1 + \sum_{j=1}^{n/2} (-1)^j \frac{n \prod_{k=0}^{2j-2} (n + 2j - 2 - 2k)}{(2j)!} \cos^{2j} \theta \right)$$

$$= (-1)^{n/2} \sum_{j=0}^{n/2} (-1)^j \frac{\prod_{k=0}^{j-1} \left( n^2 - (2k)^2 \right)}{(2j)!} \cos^{2j} \theta.$$

□

**Example 2.9** (Loney, 1928) (*Trigonometrical Series* 10)
Prove that $\forall (\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times (2\mathbb{N} - 1)$,

$$\begin{aligned}\cos n\theta &= (-1)^{(n-1)/2} n \sum_{j=0}^{(n-1)/2} (-1)^j \frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j+1)!}\cos^{2j+1}\theta \\ &= (-1)^{(n-1)/2} n \left(\cos\theta - \frac{\left(n^2-1^2\right)}{3!}\cos^3\theta + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{5!}\cos^5\theta\right. \\ &\quad \left. + \cdots + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\ldots\left(n^2-(n-2)^2\right)}{n!}\cos^n\theta\right).\end{aligned} \tag{2.19}$$

***Proof*** For $(\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times (2\mathbb{N} - 1)$, we equate the coefficients of $x^n$ in (2.17) and (2.18) to get

$$\begin{aligned}\cos n\theta &= \sum_{j=0}^{(n-1)/2} (-1)^{(n+1)/2+j} \binom{(n+1)/2+j}{2j+1} (-2\cos\theta)^{2j+1} \\ &\quad - \sum_{j=0}^{(n-1)/2} (-1)^{(n-1)/2+j} \binom{(n-1)/2+j}{2j} (-2\cos\theta)^{2j}\cos\theta \\ &= (-1)^{(n-1)/2} \sum_{j=0}^{(n-1)/2} (-1)^j 2^{2j} \\ &\quad \left(2\binom{(n+1)/2+j}{2j+1} - \binom{(n-1)/2+j}{2j}\right)\cos^{2j+1}\theta \\ &= (-1)^{(n-1)/2} \sum_{j=0}^{(n-1)/2} (-1)^j 2^{2j} \left(\frac{2\prod_{k=0}^{2j}\left(\frac{n+1}{2}+j-k\right)}{(2j+1)!}\right. \\ &\quad \left. - \frac{\prod_{k=0}^{2j-1}\left(\frac{n-1}{2}+j-k\right)}{(2j)!}\right)\cos^{2j+1}\theta \\ &= (-1)^{(n-1)/2} \sum_{j=0}^{(n-1)/2} (-1)^j 2^{2j} \left(\frac{n+1+2j}{2j+1} - 1\right) \\ &\quad \frac{\prod_{k=0}^{2j-1}\left(\frac{n-1}{2}+j-k\right)}{(2j)!}\cos^{2j+1}\theta\end{aligned}$$

$$= (-1)^{(n-1)/2} n \sum_{j=0}^{(n-1)/2} (-1)^j \frac{\prod_{k=0}^{2j-1} (n-1+2j-2k)}{(2j+1)!} \cos^{2j+1}\theta$$

$$= (-1)^{(n-1)/2} n \sum_{j=0}^{(n-1)/2} (-1)^j \frac{\prod_{k=0}^{j-1} \left(n^2-(2k+1)^2\right)}{(2j+1)!} \cos^{2j+1}\theta.$$

□

## 2.2 Trigonometrical Products

In this section, we will apply Euler's formula to factor a polynomial. Then, we will apply the factorization to express $\sin\theta$ and $\cos\theta$ as infinite products. As a corollary to the above results, we will also find the value of $\zeta(2)$.

**Example 2.10** (Loney, 1928) (*Factorization of* $x^{2n} - 2x^n \cos n\theta + 1$)
Prove that $\forall (x, \theta, n) \in \mathbb{R}^2 \times \mathbb{N}$,

$$\begin{aligned} x^{2n} - 2x^n \cos n\theta + 1 &= \prod_{j=0}^{n-1} \left( x^2 - 2x \cos\left(\theta + \frac{2j\pi}{n}\right) + 1 \right) \\ &= \left(x^2 - 2x\cos\theta + 1\right) \left( x^2 - 2x \cos\left(\theta + \frac{2\pi}{n}\right) + 1 \right) \\ &\left( x^2 - 2x \cos\left(\theta + \frac{4\pi}{n}\right) + 1 \right) \ldots \left( x^2 - 2x \cos\left(\theta + \frac{2(n-1)\pi}{n}\right) + 1 \right). \end{aligned} \tag{2.20}$$

***Proof*** Using Euler's formula, we solve the equation $x^{2n} - 2x^n \cos n\theta + 1 = 0$ as follows:

$$\begin{aligned} & x^{2n} - 2x^n \cos n\theta + 1 = 0 \\ \iff & x^{2n} - x^n \left(e^{in\theta} + e^{-in\theta}\right) + 1 = 0 \\ \implies & x^n = \frac{\left(e^{in\theta} + e^{-in\theta}\right) \pm \sqrt{\left(e^{in\theta} + e^{-in\theta}\right)^2 - 4}}{2} \\ & = \frac{\left(e^{in\theta} + e^{-in\theta}\right) \pm \left(e^{in\theta} - e^{-in\theta}\right)}{2} \\ & = e^{\pm i(n\theta + 2j\pi)}, \quad j \in [0..n-1] \\ \implies & x = e^{\pm i(\theta + 2j\pi/n)}, \quad j \in [0..n-1]. \end{aligned} \tag{2.21}$$

Now using (2.21), we can factor $x^{2n} - 2x^n \cos n\theta + 1$ as follows:

$$\begin{aligned} x^{2n} - 2x^n \cos n\theta + 1 &= \prod_{j=0}^{n-1} \left(x - e^{i(\theta+2j\pi/n)}\right)\left(x - e^{-i(\theta+2j\pi/n)}\right) \\ &= \prod_{j=0}^{n-1} \left(x^2 - x\left(e^{i(\theta+2j\pi/n)} + e^{-i(\theta+2j\pi/n)}\right) + 1\right) \\ &= \prod_{j=0}^{n-1} \left(x^2 - 2x\cos\left(\theta + \frac{2j\pi}{n}\right) + 1\right). \end{aligned}$$

□

**Example 2.11** (Loney, 1928) (*Infinite Product for* $\sin\theta$)
Prove that $\forall\theta \in \mathbb{R}$,

$$\begin{aligned} \sin\theta &= \theta \prod_{j=1}^{\infty} \left(1 - \frac{\theta^2}{j^2\pi^2}\right) \\ &= \theta\left(1 - \frac{\theta^2}{\pi^2}\right)\left(1 - \frac{\theta^2}{2^2\pi^2}\right)\left(1 - \frac{\theta^2}{3^2\pi^2}\right)\cdots \end{aligned} \tag{2.22}$$

***Proof*** Putting $x = 1$ and replacing $\theta$ with $2\theta/n$, where $n \in 2\mathbb{N}$ in (2.20), we get

$$\begin{aligned} & 2 - 2\cos 2\theta = \prod_{j=0}^{n-1} \left(2 - 2\cos\frac{2\theta + 2j\pi}{n}\right) \\ \iff & 4\sin^2\theta = \prod_{j=0}^{n-1} \left(4\sin^2\frac{\theta + j\pi}{n}\right) \\ \implies & \sin\theta = \pm 2^{n-1} \prod_{j=0}^{n-1} \sin\frac{\theta + j\pi}{n} \\ & = \pm 2^{n-1} \sin\frac{\theta}{n} \sin\frac{\theta + (n/2)\pi}{n} \prod_{j=1}^{n/2-1} \sin\frac{\theta + j\pi}{n} \sin\frac{\theta + (n-j)\pi}{n} \\ & = \pm 2^{n-1} \sin\frac{\theta}{n} \cos\frac{\theta}{n} \prod_{j=1}^{n/2-1} \left(\cos^2\frac{(n-2j)\pi}{2n} - \cos^2\frac{2\theta + n\pi}{2n}\right) \\ & = \pm 2^{n-1} \sin\frac{\theta}{n} \cos\frac{\theta}{n} \prod_{j=1}^{n/2-1} \left(\sin^2\frac{j\pi}{n} - \sin^2\frac{\theta}{n}\right) \end{aligned} \tag{2.23}$$

Dividing (2.23) by $\sin(\theta/n)$ and taking the limit as $\theta \to 0$, we get

$$\begin{aligned}\lim_{\theta\to 0}\frac{\sin\theta}{\sin\frac{\theta}{n}} &= \lim_{\theta\to 0} 2^{n-1}\cos\frac{\theta}{n}\prod_{j=1}^{n/2-1}\left(\sin^2\frac{j\pi}{n}-\sin^2\frac{\theta}{n}\right)\\ \Longleftrightarrow\ n &= 2^{n-1}\prod_{j=1}^{n/2-1}\sin^2\frac{j\pi}{n}.\end{aligned} \tag{2.24}$$

Dividing (2.23) by (2.24), we get

$$\begin{aligned}\frac{\sin\theta}{n} &= \pm\sin\frac{\theta}{n}\cos\frac{\theta}{n}\prod_{j=1}^{n/2-1}\left(1-\frac{\sin^2\frac{\theta}{n}}{\sin^2\frac{j\pi}{n}}\right)\\ \Longleftrightarrow\ \sin\theta &= \pm n\sin\frac{\theta}{n}\cos\frac{\theta}{n}\prod_{j=1}^{n/2-1}\left(1-\frac{\sin^2\frac{\theta}{n}}{\sin^2\frac{j\pi}{n}}\right).\end{aligned} \tag{2.25}$$

Taking the limit as $n\to\infty$ in (2.25), we get

$$\begin{aligned}\sin\theta &= \pm\lim_{n\to\infty} n\sin\frac{\theta}{n}\cos\frac{\theta}{n}\prod_{j=1}^{n/2-1}\left(1-\frac{\sin^2\frac{\theta}{n}}{\sin^2\frac{j\pi}{n}}\right)\\ &= \pm\theta\prod_{j=1}^{\infty}\left(1-\frac{\theta^2}{j^2\pi^2}\right).\end{aligned} \tag{2.26}$$

$\forall m\in\mathbb{N}-1$, we have

$$\begin{aligned}&\sin\theta>0,\quad 2m\pi<\theta<(2m+1)\pi\\ \Longleftrightarrow\ &\sin\theta>0,\quad 1-\left(\frac{2m}{j}\right)^2>1-\frac{\theta^2}{j^2\pi^2}>1-\left(\frac{2m+1}{j}\right)^2,\end{aligned} \tag{2.27}$$

and also

$$\begin{aligned}&\sin\theta<0,\quad (2m+1)\pi<\theta<(2m+2)\pi\\ \Longleftrightarrow\ &\sin\theta<0,\quad 1-\left(\frac{2m+1}{j}\right)^2>1-\frac{\theta^2}{j^2\pi^2}>1-\left(\frac{2m+2}{j}\right)^2.\end{aligned} \tag{2.28}$$

From (2.27), we notice that there are $2m$ negative fractions in (2.26), making its RHS positive. From (2.28), we notice that there are $2m+1$ negative fractions in (2.26), making its RHS negative. Therefore, we drop the negative sign in (2.26) to get (2.22). (2.22) is also true for $\theta\in\mathbb{R}^-$, because we have $\sin(-\theta)=-\sin\theta$, $\forall\theta\in\mathbb{R}^-$. □

**Example 2.12** (Loney, 1928) (*Infinite Product for* $\cos\theta$)
Prove that $\forall\theta \in \mathbb{R}$,

$$\begin{aligned}\cos\theta &= \prod_{j=0}^{\infty}\left(1-\frac{4\theta^2}{(2j+1)^2\pi^2}\right)\\ &= \left(1-\frac{4\theta^2}{\pi^2}\right)\left(1-\frac{4\theta^2}{3^2\pi^2}\right)\left(1-\frac{4\theta^2}{5^2\pi^2}\right)\left(1-\frac{4\theta^2}{7^2\pi^2}\right)\cdots\end{aligned} \tag{2.29}$$

***Proof*** For $\theta \neq \mathbb{Z}\pi$, using (2.22), we have

$$\begin{aligned}\cos\theta &= \frac{\sin 2\theta}{2\sin\theta}\\ &= \frac{2\theta\prod_{j=1}^{\infty}\left(1-\frac{4\theta^2}{j^2\pi^2}\right)}{2\theta\prod_{j=1}^{\infty}\left(1-\frac{\theta^2}{j^2\pi^2}\right)}\\ &= \prod_{j=0}^{\infty}\left(1-\frac{4\theta^2}{(2j+1)^2\pi^2}\right),\end{aligned}$$

Proving (2.29) to be correct. For $\theta = m\pi$, where $m \in \mathbb{Z}$, we have

$$\begin{aligned}&\prod_{j=0}^{\infty}\left(1-\frac{4m^2\pi^2}{(2j+1)^2\pi^2}\right) = \prod_{j=0}^{\infty}\frac{(2j+1)^2-4m^2}{(2j+1)^2}\\ &= \prod_{j=0}^{\infty}\frac{(2j+1+2|m|)(2j+1-2|m|)}{(2j+1)^2}\\ &= \lim_{n\to\infty}\frac{\prod_{j=0}^{n}(2j+1+2|m|)\prod_{j=2|m|}^{n}(2j+1-2|m|)\prod_{j=0}^{2|m|-1}(2j+1-2|m|)}{\prod_{j=0}^{n}(2j+1)^2}\\ &= \lim_{n\to\infty}\frac{(-1)^{|m|}\prod_{j=0}^{|m|-1}(2j+1)^2\prod_{j=|m|}^{n-2m}(2j+1)^2\prod_{j=n-2|m|+1}^{n}(2j+1+2|m|)}{\prod_{j=0}^{n}(2j+1)^2}\\ &= (-1)^m\\ &= \cos m\pi,\end{aligned}$$

once again proving (2.29) to be correct. □

**Example 2.13** (Loney, 1928) (*Value of* $\zeta(2)$)
Prove that

$$\begin{aligned}\zeta(2) &= \sum_{j=1}^{\infty} \frac{1}{j^2} \\ &= \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots \\ &= \frac{\pi^2}{6}.\end{aligned} \tag{2.30}$$

***Proof*** Using Euler's formula and (2.22), we have

$$\begin{aligned}\sin\theta &= \sum_{j=0}^{\infty} (-1)^j \frac{\theta^{2j+1}}{(2j+1)!} \\ &= \theta \prod_{j=1}^{\infty} \left(1 - \frac{\theta^2}{j^2\pi^2}\right) \\ &= \theta - \left(\sum_{j=1}^{\infty} \frac{1}{j^2\pi^2}\right) \theta^3 + \cdots\end{aligned} \tag{2.31}$$

Equating the coefficient of $\theta^3$ in (2.31), we get

$$\begin{aligned} -\frac{1}{6} &= -\sum_{j=1}^{\infty} \frac{1}{j^2\pi^2} \\ \Longleftrightarrow \sum_{j=1}^{\infty} \frac{1}{j^2} &= \frac{\pi^2}{6}, \end{aligned}$$

proving (2.30). □

## Problems

[2.1] Prove that $\forall(\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times 2\mathbb{N}$,

$$(-1)^{n/2}\frac{n\cos n\theta\sin\theta - \sin n\theta\cos\theta}{n\sin^3\theta} = \sum_{j=0}^{n/2-1}(-1)^j\frac{\prod_{k=1}^{j}\left(n^2-(2k)^2\right)}{(2j)!}\cos^{2j}\theta$$
$$= 1 - \frac{\left(n^2-2^2\right)}{2!}\cos^2\theta + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)}{4!}\cos^4\theta - \cdots$$
$$+ (-1)^{n/2-1}\frac{\left(n^2-2^2\right)\left(n^2-4^2\right)\left(n^2-6^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!}\cos^{n-2}\theta.$$

[2.2] Prove that $\forall(\theta, n) \in (\mathbb{R} - \mathbb{Z}\pi) \times (2\mathbb{N} - 1)$,

$$(-1)^{(n-1)/2}\frac{n\cos n\theta\sin\theta - \sin n\theta\cos\theta}{\sin^3\theta}$$
$$= \sum_{j=1}^{(n-1)/2}(-1)^{j-1}\frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j-1)!}\cos^{2j-1}\theta$$
$$= \frac{\left(n^2-1^2\right)}{1!}\cos\theta - \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{3!}\cos^3\theta$$
$$+ \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)}{5!}\cos^5\theta - \cdots$$
$$+ (-1)^{(n-3)/2}\frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!}\cos^{n-2}\theta.$$

[2.3] Prove that $\forall(\theta, n) \in \left(\mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}\right) \times (2\mathbb{N} - 1)$,

$$\frac{n\sin n\theta\cos\theta - \cos n\theta\sin\theta}{\cos^3\theta} = \sum_{j=1}^{(n-1)/2}(-1)^{j-1}\frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j-1)!}\sin^{2j-1}\theta$$
$$= \frac{\left(n^2-1^2\right)}{1!}\sin\theta - \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{3!}\sin^3\theta$$
$$+ \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)}{5!}\sin^5\theta - \cdots$$
$$+ (-1)^{(n-3)/2}\frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!}\sin^{n-2}\theta.$$

[2.4] Prove that $\forall(\theta, n) \in \left(\mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}\right) \times 2\mathbb{N}$,

$$\frac{n\cos n\theta\cos\theta + \sin n\theta\sin\theta}{n\cos^3\theta} = \sum_{j=0}^{n/2-1}(-1)^j \frac{\prod_{k=1}^{j}\left(n^2-(2k)^2\right)}{(2j)!}\sin^{2j}\theta$$
$$= 1 - \frac{\left(n^2-2^2\right)}{2!}\sin^2\theta + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)}{4!}\sin^4\theta - \cdots$$
$$+ (-1)^{n/2-1}\frac{\left(n^2-2^2\right)\left(n^2-4^2\right)\left(n^2-6^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!}\sin^{n-2}\theta.$$

[2.5] (Loney, 1928) Prove that $\forall x \in \mathbb{R} - \{0\} - \left\{ \frac{1}{m} \,\middle|\, m \in \mathbb{Z} - \{0\} \right\}$,

$$\frac{\pi}{x\sin\frac{\pi}{x}} = 1 + 2\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{j^2x^2-1}$$
$$= 1 + 2\left(\frac{1}{x^2-1} - \frac{1}{2^2x^2-1} + \frac{1}{3^2x^2-1} - \cdots\right).$$

[2.6] Prove that $\forall(z, \theta) \in \mathbb{Z} \times \left((4z-1)\frac{\pi}{2}, (4z+1)\frac{\pi}{2}\right)$,

$$\sec^3\theta = \sum_{j=1}^{\infty}\frac{j}{2^{2j-1}}\binom{2j}{j}\sin^{2j-2}\theta$$
$$= 1 + \frac{2\binom{4}{2}}{2^3}\sin^2\theta + \frac{3\binom{6}{3}}{2^5}\sin^4\theta + \frac{4\binom{8}{4}}{2^7}\sin^6\theta + \cdots$$
$$= \sum_{j=0}^{\infty}\frac{2j+1}{2^{2j}}\binom{2j}{j}\sin^{2j}\theta$$
$$= 1 + \frac{3\binom{2}{1}}{2^2}\sin^2\theta + \frac{5\binom{4}{2}}{2^4}\sin^4\theta + \frac{7\binom{6}{3}}{2^6}\sin^6\theta + \cdots$$

[2.7] Prove that $\forall (z,\theta) \in \mathbb{Z} \times \left((4z-1)\frac{\pi}{2}, (4z+1)\frac{\pi}{2}\right)$,

$$\begin{aligned}\sec\theta &= \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j}\theta}{2^{2j}} \\ &= 1 + \frac{\binom{2}{1}}{2^2}\sin^2\theta + \frac{\binom{4}{2}}{2^4}\sin^4\theta + \frac{\binom{6}{3}}{2^6}\sin^6\theta + \cdots\end{aligned}$$

[2.8] (Boros and Moll, 2006) Prove that $\forall \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

$$\begin{aligned}\theta &= \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+1}\theta}{2^{2j}(2j+1)} \\ &= \sin\theta + \frac{\binom{2}{1}}{2^2\cdot 3}\sin^3\theta + \frac{\binom{4}{2}}{2^4\cdot 5}\sin^5\theta + \frac{\binom{6}{3}}{2^6\cdot 7}\sin^7\theta + \cdots\end{aligned}$$

[2.9] (Lehmer, 1985; Boros and Moll, 2006) Prove that $\forall \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$$\begin{aligned}2\theta\tan\theta &= \sum_{j=1}^{\infty} \frac{2^{2j}}{j}\binom{2j}{j}^{-1}\sin^{2j}\theta \\ &= \frac{2^2}{1\binom{2}{1}}\sin^2\theta + \frac{2^4}{2\binom{4}{2}}\sin^4\theta + \frac{2^6}{3\binom{6}{3}}\sin^6\theta + \frac{2^8}{4\binom{8}{4}}\sin^8\theta + \cdots\end{aligned}$$

[2.10] (Lehmer, 1985; Boros and Moll, 2006) Prove that $\forall \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

$$\begin{aligned}2\theta^2 &= \sum_{j=1}^{\infty} \frac{2^{2j}}{j^2}\binom{2j}{j}^{-1}\sin^{2j}\theta \\ &= \frac{2^2}{1^2\binom{2}{1}}\sin^2\theta + \frac{2^4}{2^2\binom{4}{2}}\sin^4\theta + \frac{2^6}{3^2\binom{6}{3}}\sin^6\theta + \frac{2^8}{4^2\binom{8}{4}}\sin^8\theta + \cdots\end{aligned}$$

[2.11] Prove that $\forall\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

$$\theta \sin\theta + \cos\theta = 1 + \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)}$$
$$= 1 + \frac{\binom{0}{0}}{2^0 \cdot 1 \cdot 2} \sin^2\theta + \frac{\binom{2}{1}}{2^2 \cdot 3 \cdot 4} \sin^4\theta + \frac{\binom{4}{2}}{2^4 \cdot 5 \cdot 6} \sin^6\theta + \cdots$$

[2.12] Prove that $\forall\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

$$\frac{\sin\theta - \theta\cos\theta}{2} = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j(2j+1)} \binom{2j}{j}^{-1} \sin^{2j+1}\theta$$
$$= \frac{2^0}{1 \cdot 3 \cdot \binom{2}{1}} \sin^3\theta + \frac{2^2}{2 \cdot 5 \cdot \binom{4}{2}} \sin^5\theta + \frac{2^4}{3 \cdot 7 \cdot \binom{6}{3}} \sin^7\theta + \frac{2^6}{4 \cdot 9 \cdot \binom{8}{4}} \sin^9\theta + \cdots$$

[2.13] Prove that $\forall\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

$$\frac{3\sin^2\theta - \theta^2 - \theta\sin 2\theta}{8} = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j(2j+1)(2j+2)} \binom{2j}{j}^{-1} \sin^{2j+2}\theta$$
$$= \frac{2^0}{1 \cdot 3 \cdot 4 \cdot \binom{2}{1}} \sin^4\theta + \frac{2^2}{2 \cdot 5 \cdot 6 \cdot \binom{4}{2}} \sin^6\theta + \frac{2^4}{3 \cdot 7 \cdot 8 \cdot \binom{6}{3}} \sin^8\theta + \frac{2^6}{4 \cdot 9 \cdot 10 \cdot \binom{8}{4}} \sin^{10}\theta + \cdots$$

[2.14] Prove that $\forall\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$$\sec^2\theta(1 + \theta\tan\theta) = \sum_{j=0}^{\infty} 2^{2j} \binom{2j}{j}^{-1} \sin^{2j}\theta$$
$$= 1 + \frac{2^2}{\binom{2}{1}} \sin^2\theta + \frac{2^4}{\binom{4}{2}} \sin^4\theta + \frac{2^6}{\binom{6}{3}} \sin^6\theta + \cdots$$

[2.15] Prove that $\forall\theta \in \mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\frac{1+2\sin^2\theta}{\cos^5\theta} = \sum_{j=0}^{\infty} \frac{(2j+1)^2}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta$$
$$= 1 + \frac{3^2\binom{2}{1}}{2^2}\sin^2\theta + \frac{5^2\binom{4}{2}}{2^4}\sin^4\theta + \frac{7^2\binom{6}{3}}{2^6}\sin^6\theta + \cdots$$

[2.16] Prove that $\forall\theta \in \mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\frac{1+2\sin^2\theta\left(5+2\sin^2\theta\right)}{\cos^7\theta} = \sum_{j=0}^{\infty} \frac{(2j+1)^3}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta$$
$$= 1 + \frac{3^3\binom{2}{1}}{2^2}\sin^2\theta + \frac{5^3\binom{4}{2}}{2^4}\sin^4\theta + \frac{7^3\binom{6}{3}}{2^6}\sin^6\theta + \cdots$$

[2.17] Prove that $\forall\theta \in \mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\frac{1+4\sin^2\theta\left(9+15\sin^2\theta+2\sin^4\theta\right)}{\cos^9\theta} = \sum_{j=0}^{\infty} \frac{(2j+1)^4}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta$$
$$= 1 + \frac{3^4\binom{2}{1}}{2^2}\sin^2\theta + \frac{5^4\binom{4}{2}}{2^4}\sin^4\theta + \frac{7^4\binom{6}{3}}{2^6}\sin^6\theta + \cdots$$

[2.18] Prove that $\forall\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

$$\frac{\theta^2}{2}\sin\theta + \theta\cos\theta - \sin\theta = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j^2(2j+1)} \binom{2j}{j}^{-1} \sin^{2j+1}\theta$$
$$= \frac{2^0}{1^2\cdot 3\cdot\binom{2}{1}}\sin^3\theta + \frac{2^2}{2^2\cdot 5\cdot\binom{4}{2}}\sin^5\theta + \frac{2^4}{3^2\cdot 7\cdot\binom{6}{3}}\sin^7\theta$$
$$+ \frac{2^6}{4^2\cdot 9\cdot\binom{8}{4}}\sin^9\theta + \cdots$$

[2.19] Prove that $\forall\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

$$\theta \sin\theta + 2\cos\theta - \log\left(\frac{e^2}{2}(1+\cos\theta)\right) = \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)^2}$$
$$= \frac{\binom{0}{0}}{2^0 \cdot 1 \cdot 2^2}\sin^2\theta + \frac{\binom{2}{1}}{2^2 \cdot 3 \cdot 4^2}\sin^4\theta + \frac{\binom{4}{2}}{2^4 \cdot 5 \cdot 6^2}\sin^6\theta + \frac{\binom{6}{3}}{2^6 \cdot 7 \cdot 8^2}\sin^8\theta + \cdots$$

[2.20] (Loney, 1928) Prove that $\forall\theta \in \mathbb{R} - \mathbb{Z}\pi$,

$$\frac{1-\theta\cot\theta}{2\theta^2} = \sum_{j=1}^{\infty} \frac{1}{j^2\pi^2 - \theta^2}$$
$$= \frac{1}{\pi^2-\theta^2} + \frac{1}{2^2\pi^2-\theta^2} + \frac{1}{3^2\pi^2-\theta^2} + \frac{1}{4^2\pi^2-\theta^2} + \cdots$$

[2.21] (Loney, 1928) Prove that $\forall\theta \in \mathbb{R} - \{0\} \cup (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\frac{\tan\theta}{8\theta} = \sum_{j=0}^{\infty} \frac{1}{(2j+1)^2\pi^2 - 4\theta^2}$$
$$= \frac{1}{\pi^2-4\theta^2} + \frac{1}{3^2\pi^2-4\theta^2} + \frac{1}{5^2\pi^2-4\theta^2} + \frac{1}{7^2\pi^2-4\theta^2} + \cdots$$

[2.22] Prove that $\forall\theta \in \mathbb{R} - \mathbb{Z}\pi$,

$$\frac{2\theta^2 + \theta\sin 2\theta - 4\sin^2\theta}{8\theta^4\sin^2\theta} = \sum_{j=1}^{\infty} \frac{1}{\left(j^2\pi^2 - \theta^2\right)^2}$$
$$= \frac{1}{\left(\pi^2-\theta^2\right)^2} + \frac{1}{\left(2^2\pi^2-\theta^2\right)^2} + \frac{1}{\left(3^2\pi^2-\theta^2\right)^2} + \frac{1}{\left(4^2\pi^2-\theta^2\right)^2} + \cdots$$

[2.23] Prove that $\forall\theta \in \mathbb{R} - \{0\} \cup (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\frac{2\theta - \sin 2\theta}{128\theta^3\cos^2\theta} = \sum_{j=0}^{\infty} \frac{1}{\left((2j+1)^2\pi^2 - 4\theta^2\right)^2}$$
$$= \frac{1}{\left(\pi^2-4\theta^2\right)^2} + \frac{1}{\left(3^2\pi^2-4\theta^2\right)^2} + \frac{1}{\left(5^2\pi^2-4\theta^2\right)^2} + \frac{1}{\left(7^2\pi^2-4\theta^2\right)^2} + \cdots$$

[2.24] Prove that $\forall(z,\theta)\in\mathbb{Z}\times\left((4z-1)\frac{\pi}{4},(4z+1)\frac{\pi}{4}\right)$,

$$\begin{aligned}\frac{1}{4}\sin 4\theta\cos^2\theta&=\sum_{j=1}^{\infty}(-1)^{j-1}j^2\tan^{2j-1}\theta\\&=\tan\theta-2^2\tan^3\theta+3^2\tan^5\theta-4^2\tan^7\theta+\cdots\end{aligned}$$

[2.25] Prove that $\forall(z,\theta)\in\mathbb{Z}\times\left((4z-1)\frac{\pi}{4},(4z+1)\frac{\pi}{4}\right)$,

$$\begin{aligned}\frac{1}{8}\cos^2\theta\,(\sin 6\theta+\sin 2\theta\cos 4\theta)&=\sum_{j=1}^{\infty}(-1)^{j-1}j^3\tan^{2j-1}\theta\\&=\tan\theta-2^3\tan^3\theta+3^3\tan^5\theta-4^3\tan^7\theta+\cdots\end{aligned}$$

[2.26] Prove that $\forall(z,\theta)\in\mathbb{Z}\times\left((4z-1)\frac{\pi}{4},(4z+1)\frac{\pi}{4}\right)$,

$$\begin{aligned}\frac{1}{16}\cos^2\theta\,(3\sin 8\theta-2\sin 4\theta)&=\sum_{j=1}^{\infty}(-1)^{j-1}j^4\tan^{2j-1}\theta\\&=\tan\theta-2^4\tan^3\theta+3^4\tan^5\theta-4^4\tan^7\theta+\cdots\end{aligned}$$

[2.27] Prove that $\forall(\theta,n)\in(\mathbb{R}-\{0\})\times 2\mathbb{N}$,

$$\begin{aligned}&(-1)^{n/2}\frac{\sinh n\theta\cosh\theta-n\cosh n\theta\sinh\theta}{n\sinh^3\theta}\\&=\sum_{j=0}^{n/2-1}(-1)^j\frac{\prod_{k=1}^{j}\left(n^2-(2k)^2\right)}{(2j)!}\cosh^{2j}\theta\\&=1-\frac{\left(n^2-2^2\right)}{2!}\cosh^2\theta+\frac{\left(n^2-2^2\right)\left(n^2-4^2\right)}{4!}\cosh^4\theta+\cdots\\&\quad+(-1)^{n/2-1}\frac{\left(n^2-2^2\right)\left(n^2-4^2\right)\left(n^2-6^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!}\cosh^{n-2}\theta.\end{aligned}$$

[2.28] Prove that $\forall(\theta, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{N} - 1)$,

$$
\begin{aligned}
&(-1)^{(n-1)/2} \frac{\sinh n\theta \cosh\theta - n \cosh n\theta \sinh\theta}{\sinh^3\theta} \\
&= \sum_{j=1}^{(n-1)/2} (-1)^{j-1} \frac{\prod_{k=0}^{j-1} \left(n^2 - (2k+1)^2\right)}{(2j-1)!} \cosh^{2j-1}\theta \\
&= \frac{\left(n^2-1^2\right)}{1!} \cosh\theta - \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{3!} \cosh^3\theta \\
&\quad + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)}{5!} \cosh^5\theta - \cdots \\
&\quad + (-1)^{(n-3)/2} \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!} \cosh^{n-2}\theta.
\end{aligned}
$$

[2.29] Prove that $\forall(\theta, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$,

$$
\begin{aligned}
&\frac{n \sinh n\theta \cosh\theta - \cosh n\theta \sinh\theta}{\cosh^3\theta} = \sum_{j=1}^{(n-1)/2} \frac{\prod_{k=0}^{j-1} \left(n^2 - (2k+1)^2\right)}{(2j-1)!} \sinh^{2j-1}\theta \\
&= \frac{\left(n^2-1^2\right)}{1!} \sinh\theta + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)}{3!} \sinh^3\theta \\
&\quad + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)}{5!} \sinh^5\theta + \cdots \\
&\quad + \frac{\left(n^2-1^2\right)\left(n^2-3^2\right)\left(n^2-5^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!} \sinh^{n-2}\theta.
\end{aligned}
$$

[2.30] Prove that $\forall(\theta, n) \in \mathbb{R} \times 2\mathbb{N}$,

$$
\begin{aligned}
&\frac{n \cosh n\theta \cosh\theta - \sinh n\theta \sinh\theta}{n \cosh^3\theta} = \sum_{j=0}^{n/2-1} \frac{\prod_{k=1}^{j} \left(n^2 - (2k)^2\right)}{(2j)!} \sinh^{2j}\theta \\
&= 1 + \frac{\left(n^2-2^2\right)}{2!} \sinh^2\theta + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)}{4!} \sinh^4\theta + \cdots \\
&\quad + \frac{\left(n^2-2^2\right)\left(n^2-4^2\right)\left(n^2-6^2\right)\ldots\left(n^2-(n-2)^2\right)}{(n-2)!} \sinh^{n-2}\theta.
\end{aligned}
$$

[2.31] Prove that $\forall x \in \mathbb{R} - \{0\}$,

$$\begin{aligned}\frac{\pi}{x \sinh \dfrac{\pi}{x}} &= 1 - 2\sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{j^2x^2+1} \\ &= 1 - 2\left(\frac{1}{x^2+1} - \frac{1}{2^2x^2+1} + \frac{1}{3^2x^2+1} - \cdots\right).\end{aligned}$$

[2.32] Prove that $\forall \theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\begin{aligned}\operatorname{sech}^3\theta &= \sum_{j=1}^{\infty} (-1)^{j-1} \frac{j}{2^{2j-1}} \binom{2j}{j} \sinh^{2j-2}\theta \\ &= 1 - \frac{2\binom{4}{2}}{2^1}\sinh^2\theta + \frac{3\binom{6}{3}}{2^3}\sinh^4\theta - \frac{4\binom{8}{4}}{2^5}\sinh^6\theta + \cdots \\ &= \sum_{j=0}^{\infty} (-1)^{j} \frac{(2j+1)}{2^{2j}} \binom{2j}{j} \sinh^{2j}\theta \\ &= 1 - \frac{3\binom{2}{1}}{2^2}\sinh^2\theta + \frac{5\binom{4}{2}}{2^4}\sinh^4\theta - \frac{7\binom{6}{3}}{2^6}\sinh^6\theta + \cdots\end{aligned}$$

[2.33] Prove that $\forall \theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\begin{aligned}\operatorname{sech}\theta &= \sum_{j=0}^{\infty} (-1)^{j} \binom{2j}{j} \frac{\sinh^{2j}\theta}{2^{2j}} \\ &= 1 - \frac{\binom{2}{1}}{2^2}\sinh^2\theta + \frac{\binom{4}{2}}{2^4}\sinh^4\theta - \frac{\binom{6}{3}}{2^6}\sinh^6\theta + \cdots\end{aligned}$$

[2.34] Prove that $\forall \theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\begin{aligned}\theta &= \sum_{j=0}^{\infty} (-1)^{j} \binom{2j}{j} \frac{\sinh^{2j+1}\theta}{2^{2j}(2j+1)} \\ &= \sinh\theta - \frac{\binom{2}{1}}{2^2 \cdot 3}\sinh^3\theta + \frac{\binom{4}{2}}{2^4 \cdot 5}\sinh^5\theta - \frac{\binom{6}{3}}{2^6 \cdot 7}\sinh^7\theta + \cdots\end{aligned}$$

[2.35] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$2\theta\tanh\theta = \sum_{j=1}^{\infty}(-1)^{j-1}\frac{2^{2j}}{j}\binom{2j}{j}^{-1}\sinh^{2j}\theta$$
$$= \frac{2^2}{1\binom{2}{1}}\sinh^2\theta - \frac{2^4}{2\binom{4}{2}}\sinh^4\theta + \frac{2^6}{3\binom{6}{3}}\sinh^6\theta - \frac{2^8}{4\binom{8}{4}}\sinh^8\theta + \cdots$$

[2.36] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$2\theta^2 = \sum_{j=1}^{\infty}(-1)^{j-1}\frac{2^{2j}}{j^2}\binom{2j}{j}^{-1}\sinh^{2j}\theta$$
$$= \frac{2^2}{1^2\binom{2}{1}}\sinh^2\theta - \frac{2^4}{2^2\binom{4}{2}}\sinh^4\theta + \frac{2^6}{3^2\binom{6}{3}}\sinh^6\theta - \frac{2^8}{4^2\binom{8}{4}}\sinh^8\theta + \cdots$$

[2.37] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\cosh\theta - \theta\sinh\theta = 1 + \sum_{j=0}^{\infty}(-1)^{j+1}\binom{2j}{j}\frac{\sinh^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)}$$
$$= 1 - \frac{\binom{0}{0}}{2^0\cdot 1\cdot 2}\sinh^2\theta + \frac{\binom{2}{1}}{2^2\cdot 3\cdot 4}\sinh^4\theta - \frac{\binom{4}{2}}{2^4\cdot 5\cdot 6}\sinh^6\theta + \cdots$$

[2.38] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\frac{\theta\cosh\theta - \sinh\theta}{2} = \sum_{j=1}^{\infty}(-1)^{j-1}\frac{2^{2j-2}}{j(2j+1)}\binom{2j}{j}^{-1}\sinh^{2j+1}\theta$$
$$= \frac{2^0}{1\cdot 3\cdot\binom{2}{1}}\sinh^3\theta - \frac{2^2}{2\cdot 5\cdot\binom{4}{2}}\sinh^5\theta + \frac{2^4}{3\cdot 7\cdot\binom{6}{3}}\sinh^7\theta$$
$$- \frac{2^6}{4\cdot 9\cdot\binom{8}{4}}\sinh^9\theta + \cdots$$

[2.39] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\frac{\theta\sinh 2\theta + \theta^2 - 3\sinh^2\theta}{8} = \sum_{j=1}^{\infty}(-1)^{j-1}\frac{2^{2j-2}}{j(2j+1)(2j+2)}\binom{2j}{j}^{-1}\sinh^{2j+2}\theta$$
$$= \frac{2^0}{1\cdot 3\cdot 4\cdot\binom{2}{1}}\sinh^4\theta - \frac{2^2}{2\cdot 5\cdot 6\cdot\binom{4}{2}}\sinh^6\theta + \frac{2^4}{3\cdot 7\cdot 8\cdot\binom{6}{3}}\sinh^8\theta$$
$$- \frac{2^6}{4\cdot 9\cdot 10\cdot\binom{8}{4}}\sinh^{10}\theta + \cdots$$

[2.40] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\operatorname{sech}^2\theta(1-\theta\tanh\theta) = \sum_{j=0}^{\infty}(-1)^j 2^{2j}\binom{2j}{j}^{-1}\sinh^{2j}\theta$$
$$= 1 - \frac{2^2}{\binom{2}{1}}\sinh^2\theta + \frac{2^4}{\binom{4}{2}}\sinh^4\theta - \frac{2^6}{\binom{6}{3}}\sinh^6\theta + \cdots$$

[2.41] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\frac{1-2\sinh^2\theta}{\cosh^5\theta} = \sum_{j=0}^{\infty}(-1)^j\frac{(2j+1)^2}{2^{2j}}\binom{2j}{j}\sinh^{2j}\theta$$
$$= 1 - \frac{3^2\binom{2}{1}}{2^2}\sinh^2\theta + \frac{5^2\binom{4}{2}}{2^4}\sinh^4\theta - \frac{7^2\binom{6}{3}}{2^6}\sinh^6\theta + \cdots$$

[2.42] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\frac{1-2\sinh^2\theta\left(5-2\sinh^2\theta\right)}{\cosh^7\theta} = \sum_{j=0}^{\infty}(-1)^j\frac{(2j+1)^3}{2^{2j}}\binom{2j}{j}\sinh^{2j}\theta$$
$$= 1 - \frac{3^3\binom{2}{1}}{2^2}\sinh^2\theta + \frac{5^3\binom{4}{2}}{2^4}\sinh^4\theta - \frac{7^3\binom{6}{3}}{2^6}\sinh^6\theta + \cdots$$

[2.43] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\frac{1-4\sinh^2\theta\left(9-15\sinh^2\theta+2\sinh^4\theta\right)}{\cosh^9\theta} = \sum_{j=0}^{\infty}(-1)^j\frac{(2j+1)^4}{2^{2j}}\binom{2j}{j}\sinh^{2j}\theta$$
$$= 1-\frac{3^4\binom{2}{1}}{2^2}\sinh^2\theta+\frac{5^4\binom{4}{2}}{2^4}\sinh^4\theta-\frac{7^4\binom{6}{3}}{2^6}\sin^6\theta+\cdots$$

[2.44] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\frac{\theta^2}{2}\sinh\theta-\theta\cosh\theta+\sinh\theta = \sum_{j=1}^{\infty}(-1)^{j-1}\frac{2^{2j-2}}{j^2(2j+1)}\binom{2j}{j}^{-1}\sinh^{2j+1}\theta$$
$$= \frac{2^0}{1^2\cdot 3\cdot\binom{2}{1}}\sinh^3\theta-\frac{2^2}{2^2\cdot 5\cdot\binom{4}{2}}\sinh^5\theta+\frac{2^4}{3^2\cdot 7\cdot\binom{6}{3}}\sinh^7\theta$$
$$-\frac{2^6}{4^2\cdot 9\cdot\binom{8}{4}}\sinh^9\theta+\cdots$$

[2.45] Prove that $\forall\theta \in \left(-\log\left(\sqrt{2}+1\right), \log\left(\sqrt{2}+1\right)\right)$,

$$\theta\sinh\theta-2\cosh\theta+\log\left(\frac{e^2}{2}(1+\cosh\theta)\right)$$
$$= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}\frac{\sinh^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)^2}$$
$$= \frac{\binom{0}{0}}{2^0\cdot 1\cdot 2^2}\sinh^2\theta-\frac{\binom{2}{1}}{2^2\cdot 3\cdot 4^2}\sinh^4\theta+\frac{\binom{4}{2}}{2^4\cdot 5\cdot 6^2}\sinh^6\theta$$
$$-\frac{\binom{6}{3}}{2^6\cdot 7\cdot 8^2}\sinh^8\theta+\cdots$$

[2.46] Prove that $\forall\theta \in \mathbb{R} - \{0\}$,

$$\frac{\theta \coth\theta - 1}{2\theta^2} = \sum_{j=1}^{\infty} \frac{1}{j^2\pi^2 + \theta^2}$$
$$= \frac{1}{\pi^2 + \theta^2} + \frac{1}{2^2\pi^2 + \theta^2} + \frac{1}{3^2\pi^2 + \theta^2} + \frac{1}{4^2\pi^2 + \theta^2} + \cdots$$

[2.47] Prove that $\forall\theta \in \mathbb{R} - \{0\}$,

$$\frac{\tanh\theta}{8\theta} = \sum_{j=0}^{\infty} \frac{1}{(2j+1)^2\pi^2 + 4\theta^2}$$
$$= \frac{1}{\pi^2 + 4\theta^2} + \frac{1}{3^2\pi^2 + 4\theta^2} + \frac{1}{5^2\pi^2 + 4\theta^2} + \frac{1}{7^2\pi^2 + 4\theta^2} + \cdots$$

[2.48] Prove that $\forall\theta \in \mathbb{R} - \{0\}$,

$$\frac{2\theta^2 + \theta\sinh 2\theta - 4\sinh^2\theta}{8\theta^4\sinh^2\theta} = \sum_{j=1}^{\infty} \frac{1}{(j^2\pi^2 + \theta^2)^2}$$
$$= \frac{1}{(\pi^2 + \theta^2)^2} + \frac{1}{(2^2\pi^2 + \theta^2)^2} + \frac{1}{(3^2\pi^2 + \theta^2)^2} + \frac{1}{(4^2\pi^2 + \theta^2)^2} + \cdots$$

[2.49] Prove that $\forall\theta \in \mathbb{R} - \{0\}$,

$$\frac{\sinh 2\theta - 2\theta}{128\theta^3\cosh^2\theta} = \sum_{j=0}^{\infty} \frac{1}{\left((2j+1)^2\pi^2 + 4\theta^2\right)^2}$$
$$= \frac{1}{(\pi^2 + 4\theta^2)^2} + \frac{1}{(3^2\pi^2 + 4\theta^2)^2} + \frac{1}{(5^2\pi^2 + 4\theta^2)^2} + \frac{1}{(7^2\pi^2 + 4\theta^2)^2} + \cdots$$

[2.50] Prove that $\forall\theta \in \mathbb{R}$,

$$\frac{1}{4}\sinh 4\theta\cosh^2\theta = \sum_{j=1}^{\infty} j^2\tanh^{2j-1}\theta$$
$$= \tanh\theta + 2^2\tanh^3\theta + 3^2\tanh^5\theta + 4^2\tanh^7\theta + \cdots$$

[2.51] Prove that $\forall \theta \in \mathbb{R}$,

$$\frac{1}{8}\cosh^2\theta\,(\sinh 6\theta + \sinh 2\theta \cosh 4\theta) = \sum_{j=1}^{\infty} j^3 \tanh^{2j-1}\theta$$
$$= \tanh\theta + 2^3\tanh^3\theta + 3^3\tanh^5\theta + 4^3\tanh^7\theta + \cdots$$

[2.52] Prove that $\forall \theta \in \mathbb{R}$,

$$\frac{1}{16}\cosh^2\theta\,(3\sinh 8\theta - 2\sinh 4\theta) = \sum_{j=1}^{\infty} j^4 \tanh^{2j-1}\theta$$
$$= \tanh\theta + 2^4\tanh^3\theta + 3^4\tanh^5\theta + 4^4\tanh^7\theta + \cdots$$

# Chapter 3
# Bessel Functions




In this chapter, first, we will learn about Euler's constant $\gamma$ and the Gamma function $\Gamma(z)$, which is a prerequisite for studying the Bessel functions. We will learn about the Bessel coefficients $J_n(z)$ and $I_n(z)$. Then we will learn about their generalizations: the Bessel functions $J_v(z)$ and $I_v(z)$. The problems given in this chapter study some of the properties of the Bessel functions. Integral representations of the Bessel coefficients will be derived in the later chapters.

## 3.1 Euler's Constant $\gamma$

First, we estimate the sum of a *Harmonic Progression (HP)*.

**Example 3.1** Estimating the sum of the HP $\left(\frac{1}{j}\right)_{j=1}^{n}$

$\forall n \in \mathbb{N}$, let

$$\begin{aligned} H(n) &= \sum_{j=1}^{n} \frac{1}{j} \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}. \end{aligned}$$

Prove that $\forall n \in \mathbb{N} + 1$,

$$\frac{1}{n} < H(n) - \log n < 1. \tag{3.1}$$

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A. Mishra, *Special Integrals*, University Texts in the Mathematical Sciences,
https://doi.org/10.1007/978-981-97-7514-9_3

***Proof*** We have

$$\begin{aligned}
&\frac{1}{j+1} < \frac{1}{x} < \frac{1}{j}, \quad \forall x \in (j, j+1) \\
\Longrightarrow &\int_j^{j+1} \frac{dx}{j+1} < \int_j^{j+1} \frac{dx}{x} < \int_j^{j+1} \frac{dx}{j} \\
\Longrightarrow &\frac{1}{j+1} < \log(j+1) - \log j < \frac{1}{j} \\
\Longrightarrow &\sum_{j=1}^{n-1} \frac{1}{j+1} < \sum_{j=1}^{n-1} (\log(j+1) - \log j) < \sum_{j=1}^{n-1} \frac{1}{j}
\end{aligned} \tag{3.2}$$

$$\begin{aligned}
&\Longrightarrow H(n) - 1 < \log n < H(n) - \frac{1}{n} \\
&\Longrightarrow -1 < \log n - H(n) < -\frac{1}{n} \\
&\Longrightarrow \frac{1}{n} < H(n) - \log n < 1.
\end{aligned}$$

□

Now we show that the limit of (3.1) as $n \to \infty$ exists.

**Example 3.2** (*Convergence of the Sequence* $(H(j) - \log j)_{j=1}^{\infty}$)
Prove that the limit $\lim_{n\to\infty}(H(n) - \log n)$ exists.

***Proof*** From (3.2), we have

$$\begin{aligned}
&0 < \int_j^{j+1} \left(\frac{1}{j} - \frac{1}{x}\right) dx < \left(\frac{1}{j} - \frac{1}{j+1}\right) dx \\
\Longrightarrow &0 < \int_j^{j+1} \frac{(x-j)dx}{jx} < \int_j^{j+1} \frac{dx}{j(j+1)} < \int_j^{j+1} \frac{dx}{j^2} \\
\Longrightarrow &0 < \frac{1}{j} - \log(j+1) + \log j < \frac{1}{j^2}.
\end{aligned}$$

From (2.30), the series $\sum_{j=0}^{\infty} 1/j^2$ converges with sum $\pi^2/6$. Therefore, using the comparison test of convergence (Whittaker and Watson, 2016), the sum $\sum_{j=0}^{\infty}(1/j - \log(j+1) + \log j) = \lim_{n\to\infty}(H(n) - \log n)$ is convergent. □

**Definition 3.3** (*Euler's Constant* $\gamma$)
Euler's Constant $\gamma$ is defined as

$$\begin{aligned}
\gamma &= \lim_{n\to\infty} (H(n) - \log n) \\
&= \lim_{n\to\infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n\right).
\end{aligned} \tag{3.3}$$

From (3.1), we have

$$0 < \gamma < 1.$$

The decimal representation of $\gamma$ is

$$\gamma = 0.57721566490153286060651209008240243\ldots$$

## 3.2 The Gamma Function $\Gamma(z)$

**Definition 3.4** (*The Gamma Function* $\Gamma(z)$)
$\forall z \in \mathbb{C} - (\mathbb{Z} - \mathbb{N})$, the Gamma Function, $\Gamma(z)$, is defined as

$$\begin{aligned}\Gamma(z) &= \frac{e^{-\gamma z}}{z}\prod_{j=1}^{\infty} e^{z/j}\left(1+\frac{z}{j}\right)^{-1} \\ &= \frac{e^{-\gamma z}}{z} e^{z}\,(1+z)^{-1}\,e^{z/2}\left(1+\frac{z}{2}\right)^{-1} e^{z/3}\left(1+\frac{z}{3}\right)^{-1} e^{z/4}\left(1+\frac{z}{4}\right)^{-1}\ldots\end{aligned} \tag{3.4}$$

The Gamma function $\Gamma(z)$ is convergent $\forall z \in \mathbb{C} - (\mathbb{Z} - \mathbb{N})$ (Whittaker and Watson, 2016).

**Example 3.5** (*Euler's Identity for* $\Gamma(z)$)
Prove that $\forall z \in \mathbb{C} - (\mathbb{Z} - \mathbb{N})$,

$$\begin{aligned}\Gamma(z) &= \frac{1}{z}\prod_{j=1}^{\infty}\left(1+\frac{1}{j}\right)^{z}\left(1+\frac{z}{j}\right)^{-1} \\ &= \frac{2^z}{z}(1+z)^{-1}\left(\frac{3}{2}\right)^{z}\left(1+\frac{z}{2}\right)^{-1}\left(\frac{4}{3}\right)^{z}\left(1+\frac{z}{3}\right)^{-1}\left(\frac{5}{4}\right)^{z}\left(1+\frac{z}{4}\right)^{-1}\ldots\end{aligned} \tag{3.5}$$

***Proof*** From (3.3) and (3.4), we have

$$\begin{aligned}\Gamma(z) &= \frac{1}{z}\lim_{n\to\infty} e^{-\left(\sum_{j=1}^{n} 1/j - \log n\right)z}\prod_{j=1}^{n} e^{z/j}\left(1+\frac{z}{j}\right)^{-1} \\ &= \frac{1}{z}\lim_{n\to\infty} e^{-z\sum_{j=1}^{n} 1/j + z\log n} e^{z\sum_{j=1}^{n} 1/j}\prod_{j=1}^{n}\left(1+\frac{z}{j}\right)^{-1} \\ &= \frac{1}{z}\lim_{n\to\infty} n^{z}\prod_{j=1}^{n}\left(1+\frac{z}{j}\right)^{-1} \\ &= \frac{1}{z}\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{-z}\prod_{j=1}^{n}\left(1+\frac{1}{j}\right)^{z}\left(1+\frac{z}{j}\right)^{-1}\end{aligned}$$

$$= \frac{1}{z} \lim_{n\to\infty} \prod_{j=1}^{n} \left(1+\frac{1}{j}\right)^{z} \left(1+\frac{z}{j}\right)^{-1}.$$

□

**Example 3.6** (*A Recursive Formula for* $\Gamma(z)$)
Prove that $\forall z \in \mathbb{C} - (\mathbb{Z} - \mathbb{N})$,

$$\Gamma(z+1) = z\Gamma(z). \tag{3.6}$$

***Proof*** From (3.5), we have

$$\begin{aligned}
\Gamma(z+1) &= \frac{1}{z+1} \lim_{n\to\infty} \prod_{j=1}^{n} \left(1+\frac{1}{j}\right)^{z+1} \left(1+\frac{z+1}{j}\right)^{-1} \\
&= \frac{1}{z+1} \lim_{n\to\infty} \prod_{j=1}^{n} \left(1+\frac{1}{j}\right)^{z} \left(1+\frac{z+1}{j}\right)^{-1} \left(1+\frac{1}{j}\right) \\
&= \frac{1}{z+1} \lim_{n\to\infty} \prod_{j=1}^{n} \left(1+\frac{1}{j}\right)^{z} \left(\frac{j}{z+j+1}\right) \left(\frac{j+1}{j}\right) \\
&= \frac{1}{z+1} \lim_{n\to\infty} \prod_{j=1}^{n} \left(1+\frac{1}{j}\right)^{z} \left(1+\frac{z}{j+1}\right)^{-1} \\
&= \lim_{n\to\infty} \left(1+\frac{z}{n+1}\right)^{-1} \prod_{j=1}^{n} \left(1+\frac{1}{j}\right)^{z} \left(1+\frac{z}{j}\right)^{-1} \\
&= z\Gamma(z).
\end{aligned}$$

□

## 3.3 The Bessel Coefficients $J_n(z)$

**Definition 3.7** (*The Bessel Coefficients* $J_n(z)$)
$\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$, the Bessel coefficients, $J_n(z)$, are defined by the following generating function:

$$e^{z(t-1/t)/2} = \sum_{n=-\infty}^{\infty} J_n(z) t^n. \tag{3.7}$$

**Example 3.8** (*Power Series for* $J_n(z)$)
Prove that $\forall (z, n) \in \mathbb{C} \times (\mathbb{N} - 1)$,

$$\begin{aligned} J_n(z) &= \sum_{j=0}^{\infty} (-1)^j \frac{z^{n+2j}}{j!(n+j)!2^{n+2j}} \\ &= \frac{z^n}{n!2^n} - \frac{z^{n+2}}{1!(n+1)!2^{n+2}} + \frac{z^{n+4}}{2!(n+2)!2^{n+4}} - \frac{z^{n+6}}{3!(n+3)!2^{n+6}} + \cdots \end{aligned} \tag{3.8}$$

***Proof*** From (3.7), we have

$$\begin{aligned} \sum_{n=-\infty}^{\infty} J_n(z)t^n &= e^{zt/2} \cdot e^{-z/(2t)} \\ &= \sum_{k=0}^{\infty} \frac{z^k}{k!2^k} t^k \sum_{j=0}^{\infty} (-1)^j \frac{z^j}{j!2^j} t^{-j} \\ &= \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} (-1)^j \frac{z^{k+j}}{k!j!2^{k+j}} t^{k-j} \\ &= \sum_{n=-\infty}^{\infty} \left( \sum_{j=0}^{\infty} (-1)^j \frac{z^{n+2j}}{j!(n+j)!2^{n+2j}} \right) t^n. \end{aligned} \tag{3.9}$$

We get (3.8) by equating the coefficient of $t^n$ in (3.9). □

**Example 3.9** (*A Recursive Formula for* $J_n(z)$)
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$J_{n-1}(z) + J_{n+1}(z) = \frac{2n}{z} J_n(z). \tag{3.10}$$

***Proof*** Differentiating (3.7) with respect to $t$, we get

$$\begin{aligned} & \frac{z}{2}\left(1 + \frac{1}{t^2}\right) e^{z(t-1/t)/2} = \sum_{n=-\infty}^{\infty} n J_n(z) t^{n-1} \\ \Longrightarrow\ & \frac{z}{2}\left(1 + \frac{1}{t^2}\right) \sum_{n=-\infty}^{\infty} J_n(z) t^n = \sum_{n=-\infty}^{\infty} n J_n(z) t^{n-1} \\ \Longrightarrow\ & \frac{z}{2} \sum_{n=-\infty}^{\infty} \left(J_{n-1}(z) + J_{n+1}(z)\right) t^{n-1} = \sum_{n=-\infty}^{\infty} n J_n(z) t^{n-1} \\ \Longrightarrow\ & \sum_{n=-\infty}^{\infty} \left(J_{n-1}(z) + J_{n+1}(z)\right) t^{n-1} = \sum_{n=-\infty}^{\infty} \frac{2n}{z} J_n(z) t^{n-1}. \end{aligned} \tag{3.11}$$

We get (3.10) by equating the coefficients of $t^{n-1}$ in (3.11). □

**Example 3.10** (*A Recursive Formula for* $J_n'(z)$)
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$J_{n-1}(z) - J_{n+1}(z) = 2J_n'(z). \tag{3.12}$$

***Proof*** Differentiating (3.7) with respect to $z$, we get

$$\begin{aligned}
&\frac{1}{2}\left(t - \frac{1}{t}\right) e^{z(t-1/t)/2} = \sum_{n=-\infty}^{\infty} J_n'(z)t^n \\
\Longrightarrow\ &\frac{1}{2}\left(t - \frac{1}{t}\right) \sum_{n=-\infty}^{\infty} J_n(z)t^n = \sum_{n=-\infty}^{\infty} J_n'(z)t^n \\
\Longrightarrow\ &\frac{1}{2} \sum_{n=-\infty}^{\infty} \left(J_{n-1}(z) - J_{n+1}(z)\right) t^n = \sum_{n=-\infty}^{\infty} J_n'(z)t^n \\
\Longrightarrow\ &\sum_{n=-\infty}^{\infty} \left(J_{n-1}(z) - J_{n+1}(z)\right) t^n = \sum_{n=-\infty}^{\infty} 2J_n'(z)t^n.
\end{aligned} \tag{3.13}$$

We get (3.12) by equating the coefficients of $t^n$ in (3.13). □

**Example 3.11** (*Bessel's Differential Equation for* $J_n(z)$)
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$z^2 \frac{d^2 J_n(z)}{dz^2} + z \frac{dJ_n(z)}{dz} + \left(z^2 - n^2\right) J_n(z) = 0. \tag{3.14}$$

***Proof*** Replacing $n$ with $n - 1$ in [3.12], we get

$$\frac{d}{dz}\left(z^{-n+1} J_{n-1}(z)\right) = -z^{-n+1} J_n(z). \tag{3.15}$$

Multiplying both sides of [3.11] by $z^{-2n+1}$, we get

$$z^{-2n+1} \frac{d}{dz}\left(z^n J_n(z)\right) = z^{-n+1} J_{n-1}(z). \tag{3.16}$$

Differentiating (3.16) with respect to $z$ and using (3.15), we get

$$\begin{aligned}
&\frac{d}{dz}\left(z^{-2n+1} \frac{d}{dz}\left(z^n J_n(z)\right)\right) = -z^{-n+1} J_n(z) \\
\Longleftrightarrow\ &\frac{d}{dz}\left(z^{-2n+1}\left(nz^{n-1} J_n(z) + z^n \frac{dJ_n(z)}{dz}\right)\right) = -z^{-n+1} J_n(z) \\
\Longleftrightarrow\ &\frac{d}{dz}\left(nz^{-n} J_n(z) + z^{-n+1} \frac{dJ_n(z)}{dz}\right) = -z^{-n+1} J_n(z)
\end{aligned}$$

$$\begin{aligned}
&\iff n\frac{d}{dz}\left(z^{-n}J_n(z)\right)+\frac{d}{dz}\left(z^{-n+1}\frac{dJ_n(z)}{dz}\right)=-z^{-n+1}J_n(z)\\
&\iff -n^2z^{-n-1}J_n(z)+nz^{-n}\frac{dJ_n(z)}{dz}+(-n+1)z^{-n}\frac{dJ_n(z)}{dz}\\
&\qquad +z^{-n+1}\frac{d^2J_n(z)}{dz}=-z^{-n+1}J_n(z)\\
&\iff z^2\frac{d^2J_n(z)}{dz^2}+z\frac{dJ_n(z)}{dz}+\left(z^2-n^2\right)J_n(z)=0.
\end{aligned}$$

□

## 3.4 The Bessel Coefficients $I_n(z)$

**Definition 3.12** (*The Bessel Coefficients* $I_n(z)$)
$\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$, we define the Bessel coefficients, $I_n(z)$, as

$$I_n(z)=i^{-n}J_n(iz). \tag{3.17}$$

**Example 3.13** (*Generating Function for* $I_n(z)$)
$\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$, prove that the Bessel coefficients $I_n(z)$ are defined by the following generating function

$$e^{z(t+1/t)/2}=\sum_{n=-\infty}^{\infty}I_n(z)t^n. \tag{3.18}$$

***Proof*** Replacing $z$ with $iz$ and $t$ with $-it$, respectively, in (3.7), and using (3.17), we get

$$\begin{aligned}
&e^{iz(-it-1/(-it))/2}=\sum_{n=-\infty}^{\infty}J_n(iz)(-it)^n\\
&\iff e^{iz(-it-i/t)/2}=\sum_{n=-\infty}^{\infty}i^nI_n(z)(-i)^nt^n\\
&\iff e^{z(t+1/t)/2}=\sum_{n=-\infty}^{\infty}I_n(z)t^n.
\end{aligned}$$

□

**Example 3.14** (*Power Series for* $I_n(z)$)
Prove that $\forall(z,n)\in\mathbb{C}\times(\mathbb{N}-1)$,

$$\begin{aligned}
I_n(z)&=\sum_{j=0}^{\infty}\frac{z^{n+2j}}{j!(n+j)!2^{n+2j}}\\
&=\frac{z^n}{n!2^n}+\frac{z^{n+2}}{1!(n+1)!2^{n+2}}+\frac{z^{n+4}}{2!(n+2)!2^{n+4}}+\frac{z^{n+6}}{3!(n+3)!2^{n+6}}+\cdots
\end{aligned} \tag{3.19}$$

***Proof*** From (3.18), we have

$$\begin{aligned}\sum_{n=-\infty}^{\infty} I_n(z)t^n &= e^{zt/2}\cdot e^{z/(2t)}\\ &= \sum_{k=0}^{\infty}\frac{z^k}{k!2^k}t^k\sum_{j=0}^{\infty}\frac{z^j}{j!2^j}t^{-j}\\ &= \sum_{k=0}^{\infty}\sum_{j=0}^{\infty}\frac{z^{k+j}}{k!j!2^{k+j}}t^{k-j}\\ &= \sum_{n=-\infty}^{\infty}\left(\sum_{j=0}^{\infty}\frac{z^{n+2j}}{j!(n+j)!2^{n+2j}}\right)t^n.\end{aligned} \tag{3.20}$$

We get (3.19) by equating the coefficient of $t^n$ in (3.20). □

**Example 3.15** (*A Recursive Formula for* $I_n(z)$)
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$I_{n-1}(z)-I_{n+1}(z)=\frac{2n}{z}I_n(z). \tag{3.21}$$

***Proof*** Differentiating (3.18) with respect to $t$, we get

$$\begin{aligned}&\frac{z}{2}\left(1-\frac{1}{t^2}\right)e^{z(t+1/t)/2}=\sum_{n=-\infty}^{\infty}nI_n(z)t^{n-1}\\ \Longrightarrow\ &\frac{z}{2}\left(1-\frac{1}{t^2}\right)\sum_{n=-\infty}^{\infty}I_n(z)t^n=\sum_{n=-\infty}^{\infty}nI_n(z)t^{n-1}\\ \Longrightarrow\ &\frac{z}{2}\sum_{n=-\infty}^{\infty}\left(I_{n-1}(z)-I_{n+1}(z)\right)t^{n-1}=\sum_{n=-\infty}^{\infty}nI_n(z)t^{n-1}\\ \Longrightarrow\ &\sum_{n=-\infty}^{\infty}\left(I_{n-1}(z)-I_{n+1}(z)\right)t^{n-1}=\sum_{n=-\infty}^{\infty}\frac{2n}{z}I_n(z)t^{n-1}.\end{aligned} \tag{3.22}$$

We get (3.21) by equating the coefficients of $t^{n-1}$ in (3.22). □

**Example 3.16** (*A Recursive Formula for* $I_n'(z)$)
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$I_{n-1}(z)+I_{n+1}(z)=2I_n'(z). \tag{3.23}$$

***Proof*** Differentiating (3.18) with respect to $z$, we get

$$\begin{aligned}
&\frac{1}{2}\left(t+\frac{1}{t}\right)e^{z(t+1/t)/2} = \sum_{n=-\infty}^{\infty} I_n'(z)t^n \\
\Longrightarrow\ &\frac{1}{2}\left(t+\frac{1}{t}\right)\sum_{n=-\infty}^{\infty} I_n(z)t^n = \sum_{n=-\infty}^{\infty} I_n'(z)t^n \\
\Longrightarrow\ &\frac{1}{2}\sum_{n=-\infty}^{\infty}\left(I_{n-1}(z)+I_{n+1}(z)\right)t^n = \sum_{n=-\infty}^{\infty} I_n'(z)t^n \\
\Longrightarrow\ &\sum_{n=-\infty}^{\infty}\left(I_{n-1}(z)+I_{n+1}(z)\right)t^n = \sum_{n=-\infty}^{\infty} 2I_n'(z)t^n.
\end{aligned} \tag{3.24}$$

We get (3.23) by equating the coefficients of $t^n$ in (3.24). □

**Example 3.17** (*Bessel's Differential Equation for* $I_n(z)$)
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$z^2\frac{d^2I_n(z)}{dz^2}+z\frac{dI_n(z)}{dz}-\left(z^2+n^2\right)I_n(z)=0.$$

***Proof*** Replacing $n$ with $n-1$ in [3.22], we get

$$\frac{d}{dz}\left(z^{-n+1}I_{n-1}(z)\right)=z^{-n+1}I_n(z). \tag{3.25}$$

Multiplying both sides of [3.21] by $z^{-2n+1}$, we get

$$z^{-2n+1}\frac{d}{dz}\left(z^nI_n(z)\right)=z^{-n+1}I_{n-1}(z). \tag{3.26}$$

Differentiating (3.26) with respect to $z$ and using (3.25), we get

$$\begin{aligned}
&\frac{d}{dz}\left(z^{-2n+1}\frac{d}{dz}\left(z^nI_n(z)\right)\right)=z^{-n+1}I_n(z) \\
\Longleftrightarrow\ &\frac{d}{dz}\left(z^{-2n+1}\left(nz^{n-1}I_n(z)+z^n\frac{dI_n(z)}{dz}\right)\right)=z^{-n+1}I_n(z) \\
\Longleftrightarrow\ &\frac{d}{dz}\left(nz^{-n}I_n(z)+z^{-n+1}\frac{dI_n(z)}{dz}\right)=z^{-n+1}I_n(z) \\
\Longleftrightarrow\ &n\frac{d}{dz}\left(z^{-n}I_n(z)\right)+\frac{d}{dz}\left(z^{-n+1}\frac{dI_n(z)}{dz}\right)=z^{-n+1}I_n(z)
\end{aligned}$$

$$\begin{aligned}
&\Longleftrightarrow -n^2 z^{-n-1} I_n(z) + n z^{-n} \frac{dI_n(z)}{dz} + (-n+1) z^{-n} \frac{dI_n(z)}{dz} \\
&\qquad + z^{-n+1} \frac{d^2 I_n(z)}{dz} = z^{-n+1} I_n(z) \\
&\Longleftrightarrow z^2 \frac{d^2 I_n(z)}{dz^2} + z \frac{dI_n(z)}{dz} - \left(z^2 + n^2\right) I_n(z) = 0.
\end{aligned}$$

□

## 3.5 The Bessel Functions $J_v(z)$

From Sect. 3.3, we notice that the Bessel coefficients $J_n(z)$ can be defined in three ways. The original definition of $J_n(z)$ is the generating function (3.7). An alternative way of defining $J_n(z)$ is by the power series (3.8). We can also define $J_n(z)$ as a solution of Bessel's differential equation (3.14).

In this section, we will generalize the power series of $J_n(z)$ (3.8) to define the Bessel functions $J_v(z)$ for all complex values of $v$. As a special case, when $v = n$ (an integer), the Bessel functions $J_v(z)$ will become the Bessel coefficients $J_n(z)$.

**Definition 3.18** (*The Bessel Functions* $J_v(z)$)
$\forall (z, v) \in \mathbb{C} \times (\mathbb{C} - (\mathbb{Z} - \mathbb{N}))$, the Bessel functions, $J_v(z)$, are defined by the following power series:

$$\begin{aligned}
J_v(z) &= \sum_{j=0}^{\infty} (-1)^j \frac{z^{v+2j}}{j!\Gamma(v+j+1)2^{v+2j}} \\
&= \frac{z^v}{\Gamma(v+1)2^v} - \frac{z^{v+2}}{1!\Gamma(v+2)2^{v+2}} + \frac{z^{v+4}}{2!\Gamma(v+3)2^{v+4}} - \frac{z^{v+6}}{3!\Gamma(v+4)2^{v+6}} + \cdots
\end{aligned} \tag{3.27}$$

$\forall (z, v) \in \mathbb{C} \times (\mathbb{Z} - \mathbb{N})$, we define the Bessel functions, $J_v(z)$, as the corresponding Bessel coefficients $J_v(z)$.

**Example 3.19** (*A Recursive Formula for* $J_v(z)$)
Prove that $\forall (z, v) \in \mathbb{C}^2$,

$$J_{v-1}(z) + J_{v+1}(z) = \frac{2v}{z} J_v(z). \tag{3.28}$$

***Proof*** From [3.29], we have

$$\begin{aligned}
z^v J_{v-1}(z) &= \frac{d}{dz}\left(z^v J_v(z)\right) \\
&= v z^{v-1} J_v(z) + z^v J_v'(z) \\
\Longleftrightarrow J_{v-1}(z) &= \frac{v}{z} J_v(z) + J_v'(z).
\end{aligned} \tag{3.29}$$

From [3.30], we have

$$\begin{aligned} -z^{-v}J_{v+1}(z) &= \frac{d}{dz}\left(z^{-v}J_v(z)\right) \\ &= -vz^{-v-1}J_v(z) + z^{-v}J_v'(z) \\ \Longleftrightarrow J_{v+1}(z) &= \frac{v}{z}J_v(z) - J_v'(z). \end{aligned} \tag{3.30}$$

We get (3.28) by adding (3.29) and (3.30). □

**Example 3.20** (*A Recursive Formula for* $J_v'(z)$)
Prove that $\forall(z, v) \in \mathbb{C}^2$,

$$J_{v-1}(z) - J_{v+1}(z) = 2J_v'(z). \tag{3.31}$$

***Proof*** We get (3.31) by subtracting (3.30) from (3.29). □

**Example 3.21** (*Bessel's Differential Equation for* $J_v(z)$)
Prove that $\forall(z, v) \in \mathbb{C}^2$,

$$z^2\frac{d^2J_v(z)}{dz^2} + z\frac{dJ_v(z)}{dz} + \left(z^2 - v^2\right)J_v(z) = 0.$$

***Proof*** Using [3.29] and [3.30], the proof is identical to Example 3.11. □

## 3.6 The Bessel Functions $I_v(z)$

**Definition 3.22** (*The Bessel Functions* $I_v(z)$)
$\forall(z, v) \in \mathbb{C} \times (\mathbb{C} - (\mathbb{Z} - \mathbb{N}))$, the Bessel functions, $I_v(z)$, are defined by the following power series:

$$\begin{aligned} I_v(z) &= \sum_{j=0}^{\infty} \frac{z^{v+2j}}{j!\Gamma(v+j+1)2^{v+2j}} \\ &= \frac{z^v}{\Gamma(v+1)2^v} + \frac{z^{v+2}}{1!\Gamma(v+2)2^{v+2}} + \frac{z^{v+4}}{2!\Gamma(v+3)2^{v+4}} + \frac{z^{v+6}}{3!\Gamma(v+4)2^{v+6}} + \cdots \end{aligned} \tag{3.32}$$

$\forall(z, v) \in \mathbb{C} \times (\mathbb{Z} - \mathbb{N})$, we define the Bessel functions, $I_v(z)$, as the corresponding Bessel coefficients $I_v(z)$.

**Example 3.23** (*A Recursive Formula for* $I_v(z)$)
Prove that $\forall(z, v) \in \mathbb{C}^2$,

$$I_{v-1}(z) - I_{v+1}(z) = \frac{2v}{z}I_v(z). \tag{3.33}$$

***Proof*** From [3.31], we have

$$\begin{aligned} z^{v} I_{v-1}(z) &= \frac{d}{dz}\left(z^{v} I_{v}(z)\right) \\ &= v z^{v-1} I_{v}(z) + z^{v} I_{v}'(z) \\ \Longleftrightarrow I_{v-1}(z) &= \frac{v}{z} I_{v}(z) + I_{v}'(z). \end{aligned} \tag{3.34}$$

From [3.32], we have

$$\begin{aligned} z^{-v} I_{v+1}(z) &= \frac{d}{dz}\left(z^{-v} I_{v}(z)\right) \\ &= -v z^{-v-1} I_{v}(z) + z^{-v} I_{v}'(z) \\ \Longleftrightarrow I_{v+1}(z) &= -\frac{v}{z} I_{v}(z) + I_{v}'(z). \end{aligned} \tag{3.35}$$

We get (3.33) by subtracting (3.35) from (3.34). □

**Example 3.24** (*A Recursive Formula for* $I_v'(z)$)
Prove that $\forall (z, v) \in \mathbb{C}^2$,

$$I_{v-1}(z) + I_{v+1}(z) = 2 J_{v}'(z). \tag{3.36}$$

***Proof*** We get (3.36) by adding (3.34) and (3.35). □

**Example 3.25** (*Bessel's Differential Equation for* $I_v(z)$)
Prove that $\forall (z, v) \in \mathbb{C}^2$,

$$z^2 \frac{d^2 I_v(z)}{dz^2} + z \frac{d I_v(z)}{dz} - \left(z^2 + v^2\right) I_v(z) = 0.$$

***Proof*** Using [3.31] and [3.32], the proof is identical to Example 3.17. □

## Problems

[3.1] (Whittaker and Watson, 2016) Prove that $\forall n \in \mathbb{N}$,

$$H(n) = \int_0^1 \frac{1 - (1-t)^n}{t}\, dt.$$

[3.2] (Whittaker and Watson, 2016) Prove that

$$\gamma = \lim_{n\to\infty} \left( \int_0^1 \frac{1-(1-t/n)^n}{t}\, dt - \int_1^n \frac{(1-t/n)^n}{t}\, dt \right).$$

[3.3] (Whittaker and Watson, 2016) Prove that

$$\gamma = \int_0^1 \left(1-e^{-t}\right) \frac{dt}{t} - \int_1^\infty \frac{e^{-t}}{t}\, dt.$$

[3.4] (Whittaker and Watson, 2016) Prove that

$$\Gamma(1) = 1.$$

[3.5] (Whittaker and Watson, 2016) Prove that

$$\Gamma'(1) = -\gamma.$$

[3.6] (Whittaker and Watson, 2016) Prove that $\forall n \in \mathbb{N}$,

$$\Gamma(n) = (n-1)!.$$

[3.7] (Whittaker and Watson, 2016) Prove that

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

[3.8] (Whittaker and Watson, 2016) Prove that $\forall z \in \mathbb{C} - (\mathbb{Z} - \mathbb{N})$,

$$\Gamma(z) = \frac{1}{z} \lim_{n\to\infty} n^z \prod_{j=1}^{n-1} \frac{j}{z+j}.$$

[3.9] (Watson, 2011) Prove that $\forall(z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$J_{-n}(z) = (-1)^n J_n(z).$$

[3.10] (Watson, 2011) Prove that $\forall z \in \mathbb{C}$,

$$J_0'(z) = -J_1(z).$$

[3.11] (Watson, 2011) Prove that $\forall(z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\frac{d}{dz}\left(z^n J_n(z)\right) = z^n J_{n-1}(z).$$

[3.12] (Watson, 2011) Prove that $\forall(z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\frac{d}{dz}\left(z^{-n} J_n(z)\right) = -z^{-n} J_{n+1}(z).$$

[3.13] (Watson, 2011) Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\begin{aligned}\cos(z \sin\theta) &= J_0(z) + 2\sum_{j=1}^{\infty} J_{2j}(z)\cos 2j\theta \\ &= J_0(z) + 2\left(J_2(z)\cos 2\theta + J_4(z)\cos 4\theta + J_6(z)\cos 6\theta + J_8(z)\cos 8\theta + \cdots\right).\end{aligned}$$

[3.14] (Watson, 2011) Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\begin{aligned}\sin(z \sin\theta) &= 2\sum_{j=0}^{\infty} J_{2j+1}(z)\sin(2j+1)\theta \\ &= 2\left(J_1(z)\sin\theta + J_3(z)\sin 3\theta + J_5(z)\sin 5\theta + J_7(z)\sin 7\theta + \cdots\right).\end{aligned}$$

[3.15] (Watson, 2011) Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\cos(z\cos\theta) = J_0(z) + 2\sum_{j=1}^{\infty}(-1)^j J_{2j}(z)\cos 2j\theta$$
$$= J_0(z) - 2\left(J_2(z)\cos 2\theta - J_4(z)\cos 4\theta + J_6(z)\cos 6\theta - J_8(z)\cos 8\theta + \cdots\right).$$

[3.16] (Watson, 2011) Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\sin(z\cos\theta) = 2\sum_{j=0}^{\infty}(-1)^j J_{2j+1}(z)\cos(2j+1)\theta$$
$$= 2\left(J_1(z)\cos\theta - J_3(z)\cos 3\theta + J_5(z)\cos 5\theta - J_7(z)\cos 7\theta + \cdots\right).$$

[3.17] (Watson, 2011) Prove that $\forall z \in \mathbb{C}$,

$$z\sin z = 2\sum_{j=1}^{\infty}(-1)^{j-1}(2j)^2 J_{2j}(z)$$
$$= 2\left(2^2 J_2(z) - 4^2 J_4(z) + 6^2 J_6(z) - 8^2 J_8(z) + \cdots\right).$$

[3.18] (Watson, 2011) Prove that $\forall z \in \mathbb{C}$,

$$z\cos z = 2\sum_{j=0}^{\infty}(-1)^j(2j+1)^2 J_{2j+1}(z)$$
$$= 2\left(1^2 J_1(z) - 3^2 J_3(z) + 5^2 J_5(z) - 7^2 J_7(z) + \cdots\right).$$

[3.19] Prove that $\forall(z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$I_{-n}(z) = I_n(z).$$

[3.20] Prove that $\forall z \in \mathbb{C}$,

$$I_0'(z) = I_1(z).$$

[3.21] (Whittaker and Watson, 2016) Prove that $\forall(z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\frac{d}{dz}\left(z^n I_n(z)\right) = z^n I_{n-1}(z).$$

[3.22] (Whittaker and Watson, 2016) Prove that $\forall(z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\frac{d}{dz}\left(z^{-n} I_n(z)\right) = z^{-n} I_{n+1}(z).$$

[3.23] Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\begin{aligned}\cosh(z\sin\theta) &= I_0(z) + 2\sum_{j=1}^{\infty}(-1)^j I_{2j}(z)\cos 2j\theta \\ &= I_0(z) - 2\left(I_2(z)\cos 2\theta - I_4(z)\cos 4\theta + I_6(z)\cos 6\theta\ - I_8(z)\cos 8\theta + \cdots\right).\end{aligned}$$

[3.24] Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\begin{aligned}\sinh(z\sin\theta) &= 2\sum_{j=0}^{\infty}(-1)^j I_{2j+1}(z)\sin(2j+1)\theta \\ &= 2\left(I_1(z)\sin\theta - I_3(z)\sin 3\theta + I_5(z)\sin 5\theta - I_7(z)\sin 7\theta + \cdots\right).\end{aligned}$$

[3.25] Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\begin{aligned}\cosh(z\cos\theta) &= I_0(z) + 2\sum_{j=1}^{\infty} I_{2j}(z)\cos 2j\theta \\ &= I_0(z) + 2\left(I_2(z)\cos 2\theta + I_4(z)\cos 4\theta + I_6(z)\cos 6\theta\ + I_8(z)\cos 8\theta + \cdots\right).\end{aligned}$$

[3.26] Prove that $\forall(z, \theta) \in \mathbb{C} \times \mathbb{R}$,

$$\begin{aligned}\sinh(z\cos\theta) &= 2\sum_{j=0}^{\infty} I_{2j+1}(z)\cos(2j+1)\theta \\ &= 2\left(I_1(z)\cos\theta + I_3(z)\cos 3\theta + I_5(z)\cos 5\theta + I_7(z)\cos 7\theta + \cdots\right).\end{aligned}$$

[3.27] Prove that $\forall z \in \mathbb{C}$,

$$\begin{aligned} z \sinh z &= 2 \sum_{j=1}^{\infty} (2j)^2 I_{2j}(z) \\ &= 2 \left( 2^2 I_2(z) + 4^2 I_4(z) + 6^2 I_6(z) + 8^2 I_8(z) + \cdots \right). \end{aligned}$$

[3.28] Prove that $\forall z \in \mathbb{C}$,

$$\begin{aligned} z \cosh z &= 2 \sum_{j=0}^{\infty} (2j+1)^2 I_{2j+1}(z) \\ &= 2 \left( 1^2 I_1(z) + 3^2 I_3(z) + 5^2 I_5(z) + 7^2 I_7(z) + \cdots \right). \end{aligned}$$

[3.29] (Watson, 2011) Prove that $\forall (z, v) \in \mathbb{C}^2$,

$$\frac{d}{dz} \left( z^v J_v(z) \right) = z^v J_{v-1}(z).$$

[3.30] (Watson, 2011) Prove that $\forall (z, v) \in \mathbb{C}^2$,

$$\frac{d}{dz} \left( z^{-v} J_v(z) \right) = -z^{-v} J_{v+1}(z).$$

[3.31] (Watson, 2011) Prove that $\forall (z, v) \in \mathbb{C}^2$,

$$\frac{d}{dz} \left( z^v I_v(z) \right) = z^v I_{v-1}(z).$$

[3.32] (Watson, 2011) Prove that $\forall (z, v) \in \mathbb{C}^2$,

$$\frac{d}{dz} \left( z^{-v} I_v(z) \right) = z^{-v} I_{v+1}(z).$$

# Chapter 4
# Special Integrals




In this chapter, we will prove several theorems for finding the integral representation of the coefficients of a series. We will also find integral representations of various sums involving the coefficients of a series. We will also derive an integral related to the Fourier series. The problems list several special integrals and series sums derived using the theorems and corollaries proved in this chapter.

## 4.1 Special Integrals 1

To begin with, we will use the complex variables, $e^{i\theta}$ and $e^{-i\theta}$, to find an integral representation of the constant term in a series.

**Theorem 4.1** (Integral Representation of the Constant of a Series)
*Let*

$$f(z) = \sum_{j=-\infty}^{\infty} a_j z^j, \tag{4.1}$$

*be a function in which $f(z)$ can contain all possible integral powers of $z$ (positive, negative or zero). The series may be finite or infinite. In case it is infinite, $f\left(e^{i\theta}\right)$ and $f\left(e^{-i\theta}\right)$ both should be convergent for $\theta \in (0, \pi)$. Then, the constant $a_0$ is given by*

$$a_0 = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)}{2}\, d\theta.$$

***Proof*** Putting $z = e^{i\theta}$ in (4.1), we get

$$f\left(e^{i\theta}\right) = a_0 + \sum_{j=-\infty, j\neq 0}^{\infty} a_j e^{ij\theta}. \tag{4.2}$$

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Putting $z = e^{-i\theta}$ in (4.1), we get

$$f\left(e^{-i\theta}\right) = a_0 + \sum_{j=-\infty, j\neq 0}^{\infty} a_j e^{-ij\theta}. \tag{4.3}$$

Adding (4.2) and (4.3), and dividing the result by 2, we get

$$\begin{aligned} \frac{f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)}{2} &= a_0 + \sum_{j=-\infty, j\neq 0}^{\infty} a_j \frac{e^{ij\theta} + e^{-ij\theta}}{2} \\ &= a_0 + \sum_{j=-\infty, j\neq 0}^{\infty} a_j \cos j\theta. \end{aligned} \tag{4.4}$$

Integrating (4.4) with respect to $\theta$ from 0 to $\pi$, and dividing the result by $\pi$, we get

$$\begin{aligned} \frac{1}{\pi}\int_0^{\pi} \frac{f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)}{2} d\theta &= \frac{1}{\pi}\int_0^{\pi} \left(a_0 + \sum_{j=-\infty, j\neq 0}^{\infty} a_j \cos j\theta\right) d\theta \\ &= \frac{1}{\pi}\int_0^{\pi} a_0\, d\theta + \frac{1}{\pi}\sum_{j=-\infty, j\neq 0}^{\infty} \int_0^{\pi} a_j \cos j\theta d\theta \\ &= a_0 + \frac{1}{\pi}\sum_{j=-\infty, j\neq 0}^{\infty} \frac{a_j \sin j\theta}{j}\Bigg|_0^{\pi} \\ &= a_0. \end{aligned}$$

□

Now, we generalize the previous result to find three equivalent integral representations of the coefficients of a series.

**Theorem 4.2** (First Generalization of Theorem 4.1)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible integral powers of $z$ (positive, negative, or zero). The series may be finite or infinite. In case it is infinite, $f\left(e^{i\theta}\right)$ and $f\left(e^{-i\theta}\right)$ both should be convergent for $\theta \in (0, \pi)$. Then, $\forall n \in \mathbb{Z}$, the coefficient of $z^n$, $a_n$ is given by*

$$a_n = \frac{1}{\pi}\int_0^{\pi} \frac{e^{-in\theta} f\left(e^{i\theta}\right) + e^{in\theta} f\left(e^{-i\theta}\right)}{2} d\theta. \tag{4.5}$$

***Proof*** Putting $z = e^{i\theta}$ in (4.1), and multiplying by $e^{-in\theta}$, we get

$$e^{-ni\theta} f\left(e^{i\theta}\right) = a_n + \sum_{j=-\infty, j\neq n}^{\infty} a_j e^{i(j-n)\theta}. \tag{4.6}$$

Putting $z = e^{-i\theta}$ in (4.1), and multiplying by $e^{in\theta}$ we get

$$e^{in\theta} f\left(e^{-i\theta}\right) = a_n + \sum_{j=-\infty, j\neq n}^{\infty} a_j e^{-i(j-n)\theta}. \tag{4.7}$$

Adding (4.6) and (4.7), and dividing the result by 2, we get

$$\begin{aligned} \frac{e^{-in\theta} f\left(e^{i\theta}\right) + e^{in\theta} f\left(e^{-i\theta}\right)}{2} &= a_n + \sum_{j=-\infty, j\neq n}^{\infty} a_j \frac{e^{i(j-n)\theta} + e^{-i(j-n)\theta}}{2} \\ &= a_n + \sum_{j=-\infty, j\neq n}^{\infty} a_j \cos(j-n)\theta. \end{aligned} \tag{4.8}$$

Integrating (4.8) with respect to $\theta$ from 0 to $\pi$, and dividing the result by $\pi$, we get

$$\begin{aligned} &\frac{1}{\pi}\int_0^{\pi} \frac{e^{-in\theta} f\left(e^{i\theta}\right) + e^{in\theta} f\left(e^{-i\theta}\right)}{2}\, d\theta \\ &= \frac{1}{\pi}\int_0^{\pi} a_n\, d\theta + \frac{1}{\pi} \sum_{j=-\infty, j\neq n}^{\infty} \int_0^{\pi} a_j \cos(j-n)\theta\, d\theta \\ &= a_n + \frac{1}{\pi} \sum_{j=-\infty, j\neq n}^{\infty} \frac{a_j \sin(j-n)\theta}{j-n}\Bigg|_0^{\pi} \\ &= a_n. \end{aligned}$$

□

**Corollary 4.3** (Corollary to Theorem 4.2)
*Let $f(z)$ be the function given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f\left(e^{i\theta}\right)$ and $f\left(e^{-i\theta}\right)$ both should be convergent for $\theta \in (0, \pi)$. Then, $\forall n \in \mathbb{N} - 1$,*

$$\frac{f^{(n)}(0)}{n!} = \frac{1}{\pi}\int_0^{\pi} \frac{e^{-in\theta} f(e^{i\theta}) + e^{in\theta} f(e^{-i\theta})}{2}\, d\theta.$$

***Proof*** In (4.5), we have $a_n = f^{(n)}(0)/n!$ by Maclaurin's series. □

**Theorem 4.4** (Second Generalization of Theorem 4.1)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible integral powers of $z$ (positive, negative, or zero). The series may be finite or infinite. In case it is infinite, $f\left(e^{i\theta}\right)$ and $f\left(e^{-i\theta}\right)$ both should be convergent for $\theta \in (0, \pi)$. Then, $\forall n \in \mathbb{N}$,*

$$a_n + a_{-n} = \frac{2}{\pi}\int_0^{\pi} \frac{f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)}{2} \cos n\theta\, d\theta.$$

***Proof*** Multiplying (4.4) by $(2/\pi)\cos n\theta$, and integrating the result with respect to $\theta$ from 0 to $\pi$, we get

$$\begin{aligned}
&\frac{2}{\pi}\int_0^{\pi}\frac{f\left(e^{i\theta}\right)+f\left(e^{-i\theta}\right)}{2}\cos n\theta\,d\theta=\frac{2}{\pi}\int_0^{\pi}\left(\sum_{j=-\infty}^{\infty}a_j\cos j\theta\cos n\theta\right)d\theta\\
&=\frac{1}{\pi}\sum_{j=-\infty}^{\infty}a_j\int_0^{\pi}2\cos j\theta\cos n\theta\,d\theta\\
&=\frac{1}{\pi}\sum_{j=-\infty}^{\infty}a_j\int_0^{\pi}(\cos(j-n)\theta+\cos(j+n)\theta)\,d\theta\\
&=\frac{a_n}{\pi}\int_0^{\pi}(1+\cos 2n\theta)\,d\theta+\frac{a_{-n}}{\pi}\int_0^{\pi}(1+\cos(-2n\theta))\,d\theta\\
&\quad+\frac{1}{\pi}\sum_{j=-\infty,j\neq\pm n}^{\infty}a_j\int_0^{\pi}(\cos(j-n)\theta+\cos(j+n)\theta)\,d\theta\\
&=\frac{a_n}{\pi}\left(\theta+\frac{\sin 2n\theta}{2n}\right)\Big|_0^{\pi}+\frac{a_{-n}}{\pi}\left(\theta+\frac{\sin 2n\theta}{2n}\right)\Big|_0^{\pi}\\
&\quad+\sum_{j=-\infty,j\neq\pm n}^{\infty}\frac{a_j}{\pi}\left(\frac{\sin(j-n)\theta}{j-n}+\frac{\sin(j+n)\theta}{j+n}\right)\Big|_0^{\pi}\\
&=a_n+a_{-n}.
\end{aligned}$$

□

After finding integral representations of the coefficients of a series, we will focus our attention on finding sums involving the coefficients of a series. As a first step, we will find the sum of squares of the coefficients. Then, later, we will generalize the result to find a general quadratic sum involving the coefficients.

**Theorem 4.5** (Finding Sum of Squares of Terms of a Series)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f\left(e^{i\theta}\right)$ and $f\left(e^{-i\theta}\right)$ both should be convergent for $\theta\in(0,\pi)$. Then*

$$\begin{aligned}
\sum_{j=0}^{\infty}a_j^2&=a_0^2+a_1^2+a_2^2+a_3^2+\cdots\\
&=\frac{1}{\pi}\int_0^{\pi}f\left(e^{i\theta}\right)f\left(e^{-i\theta}\right)d\theta.
\end{aligned}$$

***Proof*** Multiplying (4.2) and (4.3), we get

$$\begin{aligned} f\left(e^{i\theta}\right) f\left(e^{-i\theta}\right) &= \sum_{j=-\infty}^{\infty} a_j e^{ij\theta} \sum_{k=-\infty}^{\infty} a_k e^{-ik\theta} \\ &= \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} a_j a_k e^{i(j-k)\theta} \\ &= \sum_{j=0}^{\infty} a_j^2 + \sum_{l=1}^{\infty}\sum_{j=0}^{\infty}\left(a_{l+j}a_j e^{il\theta} + a_j a_{l+j} e^{-il\theta}\right) \\ &= \sum_{j=0}^{\infty} a_j^2 + 2\sum_{l=1}^{\infty}\sum_{j=0}^{\infty} a_j a_{l+j}\cos l\theta. \end{aligned} \tag{4.9}$$

Integrating (4.9) with respect to $\theta$ from 0 to $\pi$, and dividing the result by $\pi$, we get

$$\begin{aligned} &\frac{1}{\pi}\int_0^{\pi} f\left(e^{i\theta}\right) f\left(e^{-i\theta}\right) d\theta \\ &= \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty} a_j^2\right) d\theta + \frac{2}{\pi}\int_0^{\pi}\left(\sum_{l=1}^{\infty}\sum_{j=0}^{\infty} a_j a_{l+j}\cos l\theta\right) d\theta \\ &= \sum_{j=0}^{\infty} a_j^2 + \frac{2}{\pi}\sum_{l=1}^{\infty}\sum_{j=0}^{\infty} a_j a_{l+j}\left.\frac{\sin l\theta}{l}\right|_0^{\pi} \\ &= \sum_{j=0}^{\infty} a_j^2. \end{aligned}$$

□

**Theorem 4.6** (Generalization of Theorem 4.5)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f\left(e^{i\theta}\right)$ and $f\left(e^{-i\theta}\right)$ both should be convergent for $\theta \in (0, \pi)$. Then $\forall n \in \mathbb{N} - 1$,*

$$\begin{aligned} \sum_{j=0}^{\infty} a_j a_{n+j} &= a_0 a_n + a_1 a_{n+1} + a_2 a_{n+2} + a_3 a_{n+3} + \cdots \\ &= \frac{1}{\pi}\int_0^{\pi} f\left(e^{i\theta}\right) f\left(e^{-i\theta}\right)\cos n\theta\, d\theta. \end{aligned}$$

***Proof*** The $n = 0$ case is Theorem 4.5. We will prove the theorem for $n \in \mathbb{N}$. Assuming $n \in \mathbb{N}$, multiplying (4.9) by $(\cos n\theta)/\pi$, and integrating the result with respect to $\theta$ from 0 to $\pi$, we get

$$
\begin{aligned}
&\frac{1}{\pi}\int_0^{\pi} f\left(e^{i\theta}\right) f\left(e^{-i\theta}\right)\cos n\theta\, d\theta \\
&= \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty} a_j^2\cos n\theta + \sum_{l=1}^{\infty}\sum_{j=0}^{\infty} 2a_j a_{l+j}\cos l\theta\cos n\theta\right) d\theta \\
&= \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty} a_j^2\cos n\theta\right) d\theta + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty} 2a_j a_{n+j}\cos^2 n\theta\right) d\theta \\
&\quad + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{l=1,l\neq n}^{\infty}\sum_{j=0}^{\infty} 2a_j a_{l+j}\cos l\theta\cos n\theta\right) d\theta \\
&= \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty} a_j^2\cos n\theta\right) d\theta + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty} a_j a_{n+j}(1+\cos 2n\theta)\right) d\theta \\
&\quad + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{l=1,l\neq n}^{\infty}\sum_{j=0}^{\infty} a_j a_{l+j}(\cos(l-n)\theta + \cos(l+n)\theta)\right) d\theta \\
&= \frac{1}{\pi}\sum_{j=0}^{\infty} a_j^2\frac{\sin n\theta}{n}\bigg|_0^{\pi} + \frac{1}{\pi}\sum_{j=0}^{\infty} a_j a_{n+j}\left(\theta + \frac{\sin 2n\theta}{2n}\right)\bigg|_0^{\pi} \\
&\quad + \frac{1}{\pi}\sum_{l=1,l\neq n}^{\infty}\sum_{j=0}^{\infty} a_j a_{l+j}\left(\frac{\sin(l-n)\theta}{l-n} + \frac{\sin(l+n)\theta}{l+n}\right)\bigg|_0^{\pi} \\
&= \sum_{j=0}^{\infty} a_j a_{n+j}.
\end{aligned}
$$

□

Now, we will find series sums involving the central binomial coefficients of the form $\binom{2j}{j}$. Then, later, we will find generalized sums involving the binomial coefficients of the form $\binom{n+2j}{j}$.

**Theorem 4.7** (A Series Involving Central Binomial Coefficients)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f(2x\cos\theta)$ should be convergent for $\theta \in (0, \pi)$. Let*

$$
A_0(x) = \sum_{j=0}^{\infty}\binom{2j}{j} a_{2j}x^{2j} = a_0 + \binom{2}{1}a_2x^2 + \binom{4}{2}a_4x^4 + \binom{6}{3}a_6x^6 + \cdots, \tag{4.10}
$$

*then, $A_0(x)$ is given by*

$$A_0(x) = \frac{1}{\pi}\int_0^{\pi} f\left(2x\cos\theta\right) d\theta.$$

***Proof*** Replacing $z$ with $2x\cos\theta = x\left(e^{i\theta} + e^{-i\theta}\right)$ in (4.1), we get

$$\begin{aligned}
&f(2x\cos\theta) = f\left(x\left(e^{i\theta} + e^{-i\theta}\right)\right)\\
&= \sum_{j=0}^{\infty} a_j \left(x\left(e^{i\theta} + e^{-i\theta}\right)\right)^j\\
&= \sum_{j=0}^{\infty} a_{2j} \left(x\left(e^{i\theta} + e^{-i\theta}\right)\right)^{2j} + \sum_{j=0}^{\infty} a_{2j+1} \left(x\left(e^{i\theta} + e^{-i\theta}\right)\right)^{2j+1}\\
&= \sum_{j=0}^{\infty} \binom{2j}{j} a_{2j} x^{2j} + \sum_{j=0}^{\infty} a_{2j} x^{2j} \sum_{k=0}^{j-1} \left(\binom{2j}{k} e^{ik\theta} e^{-i(2j-k)\theta}\right.\\
&\quad \left. + \binom{2j}{2j-k} e^{i(2j-k)\theta} e^{-ik\theta}\right) + \sum_{j=0}^{\infty} a_{2j+1} x^{2j+1} \sum_{k=0}^{j} \left(\binom{2j+1}{k} e^{ik\theta} e^{-i(2j+1-k)\theta}\right.\\
&\quad \left. + \binom{2j+1}{2j+1-k} e^{i(2j+1-k)\theta} e^{-ik\theta}\right)\\
&= \sum_{j=0}^{\infty} \binom{2j}{j} a_{2j} x^{2j} + \sum_{j=0}^{\infty} a_{2j} x^{2j} \sum_{k=0}^{j-1} \binom{2j}{k} \left(e^{-i(2j-2k)\theta} + e^{i(2j-2k)\theta}\right)\\
&\quad + \sum_{j=0}^{\infty} a_{2j+1} x^{2j+1} \sum_{k=0}^{j} \binom{2j+1}{k} \left(e^{-i(2j-2k+1)\theta} + e^{i(2j-2k+1)\theta}\right)\\
&= \sum_{j=0}^{\infty} \binom{2j}{j} a_{2j} x^{2j} + 2\sum_{j=0}^{\infty} a_{2j} x^{2j} \sum_{k=0}^{j-1} \binom{2j}{k} \cos(2j-2k)\theta\\
&\quad + 2\sum_{j=0}^{\infty} a_{2j+1} x^{2j+1} \sum_{k=0}^{j} \binom{2j+1}{k} \cos(2j-2k+1)\theta.
\end{aligned} \tag{4.11}$$

Dividing (4.11) by $\pi$, integrating the result with respect to $\theta$ from 0 to $\pi$, and using (4.10), we get

$$\frac{1}{\pi}\int_0^{\pi} f\,(2x\cos\theta)\;d\theta = \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\right)d\theta$$
$$+\frac{1}{\pi}\int_0^{\pi}\left(2\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}\binom{2j}{k}\cos(2j-2k)\theta\right)d\theta$$
$$+\frac{1}{\pi}\int_0^{\pi}\left(2\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0}^{j}\binom{2j+1}{k}\cos(2j-2k+1)\theta\right)d\theta$$

$$=\frac{1}{\pi}\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\int_0^{\pi}d\theta+\frac{2}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}\binom{2j}{k}\int_0^{\pi}\cos(2j-2k)\theta\,d\theta$$
$$+\frac{2}{\pi}\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0}^{j}\binom{2j+1}{k}\int_0^{\pi}\cos(2j-2k+1)\theta\,d\theta$$
$$=\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}+\frac{2}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}\binom{2j}{k}\frac{\sin(2j-2k)\theta}{2j-2k}\Bigg|_0^{\pi}$$
$$+\frac{2}{\pi}\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0}^{j}\binom{2j+1}{k}\frac{\sin(2j-2k+1)\theta}{2j-2k+1}\Bigg|_0^{\pi}$$
$$=\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}$$
$$=A_0(x).$$

□

**Theorem 4.8** (Generalization of Theorem 4.7)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f(2x\cos\theta)$ should be convergent for $\theta\in(0,\pi)$. $\forall n\in\mathbb{N}-1$, let*

$$A_n(x)=\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}$$
$$=\binom{n}{0}a_nx^n+\binom{n+2}{1}a_{n+2}x^{n+2}+\binom{n+4}{2}a_{n+4}x^{n+4}+\binom{n+6}{3}a_{n+6}x^{n+6}+\cdots, \tag{4.12}$$

*then, $A_n(x)$ is given by*

$$A_n(x) = \frac{1}{\pi}\int_0^{\pi} f\,(2x\cos\theta)\cos n\theta\,d\theta. \tag{4.13}$$

***Proof*** Multiplying (4.11) by $\cos n\theta$, we get

$$\begin{aligned}
&f\,(2x\cos\theta)\cos n\theta\\
&=\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\cos n\theta+2\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}\binom{2j}{k}\cos(2j-2k)\theta\cos n\theta\\
&\quad+2\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0}^{j}\binom{2j+1}{k}\cos(2j-2k+1)\theta\cos n\theta\\
&=\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\cos n\theta+2\sum_{k=0}^{\infty}\binom{n+2k}{k}a_{n+2k}x^{n+2k}\cos^2 n\theta\\
&\quad+2\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0,2j-2k\neq n}^{j-1}\binom{2j}{k}\cos(2j-2k)\theta\cos n\theta\\
&\quad+2\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}\binom{2j+1}{k}\cos(2j-2k+1)\theta\cos n\theta\\
&=\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\cos n\theta+\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}(1+\cos 2n\theta)\\
&\quad+\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0,2j-2k\neq n}^{j-1}\binom{2j}{k}(\cos(2j-2k+n)\theta+\cos(2j-2k-n)\theta)\\
&\quad+\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\\
&\quad\sum_{k=0,2j-2k+1\neq n}^{j}\binom{2j+1}{k}(\cos(2j-2k+1+n)\theta+\cos(2j-2k+1-n)\theta).
\end{aligned} \tag{4.14}$$

Dividing (4.14) by $\pi$, integrating the result with respect to $\theta$ from 0 to $\pi$, and using (4.12), we get

$$
\begin{aligned}
&\frac{1}{\pi}\int_0^\pi f\,(2x\cos\theta)\cos n\theta\,d\theta\\
&=\frac{1}{\pi}\int_0^\pi\left(\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\cos n\theta\right)d\theta\\
&\quad+\frac{1}{\pi}\int_0^\pi\left(\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}(1+\cos 2n\theta)\right)d\theta+\frac{1}{\pi}\int_0^\pi\left(\sum_{j=0}^{\infty}a_{2j}x^{2j}\right.\\
&\quad\left.\sum_{k=0,2j-2k\neq n}^{j-1}\binom{2j}{k}(\cos(2j-2k+n)\theta+\cos(2j-2k-n)\theta)\right)d\,\theta\\
&\quad+\frac{1}{\pi}\int_0^\pi\left(\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}\binom{2j+1}{k}\right.\\
&\quad\left.(\cos(2j-2k+1+n)\theta+\cos(2j-2k+1-n)\theta)\right)d\theta\\
&=\frac{1}{\pi}\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\int_0^\pi\cos n\theta\,d\theta\\
&\quad+\frac{1}{\pi}\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}\int_0^\pi(1+\cos 2n\theta)\,d\theta+\frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j}\\
&\quad\sum_{k=0,2j-2k\neq n}^{j-1}\binom{2j}{k}\int_0^\pi(\cos(2j-2k+n)\theta+\cos(2j-2k-n)\theta)\,d\theta\\
&\quad+\frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}\binom{2j+1}{k}\\
&\quad\int_0^\pi(\cos(2j-2k+1+n)\theta+\cos(2j-2k+1-n)\theta)\,d\theta\\
&=\frac{1}{\pi}\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\frac{\sin n\theta}{n}\bigg|_0^\pi+\frac{1}{\pi}\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}\left(\theta+\frac{\sin 2n\theta}{2n}\right)\bigg|_0^\pi\\
&+\frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0,2j-2k\neq n}^{j-1}\binom{2j}{k}\left(\frac{\sin(2j-2k+n)\theta}{2j-2k+n}+\frac{\sin(2j-2k-n)\theta}{2j-2k-n}\right)\bigg|_0^\pi\\
&\quad+\frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}\binom{2j+1}{k}\\
&\quad\left(\frac{\sin(2j-2k+1+n)\theta}{2j-2k+1+n}+\frac{\sin(2j-2k+1-n)\theta}{2j-2k+1-n}\right)\bigg|_0^\pi
\end{aligned}
$$

$$= \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}$$
$$= A_n(x).$$

□

**Corollary 4.9** (Corollary to Theorem 4.8)
*Let $f(z)$ and $A_n(x)$ be the functions as defined in Theorem 4.8. Then, $\forall y \in \mathbb{R} - \{0\}$, we have*

$$A_n(x) = \frac{y}{\pi} \int_0^{\pi/y} f\,(2x \cos y\theta) \cos ny\theta \, d\theta. \tag{4.15}$$

***Proof*** We use the substitution $\phi = y\theta$ in (4.15). As $\theta$ varies from 0 to $\pi/y$, $\phi$ varies from 0 to $\pi$. We also have $d\phi = y d\theta \implies d\theta = d\phi/y$. From (4.13), we get

$$\begin{aligned} \frac{y}{\pi} \int_0^{\pi/y} f\,(2x \cos y\theta) \cos ny\theta \, d\theta &= \frac{y}{\pi} \int_0^{\pi} f\,(2x \cos \phi) \cos n\phi \, \frac{d\phi}{y} \\ &= \frac{1}{\pi} \int_0^{\pi} f\,(2x \cos \phi) \cos n\phi \, d\phi \\ &= A_n(x). \end{aligned}$$

□

**Theorem 4.10** (Another Series Involving Central Binomial Coefficients)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f(2ix \sin\theta)$ should be convergent for $\theta \in (0, \pi)$. Let*

$$\begin{aligned} B_0(x) &= \sum_{j=0}^{\infty} (-1)^j \binom{2j}{j} a_{2j} x^{2j} \\ &= a_0 - \binom{2}{1} a_2 x^2 + \binom{4}{2} a_4 x^4 - \binom{6}{3} a_6 x^6 + \cdots, \end{aligned} \tag{4.16}$$

*then, if $f$ is an even function, then $B_0(x)$ is given by*

$$B_0(x) = \frac{1}{\pi} \int_0^{\pi} f\,(2ix \sin\theta)\, d\theta.$$

***Proof*** By replacing $z$ with $2ix \sin\theta = x\left(e^{i\theta} - e^{-i\theta}\right)$ in (4.1), we get

$$
\begin{aligned}
f(2ix\sin\theta) &= f\left(x\left(e^{i\theta}-e^{-i\theta}\right)\right) = \sum_{j=0}^{\infty} a_j \left(x\left(e^{i\theta}-e^{-i\theta}\right)\right)^j \\
&= \sum_{j=0}^{\infty} a_{2j}\left(x\left(e^{i\theta}-e^{-i\theta}\right)\right)^{2j} + \sum_{j=0}^{\infty} a_{2j+1}\left(x\left(e^{i\theta}-e^{-i\theta}\right)\right)^{2j+1} \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j} \\
&\quad + \sum_{j=0}^{\infty} a_{2j}x^{2j}\sum_{k=0}^{j-1}\left((-1)^{2j-k}\binom{2j}{k}e^{ik\theta}e^{-i(2j-k)\theta} + (-1)^k\binom{2j}{2j-k}e^{i(2j-k)\theta}e^{-ik\theta}\right) \\
&\quad + \sum_{j=0}^{\infty} a_{2j+1}x^{2j+1}\sum_{k=0}^{j}\left((-1)^{2j+1-k}\binom{2j+1}{k}e^{ik\theta}e^{-i(2j+1-k)\theta}\right. \\
&\quad \left. +(-1)^k\binom{2j+1}{2j+1-k}e^{i(2j+1-k)\theta}e^{-ik\theta}\right) \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j} + \sum_{j=0}^{\infty} a_{2j}x^{2j}\sum_{k=0}^{j-1}(-1)^k\binom{2j}{k}\left(e^{-i(2j-2k)\theta}+e^{i(2j-2k)\theta}\right) \\
&\quad + \sum_{j=0}^{\infty} a_{2j+1}x^{2j+1}\sum_{k=0}^{j}(-1)^k\binom{2j+1}{k}\left(-e^{-i(2j-2k+1)\theta}+e^{i(2j-2k+1)\theta}\right) \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j} + 2\sum_{j=0}^{\infty} a_{2j}x^{2j}\sum_{k=0}^{j-1}(-1)^k\binom{2j}{k}\cos(2j-2k)\theta \\
&\quad +2i\sum_{j=0}^{\infty} a_{2j+1}x^{2j+1}\sum_{k=0}^{j}(-1)^k\binom{2j+1}{k}\sin(2j-2k+1)\theta. \qquad (4.17)
\end{aligned}
$$

Since $f$ is an even function, we have $a_{2j+1} = 0, \forall j \in \mathbb{N} - 1$. Dividing (4.17) by $\pi$, integrating the result with respect to $\theta$ from 0 to $\pi$, and using (4.16), we get

$$\begin{aligned}
\frac{1}{\pi}\int_0^{\pi} f\,(2ix\sin\theta)\,d\theta &= \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\right)d\theta \\
&\quad+\frac{1}{\pi}\int_0^{\pi}\left(2\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}(-1)^k\binom{2j}{k}\cos(2j-2k)\theta\right)d\theta \\
&= \frac{1}{\pi}\sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\int_0^{\pi}d\theta \\
&\quad+\frac{2}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}(-1)^k\binom{2j}{k}\int_0^{\pi}\cos(2j-2k)\theta\,d\theta \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}+\frac{2}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}(-1)^k\binom{2j}{k}\frac{\sin(2j-2k)\theta}{2j-2k}\Bigg|_0^{\pi} \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j} \\
&= B_0(x).
\end{aligned}$$

□

**Theorem 4.11** (First Generalization of Theorem 4.10)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f(2ix\sin\theta)$ should be convergent for $\theta\in(0,\pi)$. $\forall n\in\mathbb{N}-1$, let*

$$\begin{aligned}
B_n(x) &= \sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j} \\
&= \binom{n}{0}a_nx^n-\binom{n+2}{1}a_{n+2}x^{n+2}+\binom{n+4}{2}a_{n+4}x^{n+4}-\binom{n+6}{3}a_{n+6}x^{n+6}+\cdots,
\end{aligned} \tag{4.18}$$

*then, if $f$ is an even function, then $B_n(x)$ is given by*

$$B_n(x)=\frac{1}{\pi}\int_0^{\pi} f\,(2ix\sin\theta)\cos n\theta d\theta.$$

***Proof*** Since $f$ is an even function, we have $a_{2j+1}=0,\forall j\in\mathbb{N}-1$. Multiplying (4.17) by $\cos n\theta$, we get

$$
\begin{aligned}
f(2ix\sin\theta)\cos n\theta &= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\cos n\theta \\
&\quad + 2\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}(-1)^k\binom{2j}{k}\cos(2j-2k)\theta\cos n\theta \\
&\quad + 2i\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0}^{j}(-1)^k\binom{2j+1}{k}\sin(2j-2k+1)\theta\cos n\theta \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\cos n\theta + 2\sum_{k=0}^{\infty}(-1)^k\binom{n+2k}{k}a_{n+2k}x^{n+2k}\cos^2 n\theta \\
&\quad + 2\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0,2j-2k\neq n}^{j-1}(-1)^k\binom{2j}{k}\cos(2j-2k)\theta\cos n\theta \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\cos n\theta + \sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}(1+\cos 2n\theta) \\
&\quad + \sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0,2j-2k\neq n}^{j-1}(-1)^k\binom{2j}{k}(\cos(2j-2k+n)\theta+\cos(2j-2k-n)\theta).
\end{aligned}
\tag{4.19}
$$

Dividing (4.19) by $\pi$, integrating the result with respect to $\theta$ from 0 to $\pi$, and using (4.18), we get

$$
\begin{aligned}
&\frac{1}{\pi}\int_0^{\pi} f\,(2ix\sin\theta)\cos n\theta\,d\theta \\
&= \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\cos n\theta\right)d\theta \\
&\quad + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}(1+\cos 2n\theta)\right)d\theta \\
&\quad + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty}a_{2j}x^{2j}\right. \\
&\qquad \left.\sum_{k=0,2j-2k\neq n}^{j-1}(-1)^k\binom{2j}{k}(\cos(2j-2k+n)\theta+\cos(2j-2k-n)\theta)\right)d\theta
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{1}{\pi}\sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\int_0^{\pi}\cos n\theta\,d\theta \\
&+ \frac{1}{\pi}\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}\int_0^{\pi}(1+\cos 2n\theta)\,d\theta \\
&+ \frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j} \\
&\sum_{k=0,2j-2k\neq n}^{j-1}(-1)^k\binom{2j}{k}\int_0^{\pi}(\cos(2j-2k+n)\theta+\cos(2j-2k-n)\theta)\,d\theta \\
&= \frac{1}{\pi}\sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\frac{\sin n\theta}{n}\Bigg|_0^{\pi} \\
&+ \frac{1}{\pi}\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}\left(\theta+\frac{\sin 2n\theta}{2n}\right)\Bigg|_0^{\pi} \\
&+ \frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j}x^{2j} \\
&\sum_{k=0,2j-2k\neq n}^{j-1}(-1)^k\binom{2j}{k}\left(\frac{\sin(2j-2k+n)\theta}{2j-2k+n}+\frac{\sin(2j-2k-n)\theta}{2j-2k-n}\right)\Bigg|_0^{\pi} \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j} \\
&= B_n(x).
\end{aligned}
$$

□

**Corollary 4.12** (Corollary to Theorem 4.11)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f(2ix\sin\theta)$ should be convergent for $\theta \in (0,\pi)$. If $f$ is an even function, then $\forall (x,n) \in \mathbb{R}\times(2\mathbb{N}-1)$,*

$$\int_0^{\pi} f\,(2ix\sin\theta)\cos n\theta = 0.$$

***Proof*** Since $f$ is an even function and $n \in 2\mathbb{N}-1$, we have $a_{n+2j} = 0, \forall j \in \mathbb{N}-1$. By Theorem 4.11, we get

$$\int_0^{\pi} f\,(2ix\sin\theta)\cos n\theta = \pi B_n(x)$$
$$= \pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}$$
$$= 0.$$

□

**Theorem 4.13** (Second Generalization of Theorem 4.10)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f(2ix\sin\theta)$ should be convergent for $\theta \in (0,\pi)$. If $f$ is an odd function, then $\forall n \in \mathbb{N}-1$, $B_n(x)$ is given by*

$$B_n(x) = \frac{1}{\pi}\int_0^{\pi} -if\,(2ix\sin\theta)\sin n\theta\, d\theta.$$

***Proof*** Since $f$ is an odd function, we have $a_{2j} = 0, \forall j \in \mathbb{N}-1$. Multiplying (4.17) by $-i\sin n\theta$, we get

$$\begin{aligned}
-if(2ix\sin\theta)\sin n\theta &= -i\sum_{j=0}^{\infty}(-1)^j\binom{2j}{j}a_{2j}x^{2j}\sin n\theta\\
&\quad -2i\sum_{j=0}^{\infty}a_{2j}x^{2j}\sum_{k=0}^{j-1}(-1)^k\binom{2j}{k}\cos(2j-2k)\theta\sin n\theta\\
&\quad +2\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0}^{j}(-1)^k\binom{2j+1}{k}\sin(2j-2k+1)\theta\sin n\theta\\
&= 2\sum_{k=0}^{\infty}(-1)^k\binom{n+2k}{k}a_{n+2k}x^{n+2k}\sin^2 n\theta\\
&\quad +2\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}(-1)^k\binom{2j+1}{k}\sin(2j-2k+1)\theta\sin n\theta\\
&= \sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}(1-\cos 2n\theta)\\
&\quad +\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}(-1)^k\binom{2j+1}{k}\\
&\quad (\cos(2j-2k+1-n)\theta-\cos(2j-2k+1+n)\theta).
\end{aligned} \tag{4.20}$$

Dividing (4.20) by $\pi$, integrating the result with respect to $\theta$ from 0 to $\pi$, and using (4.18), we get

$$\begin{aligned}
&\frac{1}{\pi}\int_0^\pi -if\,(2ix\sin\theta)\sin n\theta\,d\theta \\
&= \frac{1}{\pi}\int_0^\pi \left(\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}(1-\cos 2n\theta)\right)d\theta \\
&\quad + \frac{1}{\pi}\int_0^\pi \left(\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}(-1)^k\binom{2j+1}{k}\right. \\
&\quad \left.(\cos(2j-2k+1-n)\theta-\cos(2j-2k+1+n)\theta)\right)d\theta \\
&= \frac{1}{\pi}\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}\int_0^\pi(1-\cos 2n\theta)\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}(-1)^k\binom{2j+1}{k} \\
&\quad \int_0^\pi(\cos(2j-2k+1-n)\theta-\cos(2j-2k+1+n)\theta)\,d\theta \\
&= \frac{1}{\pi}\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}\left(\theta-\frac{\sin 2n\theta}{2n}\right)\Bigg|_0^\pi \\
&\quad + \frac{1}{\pi}\sum_{j=0}^{\infty}a_{2j+1}x^{2j+1}\sum_{k=0,2j-2k+1\neq n}^{j}(-1)^k\binom{2j+1}{k} \\
&\quad \left(\frac{\sin(2j-2k+1-n)\theta}{2j-2k+1-n}-\frac{\sin(2j-2k+1+n)\theta}{2j-2k+1+n}\right)\Bigg|_0^\pi \\
&= \sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j} \\
&= B_n(x).
\end{aligned}$$

□

**Corollary 4.14** (Corollary to Theorem 4.13)
*Let $f(z)$ be the function as given in (4.1) in which it can contain all possible non-negative integral powers of $z$ (positive or zero). The series may be finite or infinite. In case it is infinite, $f(2ix\sin\theta)$ should be convergent for $\theta\in(0,\pi)$. If $f$ is an odd function, then $\forall(x,n)\in\mathbb{R}\times(2\mathbb{N}-2)$,*

$$\int_0^\pi f\,(2ix\sin\theta)\sin n\theta = 0.$$

***Proof*** Since $f$ is an odd function and $n \in 2\mathbb{N} - 2$, we have $a_{n+2j} = 0, \forall j \in \mathbb{N} - 1$. By Theorem 4.13, we get

$$\begin{aligned}\int_0^{\pi} f\,(2ix\sin\theta)\sin n\theta &= i\pi B_n(x)\\ &= i\pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= 0.\end{aligned}$$

□

Now, we will derive a double integral representation of the integral of a product of two functions with Fourier series.

**Theorem 4.15** (A Double Integral Representation)
*Let $f$ and $g$ be any two functions that can be expanded by Fourier's series in the interval* $[0, 2\pi]$*, then*

$$2\pi\int_0^{2\pi} f(\theta)g(\theta)\,d\theta = \int_0^{2\pi}\int_0^{2\pi} f(\phi)g(\psi)\left(1 + 2\sum_{j=1}^{\infty}\cos j\,(\phi-\psi)\right)d\phi\,d\psi.$$

***Proof*** We write the Fourier series expansion of $f(\theta)$ as

$$\begin{aligned}&f(\theta)\\ &= \frac{1}{2\pi}\int_0^{2\pi} f(\phi)\,d\phi\\ &+ \frac{1}{\pi}\sum_{j=1}^{\infty}\left(\left(\int_0^{2\pi} f(\phi)\cos j\phi\,d\phi\right)\cos j\theta + \left(\int_0^{2\pi} f(\phi)\sin j\phi\,d\phi\right)\sin j\theta\right),\end{aligned} \tag{4.21}$$

and the Fourier series expansion of $g(\theta)$ as

$$\begin{aligned}&g(\theta)\\ &= \frac{1}{2\pi}\int_0^{2\pi} g(\psi)\,d\psi\\ &+ \frac{1}{\pi}\sum_{k=1}^{\infty}\left(\left(\int_0^{2\pi} g(\psi)\cos k\psi\,d\psi\right)\cos k\theta + \left(\int_0^{2\pi} g(\psi)\sin k\psi\,d\psi\right)\sin k\theta\right).\end{aligned} \tag{4.22}$$

Multiplying (4.21) and (4.22), we get

$$
\begin{aligned}
&f(\theta)g(\theta)\\
&= \frac{1}{4\pi^2}\int_0^{2\pi} f(\phi)\,d\phi \int_0^{2\pi} g(\psi)\,d\psi\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi} f(\phi)\cos j\phi\,d\phi\right)\left(\int_0^{2\pi} g(\psi)\cos k\psi\,d\psi\right)\cos j\theta\cos k\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi} f(\phi)\cos j\phi\,d\phi\right)\left(\int_0^{2\pi} g(\psi)\sin k\psi\,d\psi\right)\cos j\theta\sin k\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi} f(\phi)\sin j\phi\,d\phi\right)\left(\int_0^{2\pi} g(\psi)\cos k\psi\,d\psi\right)\sin j\theta\cos k\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi} f(\phi)\sin j\phi\,d\phi\right)\left(\int_0^{2\pi} g(\psi)\sin k\psi\,d\psi\right)\sin j\theta\sin k\theta\\
&= \frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{2\pi} f(\phi)g(\psi)\,d\phi\,d\psi\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos k\psi\,d\phi\,d\psi\right)\cos j\theta\cos k\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin k\psi\,d\phi\,d\psi\right)\cos j\theta\sin k\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos k\psi\,d\phi\,d\psi\right)\sin j\theta\cos k\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin k\psi\,d\phi\,d\psi\right)\sin j\theta\sin k\theta\\
&= \frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{2\pi} f(\phi)g(\psi)\,d\phi\,d\psi\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos j\psi\,d\phi\,d\psi\right)\cos^2 j\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin j\psi\,d\phi\,d\psi\right)\cos j\theta\sin j\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos j\psi\,d\phi\,d\psi\right)\sin j\theta\cos j\theta\\
&\quad + \frac{1}{\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin j\psi\,d\phi\,d\psi\right)\sin^2 j\theta
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos k\psi\, d\phi\, d\psi\right)\cos j\theta\cos k\theta\\
&+\frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin k\psi\, d\phi\, d\psi\right)\cos j\theta\sin k\theta\\
&+\frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos k\psi\, d\phi\, d\psi\right)\sin j\theta\cos k\theta\\
&+\frac{1}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin k\psi\, d\phi\, d\psi\right)\sin j\theta\sin k\theta\\
=&\frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{2\pi} f(\phi)g(\psi)\, d\phi\, d\psi\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos j\psi\, d\phi\, d\psi\right)(1+\cos 2j\theta)\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin j\psi\, d\phi\, d\psi\right)\sin 2j\theta\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos j\psi\, d\phi\, d\psi\right)\sin 2j\theta\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin j\psi\, d\phi\, d\psi\right)(1-\cos 2j\theta)\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos k\psi\, d\phi\, d\psi\right)\\
&(\cos(j+k)\theta+\cos(j-k)\theta)\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin k\psi\, d\phi\, d\psi\right)\\
&(\sin(j+k)\theta-\sin(j-k)\theta)\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos k\psi\, d\phi\, d\psi\right)\\
&(\sin(j+k)\theta+\sin(j-k)\theta)\\
&+\frac{1}{2\pi^2}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin k\psi\, d\phi\, d\psi\right)\\
&(\cos(j-k)\theta-\cos(j+k)\theta).
\end{aligned}
\tag{4.23}
$$

Multiplying (4.23) by $2\pi$, and integrating the result with respect to $\theta$ from 0 to $2\pi$, we get

$$
\begin{aligned}
&2\pi \int_0^{2\pi} f(\theta)g(\theta)\,d\theta \\
&= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{2\pi} f(\phi)g(\psi)\,d\phi\,d\psi \int_0^{2\pi} d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos j\psi\,d\phi\,d\psi\right)\int_0^{2\pi}(1+\cos 2j\theta)\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin j\psi\,d\phi\,d\psi\right)\int_0^{2\pi}\sin 2j\theta\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos j\psi\,d\phi\,d\psi\right)\int_0^{2\pi}\sin 2j\theta\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin j\psi\,d\phi\,d\psi\right)\int_0^{2\pi}(1-\cos 2j\theta)\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos k\psi\,d\phi\,d\psi\right) \\
&\qquad \int_0^{2\pi}(\cos(j+k)\theta+\cos(j-k)\theta)\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin k\psi\,d\phi\,d\psi\right) \\
&\qquad \int_0^{2\pi}(\sin(j+k)\theta-\sin(j-k)\theta)\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos k\psi\,d\phi\,d\psi\right) \\
&\qquad \int_0^{2\pi}(\sin(j+k)\theta+\sin(j-k)\theta)\,d\theta \\
&\quad + \frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin k\psi\,d\phi\,d\psi\right) \\
&\qquad \int_0^{2\pi}(\cos(j-k)\theta-\cos(j+k)\theta)\,d\theta \\
&= \int_0^{2\pi}\int_0^{2\pi} f(\phi)g(\psi)\,d\phi\,d\psi
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos j\psi\, d\phi\, d\psi\right)\left(\theta+\frac{\sin 2j\theta}{2j}\right)\Bigg|_0^{2\pi}\\
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin j\psi\, d\phi\, d\psi\right)\left(-\frac{\cos 2j\theta}{2j}\right)\Bigg|_0^{2\pi}\\
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos j\psi\, d\phi\, d\psi\right)\left(-\frac{\cos 2j\theta}{2j}\right)\Bigg|_0^{2\pi}\\
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin j\psi\, d\phi\, d\psi\right)\left(\theta-\frac{\sin 2j\theta}{2j}\right)\Bigg|_0^{2\pi}\\
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos k\psi\, d\phi\, d\psi\right)\\
&\quad\left(\frac{\sin(j+k)\theta}{j+k}+\frac{\sin(j-k)\theta}{j-k}\right)\Bigg|_0^{2\pi}\\
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\sin k\psi\, d\phi\, d\psi\right)\\
&\quad\left(-\frac{\cos(j+k)\theta}{j+k}+\frac{\cos(j-k)\theta}{j-k}\right)\Bigg|_0^{2\pi}\\
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\cos k\psi\, d\phi\, d\psi\right)\\
&\quad\left(-\frac{\cos(j+k)\theta}{j+k}-\frac{\cos(j-k)\theta}{j-k}\right)\Bigg|_0^{2\pi}\\
&+\frac{1}{\pi}\sum_{j=1}^{\infty}\sum_{k=1,k\neq j}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin k\psi\, d\phi\, d\psi\right)\\
&\quad\left(\frac{\sin(j-k)\theta}{j-k}-\frac{\sin(j+k)\theta}{j+k}\right)\Bigg|_0^{2\pi}\\
&=\int_0^{2\pi}\int_0^{2\pi} f(\phi)g(\psi)\, d\phi\, d\psi\\
&\quad+\left(2\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\cos j\phi g(\psi)\cos j\psi\, d\phi\, d\psi\right)\right)\\
&\quad\left(2\sum_{j=1}^{\infty}\left(\int_0^{2\pi}\int_0^{2\pi} f(\phi)\sin j\phi g(\psi)\sin j\psi\, d\phi\, d\psi\right)\right)
\end{aligned}
$$

$$= \int_0^{2\pi} \int_0^{2\pi} f(\phi) g(\psi) \left( 1 + 2 \sum_{j=1}^{\infty} \cos j \, (\phi - \psi) \right) d\phi \, d\psi.$$

□

Finally, we conclude the section by deriving another double integral representation.

**Theorem 4.16** (Another Double Integral Representation)
*Let $f$ and $g$ be any two functions which can be expanded by Maclaurin's series, such that $f\left(e^{i\theta}\right)$, $f\left(e^{-i\theta}\right)$, $g\left(e^{i\theta}\right)$, and $g\left(e^{-i\theta}\right)$ all are convergent for $\theta \in (0, \pi)$. Then*

$$\begin{aligned} & 2\pi \int_0^{\pi} \left( f\left(e^{i\theta}\right) g\left(e^{i\theta}\right) e^{-in\theta} + f\left(e^{-i\theta}\right) g\left(e^{-i\theta}\right) e^{in\theta} \right) d\theta \\ &= \sum_{j=0}^{n} \int_0^{\pi} \int_0^{\pi} \left( f\left(e^{i\phi}\right) e^{-ij\phi} + f\left(e^{-i\phi}\right) e^{ij\phi} \right) \\ & \qquad \left( g\left(e^{i\psi}\right) e^{-i(n-j)\psi} + g\left(e^{-i\psi}\right) e^{i(n-j)\phi} \right) d\phi \, d\psi. \end{aligned}$$

***Proof*** Let

$$f(z) = \sum_{j=0}^{\infty} a_j z^j, \tag{4.24}$$

and

$$g(z) = \sum_{k=0}^{\infty} b_k z^k. \tag{4.25}$$

Putting $z = e^{i\theta}$ in (4.24), we get

$$f\left(e^{i\theta}\right) = \sum_{j=0}^{\infty} a_j e^{ij\theta}. \tag{4.26}$$

Putting $z = e^{i\theta}$ in (4.25), we get

$$g\left(e^{i\theta}\right) = \sum_{k=0}^{\infty} b_k e^{ik\theta}. \tag{4.27}$$

Putting $z = e^{-i\theta}$ in (4.24), we get

$$f\left(e^{-i\theta}\right) = \sum_{j=0}^{\infty} a_j e^{-ij\theta}. \tag{4.28}$$

Putting $z = e^{-i\theta}$ in (4.25), we get

$$g\left(e^{-i\theta}\right) = \sum_{k=0}^{\infty} b_k e^{-ik\theta}. \tag{4.29}$$

Multiplying (4.26) and (4.27), we get

$$f\left(e^{i\theta}\right) g\left(e^{i\theta}\right) = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} a_j b_k e^{i(j+k)\theta}. \tag{4.30}$$

Multiplying (4.28) and (4.29), we get

$$f\left(e^{-i\theta}\right) g\left(e^{-i\theta}\right) = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} a_j b_k e^{-i(j+k)\theta}. \tag{4.31}$$

Multiplying (4.30) by $e^{-in\theta}$, multiplying (4.31) by $e^{in\theta}$, and adding the results, we get

$$\begin{aligned} &f\left(e^{i\theta}\right) g\left(e^{i\theta}\right) e^{-in\theta} + f\left(e^{-i\theta}\right) g\left(e^{-i\theta}\right) e^{in\theta} \\ &\qquad = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} a_j b_k \left(e^{i(j+k-n)\theta} + e^{-i(j+k-n)\theta}\right) \\ &\qquad = 2\sum_{j=0}^{\infty}\sum_{k=0}^{\infty} a_j b_k \cos(j+k-n)\theta. \end{aligned} \tag{4.32}$$

Multiplying (4.32) by $2\pi$, integrating the result with respect to $\theta$ from 0 to $\pi$, and using Theorem 4.2, we get

$$\begin{aligned} &2\pi \int_0^{\pi} \left(f\left(e^{-i\theta}\right) g\left(e^{-i\theta}\right) e^{-in\theta} + f\left(e^{-i\theta}\right) g\left(e^{-i\theta}\right) e^{in\theta}\right) d\theta \\ &= 4\pi \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} a_j b_k \int_0^{\pi} \cos(j+k-n)\theta \, d\theta \\ &= 4\pi \sum_{j+k=n,\, j\in[0..n],\, k\in[0..n]} a_j b_k \int_0^{\pi} d\theta \\ &\quad + 4\pi \sum_{j=0}^{\infty}\sum_{k=0,\, j+k\neq n}^{\infty} a_j b_k \int_0^{\pi} \cos(j+k-n)\theta \, d\theta \\ &= 4\pi^2 \sum_{j+k=n,\, j\in[0..n],\, k\in[0..n]} a_j b_k + 4\pi \sum_{j=0}^{\infty}\sum_{k=0,\, j+k\neq n}^{\infty} a_j b_k \frac{\sin(j+k-n)\theta}{j+k-n}\Bigg|_0^{\pi} \end{aligned}$$

$$
\begin{aligned}
&= 4\pi^2 \sum_{j+k=n,\, j\in[0..n],k\in[0..n]} a_j b_k \\
&= 4\pi^2 \sum_{j+k=n,\, j\in[0..n],k\in[0..n]} \left( \frac{1}{\pi} \int_0^\pi \frac{e^{-ij\phi} f\left(e^{i\phi}\right) + e^{ij\phi} f\left(e^{-i\phi}\right)}{2} d\phi \right) \\
&\qquad \left( \frac{1}{\pi} \int_0^\pi \frac{e^{-ik\psi} g\left(e^{i\psi}\right) + e^{ik\psi} g\left(e^{-i\psi}\right)}{2} d\psi \right) \\
&= \sum_{j=0}^{n} \int_0^\pi \int_0^\pi \left( f\left(e^{i\phi}\right) e^{-ij\phi} + f\left(e^{-i\phi}\right) e^{ij\phi} \right) \\
&\qquad \left( g\left(e^{i\psi}\right) e^{-i(n-j)\psi} + g\left(e^{-i\psi}\right) e^{i(n-j)\phi} \right) d\phi\, d\psi.
\end{aligned}
$$

□

## 4.2 Special Integrals 2

In this section, we will continue our development of the previous section. We will prove some theorems about series sums involving the binomial coefficients and several of their corollaries. We will see that Theorems 4.1, 4.4, and 4.8 can be derived as a corollary to these theorems.

We begin with an integral representation of sums involving the coefficients of a series and the binomial coefficients.

**Theorem 4.17** (A Binomial Series)
*Let*

$$
f(z) = \sum_{j=0}^{\infty} a_j z^j = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots, \tag{4.33}
$$

*be a function which can be expanded by Maclaurin's series such that* $f\left(xe^{i\theta} + ye^{-i\theta}\right)$ *and* $f\left(xe^{-i\theta} + ye^{i\theta}\right)$ *both are convergent for* $\theta \in (0, \pi)$*, and* $\forall (x, y, n) \in \mathbb{C}^2 \times \times (\mathbb{N} - 1)$*, let the function* $T_n(x, y)$ *be defined as*

$$
\begin{aligned}
T_n(x, y) &= \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\
&= \left(x^n + y^n\right) \left( \binom{n}{0} a_n + \binom{n+2}{1} a_{n+2}(xy) + \binom{n+4}{2} a_{n+4}(xy)^2 \right. \\
&\quad \left. + \binom{n+6}{3} a_{n+6}(xy)^3 + \cdots \right),
\end{aligned} \tag{4.34}
$$

*then*

$$T_n(x, y) = \frac{2}{\pi} \int_0^{\pi} \frac{f\left(xe^{i\theta} + ye^{-i\theta}\right) + f\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta. \tag{4.35}$$

***Proof*** Putting $z = xe^{i\theta} + ye^{-i\theta}$ in (4.33), we get

$$\begin{aligned} f\left(xe^{i\theta} + ye^{-i\theta}\right) &= \sum_{j=0}^{\infty} a_j \left(xe^{i\theta} + ye^{-i\theta}\right)^j \\ &= \sum_{j=0}^{\infty} a_j \sum_{k=0}^{j} \binom{j}{k} \left(xe^{i\theta}\right)^k \left(ye^{-i\theta}\right)^{j-k} \\ &= \sum_{j=0}^{\infty} a_j \sum_{k=0}^{j} \binom{j}{k} x^k y^{j-k} e^{i(2k-j)\theta}. \end{aligned} \tag{4.36}$$

Putting $z = xe^{-i\theta} + ye^{i\theta}$ in (4.33), we get

$$\begin{aligned} f\left(xe^{-i\theta} + ye^{i\theta}\right) &= \sum_{j=0}^{\infty} a_j \left(xe^{-i\theta} + ye^{i\theta}\right)^j \\ &= \sum_{j=0}^{\infty} a_j \sum_{k=0}^{j} \binom{j}{k} \left(xe^{-i\theta}\right)^k \left(ye^{i\theta}\right)^{j-k} \\ &= \sum_{j=0}^{\infty} a_j \sum_{k=0}^{j} \binom{j}{k} x^k y^{j-k} e^{-i(2k-j)\theta}. \end{aligned} \tag{4.37}$$

Adding (4.36) and (4.37), dividing the result by 2, and using (4.34) we get

$$
\begin{aligned}
&\frac{f\left(xe^{i\theta}+ye^{-i\theta}\right)+f\left(xe^{-i\theta}+ye^{i\theta}\right)}{2}\\
&=\sum_{j=0}^{\infty}a_j\sum_{k=0}^{j}\binom{j}{k}x^k y^{j-k}\frac{e^{i(2k-j)\theta}+e^{-i(2k-j)\theta}}{2}\\
&=\sum_{j=0}^{\infty}a_j\sum_{k=0}^{j}\binom{j}{k}x^k y^{j-k}\cos(2k-j)\theta\\
&=\sum_{j=0}^{\infty}a_{n+2j}\binom{n+2j}{n+j}x^{n+j}y^j\cos n\theta+\sum_{j=0}^{\infty}a_{n+2j}\binom{n+2j}{j}x^j y^{n+j}\cos(-n\theta)\\
&\quad+\sum_{j=0}^{\infty}a_j\sum_{k=0,2k-j\neq\pm n}^{j}\binom{j}{k}x^k y^{j-k}\cos(2k-j)\theta\\
&=(x^n+y^n)\sum_{j=0}^{\infty}a_{n+2j}\binom{n+2j}{j}(xy)^j\cos n\theta\\
&\quad+\sum_{j=0}^{\infty}a_j\sum_{k=0,2k-j\neq\pm n}^{j}\binom{j}{k}x^k y^{j-k}\cos(2k-j)\theta\\
&=T_n(x,y)\cos n\theta+\sum_{j=0}^{\infty}a_j\sum_{k=0,2k-j\neq\pm n}^{j}\binom{j}{k}x^k y^{j-k}\cos(2k-j)\theta.
\end{aligned}
\tag{4.38}
$$

Multiplying (4.38) by $(2/\pi)\cos n\theta$, and integrating the result with respect to $\theta$ from 0 to $\pi$, we get

$$
\begin{aligned}
&\frac{2}{\pi}\int_0^{\pi}\frac{f\left(xe^{i\theta}+ye^{-i\theta}\right)+f\left(xe^{-i\theta}+ye^{i\theta}\right)}{2}\cos n\theta\,d\theta\\
&=\frac{2}{\pi}\int_0^{\pi}T_n(x,y)\cos^2 n\theta\,d\theta\\
&\quad+\frac{2}{\pi}\int_0^{\pi}\left(\sum_{j=0}^{\infty}a_j\sum_{k=0,2k-j\neq\pm n}^{j}\binom{j}{k}x^k y^{j-k}\cos(2k-j)\theta\cos n\theta\right)d\theta\\
&=\frac{1}{\pi}\int_0^{\pi}T_n(x,y)(1+\cos 2n\theta)\,d\theta+\frac{1}{\pi}\sum_{j=0}^{\infty}a_j\\
&\qquad\sum_{k=0,2k-j\neq\pm n}^{j}\binom{j}{k}x^k y^{j-k}\int_0^{\pi}(\cos(2k-j+n)\theta+\cos(2k-j-n)\theta)\,d\theta\\
&=\frac{1}{\pi}T_n(x,y)\left(\theta+\frac{\sin 2n\theta}{2n}\right)\Bigg|_0^{\pi}
\end{aligned}
$$

$$+\frac{1}{\pi}\sum_{j=0}^{\infty}a_j\sum_{k=0,2k-j\neq\pm n}^{j}\binom{j}{k}x^k y^{j-k}\left(\frac{\sin(2k-j+n)\theta}{2k-j+n}+\frac{\sin(2k-j-n)\theta}{2k-j-n}\right)\Bigg|_0^{\pi}$$
$$=T_n(x,y).$$

□

**Corollary 4.18** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x,n)\in(\mathbb{R}-\{0\})\times(\mathbb{N}-1)$,*

$$T_n\left(x,\frac{1}{x}\right)=\left(x^n+\frac{1}{x^n}\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}$$
$$=\left(x^n+\frac{1}{x^n}\right)\left(\binom{n}{0}a_n+\binom{n+2}{1}a_{n+2}+\binom{n+4}{2}a_{n+4}+\binom{n+6}{3}a_{n+6}+\cdots\right)$$
$$=\frac{2}{\pi}\int_0^{\pi}\frac{f\left(xe^{i\theta}+x^{-1}e^{-i\theta}\right)+f\left(xe^{-i\theta}+x^{-1}e^{i\theta}\right)}{2}\cos n\theta\,d\theta.$$

***Proof*** Put $y=x^{-1}$ in (4.35). □

**Corollary 4.19** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.1. Then, $\forall(x,n)\in(\mathbb{R}-\{0\})\times(\mathbb{N}-1)$,*

$$T_n\left(x,-\frac{1}{x}\right)=\left(x^n+\frac{(-1)^n}{x^n}\right)\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}$$
$$=\left(x^n+\frac{(-1)^n}{x^n}\right)\left(\binom{n}{0}a_n-\binom{n+2}{1}a_{n+2}+\binom{n+4}{2}a_{n+4}-\binom{n+6}{3}a_{n+6}+\cdots\right)$$
$$=\frac{2}{\pi}\int_0^{\pi}\frac{f\left(xe^{i\theta}-x^{-1}e^{-i\theta}\right)+f\left(xe^{-i\theta}-x^{-1}e^{i\theta}\right)}{2}\cos n\theta\,d\theta.$$

***Proof*** Put $y=-x^{-1}$ in (4.35). □

**Corollary 4.20** (Corollary to Theorem 4.17. This is also Theorem 4.8)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x,n)\in\mathbb{R}\times(\mathbb{N}-1)$,*

$$T_n(x,x) = 2\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}$$
$$= 2\left(\binom{n}{0}a_n x^n + \binom{n+2}{1}a_{n+2}x^{n+2} + \binom{n+4}{2}a_{n+4}x^{n+4} + \binom{n+6}{3}a_{n+6}x^{n+6} + \cdots\right)$$
$$= \frac{2}{\pi}\int_0^{\pi} f(2x\cos\theta)\cos n\theta\, d\theta.$$

***Proof*** Put $y = x$ in (4.35). □

**Corollary 4.21** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x, y, n) \in \mathbb{R}^2 \times (\mathbb{N} - 1)$,*

$$T_n(ix, iy) = i^n\left(x^n + y^n\right)\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}(xy)^j$$
$$= i^n\left(x^n + y^n\right)\left(\binom{n}{0}a_n - \binom{n+2}{1}a_{n+2}(xy) + \binom{n+4}{2}a_{n+4}(xy)^2 - \binom{n+6}{3}a_{n+6}(xy)^3 + \cdots\right)$$
$$= \frac{2}{\pi}\int_0^{\pi}\frac{f\left(i\left(xe^{i\theta} + ye^{-i\theta}\right)\right) + f\left(i\left(xe^{-i\theta} + ye^{i\theta}\right)\right)}{2}\cos n\theta\, d\theta.$$

***Proof*** Replace $x$ by $ix$ and $y$ by $iy$ in (4.35). □

**Corollary 4.22** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x, y, n) \in \mathbb{R}^2 \times (\mathbb{N} - 1)$,*

$$T_n(-ix, -iy) = (-i)^n\left(x^n + y^n\right)\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}(xy)^j$$
$$= (-i)^n\left(x^n + y^n\right)\left(\binom{n}{0}a_n - \binom{n+2}{1}a_{n+2}(xy) + \binom{n+4}{2}a_{n+4}(xy)^2 - \binom{n+6}{3}a_{n+6}(xy)^3 + \cdots\right)$$
$$= \frac{2}{\pi}\int_0^{\pi}\frac{f\left(-i\left(xe^{i\theta} + ye^{-i\theta}\right)\right) + f\left(-i\left(xe^{-i\theta} + ye^{i\theta}\right)\right)}{2}\cos n\theta\, d\theta.$$

***Proof*** Replace $x$ by $-ix$ and $y$ by $-iy$ in (4.35). □

**Corollary 4.23** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x, y, n) \in \mathbb{R}^2 \times (\mathbb{N} - 1)$,*

$$T_n(ix,-iy)=i^n\left(x^n+(-y)^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}(xy)^j$$
$$=i^n\left(x^n+(-y)^n\right)\left(\binom{n}{0}a_n+\binom{n+2}{1}a_{n+2}(xy)+\binom{n+4}{2}a_{n+4}(xy)^2\right.$$
$$\left.+\binom{n+6}{3}a_{n+6}(xy)^3+\cdots\right)$$
$$=\frac{2}{\pi}\int_0^{\pi}\frac{f\left(i\left(xe^{i\theta}-ye^{-i\theta}\right)\right)+f\left(i\left(xe^{-i\theta}-ye^{i\theta}\right)\right)}{2}\cos n\theta\,d\theta.$$

***Proof*** Replace $x$ by $ix$ and $y$ by $-iy$ in (4.35). □

**Corollary 4.24** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x,y,n)\in\mathbb{R}^2\times(\mathbb{N}-1)$,*

$$T_n(-x,-y)=(-1)^n\left(x^n+y^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}(xy)^j$$
$$=(-1)^n\left(x^n+y^n\right)\left(\binom{n}{0}a_n+\binom{n+2}{1}a_{n+2}(xy)+\binom{n+4}{2}a_{n+4}(xy)^2\right.$$
$$\left.+\binom{n+6}{3}a_{n+6}(xy)^3+\cdots\right)$$
$$=\frac{2}{\pi}\int_0^{\pi}\frac{f\left(-\left(xe^{i\theta}+ye^{-i\theta}\right)\right)+f\left(-\left(xe^{-i\theta}+ye^{i\theta}\right)\right)}{2}\cos n\theta\,d\theta.$$

***Proof*** Replace $x$ by $-x$ and $y$ by $-y$ in (4.35). □

**Corollary 4.25** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x,n)\in\mathbb{R}\times(\mathbb{N}-1)$,*

$$T_n(ix,ix)=2i^n\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}$$
$$=2i^n\left(\binom{n}{0}a_nx^n-\binom{n+2}{1}a_{n+2}x^{n+2}+\binom{n+4}{2}a_{n+4}x^{n+4}\right.$$
$$\left.-\binom{n+6}{3}a_{n+6}x^{n+6}+\cdots\right)$$
$$=\frac{2}{\pi}\int_0^{\pi}f\left(2ix\cos\theta\right)\cos n\theta\,d\theta.$$

***Proof*** Replace $x$ by $ix$ and $y$ by $ix$ in (4.35). □

**Corollary 4.26** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x,n)\in\mathbb{R}\times(\mathbb{N}-1)$,*

$$T_n(ix,-ix) = i^n\left(1+(-1)^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}$$
$$= i^n\left(1+(-1)^n\right)\left(\binom{n}{0}a_n x^n + \binom{n+2}{1}a_{n+2}x^{n+2} + \binom{n+4}{2}a_{n+4}x^{n+4}\right.$$
$$\left.+\binom{n+6}{3}a_{n+6}x^{n+6}+\cdots\right)$$
$$= \frac{2}{\pi}\int_0^{\pi}\frac{f(2x\sin\theta)+f(-2x\sin\theta)}{2}\cos n\theta\, d\theta.$$

***Proof*** Replace $x$ by $ix$ and $y$ by $-ix$ in (4.35). □

**Corollary 4.27** (Corollary to Theorem 4.17)
*Let $f(z)$ be the function as defined in Theorem 4.17. Then, $\forall(x,n)\in\mathbb{R}\times(\mathbb{N}-1)$,*

$$T_n(x,-x) = \left(1+(-1)^n\right)\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}$$
$$= \left(1+(-1)^n\right)\left(\binom{n}{0}a_n x^n - \binom{n+2}{1}a_{n+2}x^{n+2} + \binom{n+4}{2}a_{n+4}x^{n+4}\right.$$
$$\left.-\binom{n+6}{3}a_{n+6}x^{n+6}+\cdots\right)$$
$$= \frac{2}{\pi}\int_0^{\pi}\frac{f(2ix\sin\theta)+f(-2ix\sin\theta)}{2}\cos n\theta\, d\theta.$$

***Proof*** Replace $y$ by $-x$ in (4.35). □

Now, we will derive an integral representation of any function in terms of its own integral involving some complex variables.

**Theorem 4.28** (Another Binomial Series)
*Let*

$$f(z) = \sum_{j=0}^{\infty}a_j z^j = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots, \tag{4.39}$$

*be a function that can be expanded by Maclaurin's series such that $f\left(x+ye^{i\theta}\right)$ and $f\left(x+ye^{-i\theta}\right)$ both are convergent for $\theta\in(0,\pi)$. Then, $\forall(x,y)\in\mathbb{C}^2$,*

$$f(x) = \frac{1}{\pi}\int_0^{\pi}\frac{f\left(x+ye^{i\theta}\right)+f\left(x+ye^{-i\theta}\right)}{2}\,d\theta. \tag{4.40}$$

***Proof*** Putting $z = x + ye^{i\theta}$ in (4.39), we get

$$\begin{aligned} f\left(x+ye^{i\theta}\right) &= \sum_{j=0}^{\infty} a_j \left(x+ye^{i\theta}\right)^j \\ &= \sum_{j=0}^{\infty} a_j x^j + \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k e^{ik\theta} \\ &= f(x) + \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k e^{ik\theta}. \end{aligned} \tag{4.41}$$

Putting $z = x + ye^{-i\theta}$ in (4.39), we get

$$\begin{aligned} f\left(x+ye^{-i\theta}\right) &= \sum_{j=0}^{\infty} a_j \left(x+ye^{-i\theta}\right)^j \\ &= \sum_{j=0}^{\infty} a_j x^j + \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k e^{-ik\theta} \\ &= f(x) + \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k e^{-ik\theta}. \end{aligned} \tag{4.42}$$

Adding (4.41) and (4.42), and dividing the result by 2, we get

$$\begin{aligned} \frac{f\left(x+ye^{i\theta}\right) + f\left(x+ye^{-i\theta}\right)}{2} &= f(x) + \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k \frac{e^{ik\theta} + e^{-ik\theta}}{2} \\ &= f(x) + \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k \cos k\theta. \end{aligned} \tag{4.43}$$

Dividing (4.43) by $\pi$, and integrating the result with respect to $\theta$ from 0 to $\pi$, we get

$$\begin{aligned} &\frac{1}{\pi}\int_0^{\pi} \frac{f\left(x+ye^{i\theta}\right) + f\left(x+ye^{-i\theta}\right)}{2}\, d\theta \\ &= \frac{1}{\pi}\int_0^{\pi} f(x)\, d\theta + \frac{1}{\pi}\int_0^{\pi} \left( \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k \cos k\theta \right) d\theta \\ &= \frac{f(x)}{\pi}\int_0^{\pi} d\theta + \frac{1}{\pi} \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k \int_0^{\pi} \cos k\theta\, d\theta \\ &= f(x) + \frac{1}{\pi} \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k \left. \frac{\sin k\theta}{k} \right|_0^{\pi} \\ &= f(x). \end{aligned}$$

□

**Corollary 4.29** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then, $\forall x \in \mathbb{R}$,*

$$f(x) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(2x\cos\frac{\theta}{2}e^{i\theta/2}\right) + f\left(2x\cos\frac{\theta}{2}e^{-i\theta/2}\right)}{2}\,d\theta.$$

***Proof*** Put $y = x$ in (4.40). □

**Corollary 4.30** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then, $\forall x \in \mathbb{R}$,*

$$f(x) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(2x\sin\frac{\theta}{2}e^{i(\pi-\theta)/2}\right) + f\left(2x\sin\frac{\theta}{2}e^{-i(\pi-\theta)/2}\right)}{2}\,d\theta.$$

***Proof*** Put $y = -x$ in (4.40). □

**Corollary 4.31** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then*

$$f(1) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(2\cos\frac{\theta}{2}e^{i\theta/2}\right) + f\left(2\cos\frac{\theta}{2}e^{-i\theta/2}\right)}{2}\,d\theta.$$

***Proof*** Put $x = y = 1$ in (4.40). □

**Corollary 4.32** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then*

$$f(1) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(2\sin\frac{\theta}{2}e^{i(\pi-\theta)/2}\right) + f\left(2\sin\frac{\theta}{2}e^{-i(\pi-\theta)/2}\right)}{2}\,d\theta.$$

***Proof*** Put $x = 1$ and $y = -1$ in (4.40). □

**Corollary 4.33** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then*

$$f(-1) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(-2\cos\frac{\theta}{2}e^{i\theta/2}\right) + f\left(-2\cos\frac{\theta}{2}e^{-i\theta/2}\right)}{2}\,d\theta.$$

***Proof*** Put $x = y = -1$ in (4.40). □

**Corollary 4.34** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then*

$$f(-1) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(-2\sin\frac{\theta}{2}e^{i(\pi-\theta)/2}\right) + f\left(-2\sin\frac{\theta}{2}e^{-i(\pi-\theta)/2}\right)}{2}\,d\theta.$$

***Proof*** Put $x = -1$ and $y = 1$ in (4.40). □

**Corollary 4.35** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then, $\forall y \in \mathbb{R}$,*

$$f(1) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(1+ye^{i\theta}\right) + f\left(1+ye^{-i\theta}\right)}{2}\,d\theta.$$

***Proof*** Put $x = 1$ in (4.40). □

**Corollary 4.36** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then, $\forall y \in \mathbb{R}$,*

$$f(-1) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(-1+ye^{i\theta}\right) + f\left(-1+ye^{-i\theta}\right)}{2}\,d\theta.$$

***Proof*** Put $x = -1$ in (4.40). □

**Corollary 4.37** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then, $\forall x \in \mathbb{R}$,*

$$f(x) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(x+e^{i\theta}\right) + f\left(x+e^{-i\theta}\right)}{2}\,d\theta.$$

***Proof*** Put $y = 1$ in (4.40). □

**Corollary 4.38** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then, $\forall x \in \mathbb{R}$,*

$$f(x) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(x-e^{i\theta}\right) + f\left(x-e^{-i\theta}\right)}{2}\,d\theta.$$

***Proof*** Put $y = -1$ in (4.40). □

**Corollary 4.39** (Corollary to Theorem 4.28)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then, $\forall y \in \mathbb{R}$,*

$$f(0) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(ye^{i\theta}\right) + f\left(ye^{-i\theta}\right)}{2}\,d\theta.$$

***Proof*** Put $x = 0$ in (4.40). □

**Corollary 4.40** (Corollary to Theorem 4.28. This is also Theorem 4.1)
*Let $f(z)$ be the function as defined in Theorem 4.28. Then*

$$f(0) = \frac{1}{\pi}\int_0^{\pi} \frac{f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)}{2}\, d\theta.$$

***Proof*** Put $x = 0$ and $y = 1$ in (4.40). □

The following theorem is a generalization of the previous theorem.

**Theorem 4.41** (Generalization of Theorem 4.28)
*Let*

$$f(z) = \sum_{j=0}^{\infty} a_j z^j = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots$$

*be a function that can be expanded by Maclaurin's series such that $f\left(x + ye^{-i\theta}\right)$ and $f\left(x + ye^{i\theta}\right)$ both are convergent for $\theta \in (0, \pi)$. $\forall (x, y, n) \in \mathbb{C}^2 \times \mathbb{N}$, let the function $S_n(x, y)$ be defined as*

$$\begin{aligned} S_n(x, y) &= y^n \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j \\ &= y^n \left( \binom{n}{0} a_n + \binom{n+1}{1} a_{n+1} x + \binom{n+2}{2} a_{n+2} x^2 + \binom{n+3}{3} a_{n+3} x^3 + \cdots \right). \end{aligned} \tag{4.44}$$

*Then, the function $S_n(x, y)$ is given by*

$$S_n(x, y) = \frac{2}{\pi}\int_0^{\pi} \frac{f\left(x + ye^{i\theta}\right) + f\left(x + ye^{-i\theta}\right)}{2} \cos n\theta \, d\theta. \tag{4.45}$$

***Proof*** Multiplying (4.43) by $(2/\pi)\cos n\theta$, integrating the result with respect to $\theta$ from 0 to $\pi$, and using (4.44), we get

$$\begin{aligned} &\frac{2}{\pi}\int_0^{\pi} \frac{f\left(x + ye^{i\theta}\right) + f\left(x + ye^{-i\theta}\right)}{2} \cos n\theta \, d\theta \\ &= \frac{2}{\pi}\int_0^{\pi} f(x) \cos n\theta \, d\theta + \frac{2}{\pi}\int_0^{\pi} \left( \sum_{j=1}^{\infty} a_j \sum_{k=1}^{j} \binom{j}{k} x^{j-k} y^k \cos k\theta \cos n\theta \right) d\theta \end{aligned}$$

$$
\begin{aligned}
&= \frac{2}{\pi} f(x) \int_0^{\pi} \cos n\theta \, d\theta + \frac{2}{\pi} \sum_{j=n}^{\infty} a_j \binom{j}{n} x^{j-n} y^n \int_0^{\pi} \cos^2 n\theta \, d\theta \\
&\quad + \frac{2}{\pi} \sum_{j=1}^{\infty} a_j \sum_{k=1, k\neq n}^{j} \binom{j}{k} x^{j-k} y^k \int_0^{\pi} \cos k\theta \cos n\theta \, d\theta \\
&= \frac{2}{\pi} f(x) \int_0^{\pi} \cos n\theta \, d\theta + \frac{1}{\pi} \sum_{j=n}^{\infty} a_j \binom{j}{n} x^{j-n} y^n \int_0^{\pi} (1 + \cos 2n\theta) \, d\theta \\
&\quad + \frac{1}{\pi} \sum_{j=1}^{\infty} a_j \sum_{k=1, k\neq n}^{j} \binom{j}{k} x^{j-k} y^k \int_0^{\pi} (\cos(k+n)\theta + \cos(k-n)\theta) \, d\theta \\
&= \frac{2}{\pi} f(x) \left. \frac{\sin n\theta}{n} \right|_0^{\pi} + \frac{1}{\pi} \sum_{j=n}^{\infty} a_j \binom{j}{n} x^{j-n} y^n \left. \left( \theta + \frac{\sin 2n\theta}{2n} \right) \right|_0^{\pi} \\
&\quad + \frac{1}{\pi} \sum_{j=1}^{\infty} a_j \sum_{k=1, k\neq n}^{j} \binom{j}{k} x^{j-k} y^k \left. \left( \frac{\sin(k+n)\theta}{k+n} + \frac{\sin(k-n)\theta}{k-n} \right) \right|_0^{\pi} \\
&= y^n \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j \\
&= S_n(x, y).
\end{aligned}
$$

□

**Corollary 4.42** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall (x, n) \in \mathbb{R} \times \mathbb{N}$,*

$$
\begin{aligned}
S_n(x, x) &= \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^{n+j} \\
&= \binom{n}{0} a_n x^n + \binom{n+1}{1} a_{n+1} x^{n+1} + \binom{n+2}{2} a_{n+2} x^{n+2} + \binom{n+3}{3} a_{n+3} x^{n+3} + \cdots \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(2x \cos \frac{\theta}{2} e^{i\theta/2}\right) + f\left(2x \cos \frac{\theta}{2} e^{-i\theta/2}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $y = x$ in (4.45). □

**Corollary 4.43** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall (x, n) \in \mathbb{R} \times \mathbb{N}$,*

$$
\begin{aligned}
S_n(x,-x) &= (-1)^n \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^{n+j} \\
&= (-1)^n \left( \binom{n}{0} a_n x^n + \binom{n+1}{1} a_{n+1} x^{n+1} + \binom{n+2}{2} a_{n+2} x^{n+2} \right. \\
&\quad \left. + \binom{n+3}{3} a_{n+3} x^{n+3} + \cdots \right) \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(2x \sin \frac{\theta}{2} e^{i(\pi-\theta)/2}\right) + f\left(2x \sin \frac{\theta}{2} e^{-i(\pi-\theta)/2}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $y = -x$ in (4.45). □

**Corollary 4.44** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall n \in \mathbb{N}$,*

$$
\begin{aligned}
S_n(1,1) &= \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} \\
&= \binom{n}{0} a_n + \binom{n+1}{1} a_{n+1} + \binom{n+2}{2} a_{n+2} + \binom{n+3}{3} a_{n+3} + \cdots \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(2 \cos \frac{\theta}{2} e^{i\theta/2}\right) + f\left(2 \cos \frac{\theta}{2} e^{-i\theta/2}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $x = y = 1$ in (4.45). □

**Corollary 4.45** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall n \in \mathbb{N}$,*

$$
\begin{aligned}
S_n(1,-1) &= (-1)^n \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} \\
&= (-1)^n \left( \binom{n}{0} a_n + \binom{n+1}{1} a_{n+1} + \binom{n+2}{2} a_{n+2} + \binom{n+3}{3} a_{n+3} + \cdots \right) \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(2 \sin \frac{\theta}{2} e^{i(\pi-\theta)/2}\right) + f\left(2 \sin \frac{\theta}{2} e^{-i(\pi-\theta)/2}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $x = 1$ and $y = -1$ in (4.45). □

**Corollary 4.46** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall n \in \mathbb{N}$,*

$$
\begin{aligned}
S_n(-1,-1) &= (-1)^n \sum_{j=0}^{\infty} (-1)^j \binom{n+j}{j} a_{n+j} \\
&= (-1)^n \left( \binom{n}{0} a_n - \binom{n+1}{1} a_{n+1} + \binom{n+2}{2} a_{n+2} - \binom{n+3}{3} a_{n+3} + \cdots \right) \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(-2\cos\frac{\theta}{2} e^{i\theta/2}\right) + f\left(-2\cos\frac{\theta}{2} e^{-i\theta/2}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $x = y = -1$ in (4.45). □

**Corollary 4.47** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall n \in \mathbb{N}$,*

$$
\begin{aligned}
S_n(-1,1) &= \sum_{j=0}^{\infty} (-1)^j \binom{n+j}{j} a_{n+j} \\
&= \binom{n}{0} a_n - \binom{n+1}{1} a_{n+1} + \binom{n+2}{2} a_{n+2} - \binom{n+3}{3} a_{n+3} + \cdots \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(-2\sin\frac{\theta}{2} e^{i(\pi-\theta)/2}\right) + f\left(-2\sin\frac{\theta}{2} e^{-i(\pi-\theta)/2}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $x = -1$ and $y = 1$ in (4.45). □

**Corollary 4.48** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall (y, n) \in \mathbb{R} \times \mathbb{N}$,*

$$
\begin{aligned}
S_n(1,y) &= y^n \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} \\
&= y^n \left( \binom{n}{0} a_n + \binom{n+1}{1} a_{n+1} + \binom{n+2}{2} a_{n+2} + \binom{n+3}{3} a_{n+3} + \cdots \right) \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(1 + ye^{i\theta}\right) + f\left(1 + ye^{-i\theta}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $x = 1$ in (4.45). □

**Corollary 4.49** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall (y, n) \in \mathbb{R} \times \mathbb{N}$,*

$$
\begin{aligned}
S_n(-1, y) &= y^n \sum_{j=0}^{\infty} (-1)^j \binom{n+j}{j} a_{n+j} \\
&= y^n \left( \binom{n}{0} a_n - \binom{n+1}{1} a_{n+1} + \binom{n+2}{2} a_{n+2} - \binom{n+3}{3} a_{n+3} + \cdots \right) \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(-1 + ye^{i\theta}\right) + f\left(-1 + ye^{-i\theta}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $x = -1$ in (4.45). □

**Corollary 4.50** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall (x, n) \in \mathbb{R} \times \mathbb{N}$,*

$$
\begin{aligned}
S_n(x, 1) &= \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j \\
&= \binom{n}{0} a_n + \binom{n+1}{1} a_{n+1} x + \binom{n+2}{2} a_{n+2} x^2 + \binom{n+3}{3} a_{n+3} x^3 + \cdots \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(x + e^{i\theta}\right) + f\left(x + e^{-i\theta}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $y = 1$ in (4.45). □

**Corollary 4.51** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall (x, n) \in \mathbb{R} \times \mathbb{N}$,*

$$
\begin{aligned}
S_n(x, -1) &= (-1)^n \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j \\
&= (-1)^n \left( \binom{n}{0} a_n + \binom{n+1}{1} a_{n+1} x + \binom{n+2}{2} a_{n+2} x^2 + \binom{n+3}{3} a_{n+3} x^3 + \cdots \right) \\
&= \frac{2}{\pi} \int_0^{\pi} \frac{f\left(x - e^{i\theta}\right) + f\left(x - e^{-i\theta}\right)}{2} \cos n\theta \, d\theta.
\end{aligned}
$$

***Proof*** Put $y = -1$ in (4.45). □

**Corollary 4.52** (Corollary to Theorem 4.41)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall (y, n) \in \mathbb{R} \times \mathbb{N}$,*

$$
S_n(0, y) = y^n a_n = \frac{2}{\pi} \int_0^{\pi} \frac{f\left(ye^{i\theta}\right) + f\left(ye^{-i\theta}\right)}{2} \cos n\theta \, d\theta.
$$

***Proof*** Put $x = 0$ in (4.45). □

**Corollary 4.53** (Corollary to Theorem 4.41. This is also Theorem 4.4)
*Let $f(z)$ be the function as defined in Theorem 4.41. Then, $\forall n \in \mathbb{N}$,*

$$S_n(0, 1) = a_n = \frac{2}{\pi} \int_0^{\pi} \frac{f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)}{2} \cos n\theta \, d\theta.$$

***Proof*** Put $x = 0$ and $y = 1$ in (4.45). □

## Problems

[4.1] Prove that

$$\int_0^{\pi} \cosh(\cos\theta) \cos(\sin\theta) \, d\theta = \pi.$$

[4.2] (Whittaker and Watson, 2016) Prove that $\forall n \in \mathbb{N} - 1$,

$$\int_0^{\pi} e^{\cos\theta} \cos(n\theta - \sin\theta) \, d\theta = \frac{\pi}{n!}.$$

[4.3] Prove that $\forall n \in -\mathbb{N}$,

$$\int_0^{\pi} e^{\cos\theta} \cos(n\theta - \sin\theta) \, d\theta = 0.$$

[4.4] Prove that $\forall n \in \mathbb{N} - 1$,

$$\int_0^{\pi} e^{-\cos\theta} \cos(n\theta + \sin\theta) \, d\theta = (-1)^n \frac{\pi}{n!}.$$

[4.5] Prove that $\forall n \in -\mathbb{N}$,

$$\int_0^{\pi} e^{-\cos\theta} \cos(n\theta + \sin\theta) \, d\theta = 0.$$

[4.6] Prove that $\forall (x, n) \in \mathbb{R} \times (\mathbb{N} - 1)$,

$$\int_0^{\pi} e^{x\cos\theta} \cos(n\theta - x\sin\theta) \, d\theta = \frac{\pi}{n!} x^n.$$

[4.7] Prove that $\forall (x, n) \in \mathbb{R} \times (-\mathbb{N})$,

$$\int_0^{\pi} e^{x \cos\theta} \cos(n\theta - x \sin\theta)\, d\theta = 0.$$

[4.8] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$,

$$\int_0^{\pi} (\cos(n\theta) \sin(x \cos\theta) \cosh(x \sin\theta) + \sin(n\theta) \cos(x \cos\theta) \sinh(x \sin\theta))\, d\theta = (-1)^{(n-1)/2} \frac{\pi}{n!} x^n.$$

[4.9] Prove that $\forall (x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 2))$,

$$\int_0^{\pi} (\cos(n\theta) \sin(x \cos\theta) \cosh(x \sin\theta) + \sin(n\theta) \cos(x \cos\theta) \sinh(x \sin\theta))\, d\theta = 0.$$

[4.10] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\int_0^{\pi} (\cos(n\theta) \cos(x \cos\theta) \cosh(x \sin\theta) - \sin(n\theta) \sin(x \cos\theta) \sinh(x \sin\theta))\, d\theta = (-1)^{n/2} \frac{\pi}{n!} x^n.$$

[4.11] Prove that $\forall (x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 1))$,

$$\int_0^{\pi} (\cos(n\theta) \cos(x \cos\theta) \cosh(x \sin\theta) - \sin(n\theta) \sin(x \cos\theta) \sinh(x \sin\theta))\, d\theta = 0.$$

[4.12] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$,

$$\int_0^{\pi} (\cos(n\theta) \sinh(x \cos\theta) \cos(x \sin\theta) + \sin(n\theta) \cosh(x \cos\theta) \sin(x \sin\theta))\, d\theta = \frac{\pi}{n!} x^n.$$

[4.13] Prove that $\forall(x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 2))$,

$$\int_0^\pi (\cos(n\theta) \sinh(x \cos\theta) \cos(x \sin\theta) + \sin(n\theta) \cosh(x \cos\theta) \sin(x \sin\theta)) \, d\theta = 0.$$

[4.14] Prove that $\forall(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\int_0^\pi (\cos(n\theta) \cosh(x \cos\theta) \cos(x \sin\theta) + \sin(n\theta) \sinh(x \cos\theta) \sin(x \sin\theta)) \, d\theta = \frac{\pi}{n!} x^n.$$

[4.15] Prove that $\forall(x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 1))$,

$$\int_0^\pi (\cos(n\theta) \cosh(x \cos\theta) \cos(x \sin\theta) + \sin(n\theta) \sinh(x \cos\theta) \sin(x \sin\theta)) \, d\theta = 0.$$

[4.16] Prove that $\forall(m, n) \in (\mathbb{N} - 1) \times [0..m]$,

$$\int_0^\pi \cos\frac{(2n - m)\theta}{2} \cos^m\frac{\theta}{2} \, d\theta = \frac{\pi}{2^m} \binom{m}{n}.$$

[4.17] Prove that $\forall(m, n) \in (\mathbb{N} - 1) \times (\mathbb{Z} - [0..m])$,

$$\int_0^\pi \cos\frac{(2n - m)\theta}{2} \cos^m\frac{\theta}{2} \, d\theta = 0.$$

[4.18] Prove that $\forall(m, n) \in (\mathbb{N} - 1) \times [0..m]$,

$$\int_0^\pi \cos\frac{(2n - m)\theta + m\pi}{2} \sin^m\frac{\theta}{2} \, d\theta = (-1)^n \frac{\pi}{2^m} \binom{m}{n}.$$

[4.19] Prove that $\forall (m, n) \in (\mathbb{N} - 1) \times (\mathbb{Z} - [0..m])$,

$$\int_0^\pi \cos \frac{(2n - m)\theta + m\pi}{2} \sin^m \frac{\theta}{2}\, d\theta = 0.$$

[4.20] Prove that $\forall (x, m, n) \in [-1, 1] \times (\mathbb{N} - 1) \times [0..m]$,

$$\int_0^\pi (1 + x^2 + 2x \cos\theta)^{m/2} \cos\left(n\theta - m \tan^{-1}\left(\frac{x \sin\theta}{1 + x\cos\theta}\right)\right) d\theta = \pi \binom{m}{n} x^n.$$

[4.21] Prove that $\forall (x, m, n) \in [-1, 1] \times (\mathbb{N} - 1) \times (\mathbb{Z} - [0..m])$,

$$\int_0^\pi (1 + x^2 + 2x \cos\theta)^{m/2} \cos\left(n\theta - m \tan^{-1}\left(\frac{x \sin\theta}{1 + x\cos\theta}\right)\right) d\theta = 0.$$

[4.22] (Watson, 2011) Prove that

$$\begin{aligned}\int_0^\pi e^{2\cos\theta}\, d\theta &= \pi I_0(2) \\ &= \pi \sum_{j=0}^{\infty} \frac{1}{(j!)^2} \\ &= \pi \left(\frac{1}{(0!)^2} + \frac{1}{(1!)^2} + \frac{1}{(2!)^2} + \frac{1}{(3!)^2} + \cdots\right).\end{aligned}$$

[4.23] Prove that $\forall n \in \mathbb{N} - 1$,

$$\begin{aligned}&\int_0^\pi e^{2\cos\theta} \cos n\theta\, d\theta = \pi I_n(2) \\ &= \pi \sum_{j=0}^{\infty} \frac{1}{j!(n + j)!} \\ &= \pi \left(\frac{1}{0!n!} + \frac{1}{1!(n+1)!} + \frac{1}{2!(n+2)!} + \frac{1}{3!(n+3)!} + \cdots\right).\end{aligned}$$

[4.24] (Watson, 2011) Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}\int_0^{\pi} e^{2x\cos\theta}\, d\theta &= \pi I_0(2x)\\ &= \pi \sum_{j=0}^{\infty} \frac{x^{2j}}{(j!)^2}\\ &= \pi\left(\frac{1}{(0!)^2} + \frac{x^2}{(1!)^2} + \frac{x^4}{(2!)^2} + \frac{x^6}{(3!)^2} + \cdots\right).\end{aligned}$$

[4.25] Prove that $\forall (x, n) \in \mathbb{R} \times (\mathbb{N} - 1)$,

$$\begin{aligned}&\int_0^{\pi} e^{2x\cos\theta}\cos n\theta\, d\theta = \pi I_n(2x)\\ &= \pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!}\\ &= \pi x^n\left(\frac{1}{0!n!} + \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} + \frac{x^6}{3!(n+3)!} + \cdots\right).\end{aligned}$$

[4.26] $\forall (\theta, n, i, a_i) \in \mathbb{R} \times (\mathbb{N} - 1) \times [0..n] \times \mathbb{R}$, let

$$\begin{aligned}C(\theta) &= \sum_{j=0}^{n} a_j \cos j\theta\\ &= a_0 + a_1\cos\theta + a_2\cos 2\theta + \cdots + a_n\cos n\theta,\end{aligned} \tag{4.46}$$

and

$$\begin{aligned}S(\theta) &= \sum_{j=1}^{n} a_j \sin j\theta\\ &= a_1\sin\theta + a_2\sin 2\theta + \cdots + a_n\sin n\theta.\end{aligned} \tag{4.47}$$

Prove that

$$\begin{aligned}\int_0^{\pi} \left(C^2(\theta) + S^2(\theta)\right) d\theta &= \pi \sum_{j=0}^{n} a_j^2\\ &= \pi\left(a_0^2 + a_1^2 + a_2^2 + \cdots + a_n^2\right).\end{aligned}$$

[4.27] Let $C(\theta)$ and $S(\theta)$ be as defined in (4.46) and (4.47) respectively. Prove that $\forall m \in \mathbb{N} - 1$,

$$\int_0^\pi \left(C^2(\theta) + S^2(\theta)\right) \cos m\theta \, d\theta = \pi \sum_{j=0}^{n-m} a_j a_{j+m}$$
$$= \pi \left(a_0 a_m + a_1 a_{1+m} + a_2 a_{2+m} + \cdots + a_{n-m} a_n\right).$$

[4.28] Prove that $\forall n \in \mathbb{N} - 1$,

$$\int_0^\pi \sin^{2n} \frac{\theta}{2} \, d\theta = \frac{\pi}{2^{2n}} \sum_{j=0}^{n} \binom{n}{j}^2$$
$$= \frac{\pi}{2^{2n}} \left( \binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \cdots + \binom{n}{n}^2 \right).$$

[4.29] Prove that $\forall (n, m) \in \mathbb{N} \times [0\,..\,n]$,

$$\int_0^\pi \sin^{2n} \frac{\theta}{2} \cos m\theta \, d\theta = (-1)^m \frac{\pi}{2^{2n}} \sum_{j=0}^{n-m} \binom{n}{j} \binom{n}{m+j}$$
$$= (-1)^m \frac{\pi}{2^{2n}} \left( \binom{n}{0} \binom{n}{m} + \binom{n}{1} \binom{n}{m+1} + \binom{n}{2} \binom{n}{m+2} \right.$$
$$\left. + \cdots + \binom{n}{n-m} \binom{n}{n} \right).$$

[4.30] Prove that $\forall (n, m) \in \mathbb{N} \times ((\mathbb{N} - 1) - [0\,..\,n])$,

$$\int_0^\pi \sin^{2n} \frac{\theta}{2} \cos m\theta \, d\theta = 0.$$

[4.31] Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}
&\int_0^\pi \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta) + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right) d\theta \\
&= \pi \sum_{j=0}^{\infty} \frac{x^{4j+2}}{((2j+1)!)^2} \\
&= \pi \left( \frac{x^2}{(1!)^2} + \frac{x^6}{(3!)^2} + \frac{x^{10}}{(5!)^2} + \frac{x^{14}}{(7!)^2} + \cdots \right).
\end{aligned}$$

[4.32] Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}
&\int_0^\pi \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta) + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right) d\theta \\
&= \pi \sum_{j=0}^{\infty} \frac{x^{4j+2}}{((2j+1)!)^2} \\
&= \pi \left( \frac{x^2}{(1!)^2} + \frac{x^6}{(3!)^2} + \frac{x^{10}}{(5!)^2} + \frac{x^{14}}{(7!)^2} + \cdots \right).
\end{aligned}$$

[4.33] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^\pi \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta) + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right) \cos n\theta \, d\theta \\
&= (-1)^{n/2}\pi x^n \sum_{j=0}^{\infty} \frac{x^{4j+2}}{(2j+1)!(n+2j+1)!} \\
&= (-1)^{n/2}\pi x^n \left( \frac{x^2}{1!(n+1)!} + \frac{x^6}{3!(n+3)!} + \frac{x^{10}}{5!(n+5)!} + \frac{x^{14}}{7!(n+7)!} + \cdots \right).
\end{aligned}$$

[4.34] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z} + 1)$,

$$\begin{aligned}
&\int_0^\pi \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta) + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right) \cos n\,\theta d\theta \\
&= 0.
\end{aligned}$$

[4.35] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^{\pi} \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta) + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\, d\theta \\
&= \pi x^n \sum_{j=0}^{\infty} \frac{x^{4j+2}}{(2j+1)!(n+2j+1)!} \\
&= \pi x^n \left(\frac{x^2}{1!(n+1)!} + \frac{x^6}{3!(n+3)!} + \frac{x^{10}}{5!(n+5)!} + \frac{x^{14}}{7!(n+7)!} + \cdots\right).
\end{aligned}$$

[4.36] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z} + 1)$,

$$\int_0^{\pi} \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta) + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\, d\theta = 0.$$

[4.37] Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}
&\int_0^{\pi} \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right) d\theta \\
&= \pi \sum_{j=0}^{\infty} \frac{x^{4j}}{((2j)!)^2} = \pi \left(\frac{1}{(0!)^2} + \frac{x^4}{(2!)^2} + \frac{x^8}{(4!)^2} + \frac{x^{12}}{(6!)^2} + \cdots\right).
\end{aligned}$$

[4.38] Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}
&\int_0^{\pi} \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right) d\theta \\
&= \pi \sum_{j=0}^{\infty} \frac{x^{4j}}{((2j)!)^2} = \pi \left(\frac{1}{(0!)^2} + \frac{x^4}{(2!)^2} + \frac{x^8}{(4!)^2} + \frac{x^{12}}{(6!)^2} + \cdots\right).
\end{aligned}$$

[4.39] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^\pi \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos n\theta\, d\theta \\
&= (-1)^{n/2}\pi x^n \sum_{j=0}^{\infty} \frac{x^{4j}}{(2j)!(n+2j)!} \\
&= (-1)^{n/2}\pi x^n \left(\frac{1}{0!n!} + \frac{x^4}{2!(n+2)!} + \frac{x^8}{4!(n+4)!} + \frac{x^{12}}{6!(n+6)!} + \cdots\right).
\end{aligned}$$

[4.40] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z} + 1)$,

$$\begin{aligned}
&\int_0^\pi \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos n\theta\, d\theta \\
&= 0.
\end{aligned}$$

[4.41] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^\pi \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\, d\theta \\
&= \pi x^n \sum_{j=0}^{\infty} \frac{x^{4j}}{(2j)!(n+2j)!} \\
&= \pi x^n \left(\frac{1}{0!n!} + \frac{x^4}{2!(n+2)!} + \frac{x^8}{4!(n+4)!} + \frac{x^{12}}{6!(n+6)!} + \cdots\right).
\end{aligned}$$

[4.42] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z} + 1)$,

$$\begin{aligned}
&\int_0^\pi \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\, d\theta \\
&= 0.
\end{aligned}$$

[4.43] Prove that $\forall n \in \mathbb{N}$,

$$\int_0^\pi \left(\cos n\theta \log\left(2\cos\frac{\theta}{2}\right) + \frac{\theta}{2}\sin n\theta\right) d\theta = (-1)^{n+1}\frac{\pi}{n}.$$

[4.44] Prove that $\forall n \in -\mathbb{N} \cup \{0\}$,

$$\int_0^\pi \left( \cos n\theta \log \left( 2 \cos \frac{\theta}{2} \right) + \frac{\theta}{2} \sin n\theta \right) d\theta = 0.$$

[4.45] Prove that $\forall (x, n) \in (\mathbb{R} - (-1, 1)) \times \mathbb{N}$,

$$\int_0^\pi \left( \cos n\theta \log \sqrt{1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta} + \sin n\theta \tan^{-1} \left( \frac{\sin\theta}{x + \cos\theta} \right) \right) d\theta$$
$$= (-1)^{n+1} \frac{\pi}{n x^n}.$$

[4.46] Prove that $\forall (x, n) \in (\mathbb{R} - (-1, 1)) \times (-\mathbb{N} \cup \{0\})$,

$$\int_0^\pi \left( \cos n\theta \log \sqrt{1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta} + \sin n\theta \tan^{-1} \left( \frac{\sin\theta}{x + \cos\theta} \right) \right) d\theta = 0.$$

[4.47] Prove that $\forall (y, n) \in \mathbb{R} \times (\mathbb{N} - 1)$,

$$\int_0^\pi \cos \frac{(2n - y)\theta}{2} \cos^y \frac{\theta}{2} \, d\theta = \frac{\pi}{2^y n!} \prod_{j=0}^{n-1} (y - j)$$
$$= \frac{\pi y (y-1)(y-2) \dots (y-n+1)}{2^y n!}.$$

[4.48] Prove that $\forall (y, n) \in \mathbb{R} \times (-\mathbb{N})$,

$$\int_0^\pi \cos \frac{(2n - y)\theta}{2} \cos^y \frac{\theta}{2} \, d\theta = 0.$$

[4.49] Prove that $\forall (x, y, n) \in (\mathbb{R} - [-1, 1]) \times \mathbb{R} \times (\mathbb{N} - 1)$,

$$\int_0^\pi \left( 1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta \right)^{y/2} \cos \left( n\theta - y \tan^{-1} \left( \frac{\sin\theta}{x + \cos\theta} \right) \right) d\theta$$
$$= \frac{\pi}{n! x^n} \prod_{j=0}^{n-1} (y - j) = \frac{\pi y (y-1)(y-2) \dots (y-n+1)}{n! x^n}.$$

[4.50] Prove that $\forall (x, y, n) \in (\mathbb{R} - [-1, 1]) \times \mathbb{R} \times (-\mathbb{N})$,

$$\int_0^\pi \left(1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta\right)^{y/2} \cos\left(n\theta - y\tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right)\right) d\theta = 0.$$

[4.51] Prove that

$$\int_0^\pi \log^2\left(2\cos\frac{\theta}{2}\right) d\theta = \int_0^\pi \log^2\left(2\sin\frac{\theta}{2}\right) d\theta = \frac{\pi^3}{12}.$$

[4.52] Prove that $\forall n \in 2\mathbb{N} - 2$,

$$\int_0^\pi \left(\log^2\left(2\cos\frac{\theta}{2}\right) + \frac{\theta^2}{4}\right)\cos n\theta\, d\theta = \pi \sum_{j=1}^{\infty} \frac{1}{j(n+j)}$$
$$= \pi\left(\frac{1}{1(n+1)} + \frac{1}{2(n+2)} + \frac{1}{3(n+3)} + \frac{1}{4(n+4)} + \cdots\right).$$

[4.53] Prove that $\forall n \in 2\mathbb{N} - 1$,

$$\int_0^\pi \left(\log^2\left(2\cos\frac{\theta}{2}\right) + \frac{\theta^2}{4}\right)\cos n\theta\, d\theta = -\pi \sum_{j=1}^{\infty} \frac{1}{j(n+j)}$$
$$= -\pi\left(\frac{1}{1(n+1)} + \frac{1}{2(n+2)} + \frac{1}{3(n+3)} + \frac{1}{4(n+4)} + \cdots\right).$$

[4.54] Prove that $\forall x \in [1, \infty)$,

$$\int_0^\pi \left(\log^2\sqrt{1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta} + \left(\tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right)\right)^2\right) d\theta = \pi \sum_{j=1}^{\infty} \frac{1}{j^2 x^{2j}}$$
$$= \pi\left(\frac{1}{x^2} + \frac{1}{2^2 x^4} + \frac{1}{3^2 x^6} + \frac{1}{4^2 x^8} + \cdots\right).$$

[4.55] Prove that $\forall (x, n) \in [1, \infty) \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^{\pi} \left( \log^2 \sqrt{1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta} + \left( \tan^{-1} \left( \frac{\sin\theta}{x + \cos\theta} \right) \right)^2 \right) \cos n\theta \, d\theta \\
&= \frac{\pi}{x^n} \sum_{j=1}^{\infty} \frac{1}{j(n+j)x^{2j}} \\
&= \frac{\pi}{x^n} \left( \frac{1}{1(n+1)x^2} + \frac{1}{2(n+2)x^4} + \frac{1}{3(n+3)x^6} + \frac{1}{4(n+4)x^8} + \cdots \right).
\end{aligned}$$

[4.56] Prove that $\forall (x, n) \in [1, \infty) \times (2\mathbb{N} - 1)$,

$$\begin{aligned}
&\int_0^{\pi} \left( \log^2 \sqrt{1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta} + \left( \tan^{-1} \left( \frac{\sin\theta}{x + \cos\theta} \right) \right)^2 \right) \cos n\theta \, d\theta \\
&= -\frac{\pi}{x^n} \sum_{j=1}^{\infty} \frac{1}{j(n+j)x^{2j}} \\
&= -\frac{\pi}{x^n} \left( \frac{1}{1(n+1)x^2} + \frac{1}{2(n+2)x^4} + \frac{1}{3(n+3)x^6} + \frac{1}{4(n+4)x^8} + \cdots \right).
\end{aligned}$$

[4.57] Prove that $\forall (x, y) \in ((1, \infty) \times \mathbb{R}) \cup (\mathbb{R} \times \mathbb{N})$,

$$\begin{aligned}
&\int_0^{\pi} \left( 1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta \right)^y d\theta = \pi \sum_{j=0}^{\infty} \left( \frac{\prod_{k=0}^{j-1} (y-k)}{j! x^j} \right)^2 \\
&= \pi \left( (1)^2 + \left( \frac{y}{x} \right)^2 + \left( \frac{y(y-1)}{2!x^2} \right)^2 + \left( \frac{y(y-1)(y-2)}{3!x^3} \right)^2 + \cdots \right).
\end{aligned}$$

[4.58] Prove that $\forall(x, y, n) \in (1, \infty) \times \mathbb{R} \times (\mathbb{N} - 1)$,

$$\int_0^\pi \left(1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta\right)^y \cos n\theta \, d\theta = \pi \sum_{j=0}^{\infty} \frac{\prod_{k=0}^{j-1}(y-k)}{j!x^j} \cdot \frac{\prod_{k=0}^{n+j-1}(y-k)}{(n+j)!x^{n+j}}$$

$$= \pi \left(1 \frac{y(y-1)(y-2)\dots(y-n+1)}{n!x^n} + \frac{y}{x} \cdot \frac{y(y-1)(y-2)\dots(y-n)}{(n+1)!x^{n+1}}\right.$$

$$+ \frac{y(y-1)}{2!x^2} \cdot \frac{y(y-1)(y-2)\dots(y-n-1)}{(n+2)!x^{n+2}}$$

$$\left. + \frac{y(y-1)(y-2)}{3!x^3} \cdot \frac{y(y-1)(y-2)\dots(y-n-2)}{(n+3)!x^{n+3}} + \cdots\right).$$

[4.59] Prove that $\forall(x, m, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{N} - 2) \times ((2\mathbb{N} - 2) \cap [0..m]) \cup (\mathbb{R} - \{0\}) \times (2\mathbb{N} - 1) \times ((2\mathbb{N} - 1) \cap [0..m])$,

$$\int_0^{\pi/x} \cos(nx\theta)\cos^m(x\theta)\, d\theta = \frac{\pi}{2^m x}\binom{m}{(m-n)/2}.$$

[4.60] Prove that $\forall(x, m, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{N} - 2) \times (2\mathbb{N} - 1) \cup (\mathbb{R} - \{0\}) \times (2\mathbb{N} - 1) \times (2\mathbb{N} - 2)$,

$$\int_0^{\pi/x} \cos(nx\theta)\cos^m(x\theta)\, d\theta = 0.$$

[4.61] Prove that $\forall(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \sin^{-1}(2x\cos\theta)\cos n\theta \, d\theta$$

$$= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2} \frac{x^{n+2j}}{(n+2j)2^{n+2j-1}}$$

$$= \pi \left( \frac{\binom{n}{0}\binom{n-1}{(n-1)/2}}{n \cdot 2^{n-1}} x^n + \frac{\binom{n+2}{1}\binom{n+1}{(n+1)/2}}{(n+2)2^{n+1}} x^{n+2} \right.$$

$$\left. + \frac{\binom{n+4}{2}\binom{n+3}{(n+3)/2}}{(n+4)2^{n+3}} x^{n+4} + \frac{\binom{n+6}{3}\binom{n+5}{(n+5)/2}}{(n+6)2^{n+5}} x^{n+6} + \cdots \right).$$

[4.62] Prove that $\forall (x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times 2\mathbb{Z}$,

$$\int_0^\pi \sin^{-1}(2x\cos\theta)\cos n\theta\, d\theta = 0.$$

[4.63] Prove that $\forall (x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$,

$$\begin{aligned}
&\int_0^\pi \sinh^{-1}(2x\sin\theta)\sin n\theta\, d\theta \\
&= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2} \frac{x^{n+2j}}{(n+2j)2^{n+2j-1}} \\
&= \pi \left( \frac{\binom{n}{0}\binom{n-1}{(n-1)/2}}{n\cdot 2^{n-1}} x^n - \frac{\binom{n+2}{1}\binom{n+1}{(n+1)/2}}{(n+2)2^{n+1}} x^{n+2} \right. \\
&\quad \left. + \frac{\binom{n+4}{2}\binom{n+3}{(n+3)/2}}{(n+4)2^{n+3}} x^{n+4} - \frac{\binom{n+6}{3}\binom{n+5}{(n+5)/2}}{(n+6)2^{n+5}} x^{n+6} + \cdots \right).
\end{aligned}$$

[4.64] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \sinh^{-1}(2x\sin\theta)\sin n\theta\, d\theta = 0.$$

[4.65] Prove that $\forall (x,n) \in \left[-\frac{1}{2},\frac{1}{2}\right] \times (2\mathbb{N}-1)$,

$$\begin{aligned}
&\int_0^\pi \sinh^{-1}(2x\cos\theta)\cos n\theta\, d\theta \\
&= (-1)^{(n-1)/2}\pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2}\frac{x^{n+2j}}{(n+2j)2^{n+2j-1}} \\
&= (-1)^{(n-1)/2}\pi \left( \frac{\binom{n}{0}\binom{n-1}{(n-1)/2}}{n\cdot 2^{n-1}}x^n - \frac{\binom{n+2}{1}\binom{n+1}{(n+1)/2}}{(n+2)2^{n+1}}x^{n+2} \right. \\
&\quad \left. + \frac{\binom{n+4}{2}\binom{n+3}{(n+3)/2}}{(n+4)2^{n+3}}x^{n+4} - \frac{\binom{n+6}{3}\binom{n+5}{(n+5)/2}}{(n+6)2^{n+5}}x^{n+6} + \cdots \right).
\end{aligned}$$

[4.66] Prove that $\forall (x,n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \sinh^{-1}(2x\cos\theta)\cos n\theta\, d\theta = 0.$$

[4.67] Prove that $\forall (x,n) \in \left[-\frac{1}{2},\frac{1}{2}\right] \times (2\mathbb{N}-1)$,

$$\begin{aligned}
&\int_0^\pi \sin^{-1}(2x\sin\theta)\sin n\theta\, d\theta \\
&= (-1)^{(n-1)/2}\pi \sum_{j=0}^{\infty} \binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2}\frac{x^{n+2j}}{(n+2j)2^{n+2j-1}} \\
&= (-1)^{(n-1)/2}\pi \left( \frac{\binom{n}{0}\binom{n-1}{(n-1)/2}}{n\cdot 2^{n-1}}x^n + \frac{\binom{n+2}{1}\binom{n+1}{(n+1)/2}}{(n+2)2^{n+1}}x^{n+2} \right. \\
&\quad \left. + \frac{\binom{n+4}{2}\binom{n+3}{(n+3)/2}}{(n+4)2^{n+3}}x^{n+4} + \frac{\binom{n+6}{3}\binom{n+5}{(n+5)/2}}{(n+6)2^{n+5}}x^{n+6} + \cdots \right).
\end{aligned}$$

[4.68] Prove that $\forall (x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times 2\mathbb{Z}$,

$$\int_0^\pi \sin^{-1}(2x \sin\theta) \sin n\theta \, d\theta = 0.$$

[4.69] Prove that $\forall (x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$,

$$\begin{aligned}\int_0^\pi \tan^{-1}(2x \cos\theta) \cos n\theta \, d\theta &= (-1)^{(n-1)/2}\pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j} \\ &= (-1)^{(n-1)/2}\pi \left( \frac{\binom{n}{0}}{n} x^n - \frac{\binom{n+2}{1}}{n+2} x^{n+2} \right. \\ &\quad \left. + \frac{\binom{n+4}{2}}{n+4} x^{n+4} - \frac{\binom{n+6}{3}}{n+6} x^{n+6} + \cdots \right).\end{aligned}$$

[4.70] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \tan^{-1}(2x \cos\theta) \cos n\theta \, d\theta = 0.$$

[4.71] Prove that $\forall (x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$,

$$\begin{aligned}\int_0^\pi \tanh^{-1}(2x \sin\theta) \sin n\theta \, d\theta &= (-1)^{(n-1)/2}\pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j} \\ &= (-1)^{(n-1)/2}\pi \left( \frac{\binom{n}{0}}{n} x^n + \frac{\binom{n+2}{1}}{n+2} x^{n+2} \right. \\ &\quad \left. + \frac{\binom{n+4}{2}}{n+4} x^{n+4} + \frac{\binom{n+6}{3}}{n+6} x^{n+6} + \cdots \right).\end{aligned}$$

[4.72] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \tanh^{-1}(2x \sin\theta) \sin n\theta \, d\theta = 0.$$

[4.73] Prove that $\forall (x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \tanh^{-1}(2x \cos\theta) \cos n\theta \, d\theta = \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j}$$
$$= \pi \left( \frac{\binom{n}{0}}{n} x^n + \frac{\binom{n+2}{1}}{n+2} x^{n+2} + \frac{\binom{n+4}{2}}{n+4} x^{n+4} + \frac{\binom{n+6}{3}}{n+6} x^{n+6} + \cdots \right).$$

[4.74] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \tanh^{-1}(2x \cos\theta) \cos n\theta \, d\theta = 0.$$

[4.75] Prove that $\forall (x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \tan^{-1}(2x \sin\theta) \sin n\theta \, d\theta = \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j}$$
$$= \pi \left( \frac{\binom{n}{0}}{n} x^n - \frac{\binom{n+2}{1}}{n+2} x^{n+2} + \frac{\binom{n+4}{2}}{n+4} x^{n+4} - \frac{\binom{n+6}{3}}{n+6} x^{n+6} + \cdots \right).$$

[4.76] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \tan^{-1}(2x \sin\theta) \sin n\theta \, d\theta = 0.$$

[4.77] (Watson, 2011) Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \sin(2x\cos\theta)\cos n\theta\, d\theta = (-1)^{(n-1)/2}\pi J_n(2x)$$
$$= (-1)^{(n-1)/2}\pi x^n \sum_{j=0}^{\infty}(-1)^j \frac{x^{2j}}{j!(n+j)!}$$
$$= (-1)^{(n-1)/2}\pi x^n \left(\frac{1}{0!n!} - \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} - \frac{x^6}{3!(n+3)!} + \cdots\right).$$

[4.78] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \sin(2x\cos\theta)\cos n\theta\, d\theta = 0.$$

[4.79] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \sinh(2x\sin\theta)\sin n\theta\, d\theta = (-1)^{(n-1)/2}\pi I_n(2x)$$
$$= (-1)^{(n-1)/2}\pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!}$$
$$= (-1)^{(n-1)/2}\pi x^n \left(\frac{1}{0!n!} + \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} + \frac{x^6}{3!(n+3)!} + \cdots\right).$$

[4.80] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \sinh(2x\sin\theta)\sin n\theta\, d\theta = 0.$$

[4.81] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \sinh(2x\cos\theta)\cos n\theta\, d\theta = \pi I_n(2x)$$
$$= \pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!}$$
$$= \pi x^n \left(\frac{1}{0!n!} + \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} + \frac{x^6}{3!(n+3)!} + \cdots\right).$$

[4.82] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \sinh(2x\cos\theta)\cos n\theta\, d\theta = 0.$$

[4.83] (Watson, 2011) Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$,

$$\begin{aligned}
&\int_0^\pi \sin(2x\sin\theta)\sin n\theta\, d\theta = \pi J_n(2x) \\
&= \pi x^n \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j}}{j!(n+j)!} \\
&= \pi x^n \left( \frac{1}{0!n!} - \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} - \frac{x^6}{3!(n+3)!} + \cdots \right).
\end{aligned}$$

[4.84] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\int_0^\pi \sin(2x\sin\theta)\sin n\theta\, d\theta = 0.$$

[4.85] (Watson, 2011) Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^\pi \cos(2x\cos\theta)\cos n\theta\, d\theta = (-1)^{n/2}\pi J_n(2x) \\
&= (-1)^{n/2}\pi x^n \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j}}{j!(n+j)!} \\
&= (-1)^{n/2}\pi x^n \left( \frac{1}{0!n!} - \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} - \frac{x^6}{3!(n+3)!} + \cdots \right).
\end{aligned}$$

[4.86] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z} + 1)$,

$$\int_0^\pi \cos(2x\cos\theta)\cos n\theta\, d\theta = 0.$$

[4.87] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^\pi \cosh(2x \sin\theta) \cos n\theta \, d\theta = (-1)^{n/2} \pi I_n(2x) \\
&= (-1)^{n/2} \pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!} \\
&= (-1)^{n/2} \pi x^n \left( \frac{1}{0!n!} + \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} + \frac{x^6}{3!(n+3)!} + \cdots \right).
\end{aligned}$$

[4.88] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z} + 1)$,

$$\int_0^\pi \cosh(2x \sin\theta) \cos n\theta \, d\theta = 0.$$

[4.89] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^\pi \cosh(2x \cos\theta) \cos n\theta \, d\theta = \pi I_n(2x) \\
&= \pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!} \\
&= \pi x^n \left( \frac{1}{0!n!} + \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} + \frac{x^6}{3!(n+3)!} + \cdots \right).
\end{aligned}$$

[4.90] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z} + 1)$,

$$\int_0^\pi \cosh(2x \cos\theta) \cos n\theta \, d\theta = 0.$$

[4.91] (Watson, 2011) Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$,

$$\begin{aligned}
&\int_0^\pi \cos(2x \sin\theta) \cos n\theta \, d\theta = \pi J_n(2x) \\
&= \pi x^n \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j}}{j!(n+j)!} \\
&= \pi x^n \left( \frac{1}{0!n!} - \frac{x^2}{1!(n+1)!} + \frac{x^4}{2!(n+2)!} - \frac{x^6}{3!(n+3)!} + \cdots \right).
\end{aligned}$$

[4.92] Prove that $\forall (x, n) \in \mathbb{R} \times (2\mathbb{Z}+1)$,

$$\int_0^\pi \cos(2x \sin\theta) \cos n\theta \, d\theta = 0.$$

[4.93] Prove that $\forall (x, y, m) \in \mathbb{R}^2 \times (\mathbb{N}-1)$,

$$\begin{aligned}\int_0^\pi (y+2x\cos\theta)^m \, d\theta &= \pi \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{2j}{j}\binom{m}{2j} y^{m-2j} x^{2j} \\ &= \pi \left( \binom{0}{0}\binom{m}{0} y^m x^0 + \binom{2}{1}\binom{m}{2} y^{m-2} x^2 + \binom{4}{2}\binom{m}{4} y^{m-4} x^4 \right. \\ &\quad \left. + \cdots + \binom{2\lfloor m/2 \rfloor}{\lfloor m/2 \rfloor}\binom{m}{2\lfloor m/2 \rfloor} y^{m-2\lfloor m/2 \rfloor} x^{2\lfloor m/2 \rfloor} \right).\end{aligned}$$

[4.94] Prove that $\forall (x, y, m, n) \in \mathbb{R}^2 \times (\mathbb{N}-1) \times [0..m]$,

$$\begin{aligned}\int_0^\pi (y+2x\cos\theta)^m \cos n\theta \, d\theta &= \pi \sum_{j=0}^{\lfloor (m-n)/2 \rfloor} \binom{n+2j}{j}\binom{m}{n+2j} y^{m-n-2j} x^{n+2j} \\ &= \pi \left( \binom{n}{0}\binom{m}{n} y^{m-n} x^n + \binom{n+2}{1}\binom{m}{n+2} y^{m-n-2} x^{n+2} \right. \\ &\quad + \binom{n+4}{2}\binom{m}{n+4} y^{m-n-4} x^{n+4} \\ &\quad + \cdots + \binom{n+2\lfloor (m-n)/2 \rfloor}{\lfloor (m-n)/2 \rfloor}\binom{m}{n+2\lfloor (m-n)/2 \rfloor} \\ &\qquad \left. y^{m-n-2\lfloor (m-n)/2 \rfloor} x^{n+2\lfloor (m-n)/2 \rfloor} \right).\end{aligned}$$

[4.95] Prove that $\forall (x, y, m, n) \in \mathbb{R}^2 \times (\mathbb{N}-1) \times ((\mathbb{N}-1) - [0..m])$,

$$\int_0^\pi (y+2x\cos\theta)^m \cos n\theta \, d\theta = 0.$$

[4.96] Prove that $\forall x \in \mathbb{R}$,

$$\int_0^\pi e^{x\cos\theta} \cos(x \sin\theta) \, d\theta = \pi.$$

[4.97] Prove that $\forall (x, n) \in \mathbb{R} \times \mathbb{N}$,

$$\int_0^\pi e^{x\cos\theta} \cos(x\sin\theta)\cos n\theta \, d\theta = \frac{\pi}{2n!} x^n.$$

[4.98] Prove that $\forall (x, n) \in \mathbb{R} \times \mathbb{N}$,

$$\int_0^\pi e^{x\cos\theta} \sin(x\sin\theta)\sin n\theta \, d\theta = \frac{\pi}{2n!} x^n.$$

[4.99] Prove that $\forall (x, n, m) \in \mathbb{R} \times (\mathbb{N} - 1) \times [0..n]$,

$$\int_0^\pi e^{x\cos\theta} \cos((m-n)\theta + x\sin\theta) \, d\theta = \frac{\pi}{(n-m)!} x^{n-m}.$$

[4.100] Prove that $\forall (x, n, m) \in \mathbb{R} \times (\mathbb{N} - 1) \times ((\mathbb{N} - 1) - [0..n])$,

$$\int_0^\pi e^{x\cos\theta} \cos((m-n)\theta + x\sin\theta) \, d\theta = 0.$$

[4.101] Prove that

$$\int_0^\pi \log\left(2\cos\frac{\theta}{2}\right) d\theta = 0.$$

[4.102] Prove that $\forall x \in \mathbb{R} - (-1, 1)$,

$$\int_0^\pi \log\sqrt{1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta}\, d\theta = 0.$$

[4.103] Prove that $\forall n \in \mathbb{N}$,

$$\int_0^\pi \log\left(2\cos\frac{\theta}{2}\right)\cos n\theta \, d\theta = (-1)^{n+1}\frac{\pi}{2n}.$$

[4.104] Prove that $\forall (x, n) \in (\mathbb{R} - (-1, 1)) \times \mathbb{N}$,

$$\int_0^\pi \cos n\theta \log \sqrt{1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta}\, d\theta = (-1)^{n+1} \frac{\pi}{2nx^n}.$$

[4.105] Prove that $\forall (x, n) \in (\mathbb{R} - (-1, 1)) \times \mathbb{N}$,

$$\int_0^\pi \sin n\theta \tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right) d\theta = (-1)^{n+1} \frac{\pi}{2nx^n}.$$

[4.106] Prove that $\forall x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$,

$$\begin{aligned}\int_0^\pi \log(1 + 2x\cos\theta)\, d\theta &= -\pi \sum_{j=1}^{\infty} \binom{2j}{j} \frac{x^{2j}}{2j} \\ &= -\pi \left( \frac{\binom{2}{1}}{2} x^2 + \frac{\binom{4}{2}}{4} x^4 + \frac{\binom{6}{3}}{6} x^6 + \frac{\binom{8}{4}}{8} x^8 + \cdots \right).\end{aligned}$$

[4.107] Prove that $\forall (x, n) \in \left(-\frac{1}{2}, \frac{1}{2}\right) \times \mathbb{N}$,

$$\begin{aligned}\int_0^\pi \log(1 + 2x\cos\theta) \cos n\theta\, d\theta &= (-1)^{n+1} \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j} \\ &= (-1)^{n+1} \pi \left( \frac{\binom{n}{0}}{n} x^n + \frac{\binom{n+2}{1}}{n+2} x^{n+2} \right. \\ &\quad \left. + \frac{\binom{n+4}{2}}{n+4} x^{n+4} + \frac{\binom{n+6}{3}}{n+6} x^{n+6} + \cdots \right).\end{aligned}$$

[4.108] Prove that $\forall (x, y) \in \left(-\frac{1}{2}, \frac{1}{2}\right) \times \mathbb{R}$,

$$\int_0^\pi (1 + 2x\cos\theta)^y \, d\theta = \pi \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\prod_{k=0}^{2j-1} (y-k)}{(2j)!} x^{2j}$$

$$= \pi \left( 1 + \frac{\binom{2}{1}}{2!} y(y-1)x^2 + \frac{\binom{4}{2}}{4!} y(y-1)(y-2)(y-3)x^4 \right.$$

$$\left. + \frac{\binom{6}{3}}{6!} y(y-1)(y-2)(y-3)(y-4)(y-5)x^6 + \cdots \right).$$

[4.109] Prove that $\forall (x, y, n) \in \left(-\frac{1}{2}, \frac{1}{2}\right) \times \mathbb{R} \times \mathbb{N}$,

$$\int_0^\pi (1 + 2x\cos\theta)^y \cos n\theta \, d\theta = \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{\prod_{k=0}^{n+2j-1} (y-k)}{(n+2j)!} x^{n+2j}$$

$$= \pi \left( \frac{\binom{n}{0}}{n!} y(y-1)(y-2)\ldots(y-n+1)x^n \right.$$

$$+ \frac{\binom{n+2}{1}}{(n+2)!} y(y-1)(y-2)\ldots(y-n-1)x^{n+2}$$

$$+ \frac{\binom{n+4}{2}}{(n+4)!} y(y-1)(y-2)\ldots(y-n-3)x^{n+4}$$

$$\left. + \frac{\binom{n+6}{3}}{(n+6)!} y(y-1)(y-2)\ldots(y-n-5)x^{n+6} + \cdots \right).$$

[4.110] Let the function $\phi(x, y, \theta)$ be defined as

$$\phi(x, y, \theta) = \tan^{-1}\left(\frac{x-y}{x+y}\tan\theta\right). \tag{4.48}$$

Prove that $\forall (x, y, m) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (2\mathbb{N} - 2)$,

$$\int_0^{\pi} \left(x^2 + y^2 + 2xy\cos 2\theta\right)^{m/2} \cos m\phi(x, y, \theta)\, d\theta = \pi \binom{m}{m/2} (xy)^{m/2}.$$

[4.111] Let the function $\phi(x, y, \theta)$ be as defined in (4.48). Prove that $\forall (x, y, m) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (2\mathbb{N} - 1)$,

$$\int_0^{\pi} \left(x^2 + y^2 + 2xy\cos 2\theta\right)^{m/2} \cos m\phi(x, y, \theta)\, d\theta = 0.$$

[4.112] Let the function $\phi(x, y, \theta)$ be as defined in (4.48). Prove that $\forall (x, y, m, n) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (\mathbb{N} - 1)^2$, if $n \leq m$ and $(m - n)/2$ is an integer, then

$$\begin{aligned}&\int_0^{\pi} \left(x^2 + y^2 + 2xy\cos 2\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta\, d\theta \\ &= \frac{\pi(x^n + y^n)}{2} \binom{m}{(m-n)/2} (xy)^{(m-n)/2}.\end{aligned}$$

[4.113] Let the function $\phi(x, y, \theta)$ be as defined in (4.48). Prove that $\forall (x, y, m, n) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (\mathbb{N} - 1)^2$, if $n > m$ or $(m - n)/2$ is not an integer, then

$$\int_0^{\pi} \left(x^2 + y^2 + 2xy\cos 2\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta\, d\theta = 0.$$

[4.114] Let the function $\phi(x, y, \theta)$ be defined as

$$\phi(x, y, \theta) = \tan^{-1}\left(\frac{y\sin\theta}{x + y\cos\theta}\right). \tag{4.49}$$

Prove that $\forall (x, y, m) \in \left(\mathbb{R}^2 - \{(0, 0)\}\right) \times (\mathbb{N} - 1)$,

$$\int_0^{\pi} \left(x^2 + y^2 + 2xy\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta)\, d\theta = \pi x^m.$$

[4.115] Let the function $\phi(x, y, \theta)$ be as defined in (4.49). Prove that $\forall (x, y, m, n) \in \left(\mathbb{R}^2 - \{(0,0)\}\right) \times \mathbb{N} \times [1..m]$,

$$\int_0^\pi \left(x^2 + y^2 + 2xy\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta \, d\theta = \frac{\pi}{2}\binom{m}{n} x^{m-n} y^n.$$

[4.116] Let the function $\phi(x, y, \theta)$ be as defined in (4.49). Prove that $\forall (x, y, m, n) \in \left(\mathbb{R}^2 - \{(0,0)\}\right) \times \mathbb{N} \times (\mathbb{N} - [1..m])$,

$$\int_0^\pi \left(x^2 + y^2 + 2xy\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta \, d\theta = 0.$$

[4.117] Let the function $\phi(x, y, \theta)$ be defined as

$$\phi(x, y, \theta) = \tan^{-1}\left(\frac{(x - y)\sin\theta}{1 + (x + y)\cos\theta}\right). \tag{4.50}$$

Prove that $\forall (x, y, m) \in \mathbb{R}^2 \times (\mathbb{N} - 1)$,

$$\begin{aligned}
&\int_0^\pi \left(1 + x^2 + y^2 + 2xy\cos 2\theta + 2(x + y)\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta) \, d\theta \\
&= \pi \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{2j}{j}\binom{m}{2j}(xy)^j \\
&= \pi\left(1 + \binom{2}{1}\binom{m}{2}xy + \binom{4}{2}\binom{m}{4}x^2y^2 + \cdots \right. \\
&\quad \left. + \binom{2\lfloor m/2 \rfloor}{\lfloor m/2 \rfloor}\binom{m}{2\lfloor m/2 \rfloor}(xy)^{\lfloor m/2 \rfloor}\right).
\end{aligned}$$

[4.118] Let the function $\phi(x, y, \theta)$ be as defined in (4.50). Prove that $\forall (x, y, m, n) \in \mathbb{R}^2 \times (\mathbb{N} - 1)^2$,

$$\begin{aligned}
&\int_0^{\pi} \left(1 + x^2 + y^2 + 2xy\cos 2\theta + 2(x + y)\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta \, d\theta \\
&= \frac{\pi\,(x^n + y^n)}{2} \sum_{j=0}^{\lfloor (m-n)/2 \rfloor} \binom{n + 2j}{j} \binom{m}{n + 2j} (xy)^j \\
&= \frac{\pi\,(x^n + y^n)}{2} \left( \binom{n}{0}\binom{m}{n} + \binom{n+2}{1}\binom{m}{n+2} xy + \binom{n+4}{2}\binom{m}{n+4} x^2y^2 \right. \\
&\quad \left. + \cdots + \binom{n + 2\lfloor (m-n)/2 \rfloor}{\lfloor (m-n)/2 \rfloor} \binom{m}{n + 2\lfloor (m-n)/2 \rfloor} (xy)^{\lfloor (m-n)/2 \rfloor} \right).
\end{aligned}$$

[4.119] Let the function $\phi(x, y, \theta)$ be defined as

$$\phi(x, y, \theta) = \tan^{-1}\left(\frac{y \sin\theta}{1 + x + y\cos\theta}\right). \tag{4.51}$$

Prove that $\forall (x, y, m) \in \left(\mathbb{R}^2 - \{\,(-1, 0)\,\}\right) \times (\mathbb{N} - 1)$,

$$\int_0^{\pi} \left((1 + x)^2 + y^2 + 2y(1 + x)\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta)\, d\theta = \pi(1 + x)^m.$$

[4.120] Let the function $\phi(x, y, \theta)$ be as defined in (4.51). Prove that $\forall (x, y, m, n) \in \left(\mathbb{R}^2 - \{\,(-1, 0)\,\}\right) \times (\mathbb{N} - 1) \times \mathbb{N}$,

$$\begin{aligned}
&\int_0^{\pi} \left((1 + x)^2 + y^2 + 2y(1 + x)\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta \, d\theta \\
&= \frac{\pi y^n}{2} \sum_{j=0}^{m-n} \binom{n + j}{j} \binom{m}{n + j} x^j \\
&= \frac{\pi y^n}{2} \left( \binom{n}{0}\binom{m}{n} + \binom{n+1}{1}\binom{m}{n+1} x + \binom{n+2}{2}\binom{m}{n+2} x^2 \right. \\
&\quad \left. + \cdots + \binom{m}{m-n}\binom{m}{m} x^{m-n} \right) \\
&= \frac{\pi}{2}\binom{m}{n}(1 + x)^{m-n} y^n.
\end{aligned}$$

[4.121] Prove that $\forall (x, y) \in \mathbb{R}^2$, such that $|x| + |y| \leq 1$,

$$\int_0^\pi \log\left(1 + x^2 + y^2 + 2(x + y)\cos\theta + 2xy\cos 2\theta\right) d\theta = -\pi \sum_{j=1}^{\infty} \binom{2j}{j} \frac{(xy)^j}{j}$$
$$= -\pi \left( \frac{\binom{2}{1}}{1} xy + \frac{\binom{4}{2}}{2} x^2 y^2 + \frac{\binom{6}{3}}{3} x^3 y^3 + \frac{\binom{8}{4}}{4} x^4 y^4 + \cdots \right).$$

[4.122] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times \mathbb{N}$, such that $|x| + |y| \leq 1$,

$$\int_0^\pi \log\left(1 + x^2 + y^2 + 2(x + y)\cos\theta + 2xy\cos 2\theta\right) \cos n\theta \, d\theta$$
$$= (-1)^{n+1} \pi (x^n + y^n) \sum_{j=0}^{\infty} \binom{n + 2j}{j} \frac{(xy)^j}{n + 2j}$$
$$= (-1)^{n+1} \pi (x^n + y^n) \left( \frac{\binom{n}{0}}{n} + \frac{\binom{n+2}{1}}{n+2} xy + \frac{\binom{n+4}{2}}{n+4} x^2 y^2 \right.$$
$$\left. + \frac{\binom{n+6}{3}}{n+6} x^3 y^3 + \cdots \right).$$

[4.123] Prove that $\forall x \in [-1, 1]$,

$$\int_0^\pi \log\left(1 + x^2 + 2x\cos\theta\right) d\theta = 0.$$

[4.124] Prove that $\forall (x, n) \in [-1, 1] \times \mathbb{N}$,

$$\int_0^\pi \log\left(1 + x^2 + 2x\cos\theta\right) \cos n\theta \, d\theta = (-1)^{n+1} \frac{\pi}{n} x^n.$$

[4.125] Prove that $\forall (x, y) \in \mathbb{R}^2$,

$$\int_0^\pi \sin(x + y\cos\theta)\cosh(y\sin\theta)\,d\theta = \pi \sin x.$$

[4.126] Prove that $\forall (x, y) \in \mathbb{R}^2$,

$$\int_0^\pi \sinh(x + y\cos\theta)\cos(y\sin\theta)\,d\theta = \pi \sinh x.$$

[4.127] Prove that $\forall (x, y) \in \mathbb{R}^2$,

$$\int_0^\pi \cos(x + y\cos\theta)\cosh(y\sin\theta)\,d\theta = \pi \cos x.$$

[4.128] Prove that $\forall (x, y) \in \mathbb{R}^2$,

$$\int_0^\pi \cosh(x + y\cos\theta)\cos(y\sin\theta)\,d\theta = \pi \cosh x.$$

[4.129] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \sin(x + y\cos\theta)\cosh(y\sin\theta)\cos n\theta\,d\theta = (-1)^{(n-1)/2}\frac{\pi y^n}{2n!}\cos x.$$

[4.130] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{N}$,

$$\int_0^\pi \sin(x + y\cos\theta)\cosh(y\sin\theta)\cos n\theta\,d\theta = (-1)^{n/2}\frac{\pi y^n}{2n!}\sin x.$$

[4.131] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \sinh(x + y\cos\theta)\cos(y\sin\theta)\cos n\theta\,d\theta = \frac{\pi y^n}{2n!}\cosh x.$$

[4.132] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{N}$,

$$\int_0^\pi \sinh(x + y\cos\theta)\cos(y\sin\theta)\cos n\theta\, d\theta = \frac{\pi y^n}{2n!}\sinh x.$$

[4.133] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \cos(x + y\cos\theta)\cosh(y\sin\theta)\cos n\theta\, d\theta = (-1)^{(n+1)/2}\frac{\pi y^n}{2n!}\sin x.$$

[4.134] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{N}$,

$$\int_0^\pi \cos(x + y\cos\theta)\cosh(y\sin\theta)\cos n\theta\, d\theta = (-1)^{n/2}\frac{\pi y^n}{2n!}\cos x.$$

[4.135] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$,

$$\int_0^\pi \cosh(x + y\cos\theta)\cos(y\sin\theta)\cos n\theta\, d\theta = \frac{\pi y^n}{2n!}\sinh x.$$

[4.136] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{N}$,

$$\int_0^\pi \cosh(x + y\cos\theta)\cos(y\sin\theta)\cos n\theta\, d\theta = \frac{\pi y^n}{2n!}\cosh x.$$

[4.137] Let the function $\phi(x, \theta)$ be given by

$$\phi(x, \theta) = \tan^{-1}\left(\frac{x^2 - 1}{x^2 + 1}\tan\theta\right). \tag{4.52}$$

Prove that $\forall (x, m) \in (\mathbb{R} - \{0\}) \times (2\mathbb{N} - 2)$,

$$\int_0^\pi \left(x^2 + \frac{1}{x^2} + 2\cos 2\theta\right)^{m/2}\cos m\phi(x, \theta)\, d\theta = \pi\binom{m}{m/2}.$$

[4.138] Let the function $\phi(x,\theta)$ be given by (4.52). Prove that $\forall(x,m)\in(\mathbb{R}-\{0\})\times(2\mathbb{N}-1)$,

$$\int_0^\pi \left(x^2+\frac{1}{x^2}+2\cos 2\theta\right)^{m/2}\cos m\phi(x,\theta)\,d\theta=0.$$

[4.139] Let the function $\phi(x,\theta)$ be given by

$$\phi(x,\theta)=\tan^{-1}\left(\frac{x\sin\theta}{1+x\cos\theta}\right).$$

Prove that $\forall(x,m)\in\mathbb{R}\times(\mathbb{N}-1)$,

$$\int_0^\pi \left(1+x^2+2x\cos\theta\right)^{m/2}\cos m\phi(x,\theta)\,d\theta=\pi.$$

[4.140] Let the function $\phi(x,\theta)$ be given by

$$\phi(x,\theta)=\tan^{-1}\left(\frac{\sin\theta}{x^2+\cos\theta}\right).$$

Prove that $\forall(x,m)\in(\mathbb{R}-\{0\})\times(\mathbb{N}-1)$,

$$\int_0^\pi \left(x^2+\frac{1}{x^2}+2\cos\theta\right)^{m/2}\cos m\phi(x,\theta)\,d\theta=\pi x^m.$$

[4.141] Let the function $\phi(\theta)$ be given by

$$\phi(\theta)=\tan^{-1}\left(\frac{\sin\theta}{2+\cos\theta}\right). \tag{4.53}$$

Prove that $\forall m\in\mathbb{N}-1$,

$$\int_0^\pi (5+4\cos\theta)^{m/2}\cos m\phi(\theta)\,d\theta=2^m\pi.$$

[4.142] Let the function $\phi(\theta)$ be given by (4.53). Prove that $\forall (m, n) \in (\mathbb{N} - 1) \times \mathbb{N}$,

$$\int_0^\pi (5 + 4\cos\theta)^{m/2} \cos m\phi(\theta) \cos n\theta \, d\theta = \frac{\pi}{2} \sum_{j=0}^{m-n} \binom{n+j}{j} \binom{m}{n+j}$$
$$= \frac{\pi}{2} \left( \binom{n}{0}\binom{m}{n} + \binom{n+1}{1}\binom{m}{n+1} + \binom{n+2}{2}\binom{m}{n+2} + \cdots \right.$$
$$\left. + \binom{m}{m-n}\binom{m}{m} \right).$$

[4.143] (Boros and Moll, 2006) Prove that $\forall m \in 2\mathbb{N} - 2$,

$$\int_0^\pi \cos^m \theta \, d\theta = \frac{\pi}{2^m} \binom{m}{m/2}.$$

[4.144] Prove that $\forall m \in 2\mathbb{N} - 1$,

$$\int_0^\pi \cos^m \theta \, d\theta = 0.$$

[4.145] Prove that $\forall m \in \mathbb{N} - 1$,

$$\int_0^\pi \cos^m \frac{\theta}{2} \cos \frac{m\theta}{2} \, d\theta = \frac{\pi}{2^m}.$$

[4.146] Prove that

$$\int_0^\pi \log(5 + 4\cos\theta) \, d\theta = 2\pi \log 2.$$

[4.147] (Boros and Moll, 2006) Prove that

$$\int_0^\pi \log \sin\theta \, d\theta = -\pi \log 2.$$

[4.148] Prove that $\forall n \in \mathbb{N}$,

$$\int_0^\pi \log\left(\cos\frac{\theta}{2}\right) \cos n\theta \, d\theta = (-1)^{n+1} \frac{\pi}{2n}.$$

[4.149] Prove that

$$\int_0^{\pi} \log\left(e^2 + \frac{1}{e^2} + 2\cos\theta\right) d\theta = 2\pi.$$

[4.150] Prove that $\forall x \in \left[-1, -\frac{1}{4}\right] \cup \left[\frac{1}{4}, 1\right]$,

$$\begin{aligned}
&\int_0^{\pi} \log\left(1 + x^2 + \frac{1}{16x^2} + 2\left(x - \frac{1}{4x}\right)\cos\theta - \frac{\cos 2\theta}{2}\right) d\theta \\
&= \pi \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{2^{2j} j} \binom{2j}{j} \\
&= \pi \left( \frac{\binom{2}{1}}{1 \cdot 2^2} - \frac{\binom{4}{2}}{2 \cdot 2^4} + \frac{\binom{6}{3}}{3 \cdot 2^6} - \frac{\binom{8}{4}}{4 \cdot 2^8} + \cdots \right).
\end{aligned}$$

[4.151] Prove that $\forall (x, n) \in \left(\left[-1, -\frac{1}{4}\right] \cup \left[\frac{1}{4}, 1\right]\right) \times \mathbb{N}$,

$$\begin{aligned}
&\int_0^{\pi} \log\left(1 + x^2 + \frac{1}{16x^2} + 2\left(x - \frac{1}{4x}\right)\cos\theta - \frac{\cos 2\theta}{2}\right) \cos n\theta \, d\theta \\
&= (-1)^{n+1} \left(x^n + \frac{(-1)^n}{4^n x^n}\right) \pi \sum_{j=0}^{\infty} \frac{(-1)^j}{2^{2j}(n+2j)} \binom{n+2j}{j} \\
&= (-1)^{n+1} \left(x^n + \frac{(-1)^n}{4^n x^n}\right) \left( \frac{\binom{n}{0}}{n} - \frac{\binom{n+2}{1}}{2^2(n+2)} + \frac{\binom{n+4}{2}}{2^4(n+4)} - \frac{\binom{n+6}{3}}{2^6(n+6)} \right. \\
&\quad \left. + \cdots \right).
\end{aligned}$$

[4.152] Prove that

$$\begin{aligned}
\int_0^{\pi} \log\left(1 + \sin^2\theta\right) d\theta &= \pi \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{2^{2j} j} \binom{2j}{j} \\
&= \pi \left( \frac{\binom{2}{1}}{1 \cdot 2^2} - \frac{\binom{4}{2}}{2 \cdot 2^4} + \frac{\binom{6}{3}}{3 \cdot 2^6} - \frac{\binom{8}{4}}{4 \cdot 2^8} + \cdots \right).
\end{aligned}$$

[4.153] Prove that $\forall n \in 2\mathbb{N}$,

$$\int_0^\pi \log\left(1+\sin^2\theta\right)\cos n\theta\, d\theta = -\frac{\pi}{2^{n-1}}\sum_{j=0}^{\infty}\frac{(-1)^j}{2^{2j}(n+2j)}\binom{n+2j}{j}$$
$$= -\frac{\pi}{2^{n-1}}\left(\frac{\binom{n}{0}}{n} - \frac{\binom{n+2}{1}}{2^2(n+2)} + \frac{\binom{n+4}{2}}{2^4(n+4)} - \frac{\binom{n+6}{3}}{2^6(n+6)} + \cdots\right).$$

[4.154] Prove that $\forall n \in 2\mathbb{N}-1$,

$$\int_0^\pi \log\left(1+\sin^2\theta\right)\cos n\theta\, d\theta = 0.$$

[4.155] Let the function $\phi(x,\theta)$ be given by

$$\phi(x,\theta) = \tan^{-1}\left(\frac{(x^2-1)\sin\theta}{x+(x^2+1)\cos\theta}\right).$$

Prove that $\forall (x,m) \in (\mathbb{R}-\{0\})\times(\mathbb{N}-1)$,

$$\int_0^\pi \left(1+x^2+\frac{1}{x^2}+2\cos 2\theta+2\left(x+\frac{1}{x}\right)\cos\theta\right)^{m/2}\cos m\phi(x,\theta)\, d\theta$$
$$= \pi\sum_{j=0}^{\lfloor m/2\rfloor}\binom{2j}{j}\binom{m}{2j}$$
$$= \pi\left(1+\binom{2}{1}\binom{m}{2}+\binom{4}{2}\binom{m}{4}+\cdots+\binom{2\lfloor m/2\rfloor}{\lfloor m/2\rfloor}\binom{m}{2\lfloor m/2\rfloor}\right).$$

[4.156] Let the function $\phi(x,\theta)$ be given by

$$\phi(x,\theta) = \tan^{-1}\left(\frac{(x^2+1)\sin\theta}{x+(x^2-1)\cos\theta}\right).$$

Prove that $\forall (x,m) \in (\mathbb{R}-\{0\})\times(\mathbb{N}-1)$,

$$\int_0^\pi \left(1+x^2+\frac{1}{x^2}-2\cos 2\theta+2\left(x-\frac{1}{x}\right)\cos\theta\right)^{m/2}\cos m\phi(x,\theta)\,d\theta$$
$$=\pi\sum_{j=0}^{\lfloor m/2\rfloor}(-1)^j\binom{2j}{j}\binom{m}{2j}$$
$$=\pi\left(1-\binom{2}{1}\binom{m}{2}+\binom{4}{2}\binom{m}{4}-\cdots+(-1)^{\lfloor m/2\rfloor}\binom{2\lfloor m/2\rfloor}{\lfloor m/2\rfloor}\binom{m}{2\lfloor m/2\rfloor}\right).$$

[4.157] Prove that $\forall m\in\mathbb{N}-1$,

$$\int_0^\pi (3+4\cos\theta+2\cos 2\theta)^{m/2}\,d\theta=\pi\sum_{j=0}^{\lfloor m/2\rfloor}\binom{2j}{j}\binom{m}{2j}$$
$$=\pi\left(1+\binom{2}{1}\binom{m}{2}+\binom{4}{2}\binom{m}{4}+\cdots+\binom{2\lfloor m/2\rfloor}{\lfloor m/2\rfloor}\binom{m}{2\lfloor m/2\rfloor}\right).$$

[4.158] Prove that $\forall(m,n)\in(\mathbb{N}-1)^2$,

$$\int_0^\pi (3+4\cos\theta+2\cos 2\theta)^{m/2}\cos n\theta\,d\theta=\pi\sum_{j=0}^{\lfloor (m-n)/2\rfloor}\binom{n+2j}{j}\binom{m}{n+2j}$$
$$=\pi\left(\binom{n}{0}\binom{m}{n}+\binom{n+2}{1}\binom{m}{n+2}+\binom{n+4}{2}\binom{m}{n+4}\right.$$
$$\left.+\cdots+\binom{n+2\lfloor (m-n)/2\rfloor}{\lfloor (m-n)/2\rfloor}\binom{m}{n+2\lfloor (m-n)/2\rfloor}\right).$$

[4.159] Let the function $\phi(\theta)$ be given by

$$\phi(\theta)=\tan^{-1}(2\sin\theta)\,. \tag{4.54}$$

Prove that $\forall m\in\mathbb{N}-1$,

$$\int_0^\pi (3-2\cos 2\theta)^{m/2}\cos m\phi(\theta)\,d\theta=\pi\sum_{j=0}^{\lfloor m/2\rfloor}(-1)^j\binom{2j}{j}\binom{m}{2j}$$
$$=\pi\left(1-\binom{2}{1}\binom{m}{2}+\binom{4}{2}\binom{m}{4}-\cdots+(-1)^{\lfloor m/2\rfloor}\binom{2\lfloor m/2\rfloor}{\lfloor m/2\rfloor}\binom{m}{2\lfloor m/2\rfloor}\right).$$

[4.160] Let the function $\phi(\theta)$ be given by (4.54). Prove that $\forall(m,n) \in (\mathbb{N}-1)\times(2\mathbb{N}-2)$,

$$\int_0^\pi (3-2\cos 2\theta)^{m/2}\cos m\phi(\theta)\cos n\theta\,d\theta = \pi \sum_{j=0}^{\lfloor (m-n)/2\rfloor}(-1)^j \binom{n+2j}{j}\binom{m}{n+2j}$$
$$= \pi\left(\binom{n}{0}\binom{m}{n} - \binom{n+2}{1}\binom{m}{n+2} + \binom{n+4}{2}\binom{m}{n+4}\right.$$
$$\left. -\cdots+(-1)^{\lfloor (m-n)/2\rfloor}\binom{n+2\lfloor (m-n)/2\rfloor}{\lfloor (m-n)/2\rfloor}\binom{m}{n+2\lfloor (m-n)/2\rfloor}\right).$$

[4.161] Let the function $\phi(\theta)$ be given by (4.54). Prove that $\forall(m,n) \in (\mathbb{N}-1)\times(2\mathbb{N}-1)$,

$$\int_0^\pi (3-2\cos 2\theta)^{m/2}\cos m\phi(\theta)\cos n\theta\,d\theta = 0.$$

[4.162] Let the function $\phi(\theta)$ be defined as

$$\phi(\theta) = \tan^{-1}\left(\frac{\sin\theta}{2+\cos\theta}\right). \tag{4.55}$$

Prove that $\forall m \in \mathbb{N}-1$,

$$\int_0^\pi (5+4\cos\theta)^{m/2}\cos m\phi(\theta)\,d\theta = 2^m\pi.$$

[4.163] Let the function $\phi(\theta)$ be defined as in (4.55). Prove that $\forall(m,n) \in (\mathbb{N}-1)\times\mathbb{N}$,

$$\int_0^\pi (5+4\cos\theta)^{m/2}\cos m\phi(\theta)\cos n\theta\,d\theta = \frac{\pi}{2}\sum_{j=0}^{m-n}\binom{n+j}{j}\binom{m}{n+j}$$
$$= \frac{\pi}{2}\left(\binom{n}{0}\binom{m}{n} + \binom{n+1}{1}\binom{m}{n+1} + \binom{n+2}{2}\binom{m}{n+2}\right.$$
$$\left. +\cdots+\binom{m}{m-n}\binom{m}{m}\right).$$

# Chapter 5
# Generalized Bessel Functions




The most common elementary complex functions are the exponential, circular trigonometrical, and hyperbolic trigonometrical functions. $\forall z \in \mathbb{C}$, the exponential function, $e^z$, is defined as

$$e^z = \sum_{j=0}^{\infty} \frac{z^j}{j!} = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots \tag{5.1}$$

$\forall z \in \mathbb{C}$, the circular trigonometrical functions, $\sin z$ and $\cos z$, are defined as

$$\sin z = \sum_{j=0}^{\infty} (-1)^j \frac{z^{2j+1}}{(2j+1)!} = \frac{z}{1!} - \frac{z^3}{3!} + \frac{z^5}{5!} + \frac{z^7}{7!} + \cdots, \tag{5.2}$$

and

$$\cos z = \sum_{j=0}^{\infty} (-1)^j \frac{z^{2j}}{(2j)!} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \cdots, \tag{5.3}$$

respectively. The circular trigonometrical functions are related to the exponential function by Euler's formula, which is equivalent to

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i},$$

and

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

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$\forall z \in \mathbb{C}$, the hyperbolic trigonometrical functions, $\sinh z$ and $\cosh z$, are defined as

$$\sinh z = \sum_{j=0}^{\infty} \frac{z^{2j+1}}{(2j+1)!} = \frac{z}{1!} + \frac{z^3}{3!} + \frac{z^5}{5!} + \frac{z^7}{7!} + \cdots, \tag{5.4}$$

and

$$\cosh z = \sum_{j=0}^{\infty} \frac{z^{2j}}{(2j)!} = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \cdots, \tag{5.5}$$

respectively. We can easily see that the hyperbolic trigonometrical functions are also related to the exponential functions by the equations

$$\sinh z = \frac{e^z - e^{-z}}{2},$$

and

$$\cosh z = \frac{e^z + e^{-z}}{2}.$$

Finally, the circular and hyperbolic trigonometrical functions are also related to each other by the equations

$$\sin(iz) = i \sinh z,$$

and

$$\cos(iz) = \cosh z.$$

Collectively, in this text, the above functions are called the elementary exponential functions.

In this chapter, first we will define the Mittag-Leffler functions $E_\alpha(z)$ (Mittag-Leffler, 1903; Erdélyi, 1955) and their various generalizations: $E_{\alpha,\beta}(z)$ by Wiman (1905), $E^{\gamma}_{\alpha,\beta}(z)$ by Prabhakar (1971), and $E^{\gamma,q}_{\alpha,\beta}(z)$ by Shukla and Prajapati (2007). Then, we will define the generalized Bessel functions $\phi(\rho, \beta; z)$ as defined by Wright (1933). Finally, we will define the *Pseudo-Exponential* functions, which look very similar to the above functions. Using the theorems proved in the previous chapter, we can derive the integral representations of the pseudo-exponential functions. We will also find the derivatives and integrals of the pseudo-exponential functions and see that they are very similar to the elementary exponential functions. The problems listed in this chapter show the relations between the pseudo-exponential functions and the elementary exponential functions. In particular, we will see that the pseudo-exponential functions are the special integrals of the elementary exponential functions. We will also see that many special integrals can be simplified using the pseudo-exponential functions.

## 5.1 The Mittag-Leffler Functions $E_{\alpha}(z)$

In this section, we will give a generalization of the exponential function (5.1), called the Mittag-Leffler functions, $E_{\alpha}(z)$ (Mittag-Leffler, 1903).

**Definition 5.1** *(The Mittag-Leffler Functions $E_{\alpha}(z)$)* (Mittag-Leffler, 1903)
$\forall(z, \alpha) \in \mathbb{C} \times \Re(\mathbb{C})^{+}$, the Mittag-Leffler functions, $E_{\alpha}(z)$, are defined as

$$\begin{aligned} E_{\alpha}(z) &= \sum_{j=0}^{\infty} \frac{z^{j}}{\Gamma(\alpha j + 1)} \\ &= 1 + \frac{z}{\Gamma(\alpha + 1)} + \frac{z^{2}}{\Gamma(2\alpha + 1)} + \frac{z^{3}}{\Gamma(3\alpha + 1)} + \cdots \end{aligned} \tag{5.6}$$

**Example 5.2** *(The Exponential Function as a Special Case of $E_{\alpha}(z)$)* (Erdélyi, 1955)
Prove that $\forall z \in \mathbb{C}$,

$$e^{z} = E_{1}(z).$$

***Proof*** Putting $\alpha = 1$ in (5.6), and using (5.1) and [3.6], we get

$$\begin{aligned} E_{1}(z) &= \sum_{j=0}^{\infty} \frac{z^{j}}{\Gamma(j + 1)} \\ &= \sum_{j=0}^{\infty} \frac{z^{j}}{j!} \\ &= e^{z}. \end{aligned}$$

□

## 5.2 Wiman's Generalized Mittag-Leffler Functions $E_{\alpha,\beta}(z)$

In this section, we will give a generalization of the Mittag-Leffler functions $E_{\alpha}(z)$ (5.6), called the Wiman's generalized Mittag-Leffler functions, $E_{\alpha,\beta}(z)$ (Wiman, 1905).

**Definition 5.3** *(Wiman's Generalized Mittag-Leffler Functions $E_{\alpha,\beta}(z)$)* (Wiman, 1905)
$\forall(z, \alpha, \beta) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^{2}$, the Wiman's Generalized Mittag-Leffler functions, $E_{\alpha,\beta}(z)$, are defined as

$$E_{\alpha,\beta}(z) = \sum_{j=0}^{\infty} \frac{z^j}{\Gamma(\alpha j + \beta)} = \frac{1}{\Gamma(\beta)} + \frac{z}{\Gamma(\alpha+\beta)} + \frac{z^2}{\Gamma(2\alpha+\beta)} + \frac{z^3}{\Gamma(3\alpha+\beta)} + \cdots \tag{5.7}$$

From (5.6) and (5.7), it is obvious that (Erdélyi, 1955)

$$E_{\alpha}(z) = E_{\alpha,1}(z).$$

## 5.3 Prabhakar's Generalized Mittag-Leffler Functions $E^{\gamma}_{\alpha,\beta}(z)$

In this section, we will give a generalization of the Wiman's generalized Mittag-Leffler functions $E_{\alpha,\beta}(z)$ (5.7), called the Prabhakar's Generalized Mittag-Leffler functions, $E^{\gamma}_{\alpha,\beta}(z)$ (Prabhakar, 1971).

**Definition 5.4** *(Prabhakar's Generalized Mittag-Leffler Functions $E^{\gamma}_{\alpha,\beta}(z)$)* (Prabhakar, 1971)
$\forall(z, \alpha, \beta, \gamma) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^3$, the Prabhakar's Generalized Mittag-Leffler functions, $E^{\gamma}_{\alpha,\beta}(z)$, are defined as

$$E^{\gamma}_{\alpha,\beta}(z) = \sum_{j=0}^{\infty} \frac{\prod_{k=0}^{j-1}(\gamma+k)}{j!\Gamma(\alpha j+\beta)} z^j = \frac{1}{\Gamma(\beta)} + \frac{\gamma}{1!\Gamma(\alpha+\beta)} z + \frac{\gamma(\gamma+1)}{2!\Gamma(2\alpha+\beta)} z^2 + \frac{\gamma(\gamma+1)(\gamma+2)}{3!\Gamma(3\alpha+\beta)} z^3 + \cdots \tag{5.8}$$

From (5.6), (5.7), and (5.8), it is obvious that (Prabhakar, 1971)

$$E_{\alpha}(z) = E^{1}_{\alpha,1}(z),$$

and

$$E_{\alpha,\beta}(z) = E^{1}_{\alpha,\beta}(z).$$

## 5.4 The Shukla-Prajapati Generalized Mittag-Leffler Functions $E^{\gamma,q}_{\alpha,\beta}(z)$

In this section, we will give a generalization of Prabhakar's generalized Mittag-Leffler functions $E^{\gamma}_{\alpha,\beta}(z)$ (5.8), called the Shukla-Prajapati generalized Mittag-Leffler functions, $E^{\gamma,q}_{\alpha,\beta}(z)$ (Shukla and Prajapati, 2007).

**Definition 5.5** *(The Shukla-Prajapati Generalized Mittag-Leffler Functions $E^{\gamma,q}_{\alpha,\beta}(z)$)* (Shukla and Prajapati, 2007)
$\forall (z, \alpha, \beta, \gamma, q) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^{3} \times ((0, 1) \cup \mathbb{N})$, the Shukla-Prajapati Generalized Mittag-Leffler functions, $E^{\gamma,q}_{\alpha,\beta}(z)$, are defined as

$$\begin{aligned} E^{\gamma,q}_{\alpha,\beta}(z) &= \sum_{j=0}^{\infty} \frac{\prod_{k=0}^{qj-1}(\gamma + k)}{j!\Gamma(\alpha j + \beta)} z^{j} \\ &= \frac{1}{\Gamma(\beta)} + \frac{\prod_{k=0}^{q-1}(\gamma + k)}{1!\Gamma(\alpha + \beta)} z + \frac{\prod_{k=0}^{2q-1}(\gamma + k)}{2!\Gamma(2\alpha + \beta)} z^{2} + \frac{\prod_{k=0}^{3q-1}(\gamma + k)}{3!\Gamma(3\alpha + \beta)} z^{3} + \cdots \end{aligned} \tag{5.9}$$

From (5.8) and (5.9), it is obvious that

$$E^{\gamma}_{\alpha,\beta}(z) = E^{\gamma,1}_{\alpha,\beta}(z).$$

## 5.5 Wright's Generalized Bessel Functions $\phi(\alpha, \beta; z)$

In this section, we will give a generalization of Bessel functions $J_v(z)$ (3.27), called the Wright's generalized Bessel functions, $\phi(\alpha, \beta; z)$ (Wright, 1933).

**Definition 5.6** *(Wright's Generalized Bessel Functions $\phi(\alpha, \beta; z)$)* (Wright, 1933)
$\forall (z, \alpha, \beta) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^{2}$, the Wright's Generalized Bessel functions, $\phi(\alpha, \beta; z)$, are defined as

$$\begin{aligned} \phi(\alpha, \beta; z) &= \sum_{j=0}^{\infty} \frac{z^{j}}{j!\Gamma(\alpha j + \beta)} \\ &= \frac{1}{\Gamma(\beta)} + \frac{z}{1!\Gamma(\alpha + \beta)} + \frac{z^{2}}{\Gamma(2\alpha + \beta)} + \frac{z^{3}}{3!\Gamma(3\alpha + \beta)} + \cdots \end{aligned} \tag{5.10}$$

**Example 5.7** *(The Bessel Functions $J_v(z)$ as a Special Case of $\phi(\rho, \beta; z)$)* (Erdélyi, 1955)
Prove that $\forall (z, v) \in \mathbb{C} \times (\mathbb{C} - (\mathbb{Z} - \mathbb{N}))$,

$$J_v(z) = \left(\frac{z}{2}\right)^v \phi\left(1, v+1; -\frac{z^2}{4}\right).$$

***Proof*** Putting $\alpha = 1$ and $\beta = v + 1$ in (5.10), and using (3.27), we get

$$\begin{aligned}\left(\frac{z}{2}\right)^v \phi\left(1, v+1; -\frac{z^2}{4}\right) &= \left(\frac{z}{2}\right)^v \sum_{j=0}^{\infty} \frac{\left(-\frac{z^2}{4}\right)^j}{j!\Gamma(j+v+1)} \\ &= \sum_{j=0}^{\infty} (-1)^j \frac{z^{v+2j}}{j!\Gamma(v+j+1)2^{v+2j}} \\ &= J_v(z).\end{aligned}$$

□

## 5.6 The Pseudo-Exponential Functions

In this section, we will define the Pseudo-Exponential functions as a special case of Wright's generalized Bessel functions $\phi(\alpha, \beta; z)$ (5.10).

**Definition 5.8** *(The Pseudo-Exponential Functions)*
$\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$, we define the Pseudo-Exponential function, $e(z, n)$, as

$$\begin{aligned} e(z,n) &= z^{\max\{0,-n\}} \phi(1, |n|+1; z) \\ &= \sum_{j=\max\{0,-n\}}^{\infty} \frac{z^j}{j!(n+j)!} \\ &= \begin{cases} \dfrac{1}{0!n!} + \dfrac{z}{1!(n+1)!} + \dfrac{z^2}{2!(n+2)!} + \dfrac{z^3}{3!(n+3)!} + \cdots, & \text{if } n \in \mathbb{N} - 1; \\ \dfrac{z^{-n}}{0!(-n)!} + \dfrac{z^{-n+1}}{1!(-n+1)!} + \dfrac{z^{-n+2}}{2!(-n+2)!} & \\ \quad + \dfrac{z^{-n+3}}{3!(-n+3)!} + \cdots, & \text{if } n \in -\mathbb{N}. \end{cases} \end{aligned} \tag{5.11}$$

**Example 5.9** *($e(z, n)$ for negative values of $n$)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$e(z, -n) = z^n e(z, n). \tag{5.12}$$

***Proof*** The result follows from (5.11). □

**Example 5.10** *(Relation Between $e(z, n)$ and $I_n(z)$)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$e(z, n) = z^{-n/2} I_n \left(2\sqrt{z}\right).$$

***Proof*** Using (5.11) and (3.19), $\forall (z, n) \in \mathbb{C} \times (\mathbb{N} - 1)$, we have

$$\begin{aligned} z^{-n/2} I_n \left(2\sqrt{z}\right) &= z^{-n/2} \sum_{j=0}^{\infty} \frac{\left(2\sqrt{z}\right)^{n+2j}}{j!(n+j)!2^{n+2j}} \\ &= \sum_{j=0}^{\infty} \frac{z^j}{j!(n+j)!} \\ &= e(z, n). \end{aligned} \tag{5.13}$$

Using (5.12), (5.13), and [3.19], $\forall (z, n) \in \mathbb{C} \times (-\mathbb{N})$, we have

$$\begin{aligned} e(z, n) &= z^{-n} e(z, -n) \\ &= z^{-n} \cdot z^{n/2} I_{-n} \left(2\sqrt{z}\right) \\ &= z^{-n/2} I_n \left(2\sqrt{z}\right). \end{aligned}$$

□

**Definition 5.11** *(The Pseudo-Sine and Pseudo-Cosine Functions)*
$\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$, we define the Pseudo-Sine function, $sin(z, n)$, and the Pseudo-Cosine function, $cos(z, n)$, related by the equation

$$e(iz, n) = \cos(z, n) + i \sin(z, n), \tag{5.14}$$

as

$$\begin{aligned} \sin(z, n) &= \sum_{j=\max\{0, \lfloor -n/2 \rfloor\}}^{\infty} (-1)^j \frac{z^{2j+1}}{(2j+1)!(n+2j+1)!} \\ &= \begin{cases} \dfrac{z}{1!(n+1)!} - \dfrac{z^3}{3!(n+3)!} + \dfrac{z^5}{5!(n+5)!} \\ \quad - \dfrac{z^7}{7!(n+7)!} + \cdots, & \text{if } n \in \mathbb{N} - 1; \\ (-1)^{-n/2} \left( \dfrac{z^{-n+1}}{1!(-n+1)!} - \dfrac{z^{-n+3}}{3!(-n+3)!} \right. \\ \quad \left. + \dfrac{z^{-n+5}}{5!(-n+5)!} - \dfrac{z^{-n+7}}{7!(-n+7)!} + \cdots \right), & \text{if } n \in -2\mathbb{N}; \\ (-1)^{(-n-1)/2} \left( \dfrac{z^{-n}}{0!(-n)!} - \dfrac{z^{-n+2}}{2!(-n+2)!} \right. \\ \quad \left. + \dfrac{z^{-n+4}}{4!(-n+4)!} - \dfrac{z^{-n+6}}{6!(-n+6)!} + \cdots \right), & \text{if } n \in -(2\mathbb{N} - 1); \end{cases} \end{aligned} \tag{5.15}$$

and

$$\cos(z,n) = \sum_{j=\max\{0,\lceil -n/2\rceil\}}^{\infty} (-1)^j \frac{z^{2j}}{(2j)!(n+2j)!}$$
$$= \begin{cases} \dfrac{1}{0!n!} - \dfrac{z^2}{2!(n+2)!} + \dfrac{z^4}{4!(n+4)!} \\ \quad - \dfrac{z^6}{6!(n+6)!} + \cdots, & \text{if } n \in \mathbb{N}-1; \\ (-1)^{-n/2}\left( \dfrac{z^{-n}}{0!(-n)!} - \dfrac{z^{-n+2}}{2!(-n+2)!} \right. \\ \quad \left. + \dfrac{z^{-n+4}}{4!(-n+4)!} - \dfrac{z^{-n+6}}{6!(-n+6)!} + \cdots \right), & \text{if } n \in -2\mathbb{N}; \\ (-1)^{(-n+1)/2}\left( \dfrac{z^{-n+1}}{1!(-n+1)!} - \dfrac{z^{-n+3}}{3!(-n+3)!} \right. \\ \quad \left. + \dfrac{z^{-n+5}}{5!(-n+5)!} - \dfrac{z^{-n+7}}{7!(-n+7)!} + \cdots \right), & \text{if } n \in -(2\mathbb{N}-1); \end{cases} \tag{5.16}$$

respectively.

**Definition 5.12** *(The Pseudo-Hyperbolic Sine and Pseudo-Hyperbolic Cosine Functions)*
$\forall (z,n) \in \mathbb{C} \times \mathbb{Z}$, we define the Pseudo-Hyperbolic Sine function, $sinh(z,n)$, and the Pseudo-Hyperbolic Cosine function, $cosh(z,n)$, defined using the equations

$$\sinh(z,n) = -i \sin(iz,n), \tag{5.17}$$

and

$$\cosh(z,n) = \cos(iz,n), \tag{5.18}$$

respectively, as

$$\sinh(z,n) = \sum_{j=\max\{0,\lfloor -n/2\rfloor\}}^{\infty} \frac{z^{2j+1}}{(2j+1)!(n+2j+1)!}$$
$$= \begin{cases} \dfrac{z}{1!(n+1)!} + \dfrac{z^3}{3!(n+3)!} + \dfrac{z^5}{5!(n+5)!} \\ \quad + \dfrac{z^7}{7!(n+7)!} + \cdots, & \text{if } n \in \mathbb{N}-1; \\ \dfrac{z^{-n+1}}{1!(-n+1)!} + \dfrac{z^{-n+3}}{3!(-n+3)!} \\ \quad + \dfrac{z^{-n+5}}{5!(-n+5)!} + \dfrac{z^{-n+7}}{7!(-n+7)!} + \cdots, & \text{if } n \in -2\mathbb{N}; \\ \dfrac{z^{-n}}{0!(-n)!} + \dfrac{z^{-n+2}}{2!(-n+2)!} \\ \quad + \dfrac{z^{-n+4}}{4!(-n+4)!} + \dfrac{z^{-n+6}}{6!(-n+6)!} + \cdots, & \text{if } n \in -(2\mathbb{N}-1); \end{cases} \tag{5.19}$$

and

$$\cosh(z,n)=\sum_{j=\max\{0,\lceil -n/2\rceil\}}^{\infty}\frac{z^{2j}}{(2j)!(n+2j)!}$$
$$=\begin{cases}\dfrac{1}{0!n!}+\dfrac{z^2}{2!(n+2)!}+\dfrac{z^4}{4!(n+4)!}\\ \quad+\dfrac{z^6}{6!(n+6)!}+\cdots, & \text{if } n\in\mathbb{N}-1;\\ \dfrac{z^{-n}}{0!(-n)!}+\dfrac{z^{-n+2}}{2!(-n+2)!}\\ \quad+\dfrac{z^{-n+4}}{4!(-n+4)!}+\dfrac{z^{-n+6}}{6!(-n+6)!}+\cdots, & \text{if } n\in-2\mathbb{N};\\ \dfrac{z^{-n+1}}{1!(-n+1)!}+\dfrac{z^{-n+3}}{3!(-n+3)!}\\ \quad+\dfrac{z^{-n+5}}{5!(-n+5)!}+\dfrac{z^{-n+7}}{7!(-n+7)!}+\cdots, & \text{if } n\in-(2\mathbb{N}-1);\end{cases}\tag{5.20}$$

respectively.

Before proceeding to derive the properties of the pseudo-exponential functions, we should plot their graphs and compare them with the elementary exponential functions.

**Example 5.13** *(The Plot of $y=e(x,0)$)*
The plot of $y=e(x,0)$ in $[-6\pi,\pi]$ is shown in Fig. 5.1. As compared to the function $y=e^x$, we notice that their structures are similar in the $x\geq 0$ region. Both tend to infinity. However, we notice a difference for the $x<0$ region. $y=e(x,0)$ is taking both positive and negative values (including 0), but $y=e^x$ takes only the positive values.

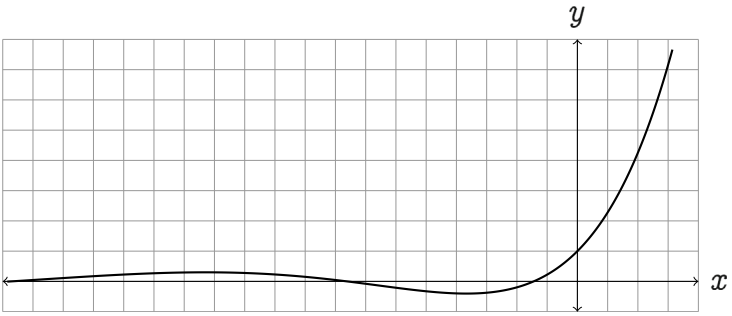



**Fig. 5.1** The plot of $y=e(x,0)$ in $[-6\pi,\pi]$

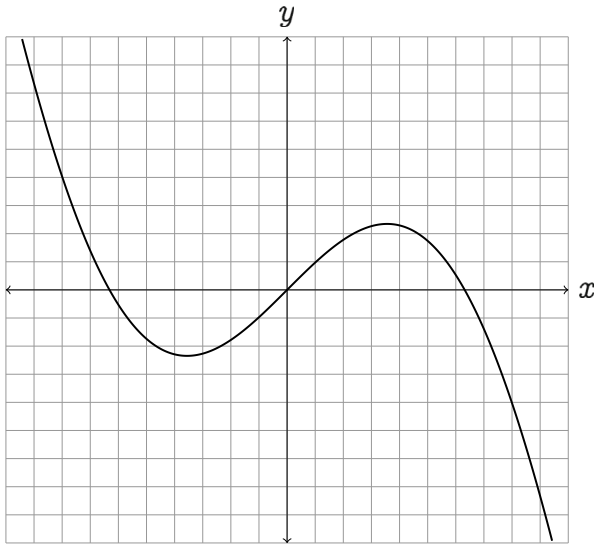



**Fig. 5.2** The plot of $y = \sin(x, 0)$ in $[-3\pi, 3\pi]$

**Example 5.14** *(The plot of* $y = \sin(x, 0)$*)*
The plot of $y = \sin(x, 0)$ in $[-3\pi, 3\pi]$ is shown in Fig. 5.2. As before, we notice that both $y = \sin(x, 0)$ and $y = \sin x$ have similar structures in the plotted region. The difference may be in the periodicity of the functions. $y = \sin x$ is periodic with period $2\pi$, while $y = \sin(x, 0)$ is not seen to be periodic. Does $y = \sin(x, 0)$ takes the value 0 infinitely often, similar to $y = \sin x$? We have to find the answer.

**Example 5.15** *(The plot of* $y = \cos(x, 0)$*)*
The plot of $y = \cos(x, 0)$ in $[-9\pi/2, 9\pi/2]$ is shown in Fig. 5.3. We notice that both $y = \cos(x, 0)$ and $y = \cos x$ have similar structures in the plotted region. The difference may be in the periodicity of the functions. $y = \cos x$ is periodic with period $2\pi$, while $y = \cos(x, 0)$ is not seen to be periodic. Does $y = \cos(x, 0)$ takes the value 0 infinitely often, similar to $y = \cos x$? We have to find the answer.

**Example 5.16** *(The plot of* $y = \sinh(x, 0)$*)*
The plot of $y = \sinh(x, 0)$ in $[-2\pi, 2\pi]$ is shown in Fig. 5.4. As before, both $y = \sinh(x, 0)$ and $y = \sinh x$ have similar structures in the plotted region.

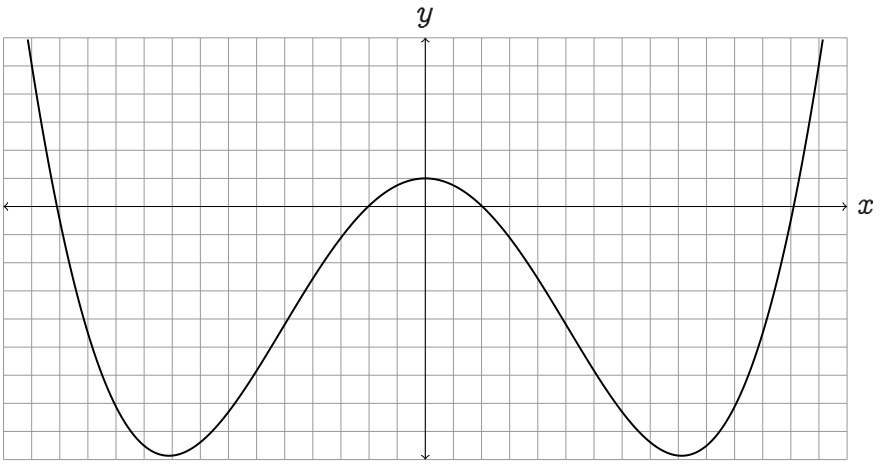
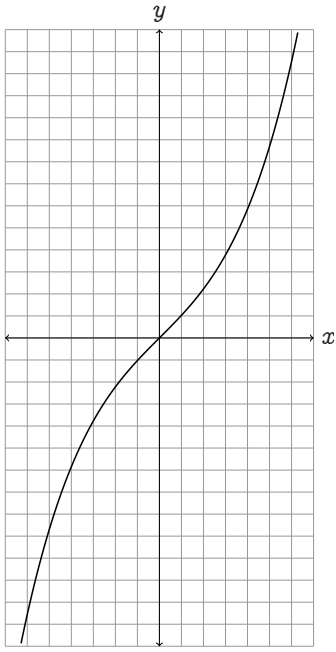



**Fig. 5.3** The plot of $y = \cos(x, 0)$ in $[-9\pi/2, 9\pi/2]$

**Fig. 5.4** The plot of $y = \sinh(x, 0)$ in $[-2\pi, 2\pi]$




**Example 5.17** *(The plot of* $y = \cosh(x, 0)$*)*
The plot of $y = \cosh(x, 0)$ in $[-2\pi, 2\pi]$ is shown in Fig. 5.5. Both $y = \cosh(x, 0)$ and $y = \cosh x$ have similar structures in the plotted region.

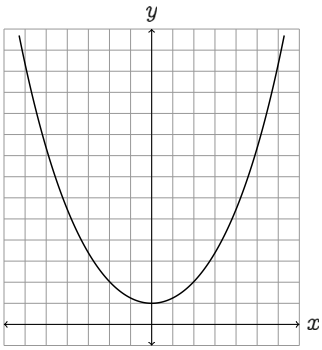



**Fig. 5.5** The plot of $y = \cosh(x, 0)$ in $[-2\pi, 2\pi]$

## 5.7 Derivatives of the Pseudo-Exponential Functions

Recall the derivatives of the elementary exponential functions. We have

$$\begin{aligned}\frac{de^z}{dz} &= e^z,\\ \frac{d\sin z}{dz} &= \cos z,\\ \frac{d\cos z}{dz} &= -\sin z,\\ \frac{d\sinh z}{dz} &= \cosh z,\end{aligned}$$

and

$$\frac{d\cosh z}{dz} = \sinh z.$$

We will see that the derivatives of the pseudo-exponential functions also exhibit similar properties. We will also observe that differentiating a pseudo-exponential function increments its level by 1 (the parameter $n$).

**Example 5.18** *(Derivative of $e(z, n)$)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\frac{de(z, n)}{dz} = e(z, n + 1). \tag{5.21}$$

***Proof*** From (5.11), we get

$$\begin{aligned}\frac{de(z,n)}{dz} &= \frac{d}{dz}\left(\sum_{j=\max\{0,-n\}}^{\infty}\frac{z^j}{j!(n+j)!}\right)\\ &= \sum_{j=\max\{1,-n\}}^{\infty}\frac{jz^{j-1}}{j!(n+j)!}\\ &= \sum_{j=\max\{1,-n\}}^{\infty}\frac{z^{j-1}}{(j-1)!(n+j)!}\\ &= \sum_{j=\max\{0,-(n+1)\}}^{\infty}\frac{z^j}{j!(n+1+j)!}\\ &= e(z,n+1).\end{aligned}$$

□

**Example 5.19** *(Derivative of* $\sin(z,n)$*)*
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$\frac{d\sin(z,n)}{dz} = \cos(z,n+1). \tag{5.22}$$

***Proof*** From (5.15) and (5.16), we get

$$\begin{aligned}\frac{d\sin(z,n)}{dz} &= \frac{d}{dz}\left(\sum_{j=\max\{0,\lfloor -n/2\rfloor\}}^{\infty}(-1)^j\frac{z^{2j+1}}{(2j+1)!(n+2j+1)!}\right)\\ &= \sum_{j=\max\{0,\lfloor -n/2\rfloor\}}^{\infty}(-1)^j\frac{(2j+1)z^{2j}}{(2j+1)!(n+2j+1)!}\\ &= \sum_{j=\max\{0,\lceil -(n+1)/2\rceil\}}^{\infty}(-1)^j\frac{z^{2j}}{(2j)!(2j+n+1)!}\\ &= \cos(z,n+1).\end{aligned}$$

□

**Example 5.20** *(Derivative of* $\cos(z,n)$*)*
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$\frac{d\cos(z,n)}{dz} = -\sin(z,n+1). \tag{5.23}$$

***Proof*** From (5.16) and (5.15), we get

$$\begin{aligned}\frac{d\cos(z,n)}{dz} &= \frac{d}{dz}\left(\sum_{j=\max\{0,\lceil -n/2\rceil\}}^{\infty}(-1)^j\frac{z^{2j}}{(2j)!(n+2j)!}\right)\\ &= \sum_{j=\max\{1,\lceil -n/2\rceil\}}^{\infty}(-1)^j\frac{(2j)z^{2j-1}}{(2j)!(n+2j)!}\\ &= \sum_{j=\max\{1,\lceil -n/2\rceil\}}^{\infty}(-1)^j\frac{z^{2j-1}}{(2j-1)!(n+2j)!}\\ &= -\sum_{j=\max\{0,\lfloor -(n+1)/2\rfloor\}}^{\infty}(-1)^j\frac{z^{2j+1}}{(2j+1)!(n+1+2j+1)!}\\ &= -\sin(z,n+1).\end{aligned}$$

□

**Example 5.21** *(Derivative of* $\sinh(z,n)$*)*
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$\frac{d\sinh(z,n)}{dz} = \cosh(z,n+1). \tag{5.24}$$

***Proof*** From (5.19) and (5.20), we get

$$\begin{aligned}\frac{d\sinh(z,n)}{dz} &= \frac{d}{dz}\left(\sum_{j=\max\{0,\lfloor -n/2\rfloor\}}^{\infty}\frac{z^{2j+1}}{(2j+1)!(n+2j+1)!}\right)\\ &= \sum_{j=\max\{0,\lfloor -n/2\rfloor\}}^{\infty}\frac{(2j+1)z^{2j}}{(2j+1)!(n+2j+1)!}\\ &= \sum_{j=\max\{0,\lceil -(n+1)/2\rceil\}}^{\infty}\frac{z^{2j}}{(2j)!(n+1+2j)!}\\ &= \cosh(z,n+1).\end{aligned}$$

□

**Example 5.22** *(Derivative of* $\cosh(z,n)$*)*
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$\frac{d\cosh(z,n)}{dz} = \sinh(z,n+1). \tag{5.25}$$

***Proof*** From (5.20) and (5.19), we get

$$\begin{aligned}\frac{d\cosh(z,n)}{dz} &= \frac{d}{dz}\left(\sum_{j=\max\{0,\lceil -n/2\rceil\}}^{\infty}\frac{z^{2j}}{(2j)!(n+2j)!}\right)\\ &= \sum_{j=\max\{1,\lceil -n/2\rceil\}}^{\infty}\frac{2jz^{2j-1}}{(2j)!(n+2j)!}\\ &= \sum_{j=\max\{1,\lceil -n/2\rceil\}}^{\infty}\frac{z^{2j-1}}{(2j-1)!(n+2j)!}\\ &= \sum_{j=\max\{0,\lfloor -(n+1)/2\rfloor\}}^{\infty}\frac{z^{2j+1}}{(2j+1)!(n+1+2j+1)!}\\ &= \sinh(z,n+1).\end{aligned}$$

□

## 5.8 Integrals of the Pseudo-Exponential Functions

As before, we revisit the integrals of the elementary exponential functions. We have

$$\int e^z\,dz = e^z,$$

$$\int \sin z\,dz = -\cos z,$$

$$\int \cos z\,dz = \sin z,$$

$$\int \sinh z\,dz = \cosh z,$$

and

$$\int \cosh z\,dz = \sinh z.$$

We will see that the integrals of the pseudo-exponential functions show similar properties and that integrating a pseudo-exponential function decrements its level by 1 (the parameter $n$).

**Example 5.23** *(Integral of $e(z,n)$)*
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$\int e(z,n)\,dz = e(z,n-1).$$

***Proof*** From (5.21), we get the above result. □

**Example 5.24** *(Integral of* $\sin(z, n)$*)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\int \sin(z, n)\, dz = -\cos(z, n-1).$$

***Proof*** From (5.23), we get the above result. □

**Example 5.25** *(Integral of* $\cos(z, n)$*)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\int \cos(z, n)\, dz = \sin(z, n-1).$$

***Proof*** From (5.22), we get the above result. □

**Example 5.26** *(Integral of* $\sinh(z, n)$*)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\int \sinh(z, n)\, dz = \cosh(z, n-1).$$

***Proof*** From (5.25), we get the above result. □

**Example 5.27** *(Integral of* $\cosh(z, n)$*)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\int \cosh(z, n)\, dz = \sinh(z, n-1).$$

***Proof*** From (5.24), we get the above result. □

## Problems

[5.1] (Erdélyi, 1955) Prove that $\forall z \in \mathbb{C}$,

$$\cosh z = E_2(z^2).$$

[5.2] (Erdélyi, 1955) Prove that $\forall (z, \alpha, \beta) \in \mathbb{C} \times \left(\Re(\mathbb{C})^+\right)^2$,

$$E_{\alpha,\beta}(z) = \frac{1}{\Gamma(\beta)} + zE_{\alpha,\alpha+\beta}(z).$$

[5.3] (Shukla and Prajapati, 2007) Prove that $\forall (z, \alpha, \beta, \gamma) \in \mathbb{C} \times \left(\Re(\mathbb{C})^+\right)^3$,

$$E^{\gamma}_{\alpha,\beta-\alpha}(z) - E^{\gamma-1}_{\alpha,\beta-\alpha}(z) = zE^{\gamma}_{\alpha,\beta}(z).$$

[5.4] (Shukla and Prajapati, 2007) Prove that $\forall(z, \alpha, \beta, \gamma, q) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^{3} \times ((0, 1) \cup \mathbb{N})$,

$$E_{\alpha,\beta}^{\gamma,q}(z) = \beta E_{\alpha,\beta+1}^{\gamma,q}(z) + \alpha z \frac{d}{dz}\left(E_{\alpha,\beta+1}^{\gamma,q}(z)\right).$$

[5.5] (Erdélyi, 1955) Prove that $\forall(z, \alpha, \beta) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^{2}$,

$$\alpha z \phi(\alpha, \alpha + \beta; z) = \phi(\alpha, \beta - 1; z) + (1 - \beta)\phi(\alpha, \beta; z).$$

[5.6] (Erdélyi, 1955) Prove that $\forall(z, \alpha, \beta) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^{2}$,

$$\frac{d\phi(\alpha, \beta; z)}{dz} = \phi(\alpha, \alpha + \beta; z).$$

[5.7] (Erdélyi, 1955) Prove that $\forall(z, \alpha, \beta) \in \mathbb{C} \times \left(\Re(\mathbb{C})^{+}\right)^{2}$,

$$\alpha z \frac{d\phi(\alpha, \beta; z)}{dz} = \phi(\alpha, \beta - 1; z) + (1 - \beta)\phi(\alpha, \beta; z).$$

[5.8] Prove that

$$e(1, 0) = \frac{1}{\pi}\int_{0}^{\pi} e^{2\cos\theta} d\,\theta.$$

[5.9] Prove that $\forall n \in \mathbb{Z}$,

$$e(1, n) = \frac{1}{\pi}\int_{0}^{\pi} e^{2\cos\theta}\cos n\theta\, d\theta.$$

[5.10] Prove that $\forall x \in \mathbb{R}$,

$$e\left(x^2, 0\right) = \frac{1}{\pi}\int_{0}^{\pi} e^{2x\cos\theta}\, d\theta.$$

[5.11] Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times \mathbb{Z}$,

$$e\left(x^2, n\right) = \frac{1}{\pi x^n}\int_{0}^{\pi} e^{2x\cos\theta}\cos n\theta\, d\theta.$$

[5.12] (Watson, 2011) Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$e\left(-x^2, n\right) = \frac{J_n(2x)}{x^n} = \frac{(-1)^{(n-1)/2}}{\pi x^n}\int_{0}^{\pi} \sin(2x\cos\theta)\cos n\theta\, d\theta.$$

[5.13] (Watson, 2011) Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$e\left(-x^2, n\right) = \frac{J_n(2x)}{x^n} = \frac{(-1)^{n/2}}{\pi x^n} \int_0^\pi \cos(2x\cos\theta)\cos n\theta \, d\theta.$$

[5.14] Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$e\left(x^2, n\right) = \frac{I_n(2x)}{x^n} = \frac{(-1)^{(n-1)/2}}{\pi x^n} \int_0^\pi \sinh(2x\sin\theta)\sin n\theta \, d\theta.$$

[5.15] Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$e\left(x^2, n\right) = \frac{I_n(2x)}{x^n} = \frac{(-1)^{n/2}}{\pi x^n} \int_0^\pi \cosh(2x\sin\theta)\cos n\theta \, d\theta.$$

[5.16] Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$e\left(x^2, n\right) = \frac{I_n(2x)}{x^n} = \frac{1}{\pi x^n} \int_0^\pi \sinh(2x\cos\theta)\cos n\theta \, d\theta.$$

[5.17] Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$e\left(x^2, n\right) = \frac{I_n(2x)}{x^n} = \frac{1}{\pi x^n} \int_0^\pi \cosh(2x\cos\theta)\cos n\theta \, d\theta.$$

[5.18] (Watson, 2011) Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$e\left(-x^2, n\right) = \frac{J_n(2x)}{x^n} = \frac{1}{\pi x^n} \int_0^\pi \sin(2x\sin\theta)\sin n\theta \, d\theta.$$

[5.19] (Watson, 2011) Prove that $\forall(x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$e\left(-x^2, n\right) = \frac{J_n(2x)}{x^n} = \frac{1}{\pi x^n} \int_0^\pi \cos(2x\sin\theta)\cos n\theta \, d\theta.$$

[5.20] Prove that $\forall x \in \mathbb{R}$,

$$e(x, 0) = \frac{1}{\pi} \int_0^\pi e^{(x+1)\cos\theta} \cos((x-1)\sin\theta) \, d\theta.$$

[5.21] Prove that $\forall(x, n) \in \mathbb{R} \times \mathbb{Z} - \{-1\} \times (2\mathbb{Z} + 1)$,

$$e(x, n) = \frac{2}{\pi(1 + x^n)} \int_0^\pi e^{(x+1)\cos\theta} \cos((x-1)\sin\theta)\cos n\theta \, d\theta.$$

[5.22] Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$e(x, n) = \frac{2}{\pi(1 + x^n)} \int_0^\pi \cosh\left((x + 1)\cos\theta\right) \cos\left((x - 1)\sin\theta\right) \cos n\theta \, d\theta.$$

[5.23] Prove that $\forall (x, n) \in (\mathbb{R} - \{-1\}) \times (2\mathbb{Z} + 1)$,

$$e(x, n) = \frac{2}{\pi(1 + x^n)} \int_0^\pi \sinh\left((x + 1)\cos\theta\right) \cos\left((x - 1)\sin\theta\right) \cos n\theta \, d\theta.$$

[5.24] Prove that $\forall x \in \mathbb{R}$,

$$e(x, 0) = -1 + \frac{2}{\pi} \int_0^\pi e^{(x+1)\cos\theta} \cos\left(\sin\theta\right) \cos\left(x \sin\theta\right) \, d\theta.$$

[5.25] Prove that $\forall x \in \mathbb{R}$,

$$e(x, 0) = 1 + \frac{2}{\pi} \int_0^\pi e^{(x+1)\cos\theta} \sin\left(\sin\theta\right) \sin\left(x \sin\theta\right) \, d\theta.$$

[5.26] Prove that $\forall x \in \mathbb{R}$,

$$\int_0^\pi e^{(x+1)\cos\theta} \cos\left((x + 1)\sin\theta\right) \, d\theta = \pi.$$

[5.27] Prove that $\forall x \in \mathbb{R}$,

$$\sinh\left(x^2, 0\right) = \frac{1}{\pi} \int_0^\pi \left(\sinh^2\left(x \cos\theta\right) \cos^2\left(x \sin\theta\right) + \cosh^2\left(x \cos\theta\right) \sin^2\left(x \sin\theta\right)\right) \, d\theta.$$

[5.28] Prove that $\forall x \in \mathbb{R}$,

$$\sinh\left(x^2, 0\right) = \frac{1}{\pi} \int_0^\pi \left(\sin^2\left(x \cos\theta\right) \cosh^2\left(x \sin\theta\right) + \cos^2\left(x \cos\theta\right) \sinh^2\left(x \sin\theta\right)\right) \, d\theta.$$

[5.29] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\sinh\left(x^2, n\right) = \frac{(-1)^{n/2}}{\pi x^n} \int_0^\pi \left(\sin^2\left(x \cos\theta\right) \cosh^2\left(x \sin\theta\right) + \cos^2\left(x \cos\theta\right) \sinh^2\left(x \sin\theta\right)\right) \cos n\theta \, d\theta.$$

[5.30] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\sinh\left(x^2, n\right) = \frac{1}{\pi x^n} \int_0^\pi \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta) + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\, d\theta.$$

[5.31] Prove that $\forall x \in \mathbb{R}$,

$$\cosh\left(x^2, 0\right) = \frac{1}{\pi} \int_0^\pi \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\, d\theta.$$

[5.32] Prove that $\forall x \in \mathbb{R}$,

$$\cosh\left(x^2, 0\right) = \frac{1}{\pi} \int_0^\pi \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\, d\theta.$$

[5.33] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\cosh\left(x^2, n\right) = \frac{(-1)^{n/2}}{\pi x^n} \int_0^\pi \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos n\theta\, d\theta.$$

[5.34] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\cosh\left(x^2, n\right) = \frac{1}{\pi x^n} \int_0^\pi \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\, d\theta.$$

[5.35] Let $S_1(x, \theta)$ be given by

$$S_1(x, \theta) = \sinh(x\sin\theta)\cos(x\cos\theta)\sin(\sin\theta) + \cosh(x\sin\theta)\sin(x\cos\theta)\cos(\sin\theta). \tag{5.26}$$

Prove that $\forall (x, n) \in \mathbb{R} \times 2\mathbb{Z} - \{-1, 1\} \times 2(2\mathbb{Z} + 1)$,

$$\sin(x, n) = \frac{2}{\pi\left(1 + (-1)^{n/2} x^n\right)} \int_0^\pi S_1(x, \theta) e^{\cos\theta} \cos n\theta\, d\theta.$$

[5.36] Let $S_1(x,\theta)$ be given by (5.26). Prove that $\forall (x,n) \in \{-1,1\} \times 2(2\mathbb{Z}+1)$,

$$\int_0^\pi S_1(x,\theta) e^{\cos\theta} \cos n\theta \, d\theta = 0.$$

[5.37] Let $S_1(x,\theta)$ be given by (5.26), and $C_1(x,\theta)$ be given by

$$\begin{aligned} C_1(x,\theta) = &\cosh(x\sin\theta)\cos(x\cos\theta)\cos(\sin\theta) \\ &- \sinh(x\sin\theta)\sin(x\cos\theta)\sin(\sin\theta)\,. \end{aligned} \tag{5.27}$$

Prove that $\forall (x,n) \in \mathbb{R} \times (2\mathbb{Z}+1)$,

$$\begin{aligned} \sin(x,n) = &\frac{2}{\pi\left(1+x^{2n}\right)} \int_0^\pi S_1(x,\theta) e^{\cos\theta} \cos n\theta \, d\theta \\ &+ (-1)^{(n+1)/2} \frac{2x^n}{\pi\left(1+x^{2n}\right)} \int_0^\pi C_1(x,\theta) e^{\cos\theta} \cos n\theta \, d\theta. \end{aligned}$$

[5.38] Let $C_1(x,\theta)$ be given by (5.27). Prove that $\forall (x,n) \in \mathbb{R} \times 2\mathbb{Z} - \{-1,1\} \times 2(2\mathbb{Z}+1)$,

$$\cos(x,n) = \frac{2}{\pi\left(1+(-1)^{n/2}x^n\right)} \int_0^\pi C_1(x,\theta) e^{\cos\theta} \cos n\theta \, d\theta.$$

[5.39] Let $C_1(x,\theta)$ be given by (5.27). Prove that $\forall (x,n) \in \{-1,1\} \times 2(2\mathbb{Z}+1)$,

$$\int_0^\pi C_1(x,\theta) e^{\cos\theta} \cos n\theta \, d\theta = 0.$$

[5.40] Let $C_1(x,\theta)$ be given by (5.27), and $S_1(x,\theta)$ be given by (5.26). Prove that $\forall (x,n) \in \mathbb{R} \times (2\mathbb{Z}+1)$,

$$\begin{aligned} \cos(x,n) = &\frac{2}{\pi\left(1+x^{2n}\right)} \int_0^\pi C_1(x,\theta) e^{\cos\theta} \cos n\theta \, d\theta \\ &+ (-1)^{(n-1)/2} \frac{2x^n}{\pi\left(1+x^{2n}\right)} \int_0^\pi S_1(x,\theta) e^{\cos\theta} \cos n\theta \, d\theta. \end{aligned}$$

[5.41] Let $S_2(x,\theta)$ be given by

$$\begin{aligned} S_2(x,\theta) = &\cosh(x\cos\theta)\sin(x\sin\theta)\sin(\sin\theta) \\ &+ \sinh(x\cos\theta)\cos(x\sin\theta)\cos(\sin\theta)\,. \end{aligned} \tag{5.28}$$

Prove that $\forall (x,n) \in \mathbb{R} \times 2\mathbb{Z}$,

$$\sinh(x,n)=\frac{2}{\pi\,(1+x^n)}\int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta.$$

[5.42] Let $S_2(x,\theta)$ be given by (5.28), and $C_2(x,\theta)$ be given by

$$\begin{aligned}C_2(x,\theta)&=\cosh(x\cos\theta)\cos(x\sin\theta)\cos(\sin\theta)\\&\quad+\sinh(x\cos\theta)\sin(x\sin\theta)\sin(\sin\theta)\,.\end{aligned}\tag{5.29}$$

Prove that $\forall(x,n)\in(\mathbb{R}-\{-1,1\})\times(2\mathbb{Z}+1)$,

$$\begin{aligned}\sinh(x,n)&=\frac{2}{\pi\left(1-x^{2n}\right)}\int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta\\&\quad-\frac{2x^n}{\pi\left(1-x^{2n}\right)}\int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta.\end{aligned}$$

[5.43] Let $S_2(x,\theta)$ be given by (5.28). Prove that $\forall(x,n)\in\{-1,1\}\times(2\mathbb{Z}+1)$,

$$\int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta=x^n\int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta.$$

[5.44] Let $C_2(x,\theta)$ be given by (5.29). Prove that $\forall(x,n)\in\mathbb{R}\times2\mathbb{Z}$,

$$\cosh(x,n)=\frac{2}{\pi\,(1+x^n)}\int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta.$$

[5.45] Let $C_2(x,\theta)$ be given by (5.29), and $S_2(x,\theta)$ be given by (5.28). Prove that $\forall(x,n)\in(\mathbb{R}-\{-1,1\})\times(2\mathbb{Z}+1)$,

$$\begin{aligned}\cosh(x,n)&=\frac{2}{\pi\left(1-x^{2n}\right)}\int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta\\&\quad-\frac{2x^n}{\pi\left(1-x^{2n}\right)}\int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta.\end{aligned}$$

[5.46] Let $C_2(x,\theta)$ be given by (5.29), and $S_2(x,\theta)$ be given by (5.28). Prove that $\forall(x,n)\in\{-1,1\}\times(2\mathbb{Z}+1)$,

$$\int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta=x^n\int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\,d\theta.$$

[5.47] Prove that $\forall(x,y,n)\in\mathbb{R}^2\times\mathbb{Z}$,

$$\int_0^\pi e^{(x+y)\cos\theta}\cos((x-y)\sin\theta)\cos n\theta\,d\theta=\frac{\pi}{2}\left(x^n+y^n\right)e(xy,n).$$

[5.48] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{Z}+1)$,

$$\int_0^\pi \sin((x+y)\cos\theta)\cosh((x-y)\sin\theta)\cos n\theta\, d\theta$$
$$= (-1)^{(n-1)/2}\frac{\pi}{2}\left(x^n + y^n\right)e(-xy, n).$$

[5.49] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{Z}$,

$$\int_0^\pi \sin((x+y)\cos\theta)\cosh((x-y)\sin\theta)\cos n\theta\, d\theta = 0.$$

[5.50] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{Z}$,

$$\int_0^\pi \cos((x+y)\cos\theta)\cosh((x-y)\sin\theta)\cos n\theta\, d\theta$$
$$= (-1)^{n/2}\frac{\pi}{2}\left(x^n + y^n\right)e(-xy, n).$$

[5.51] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{Z}+1)$,

$$\int_0^\pi \cos((x+y)\cos\theta)\cosh((x-y)\sin\theta)\cos n\theta\, d\theta = 0.$$

[5.52] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{Z}+1)$,

$$\int_0^\pi \sinh((x+y)\cos\theta)\cos((x-y)\sin\theta)\cos n\theta\, d\theta = \frac{\pi}{2}\left(x^n + y^n\right)e(xy, n).$$

[5.53] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{Z}$,

$$\int_0^\pi \sinh((x+y)\cos\theta)\cos((x-y)\sin\theta)\cos n\theta\, d\theta = 0.$$

[5.54] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times 2\mathbb{Z}$,

$$\int_0^\pi \cosh((x+y)\cos\theta)\cos((x-y)\sin\theta)\cos n\theta\, d\theta = \frac{\pi}{2}\left(x^n + y^n\right)e(xy, n).$$

[5.55] Prove that $\forall (x, y, n) \in \mathbb{R}^2 \times (2\mathbb{Z}+1)$,

$$\int_0^\pi \cosh((x+y)\cos\theta)\cos((x-y)\sin\theta)\cos n\theta\, d\theta = 0.$$

[5.56] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times \mathbb{Z}$,

$$\int_0^\pi e^{(x+1/x)\cos\theta} \cos\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta \, d\theta = \frac{\pi}{2}\left(x^n + \frac{1}{x^n}\right) e(1, n).$$

[5.57] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$\int_0^\pi \sin\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cosh\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta \, d\theta$$
$$= (-1)^{(n-1)/2}\frac{\pi}{2}\left(x^n + \frac{1}{x^n}\right) e(-1, n).$$

[5.58] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\int_0^\pi \sin\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cosh\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta \, d\theta = 0.$$

[5.59] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\int_0^\pi \cos\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cosh\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta \, d\theta$$
$$= (-1)^{n/2}\frac{\pi}{2}\left(x^n + \frac{1}{x^n}\right) e(-1, n).$$

[5.60] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$\int_0^\pi \cos\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cosh\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta \, d\theta = 0.$$

[5.61] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$\int_0^\pi \sinh\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cos\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta \, d\theta$$
$$= \frac{\pi}{2}\left(x^n + \frac{1}{x^n}\right) e(1, n).$$

[5.62] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\int_0^\pi \sinh\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cos\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta \, d\theta = 0.$$

[5.63] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times 2\mathbb{Z}$,

$$\int_0^\pi \cosh\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cos\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta\, d\theta = \frac{\pi}{2}\left(x^n + \frac{1}{x^n}\right)e(1, n).$$

[5.64] Prove that $\forall (x, n) \in (\mathbb{R} - \{0\}) \times (2\mathbb{Z} + 1)$,

$$\int_0^\pi \cosh\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cos\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\cos n\theta\, d\theta = 0.$$

[5.65] Prove that $\forall x \in \mathbb{R} - \{0\}$,

$$\int_0^\pi e^{(x+1/x)\cos\theta}\cos\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\, d\theta = \pi e(1, 0).$$

[5.66] Prove that $\forall x \in \mathbb{R} - \{0\}$,

$$\int_0^\pi \cos\left(\left(x - \frac{1}{x}\right)\cos\theta\right)\cosh\left(\left(x + \frac{1}{x}\right)\sin\theta\right)\, d\theta = \pi e(1, 0).$$

[5.67] Prove that $\forall x \in \mathbb{R} - \{0\}$,

$$\int_0^\pi \cosh\left(\left(x + \frac{1}{x}\right)\cos\theta\right)\cos\left(\left(x - \frac{1}{x}\right)\sin\theta\right)\, d\theta = \pi e(1, 0).$$

[5.68] Prove that $\forall n \in \mathbb{Z}$,

$$\int_0^\pi e^{2\cos\theta}\cos n\theta\, d\theta = \pi e(1, n).$$

[5.69] Prove that $\forall n \in 2\mathbb{Z} + 1$,

$$\int_0^\pi \sin(2\cos\theta)\cos n\theta\, d\theta = (-1)^{(n-1)/2}\pi e(-1, n).$$

[5.70] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \sin(2\cos\theta)\cos n\theta\, d\theta = 0.$$

[5.71] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \cos(2\cos\theta)\cos n\theta\, d\theta = (-1)^{n/2}\pi e(-1, n).$$

[5.72] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \cos(2\cos\theta)\cos n\theta\, d\theta = 0.$$

[5.73] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \sinh(2\cos\theta)\cos n\theta\, d\theta = \pi e(1, n).$$

[5.74] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \sinh(2\cos\theta)\cos n\theta\, d\theta = 0.$$

[5.75] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \cosh(2\cos\theta)\cos n\theta\, d\theta = \pi e(1, n).$$

[5.76] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \cosh(2\cos\theta)\cos n\theta\, d\theta = 0.$$

[5.77] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \left(\sin^2(\cos\theta)\cosh^2(\sin\theta) + \cos^2(\cos\theta)\sinh^2(\sin\theta)\right)\cos n\theta\, d\theta$$
$$= (-1)^{n/2}\pi \sinh(1, n).$$

[5.78] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \left(\sin^2(\cos\theta)\cosh^2(\sin\theta) + \cos^2(\cos\theta)\sinh^2(\sin\theta)\right)\cos n\theta\, d\theta = 0.$$

[5.79] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \left(\sinh^2(\cos\theta)\cos^2(\sin\theta) + \cosh^2(\cos\theta)\sin^2(\sin\theta)\right)\cos n\theta\, d\theta$$
$$= \pi \sinh(1, n).$$

[5.80] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \left(\sinh^2(\cos\theta)\cos^2(\sin\theta)+\cosh^2(\cos\theta)\sin^2(\sin\theta)\right)\cos n\theta\, d\theta = 0.$$

[5.81] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \left(\cos^2(\cos\theta)\cosh^2(\sin\theta)+\sin^2(\cos\theta)\sinh^2(\sin\theta)\right)\cos n\theta\, d\theta$$
$$= (-1)^{n/2}\pi\cosh(1,n).$$

[5.82] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \left(\cos^2(\cos\theta)\cosh^2(\sin\theta)+\sin^2(\cos\theta)\sinh^2(\sin\theta)\right)\cos n\theta\, d\theta = 0.$$

[5.83] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^\pi \left(\cosh^2(\cos\theta)\cos^2(\sin\theta)+\sinh^2(\cos\theta)\sin^2(\sin\theta)\right)\cos n\theta\, d\theta$$
$$= \pi\cosh(1,n).$$

[5.84] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \left(\cosh^2(\cos\theta)\cos^2(\sin\theta)+\sinh^2(\cos\theta)\sin^2(\sin\theta)\right)\cos n\theta\, d\theta = 0.$$

[5.85] Prove that

$$\int_0^\pi e^{\cos\theta}\cos(\sin\theta)\; d\theta = \pi.$$

[5.86] Prove that $\forall n \in \mathbb{N}$,

$$\int_0^\pi e^{\cos\theta}\cos(\sin\theta)\cos n\theta\, d\theta = \frac{\pi}{2n!}.$$

[5.87] Prove that $\forall n \in 2\mathbb{N}$,

$$\int_0^\pi \cosh(\cos\theta)\cos(\sin\theta)\cos n\theta\, d\theta = \frac{\pi}{2n!}.$$

[5.88] Prove that $\forall n \in 2\mathbb{Z}+1$,

$$\int_0^\pi \cosh(\cos\theta)\cos(\sin\theta)\cos n\theta\, d\theta = 0.$$

[5.89] Prove that $\forall n \in 2\mathbb{N} - 1$,

$$\int_0^{\pi} \sinh(\cos\theta)\cos(\sin\theta)\cos n\theta \, d\theta = \frac{\pi}{2n!}.$$

[5.90] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^{\pi} \sinh(\cos\theta)\cos(\sin\theta)\cos n\theta \, d\theta = 0.$$

[5.91] Prove that $\forall n \in 2\mathbb{Z}$,

$$\int_0^{\pi} \cos(2\sin\theta)\cos n\theta \, d\theta = \pi e(-1, n).$$

[5.92] Prove that $\forall n \in 2\mathbb{Z} + 1$,

$$\int_0^{\pi} \cos(2\sin\theta)\cos n\theta \, d\theta = 0.$$

[5.93] Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\sin(z, n) = \frac{e(iz, n) - e(-iz, n)}{2i}.$$

[5.94] Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\cos(z, n) = \frac{e(iz, n) + e(-iz, n)}{2}.$$

[5.95] Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\sinh(z, n) = \frac{e(z, n) - e(-z, n)}{2}.$$

[5.96] Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\cosh(z, n) = \frac{e(z, n) + e(-z, n)}{2}.$$

# Chapter 6
# Series Sums Using Special Integrals




In this chapter, we will explore some more interesting properties of the pseudo-exponential functions. First, we will prove that the pseudo-exponential functions are expressible as a finite-polynomial combination of each other. Next, we will derive the differential equations satisfied by the pseudo-exponential functions. Finally, we will define the *star product* of series and find its integral representation. In the problems listed in this chapter, we will see that several interesting series can be summed using the special integral of [5.21]. We will also evaluate special integrals and series sums derived using the star product identity.

## 6.1 Polynomial Dependence in the Pseudo-Exponential Functions

Recall the concept of linear dependence in *Linear Algebra* (Hoffman and Kunje, 2015). We say that the functions $\left(f_j(x)\right)_{j=1}^{n}$ are linearly dependent if there exist constants $\left(\alpha_j\right)_{j=1}^{n}$ such that $\sum_{j=1}^{n}\alpha_j f_j(x)=0$. We generalize the concept of linear dependence of functions into polynomial dependence, in which the constants $\left(\alpha_j\right)_{j=1}^{n}$ are replaced by finite-degree polynomials $\left(\alpha_j(x)\right)_{j=1}^{n}$.

**Definition 6.1** *(Polynomially Dependent Functions)*

We say that the functions $\left(f_j(x)\right)_{j=1}^{n}$ are polynomially dependent if there exist finite-degree polynomials $\left(\alpha_j(x)\right)_{j=1}^{n}$ such that

$$\sum_{j=1}^{n}\alpha_j(x)f_j(x)=0.$$

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A. Mishra, *Special Integrals*, University Texts in the Mathematical Sciences,
https://doi.org/10.1007/978-981-97-7514-9_6

Now, we will prove polynomial dependence among the pseudo-exponential functions in the examples that follow.

**Example 6.2** *(Polynomial Dependence of $e(z, n)$)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$e(z, n) = (n + 1)e(z, n + 1) + ze(z, n + 2). \tag{6.1}$$

***Proof*** Using (5.11), we get

$$\begin{aligned}
&(n + 1)e(z, n + 1) + ze(z, n + 2) \\
&= (n + 1) \sum_{j=\max\{0,-(n+1)\}}^{\infty} \frac{z^j}{j!(n + 1 + j)!} + z \sum_{j=\max\{0,-(n+2)\}}^{\infty} \frac{z^j}{j!(n + 2 + j)!} \\
&= (n + 1) \sum_{j=\max\{0,-(n+1)\}}^{\infty} \frac{z^j}{j!(n + 1 + j)!} + \sum_{j=\max\{0,-(n+2)\}}^{\infty} \frac{z^{j+1}}{j!(n + 2 + j)!} \\
&= (n + 1) \sum_{j=\max\{0,-(n+1)\}}^{\infty} \frac{z^j}{j!(n + 1 + j)!} + \sum_{j=\max\{1,-(n+1)\}}^{\infty} \frac{z^j}{(j - 1)!(n + 1 + j)!} \\
&= \frac{[n \geq 0](n + 1)}{0!(n + 1)!} + \sum_{j=\max\{1,-(n+1)\}}^{\infty} \left( \frac{n + 1}{j!(n + 1 + j)!} + \frac{1}{(j - 1)!(n + 1 + j)!} \right) z^j \\
&= \frac{[n \geq 0]}{0!n!} + \sum_{j=\max\{1,-(n+1)\}}^{\infty} \left( \frac{n + 1}{j} + 1 \right) \frac{z^j}{(j - 1)!(n + 1 + j)!} \\
&= \frac{[n \geq 0]}{0!n!} + \sum_{j=\max\{1,-n\}}^{\infty} \frac{z^j}{j!(n + j)!} \\
&= \sum_{j=\max\{0,-n\}}^{\infty} \frac{z^j}{j!(n + j)!} \\
&= e(z, n).
\end{aligned}$$

□

**Example 6.3** *(Polynomial Dependence of $\sin(z, n)$)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$,

$$\sin(z, n) = (n + 1) \sin(z, n + 1) + z \cos(z, n + 2). \tag{6.2}$$

***Proof*** Using (5.15) and (5.16), we get

$$\begin{aligned}
&(n + 1) \sin(z, n + 1) + z \cos(z, n + 2) \\
&= (n + 1) \sum_{j=\max\{0,\lfloor -(n+1)/2 \rfloor\}}^{\infty} (-1)^j \frac{z^{2j+1}}{(2j + 1)!(n + 1 + 2j + 1)!}
\end{aligned}$$

$$
\begin{aligned}
&+z\sum_{j=\max\{0,\lceil -(n+2)/2\rceil\}}^{\infty}(-1)^j\frac{z^{2j}}{(2j)!(n+2+2j)!}\\
&=(n+1)\sum_{j=\max\{0,\lfloor -(n+1)/2\rfloor\}}^{\infty}(-1)^j\frac{z^{2j+1}}{(2j+1)!(n+1+2j+1)!}\\
&\quad+\sum_{j=\max\{0,\lceil -(n+2)/2\rceil\}}^{\infty}(-1)^j\frac{z^{2j+1}}{(2j)!(n+2+2j)!}\\
&=\sum_{j=\max\{0,\lfloor -(n+1)/2\rfloor\}}^{\infty}(-1)^j\left(\frac{n+1}{(2j+1)!(n+1+2j+1)!}\right.\\
&\quad\left.+\frac{1}{(2j)!(n+2+2j)!}\right)z^{2j+1}\\
&=\sum_{j=\max\{0,\lfloor -(n+1)/2\rfloor\}}^{\infty}(-1)^j\left(\frac{n+1}{2j+1}+1\right)\frac{z^{2j+1}}{(2j)!(n+2+2j)!}\\
&=\sum_{j=\max\{0,\lfloor -(n+1)/2\rfloor\}}^{j=\max\{0,\lfloor -(n+1)/2\rfloor\}}(-1)^j\left(\frac{n+2j+2}{2j+1}\right)\frac{z^{2j+1}}{(2j)!(n+2+2j)!}\\
&\quad+\sum_{j=\max\{1,\lfloor -(n-1)/2\rfloor\}}^{\infty}(-1)^j\frac{z^{2j+1}}{(2j+1)!(n+2j+1)!}\\
&=\sum_{j=\max\{0,\lfloor -n/2\rfloor\}}^{\infty}(-1)^j\frac{z^{2j+1}}{(2j+1)!(n+2j+1)!}\\
&=\sin(z,n).
\end{aligned}
$$

□

**Example 6.4** *(Polynomial Dependence of* $\cos(z,n)$*)*
Prove that $\forall(z,n)\in\mathbb{C}\times\mathbb{Z}$,

$$\cos(z,n)=(n+1)\cos(z,n+1)-z\sin(z,n+2). \tag{6.3}$$

***Proof*** Using (5.16) and (5.15), we get

$$
\begin{aligned}
&(n+1)\cos(z,n+1)-z\sin(z,n+2)\\
&=(n+1)\sum_{j=\max\{0,\lceil -(n+1)/2\rceil\}}^{\infty}(-1)^j\frac{z^{2j}}{(2j)!(n+1+2j)!}
\end{aligned}
$$

$$
\begin{aligned}
&-z \sum_{j=\max\{0,\lfloor -(n+2)/2\rfloor\}}^{\infty} (-1)^j \frac{z^{2j+1}}{(2j+1)!(n+2+2j+1)!} \\
&= (n+1) \sum_{j=\max\{0,\lceil -(n+1)/2\rceil\}}^{\infty} (-1)^j \frac{z^{2j}}{(2j)!(n+1+2j)!} \\
&\quad - \sum_{j=\max\{0,\lfloor -(n+2)/2\rfloor\}}^{\infty} (-1)^j \frac{z^{2j+2}}{(2j+1)!(n+2+2j+1)!} \\
&= (n+1) \sum_{j=\max\{0,\lceil -(n+1)/2\rceil\}}^{\infty} (-1)^j \frac{z^{2j}}{(2j)!(n+1+2j)!} \\
&\quad + \sum_{j=\max\{1,\lfloor -n/2\rfloor\}}^{\infty} (-1)^j \frac{z^{2j}}{(2j-1)!(n+1+2j)!}
\end{aligned}
$$

$$
\begin{aligned}
&= [n \geq 0] \frac{n+1}{0!(n+1)!} + \sum_{j=\max\{1,\lfloor -n/2\rfloor\}}^{\infty} (-1)^j \left(\frac{n+1}{2j}+1\right) \frac{z^{2j}}{(2j-1)!(n+1+2j)!} \\
&= [n \geq 0] \frac{n+1}{0!(n+1)!} + \sum_{j=\max\{1,\lceil -n/2\rceil\}}^{\infty} (-1)^j \frac{z^{2j}}{(2j)!(n+2j)!} \\
&= \sum_{j=\max\{0,\lceil -n/2\rceil\}}^{\infty} (-1)^j \frac{z^{2j}}{(2j)!(n+2j)!} \\
&= \cos(z,n).
\end{aligned}
$$

□

**Example 6.5** *(Polynomial Dependence of* $\sinh(z,n)$*)*
Prove that $\forall (z,n) \in \mathbb{C} \times \mathbb{Z}$,

$$\sinh(z,n) = (n+1)\sinh(z,n+1) + z\cosh(z,n+2). \tag{6.4}$$

***Proof*** Replacing $z$ with $iz$ in (6.2) and using (5.17) and (5.18), we get

$$
\begin{aligned}
&\sin(iz,n) = (n+1)\sin(iz,n+1) + iz\cos(iz,n+2) \\
\iff &i\sinh(z,n) = (n+1)i\sinh(z,n+1) + iz\cosh(z,n+2) \\
\iff &\sinh(z,n) = (n+1)\sinh(z,n+1) + z\cosh(z,n+2).
\end{aligned}
$$

□

**Example 6.6** *(Polynomial Dependence of* $\cosh(z,n)$*)*
Prove that $\forall (z,n) \in \mathbb{C} \times \mathbb{Z}$,

$$\cosh(z,n) = (n+1)\cosh(z,n+1) + z\sinh(z,n+2). \tag{6.5}$$

***Proof*** Replacing $z$ with $iz$ in (6.3) and using (5.18) and (5.17), we get

$$\begin{aligned} &\cos(iz, n) = (n+1)\cos(iz, n+1) - iz\sin(iz, n+2) \\ \Longleftrightarrow\ &\cosh(z, n) = (n+1)\cosh(z, n+1) + z\sinh(z, n+2). \end{aligned}$$

□

## 6.2 Differential Equations Satisfied by the Pseudo-Exponential Functions

We know that the elementary exponential functions satisfy simple differential equations.

$$\begin{aligned} &\frac{dy}{dz} - y = 0 \implies y = e^z, \\ &\frac{d^2y}{dz^2} + y = 0 \implies y = \sin z, \quad y = \cos z, \\ &\frac{d^2y}{dz^2} - y = 0 \implies y = \sinh z, \quad y = \cosh z. \end{aligned}$$

From the above, we can see that the elementary exponential functions satisfy up to second-order differential equations. Now, we will derive the differential equations satisfied by the pseudo-exponential functions and see that they satisfy up to fourth-order differential equations.

**Example 6.7** *(Differential Equation Satisfied by $e(z, n)$)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$, $y = e(z, n)$ satisfies the differential equation

$$z\frac{d^2y}{dz^2} + (n+1)\frac{dy}{dz} - y = 0.$$

***Proof*** From (5.21), we have

$$\begin{aligned} &y = e(z, n) \\ \Longrightarrow\ &\frac{dy}{dz} = e(z, n+1) \\ \Longrightarrow\ &\frac{d^2y}{dz^2} = e(z, n+2). \end{aligned} \tag{6.6}$$

From (6.6) and (6.1), we get

$$y = (n+1)\frac{dy}{dz} + z\frac{d^2y}{dz^2}$$

$$\iff z\frac{d^2y}{dz^2} + (n+1)\frac{dy}{dz} - y = 0.$$

□

**Example 6.8** *(Differential Equation Satisfied by* $\sin(z, n)$*)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$, $y = \sin(z, n)$ satisfies the differential equation

$$z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} + y = 0.$$

***Proof*** From (5.22) and (5.23), we have

$$\begin{aligned} & y = \sin(z, n) \\ \implies & \frac{dy}{dz} = \cos(z, n+1) \\ \implies & \frac{d^2y}{dz^2} = -\sin(z, n+2). \end{aligned} \tag{6.7}$$

From (6.7) and (6.3), we get

$$\cos(z, n) = (n+1)\frac{dy}{dz} + z\frac{d^2y}{dz^2}. \tag{6.8}$$

Differentiating (6.8) with respect to $z$, and using (5.23), we get

$$-\sin(z, n+1) = (n+2)\frac{d^2y}{dz^2} + z\frac{d^3y}{dz^3}. \tag{6.9}$$

Differentiating (6.9) with respect to $z$, and using (5.22), we get

$$-\cos(z, n+2) = (n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}. \tag{6.10}$$

From (6.2), (6.7), (6.9), and (6.10), we get

$$\begin{aligned} & z\left((n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}\right) + (n+1)\left((n+2)\frac{d^2y}{dz^2} + z\frac{d^3y}{dz^3}\right) + y = 0 \\ \iff & z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} + y = 0. \end{aligned}$$

□

**Example 6.9** *(Differential Equation Satisfied by* $\cos(z, n)$*)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$, $y = \cos(z, n)$ satisfies the differential equation

$$z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} + y = 0.$$

***Proof*** From (5.23) and (5.22), we have

$$\begin{aligned} & y = \cos(z, n) \\ \Longrightarrow\ & \frac{dy}{dz} = -\sin(z, n+1) \\ \Longrightarrow\ & \frac{d^2y}{dz^2} = -\cos(z, n+2). \end{aligned} \tag{6.11}$$

From (6.11) and (6.2), we get

$$\sin(z, n) = -(n+1)\frac{dy}{dz} - z\frac{d^2y}{dz^2}. \tag{6.12}$$

Differentiating (6.12) with respect to $z$, and using (5.22), we get

$$\cos(z, n+1) = -(n+2)\frac{d^2y}{dz^2} - z\frac{d^3y}{dz^3}. \tag{6.13}$$

Differentiating (6.13) with respect to $z$, and using (5.23), we get

$$\begin{aligned} -\sin(z, n+2) &= -(n+3)\frac{d^3y}{dz^3} - z\frac{d^4y}{dz^4} \\ \Longleftrightarrow\ \sin(z, n+2) &= (n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}. \end{aligned} \tag{6.14}$$

From (6.3), (6.11), (6.13), and (6.14), we get

$$\begin{aligned} & z\left((n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}\right) + (n+1)\left((n+2)\frac{d^2y}{dz^2} + z\frac{d^3y}{dz^3}\right) + y = 0 \\ \Longleftrightarrow\ & z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} + y = 0. \end{aligned}$$

□

**Example 6.10** *(Differential Equation Satisfied by* $\sinh(z, n)$*)*
Prove that $\forall (z, n) \in \mathbb{C} \times \mathbb{Z}$, $y = \sinh(z, n)$ satisfies the differential equation

$$z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} - y = 0.$$

***Proof*** From (5.24) and (5.25), we have

$$\begin{aligned} & y = \sinh(z, n) \\ \Longrightarrow & \frac{dy}{dz} = \cosh(z, n+1) \\ \Longrightarrow & \frac{d^2y}{dz^2} = \sinh(z, n+2). \end{aligned} \tag{6.15}$$

From (6.15) and (6.5), we get

$$\cosh(z, n) = (n+1)\frac{dy}{dz} + z\frac{d^2y}{dz^2}. \tag{6.16}$$

Differentiating (6.16) with respect to $z$, and using (5.25), we get

$$\sinh(z, n+1) = (n+2)\frac{d^2y}{dz^2} + z\frac{d^3y}{dz^3}. \tag{6.17}$$

Differentiating (6.17) with respect to $z$, and using (5.24), we get

$$\cosh(z, n+2) = (n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}. \tag{6.18}$$

From (6.4), (6.15), (6.17), and (6.18), we get

$$\begin{aligned} & z\left((n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}\right) + (n+1)\left((n+2)\frac{d^2y}{dz^2} + z\frac{d^3y}{dz^3}\right) - y = 0 \\ \Longleftrightarrow\ & z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} - y = 0. \end{aligned}$$

□

**Example 6.11** *(Differential Equation Satisfied by* $\cosh(z, n)$*)*
Prove that $\forall(z, n) \in \mathbb{C} \times \mathbb{Z}$, $y = \cosh(z, n)$ satisfies the differential equation

$$z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} - y = 0.$$

***Proof*** From (5.25) and (5.24), we have

$$\begin{aligned} & y = \cosh(z, n) \\ \Longrightarrow & \frac{dy}{dz} = \sinh(z, n+1) \\ \Longrightarrow & \frac{d^2y}{dz^2} = \cosh(z, n+2). \end{aligned} \tag{6.19}$$

From (6.19) and (6.4), we get

$$\sinh(z, n) = (n+1)\frac{dy}{dz} + z\frac{d^2y}{dz^2}. \tag{6.20}$$

Differentiating (6.20) with respect to $z$, and using (5.24), we get

$$\cosh(z, n+1) = (n+2)\frac{d^2y}{dz^2} + z\frac{d^3y}{dz^3}. \tag{6.21}$$

Differentiating (6.21) with respect to $z$, and using (5.25), we get

$$\sinh(z, n+2) = (n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}. \tag{6.22}$$

From (6.5), (6.19), (6.21), and (6.22), we get

$$\begin{aligned} & z\left((n+3)\frac{d^3y}{dz^3} + z\frac{d^4y}{dz^4}\right) + (n+1)\left((n+2)\frac{d^2y}{dz^2} + z\frac{d^3y}{dz^3}\right) - y = 0 \\ \iff & z^2\frac{d^4y}{dz^4} + 2(n+2)z\frac{d^3y}{dz^3} + (n+1)(n+2)\frac{d^2y}{dz^2} - y = 0. \end{aligned}$$

□

## 6.3 Series Transformations

In Theorem 4.5, we computed the sum of squares of terms in a series and generalized it to find a quadratic sum of terms in Theorem 4.6. Similarly, we computed several series sums involving the binomial coefficients in Theorems 4.7, 4.8, 4.10, 4.11, and 4.13. We computed some more series sums mainly involving the binomial coefficients in Theorems 4.17, 4.28, and 4.41. We can view the above series sums as an instance of the *star product* of two series that we will define below. In this section, we will find a special integral to compute the star product of two functions. The above-mentioned theorems can easily be proved using the star product special integral.

**Definition 6.12** *(Star Product of Series)*
Let

$$f(z) = \sum_{j=0}^{\infty} a_j z^j = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots \tag{6.23}$$

and

$$g(z) = \sum_{j=0}^{\infty} b_j z^j = b_0 + b_1 z + b_2 z^2 + b_3 z^3 + \cdots \tag{6.24}$$

be functions that can be expanded by Maclaurin's series. Then, the star product, $f \star g$, of $f$ and $g$ is defined by

$$(f \star g)(z) = \sum_{j=0}^{\infty} a_j b_j z^j = a_0 b_0 + a_1 b_1 z + a_2 b_2 z^2 + a_3 b_3 z^3 + \cdots \tag{6.25}$$

We also write $(f \star g)(xy)$ as $f(x) \star g(y)$.

**Theorem 6.13** (A Star Product Identity)
*Let the functions $f$ and $g$ be defined as in Definition 6.12. Assuming that $f\left(xe^{i\theta}\right)$, $f\left(xe^{-i\theta}\right)$, $g\left(ye^{i\theta}\right)$, and $g\left(ye^{-i\theta}\right)$ are all convergent, the star product, $f \star g$, of $f$ and $g$ is given by*

$$\begin{aligned}(f \star g)(z) = &- f(0)g(0) + \frac{1}{2\pi}\int_0^{\pi} \left(f\left(xe^{i\theta}\right) + f\left(xe^{-i\theta}\right)\right)\left(g\left(ye^{i\theta}\right)\right. \\ &\left. + g\left(ye^{-i\theta}\right)\right)\, d\theta,\end{aligned}$$

*where $z = xy$.*

***Proof*** Using (6.23) and (6.24), we get

$$\begin{aligned}&\left(f\left(xe^{i\theta}\right) + f\left(xe^{-i\theta}\right)\right)\left(g\left(ye^{i\theta}\right) + g\left(ye^{-i\theta}\right)\right) \\ &= \left(\sum_{k=0}^{\infty} a_k\left(e^{ik\theta} + e^{-ik\theta}\right)x^k\right)\left(\sum_{j=0}^{\infty} b_j\left(e^{ij\theta} + e^{-ij\theta}\right)y^j\right) \\ &= 4\left(\sum_{k=0}^{\infty} a_k x^k \cos k\theta\right)\left(\sum_{j=0}^{\infty} b_j y^j \cos j\theta\right) \\ &= 4\sum_{k=0}^{\infty} a_k b_k x^k y^k \cos^2 k\theta + 4\sum_{k=0}^{\infty}\sum_{j=0, j\neq k}^{\infty} a_k b_j x^k y^j \cos k\theta \cos j\theta \\ &= 2\sum_{k=0}^{\infty} a_k b_k (1 + \cos 2k\theta) z^k + 2\sum_{k=0}^{\infty}\sum_{j=0, j\neq k}^{\infty} a_k b_j (\cos(k+j)\theta + \cos(k-j)\theta) x^k y^j.\end{aligned} \tag{6.26}$$

Integrating (6.26) with respect to $\theta$ from 0 to $\pi$, dividing the result by $2\pi$, subtracting by $f(0)g(0)$, and using (6.25), we get

$$-f(0)g(0) + \frac{1}{2\pi}\int_0^{\pi} \left(f\left(xe^{i\theta}\right) + f\left(xe^{-i\theta}\right)\right)\left(g\left(ye^{i\theta}\right) + g\left(ye^{-i\theta}\right)\right)\, d\theta$$

$$
\begin{aligned}
&= -a_0b_0 + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{k=0}^{\infty} a_k b_k (1+\cos 2k\theta) z^k\right) d\theta \\
&\quad + \frac{1}{\pi}\int_0^{\pi}\left(\sum_{k=0}^{\infty}\sum_{j=0, j\neq k}^{\infty} a_k b_j (\cos(k+j)\theta + \cos(k-j)\theta) x^k y^j\right) d\theta \\
&= -a_0b_0 + \frac{1}{\pi}\sum_{k=0}^{\infty} a_k b_k z^k \int_0^{\pi}(1+\cos 2k\theta) d\theta \\
&\quad + \frac{1}{\pi}\sum_{k=0}^{\infty}\sum_{j=0, j\neq k}^{\infty} a_k b_j x^k y^j \int_0^{\pi}(\cos(k+j)\theta + \cos(k-j)\theta)\, d\theta \\
&= -a_0b_0 + \frac{a_0b_0}{\pi}(2\theta)\Big|_0^{\pi} + \frac{1}{\pi}\sum_{k=1}^{\infty} a_k b_k z^k \left(\theta + \frac{\sin 2k\theta}{2k}\right)\Big|_0^{\pi} \\
&\quad + \frac{1}{\pi}\sum_{k=0}^{\infty}\sum_{j=0, j\neq k}^{\infty} a_k b_j x^k y^j \left(\frac{\sin(k+j)\theta}{k+j} + \frac{\sin(k-j)\theta}{k-j}\right)\Big|_0^{\pi} \\
&= \sum_{k=0}^{\infty} a_k b_k z^k \\
&= (f \star g)(z).
\end{aligned}
$$

□

**Theorem 6.14** (Another Star Product Identity)
*Let the functions $f$ and $g$ be defined as in Definition 6.12. Assuming that $f\left(xe^{i\theta}\right)$, $f\left(xe^{-i\theta}\right)$, $g\left(ye^{i\theta}\right)$, and $g\left(ye^{-i\theta}\right)$ are all convergent, the star product, $f \star g$, of $f$ and $g$ is given by*

$$
(f \star g)(z) = f(0)g(0) - \frac{1}{2\pi}\int_0^{\pi}\left(f\left(xe^{i\theta}\right) - f\left(xe^{-i\theta}\right)\right)\left(g\left(ye^{i\theta}\right) - g\left(ye^{-i\theta}\right)\right) d\theta,
$$

*where $z = xy$.*

***Proof*** Using (6.23) and (6.24), we get

$$
\begin{aligned}
&\left(f\left(xe^{i\theta}\right)-f\left(xe^{-i\theta}\right)\right)\left(g\left(ye^{i\theta}\right)-g\left(ye^{-i\theta}\right)\right)\\
&=\left(\sum_{k=0}^{\infty}a_k\left(e^{ik\theta}-e^{-ik\theta}\right)x^k\right)\left(\sum_{j=0}^{\infty}b_j\left(e^{ij\theta}-e^{-ij\theta}\right)y^j\right)\\
&=-4\left(\sum_{k=0}^{\infty}a_kx^k\sin k\theta\right)\left(\sum_{j=0}^{\infty}b_jy^j\sin j\theta\right)\\
&=-4\sum_{k=0}^{\infty}a_kb_kx^ky^k\sin^2 k\theta-4\sum_{k=0}^{\infty}\sum_{j=0,j\neq k}^{\infty}a_kb_jx^ky^j\sin k\theta\sin j\theta\\
&=-2\sum_{k=0}^{\infty}a_kb_k(1-\cos 2k\theta)z^k\\
&\quad+2\sum_{k=0}^{\infty}\sum_{j=0,j\neq k}^{\infty}a_kb_j(\cos(k+j)\theta-\cos(k-j)\theta)x^ky^j.
\end{aligned}
\tag{6.27}
$$

Integrating (6.27) with respect to $\theta$ from 0 to $\pi$, dividing the result by $2\pi$, subtracting from $f(0)g(0)$, and using (6.25), we get

$$
\begin{aligned}
&f(0)g(0)-\frac{1}{2\pi}\int_0^{\pi}\left(f\left(xe^{i\theta}\right)-f\left(xe^{-i\theta}\right)\right)\left(g\left(ye^{i\theta}\right)-g\left(ye^{-i\theta}\right)\right)d\theta\\
&=a_0b_0+\frac{1}{\pi}\int_0^{\pi}\left(\sum_{k=0}^{\infty}a_kb_k(1-\cos 2k\theta)z^k\right)d\theta\\
&\quad-\frac{1}{\pi}\int_0^{\pi}\left(\sum_{k=0}^{\infty}\sum_{j=0,j\neq k}^{\infty}a_kb_j(\cos(k+j)\theta-\cos(k-j)\theta)x^ky^j\right)d\theta\\
&=a_0b_0+\frac{1}{\pi}\sum_{k=0}^{\infty}a_kb_kz^k\int_0^{\pi}(1-\cos 2k\theta)d\theta\\
&\quad-\frac{1}{\pi}\sum_{k=0}^{\infty}\sum_{j=0,j\neq k}^{\infty}a_kb_jx^ky^j\int_0^{\pi}(\cos(k+j)\theta-\cos(k-j)\theta)\,d\theta\\
&=a_0b_0+\frac{1}{\pi}\sum_{k=1}^{\infty}a_kb_kz^k\left(\theta-\frac{\sin 2k\theta}{2k}\right)\Bigg|_0^{\pi}\\
&\quad-\frac{1}{\pi}\sum_{k=0}^{\infty}\sum_{j=0,j\neq k}^{\infty}a_kb_jx^ky^j\left(\frac{\sin(k+j)\theta}{k+j}-\frac{\sin(k-j)\theta}{k-j}\right)\Bigg|_0^{\pi}\\
&=\sum_{k=0}^{\infty}a_kb_kz^k=(f\star g)(z).
\end{aligned}
$$

□

## Problems

[6.1] Let the function $C_3(x,\theta)$ be defined as

$$\begin{aligned} C_3(x,\theta) = &\cosh(x\sin\theta)\cos(x\cos\theta)\cos(2\sin\theta) \\ &- \sinh(x\sin\theta)\sin(x\cos\theta)\sin(2\sin\theta)\,, \end{aligned} \tag{6.28}$$

and the function $S_3(x,\theta)$ be defined as

$$\begin{aligned} S_3(x,\theta) = &\sinh(x\sin\theta)\cos(x\cos\theta)\sin(2\sin\theta) \\ &+ \cosh(x\sin\theta)\sin(x\cos\theta)\cos(2\sin\theta)\,. \end{aligned} \tag{6.29}$$

Let the function $\phi(x)$ be defined as $\phi(x) = \tan^{-1} x$. Prove that $\forall (x,n) \in \mathbb{R} \times (\mathbb{N}-1) - \{0\} \times (2\mathbb{N}-1)$,

$$\begin{aligned} &\sum_{j=0}^{\infty} \frac{(1+x^2)^j \cos 2j\phi(x)}{(2j)!(n+2j)!} - \sqrt{1+x^2} \sum_{j=0}^{\infty} \frac{(1+x^2)^j \cos(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!} \\ &= \left( \frac{1}{0!n!} + \frac{(1+x^2)\cos 2\phi(x)}{2!(n+2)!} + \frac{(1+x^2)^2 \cos 4\phi(x)}{4!(n+4)!} \right. \\ &\quad \left. + \frac{(1+x^2)^3 \cos 6\phi(x)}{6!(n+6)!} + \cdots \right) - \sqrt{1+x^2} \left( \frac{\cos\phi(x)}{1!(n+1)!} \right. \\ &\quad \left. + \frac{(1+x^2)\cos 3\phi(x)}{3!(n+3)!} + \frac{(1+x^2)^2 \cos 5\phi(x)}{5!(n+5)!} + \frac{(1+x^2)^3 \cos 7\phi(x)}{7!(n+7)!} + \cdots \right) \\ &= \frac{2}{\pi} \cdot \frac{1 + (1+x^2)^{n/2} \cos n(\pi - \phi(x))}{1 + (1+x^2)^n + 2(1+x^2)^{n/2} \cos n(\pi - \phi(x))} \int_0^{\pi} C_3(x,\theta) \cos n\theta \, d\theta \\ &+ \frac{2}{\pi} \cdot \frac{(1+x^2)^{n/2} \sin n(\pi - \phi(x))}{1 + (1+x^2)^n + 2(1+x^2)^{n/2} \cos n(\pi - \phi(x))} \int_0^{\pi} S_3(x,\theta) \cos n\theta \, d\theta. \end{aligned}$$

[6.2] Let the function $\phi(x)$ be defined as $\phi(x) = \tan^{-1} x$ and $C_3(x, \theta)$ be given by (6.28). Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}
&\sum_{j=0}^{\infty} \frac{(1+x^2)^j \cos 2j\phi(x)}{((2j)!)^2} - \sqrt{1+x^2} \sum_{j=0}^{\infty} \frac{(1+x^2)^j \cos(2j+1)\phi(x)}{((2j+1)!)^2} \\
&= \left( \frac{1}{(0!)^2} + \frac{(1+x^2)\cos 2\phi(x)}{(2!)^2} + \frac{(1+x^2)^2 \cos 4\phi(x)}{(4!)^2} \right. \\
&\quad \left. + \frac{(1+x^2)^3 \cos 6\phi(x)}{(6!)^2} + \cdots \right) \\
&\quad - \sqrt{1+x^2} \left( \frac{\cos\phi(x)}{(1!)^2} + \frac{(1+x^2)\cos 3\phi(x)}{(3!)^2} + \frac{(1+x^2)^2 \cos 5\phi(x)}{(5!)^2} \right. \\
&\quad \left. + \frac{(1+x^2)^3 \cos 7\phi(x)}{(7!)^2} + \cdots \right) \\
&= \frac{1}{\pi} \int_0^{\pi} C_3(x, \theta)\, d\theta.
\end{aligned}$$

[6.3] Let the function $\phi(x)$ be defined as $\phi(x) = \tan^{-1} x$, $S_3(x, \theta)$ be given by (6.29), and $C_3(x, \theta)$ be given by (6.28). Prove that $\forall (x, n) \in \mathbb{R} \times (\mathbb{N} - 1) - \{0\} \times (2\mathbb{N} - 1)$,

$$\begin{aligned}
&\sqrt{1+x^2} \sum_{j=0}^{\infty} \frac{(1+x^2)^j \sin(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!} - \sum_{j=1}^{\infty} \frac{(1+x^2)^j \sin 2j\phi(x)}{(2j)!(n+2j)!} \\
&= \sqrt{1+x^2} \left( \frac{\sin\phi(x)}{1!(n+1)!} + \frac{(1+x^2)\sin 3\phi(x)}{3!(n+3)!} + \frac{(1+x^2)^2 \sin 5\phi(x)}{5!(n+5)!} \right. \\
&\quad \left. + \frac{(1+x^2)^3 \sin 7\phi(x)}{7!(n+7)!} + \cdots \right) \\
&\quad - \left( \frac{(1+x^2)\sin 2\phi(x)}{2!(n+2)!} + \frac{(1+x^2)^2 \sin 4\phi(x)}{4!(n+4)!} + \frac{(1+x^2)^3 \sin 6\phi(x)}{6!(n+6)!} \right. \\
&\quad \left. + \frac{(1+x^2)^4 \sin 8\phi(x)}{8!(n+8)!} + \cdots \right) \\
&= \frac{2}{\pi} \cdot \frac{1 + (1+x^2)^{n/2} \cos n(\pi - \phi(x))}{1 + (1+x^2)^n + 2(1+x^2)^{n/2} \cos n(\pi - \phi(x))} \int_0^{\pi} S_3(x, \theta) \cos n\theta \, d\theta \\
&\quad - \frac{2}{\pi} \cdot \frac{(1+x^2)^{n/2} \sin n(\pi - \phi(x))}{1 + (1+x^2)^n + 2(1+x^2)^{n/2} \cos n(\pi - \phi(x))} \int_0^{\pi} C_3(x, \theta) \cos n\theta \, d\theta.
\end{aligned}$$

[6.4] Let the function $\phi(x)$ be defined as $\phi(x) = \tan^{-1} x$ and $S_3(x, \theta)$ be given by (6.29). Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}
&\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j \sin(2j+1)\phi(x)}{((2j+1)!)^2} - \sum_{j=1}^{\infty}\frac{(1+x^2)^j \sin 2j\phi(x)}{((2j)!)^2} \\
&= \sqrt{1+x^2}\left(\frac{\sin\phi(x)}{(1!)^2} + \frac{(1+x^2)\sin 3\phi(x)}{(3!)^2} + \frac{(1+x^2)^2 \sin 5\phi(x)}{(5!)^2}\right. \\
&\quad \left. + \frac{(1+x^2)^3 \sin 7\phi(x)}{(7!)^2} + \cdots\right) \\
&\quad - \left(\frac{(1+x^2)\sin 2\phi(x)}{(2!)^2} + \frac{(1+x^2)^2 \sin 4\phi(x)}{(4!)^2} + \frac{(1+x^2)^3 \sin 6\phi(x)}{(6!)^2}\right. \\
&\quad \left. + \frac{(1+x^2)^4 \sin 8\phi(x)}{(8!)^2} + \cdots\right) \\
&= \frac{1}{\pi}\int_0^{\pi} S_3(x, \theta)\, d\theta.
\end{aligned}$$

[6.5] Let $S_3(1, \theta)$ be given by (6.29). Prove that

$$\begin{aligned}
&\sum_{j=0}^{\infty}(-1)^j\left(\frac{2^{2j}}{((4j+1)!)^2} + \frac{2^{2j+1}}{((4j+3)!)^2}\right) - \sum_{j=0}^{\infty}(-1)^j\frac{2^{2j+1}}{((4j+2)!)^2} \\
&= \left(\frac{1}{(1!)^2} + \frac{2}{(3!)^2} - \frac{2^2}{(5!)^2} - \frac{2^3}{(7!)^2} + \cdots\right) \\
&\quad - \left(\frac{2}{(2!)^2} - \frac{2^3}{(6!)^2} + \frac{2^5}{(10!)^2} - \frac{2^7}{(14!)^2} + \cdots\right) \\
&= \frac{1}{\pi}\int_0^{\pi} S_3(1, \theta)\, d\theta.
\end{aligned}$$

[6.6] Let $C_3(1, \theta)$ be given by (6.28). Prove that

$$\begin{aligned}
&\sum_{j=0}^{\infty}(-1)^j\frac{2^{2j}}{((4j)!)^2} - \sum_{j=0}^{\infty}(-1)^j\left(\frac{2^{2j}}{((4j+1)!)^2} - \frac{2^{2j+1}}{((4j+3)!)^2}\right) \\
&= \left(\frac{1}{(0!)^2} - \frac{2^2}{(4!)^2} + \frac{2^4}{(8!)^2} - \frac{2^6}{(12!)^2} + \cdots\right) \\
&\quad - \left(\frac{1}{(1!)^2} - \frac{2^1}{(3!)^2} - \frac{2^2}{(5!)^2} + \frac{2^3}{(7!)^2} + \cdots\right) \\
&= \frac{1}{\pi}\int_0^{\pi} C_3(1, \theta)\, d\theta.
\end{aligned}$$

[6.7] Let the function $C_4(x,\theta)$ be defined as

$$\begin{aligned} C_4(x,\theta) &= \cosh(x\cos\theta)\cos(x\sin\theta)\cos(2\sin\theta) \\ &\quad + \sinh(x\cos\theta)\sin(x\sin\theta)\sin(2\sin\theta)\,, \end{aligned} \tag{6.30}$$

and the function $S_4(x,\theta)$ be defined as

$$\begin{aligned} S_4(x,\theta) &= \cosh(x\cos\theta)\sin(x\sin\theta)\sin(2\sin\theta) \\ &\quad + \sinh(x\cos\theta)\cos(x\sin\theta)\cos(2\sin\theta)\,. \end{aligned} \tag{6.31}$$

Let the function $\phi(x)$ be defined as $\phi(x) = \tanh^{-1} x$. Prove that $\forall (x,n) \in (-1,1)\times(\mathbb{N}-1) - \{0\}\times(2\mathbb{N}-1)$,

$$\begin{aligned} &\sum_{j=0}^{\infty} \frac{(1-x^2)^j \cosh 2j\phi(x)}{(2j)!(n+2j)!} - \sqrt{1-x^2}\sum_{j=0}^{\infty} \frac{(1-x^2)^j \cosh(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!} \\ &= \left(\frac{1}{0!n!} + \frac{(1-x^2)\cosh 2\phi(x)}{2!(n+2)!} + \frac{(1-x^2)^2\cosh 4\phi(x)}{4!(n+4)!}\right. \\ &\quad \left. + \frac{(1-x^2)^3\cosh 6\phi(x)}{6!(n+6)!} + \cdots\right) \\ &\quad - \sqrt{1-x^2}\left(\frac{\cosh\phi(x)}{1!(n+1)!} + \frac{(1-x^2)\cosh 3\phi(x)}{3!(n+3)!} + \frac{(1-x^2)^2\cosh 5\phi(x)}{5!(n+5)!}\right. \\ &\quad \left. + \frac{(1-x^2)^3\cosh 7\phi(x)}{7!(n+7)!} + \cdots\right) \\ &= \frac{2}{\pi}\cdot\frac{1+(-1)^n(1-x^2)^{n/2}\cosh n\phi(x)}{1+(1-x^2)^n+(-1)^n 2(1-x^2)^{n/2}\cosh n\phi(x)}\int_0^{\pi} C_4(x,\theta)\cos n\theta\, d\theta \\ &\quad + \frac{2}{\pi}\cdot\frac{(-1)^n(1-x^2)^{n/2}\sinh n\phi(x)}{1+(1-x^2)^n+(-1)^n 2(1-x^2)^{n/2}\cosh n\phi(x)}\int_0^{\pi} S_4(x,\theta)\cos n\theta\, d\theta. \end{aligned}$$

[6.8] Let the function $\phi(x)$ be defined as $\phi(x) = \tanh^{-1} x$ and $C_4(x,\theta)$ be given by (6.30). Prove that $\forall x \in (-1,1)$,

$$\begin{aligned} &\sum_{j=0}^{\infty} \frac{(1-x^2)^j \cosh 2j\phi(x)}{((2j)!)^2} - \sqrt{1-x^2}\sum_{j=0}^{\infty} \frac{(1-x^2)^j \cosh(2j+1)\phi(x)}{((2j+1)!)^2} \\ &= \left(\frac{1}{(0!)^2} + \frac{(1-x^2)\cosh 2\phi(x)}{(2!)^2} + \frac{(1-x^2)^2\cosh 4\phi(x)}{(4!)^2}\right. \\ &\quad \left. + \frac{(1-x^2)^3\cosh 6\phi(x)}{(6!)^2} + \cdots\right) \end{aligned}$$

$$
\begin{aligned}
&-\sqrt{1-x^2}\left(\frac{\cosh\phi(x)}{(1!)^2}+\frac{(1-x^2)\cosh 3\phi(x)}{(3!)^2}+\frac{(1-x^2)^2\cosh 5\phi(x)}{(5!)^2}\right.\\
&\left.+\frac{(1-x^2)^3\cosh 7\phi(x)}{(7!)^2}+\cdots\right)\\
&=\frac{1}{\pi}\int_0^{\pi} C_4(x,\theta)\,d\theta.
\end{aligned}
$$

[6.9] Let the function $\phi(x)$ be defined as $\phi(x)=\tanh^{-1}x$, $S_4(x,\theta)$ be given by (6.31), and $C_4(x,\theta)$ be given by (6.30). Prove that $\forall(x,n)\in(-1,1)\times(\mathbb{N}-1)-\{0\}\times(2\mathbb{N}-1)$,

$$
\begin{aligned}
&\sqrt{1-x^2}\sum_{j=0}^{\infty}\frac{(1-x^2)^j\sinh(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}-\sum_{j=1}^{\infty}\frac{(1-x^2)^j\sinh 2j\phi(x)}{(2j)!(n+2j)!}\\
&=\sqrt{1-x^2}\left(\frac{\sinh\phi(x)}{1!(n+1)!}+\frac{(1-x^2)\sinh 3\phi(x)}{3!(n+3)!}+\frac{(1-x^2)^2\sinh 5\phi(x)}{5!(n+5)!}\right.\\
&\left.+\frac{(1-x^2)^3\sinh 7\phi(x)}{7!(n+7)!}+\cdots\right)\\
&-\left(\frac{(1-x^2)\sinh 2\phi(x)}{2!(n+2)!}+\frac{(1-x^2)^2\sinh 4\phi(x)}{4!(n+4)!}+\frac{(1-x^2)^3\sinh 6\phi(x)}{6!(n+6)!}\right.\\
&\left.+\frac{(1-x^2)^4\sinh 8\phi(x)}{8!(n+8)!}+\cdots\right)\\
&=\frac{2}{\pi}\cdot\frac{1+(-1)^n(1-x^2)^{n/2}\cosh n\phi(x)}{1+(1-x^2)^n+(-1)^n2(1-x^2)^{n/2}\cosh n\phi(x)}\int_0^{\pi}S_4(x,\theta)\cos n\theta\,d\theta\\
&+\frac{2}{\pi}\cdot\frac{(-1)^n(1-x^2)^{n/2}\sinh n\phi(x)}{1+(1-x^2)^n+(-1)^n2(1-x^2)^{n/2}\cosh n\phi(x)}\int_0^{\pi}C_4(x,\theta)\cos n\theta\,d\theta.
\end{aligned}
$$

[6.10] Let the function $\phi(x)$ be defined as $\phi(x)=\tanh^{-1}x$ and $S_4(x,\theta)$ be given by (6.31). Prove that $\forall x\in(-1,1)$,

$$
\begin{aligned}
&\sqrt{1-x^2}\sum_{j=0}^{\infty}\frac{(1-x^2)^j\sinh(2j+1)\phi(x)}{((2j+1)!)^2}-\sum_{j=1}^{\infty}\frac{(1-x^2)^j\sinh 2j\phi(x)}{((2j)!)^2}\\
&=\sqrt{1-x^2}\left(\frac{\sinh\phi(x)}{(1!)^2}+\frac{(1-x^2)\sinh 3\phi(x)}{(3!)^2}+\frac{(1-x^2)^2\sinh 5\phi(x)}{(5!)^2}\right.\\
&\left.+\frac{(1-x^2)^3\sinh 7\phi(x)}{(7!)^2}+\cdots\right)\\
&-\left(\frac{(1-x^2)\sinh 2\phi(x)}{(2!)^2}+\frac{(1-x^2)^2\sinh 4\phi(x)}{(4!)^2}+\frac{(1-x^2)^3\sinh 6\phi(x)}{(6!)^2}\right.
\end{aligned}
$$

$$+\frac{(1-x^2)^4\sinh 8\phi(x)}{(8!)^2}+\cdots\Bigg)$$
$$=\frac{1}{\pi}\int_0^{\pi} S_4(x,\theta)\,d\theta.$$

[6.11] Let the function $\phi(x)$ be defined as $\phi(x)=\tan^{-1}x$. Prove that $\forall(x,n)\in\mathbb{R}\times(\mathbb{N}-1)$,

$$\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos 2j\phi(x)}{(2j)!(n+2j)!}+\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}$$
$$=\left(\frac{1}{0!n!}+\frac{(1+x^2)\cos 2\phi(x)}{2!(n+2)!}+\frac{(1+x^2)^2\cos 4\phi(x)}{4!(n+4)!}\right.$$
$$\left.+\frac{(1+x^2)^3\cos 6\phi(x)}{6!(n+6)!}+\cdots\right)+\sqrt{1+x^2}\left(\frac{\cos\phi(x)}{1!(n+1)!}\right.$$
$$\left.+\frac{(1+x^2)\cos 3\phi(x)}{3!(n+3)!}+\frac{(1+x^2)^2\cos 5\phi(x)}{5!(n+5)!}+\frac{(1+x^2)^3\cos 7\phi(x)}{7!(n+7)!}+\cdots\right)$$
$$=\frac{2}{\pi}\cdot\frac{1+(1+x^2)^{n/2}\cos n\phi(x)}{1+(1+x^2)^n+2(1+x^2)^{n/2}\cos n\phi(x)}$$
$$\int_0^{\pi}e^{2\cos\theta}\cos(x\cos\theta)\cosh(x\sin\theta)\cos n\theta\,d\theta$$
$$+\frac{2}{\pi}\cdot\frac{(1+x^2)^{n/2}\sin n\phi(x)}{1+(1+x^2)^n+2(1+x^2)^{n/2}\cos n\phi(x)}$$
$$\int_0^{\pi}e^{2\cos\theta}\sin(x\cos\theta)\cosh(x\sin\theta)\cos n\theta\,d\theta.$$

[6.12] Let the function $\phi(x)$ be defined as $\phi(x)=\tan^{-1}x$. Prove that $\forall x\in\mathbb{R}$,

$$\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos 2j\phi(x)}{((2j)!)^2}+\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos(2j+1)\phi(x)}{((2j+1)!)^2}$$
$$=\left(\frac{1}{(0!)^2}+\frac{(1+x^2)\cos 2\phi(x)}{(2!)^2}+\frac{(1+x^2)^2\cos 4\phi(x)}{(4!)^2}\right.$$
$$\left.+\frac{(1+x^2)^3\cos 6\phi(x)}{(6!)^2}+\cdots\right)+\sqrt{1+x^2}\left(\frac{\cos\phi(x)}{(1!)^2}\right.$$
$$+\frac{(1+x^2)\cos 3\phi(x)}{(3!)^2}+\frac{(1+x^2)^2\cos 5\phi(x)}{(5!)^2}$$
$$\left.+\frac{(1+x^2)^3\cos 7\phi(x)}{(7!)^2}+\cdots\right)$$

$$= \frac{1}{\pi} \int_0^{\pi} e^{2\cos\theta} \cos(x\cos\theta) \cosh(x\sin\theta)\, d\theta.$$

[6.13] Let the function $\phi(x)$ be defined as $\phi(x) = \tan^{-1} x$. Prove that $\forall (x, n) \in \mathbb{R} \times (\mathbb{N} - 1)$,

$$\begin{aligned}
&\sum_{j=1}^{\infty} \frac{(1+x^2)^j \sin 2j\phi(x)}{(2j)!(n+2j)!} + \sqrt{1+x^2} \sum_{j=0}^{\infty} \frac{(1+x^2)^j \sin(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!} \\
&= \left( \frac{(1+x^2)\sin 2\phi(x)}{2!(n+2)!} + \frac{(1+x^2)^2 \sin 4\phi(x)}{4!(n+4)!} + \frac{(1+x^2)^3 \sin 6\phi(x)}{6!(n+6)!} \right. \\
&\quad \left. + \frac{(1+x^2)^4 \sin 8\phi(x)}{8!(n+8)!} + \cdots \right) \\
&\quad + \sqrt{1+x^2} \left( \frac{\sin\phi(x)}{1!(n+1)!} + \frac{(1+x^2)\sin 3\phi(x)}{3!(n+3)!} + \frac{(1+x^2)^2 \sin 5\phi(x)}{5!(n+5)!} \right. \\
&\quad \left. + \frac{(1+x^2)^3 \sin 7\phi(x)}{7!(n+7)!} + \cdots \right) \\
&= \frac{2}{\pi} \cdot \frac{1 + (1+x^2)^{n/2} \cos n\phi(x)}{1 + (1+x^2)^n + 2(1+x^2)^{n/2} \cos n\phi(x)} \\
&\quad \int_0^{\pi} e^{2\cos\theta} \sin(x\cos\theta) \cosh(x\sin\theta) \cos n\theta\, d\theta \\
&\quad - \frac{2}{\pi} \cdot \frac{(1+x^2)^{n/2} \sin n\phi(x)}{1 + (1+x^2)^n + 2(1+x^2)^{n/2} \cos n\phi(x)} \\
&\quad \int_0^{\pi} e^{2\cos\theta} \cos(x\cos\theta) \cosh(x\sin\theta) \cos n\theta\, d\theta.
\end{aligned}$$

[6.14] Let the function $\phi(x)$ be defined as $\phi(x) = \tan^{-1} x$. Prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned}
&\sum_{j=1}^{\infty} \frac{(1+x^2)^j \sin 2j\phi(x)}{((2j)!)^2} + \sqrt{1+x^2} \sum_{j=0}^{\infty} \frac{(1+x^2)^j \sin(2j+1)\phi(x)}{((2j+1)!)^2} \\
&= \left( \frac{(1+x^2)\sin 2\phi(x)}{(2!)^2} + \frac{(1+x^2)^2 \sin 4\phi(x)}{(4!)^2} + \frac{(1+x^2)^3 \sin 6\phi(x)}{(6!)^2} \right. \\
&\quad \left. + \frac{(1+x^2)^4 \sin 8\phi(x)}{(8!)^2} + \cdots \right) \\
&\quad + \sqrt{1+x^2} \left( \frac{\sin\phi(x)}{(1!)^2} + \frac{(1+x^2)\sin 3\phi(x)}{(3!)^2} + \frac{(1+x^2)^2 \sin 5\phi(x)}{(5!)^2} \right. \\
&\quad \left. + \frac{(1+x^2)^3 \sin 7\phi(x)}{(7!)^2} + \cdots \right) \\
&= \frac{1}{\pi} \int_0^{\pi} e^{2\cos\theta} \sin(x\cos\theta) \cosh(x\sin\theta)\, d\theta.
\end{aligned}$$

[6.15] Prove that

$$\begin{aligned}
&\sum_{j=0}^{\infty}(-1)^j\frac{2^{2j}}{((4j)!)^2}+\sum_{j=0}^{\infty}(-1)^j\left(\frac{2^{2j}}{((4j+1)!)^2}-\frac{2^{2j+1}}{((4j+3)!)^2}\right)\\
&=\left(\frac{1}{(0!)^2}-\frac{2^2}{(4!)^2}+\frac{2^4}{(8!)^2}-\frac{2^6}{(12!)^2}+\cdots\right)\\
&\quad+\left(\frac{1}{(1!)^2}-\frac{2}{(3!)^2}-\frac{2^2}{(5!)^2}+\frac{2^3}{(7!)^2}+\cdots\right)\\
&=\frac{1}{\pi}\int_0^{\pi}e^{2\cos\theta}\cos(\cos\theta)\cosh(\sin\theta)\,d\theta.
\end{aligned}$$

[6.16] Prove that

$$\begin{aligned}
&\sum_{j=0}^{\infty}(-1)^j\frac{2^{2j+1}}{((4j+2)!)^2}+\sum_{j=0}^{\infty}(-1)^j\left(\frac{2^{2j}}{((4j+1)!)^2}+\frac{2^{2j+1}}{((4j+3)!)^2}\right)\\
&=\left(\frac{2}{(2!)^2}-\frac{2^3}{(6!)^2}+\frac{2^5}{(10!)^2}-\frac{2^7}{(14!)^2}+\cdots\right)\\
&\quad+\left(\frac{1}{(1!)^2}+\frac{2}{(3!)^2}-\frac{2^2}{(5!)^2}-\frac{2^3}{(7!)^2}+\cdots\right)\\
&=\frac{1}{\pi}\int_0^{\pi}e^{2\cos\theta}\sin(\cos\theta)\cosh(\sin\theta)\,d\theta.
\end{aligned}$$

[6.17] Let the function $\phi(x)$ be defined as $\phi(x)=\tanh^{-1}x$. Prove that $\forall(x,n)\in(-1,1)\times(\mathbb{N}-1)$,

$$\begin{aligned}
&\sum_{j=0}^{\infty}\frac{(1-x^2)^j\cosh 2j\phi(x)}{(2j)!(n+2j)!}+\sqrt{1-x^2}\sum_{j=0}^{\infty}\frac{(1-x^2)^j\cosh(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}\\
&=\left(\frac{1}{0!n!}+\frac{(1-x^2)\cosh 2\phi(x)}{2!(n+2)!}+\frac{(1-x^2)^2\cosh 4\phi(x)}{4!(n+4)!}\right.\\
&\quad\left.+\frac{(1-x^2)^3\cosh 6\phi(x)}{6!(n+6)!}+\cdots\right)\\
&\quad+\sqrt{1-x^2}\left(\frac{\cosh\phi(x)}{1!(n+1)!}+\frac{(1-x^2)\cosh 3\phi(x)}{3!(n+3)!}+\frac{(1-x^2)^2\cosh 5\phi(x)}{5!(n+5)!}\right.\\
&\quad\left.+\frac{(1-x^2)^3\cosh 7\phi(x)}{7!(n+7)!}+\cdots\right)
\end{aligned}$$

$$= \frac{2}{\pi} \cdot \frac{1 + (1-x^2)^{n/2} \cosh n\phi(x)}{1 + (1-x^2)^n + 2(1-x^2)^{n/2} \cosh n\phi(x)} \int_0^\pi e^{2\cos\theta} \cosh(x\cos\theta) \cos(x\sin\theta) \cos n\theta \, d\theta - \frac{2}{\pi} \cdot \frac{(1-x^2)^{n/2} \sinh n\phi(x)}{1 + (1-x^2)^n + 2(1-x^2)^{n/2} \cosh n\phi(x)} \int_0^\pi e^{2\cos\theta} \sinh(x\cos\theta) \cos(x\sin\theta) \cos n\theta \, d\theta.$$

[6.18] Let the function $\phi(x)$ be defined as $\phi(x) = \tanh^{-1} x$. Prove that $\forall x \in (-1, 1)$,

$$\begin{aligned}
&\sum_{j=0}^{\infty} \frac{(1-x^2)^j \cosh 2j\phi(x)}{((2j)!)^2} + \sqrt{1-x^2} \sum_{j=0}^{\infty} \frac{(1-x^2)^j \cosh(2j+1)\phi(x)}{((2j+1)!)^2} \\
&= \left( \frac{1}{(0!)^2} + \frac{(1-x^2)\cosh 2\phi(x)}{(2!)^2} + \frac{(1-x^2)^2 \cosh 4\phi(x)}{(4!)^2} \right. \\
&\quad \left. + \frac{(1-x^2)^3 \cosh 6\phi(x)}{(6!)^2} + \cdots \right) \\
&\quad + \sqrt{1-x^2} \left( \frac{\cosh\phi(x)}{(1!)^2} + \frac{(1-x^2)\cosh 3\phi(x)}{(3!)^2} + \frac{(1-x^2)^2 \cosh 5\phi(x)}{(5!)^2} \right. \\
&\quad \left. + \frac{(1-x^2)^3 \cosh 7\phi(x)}{(7!)^2} + \cdots \right) \\
&= \frac{1}{\pi} \int_0^\pi e^{2\cos\theta} \cosh(x\cos\theta) \cos(x\sin\theta) \, d\theta.
\end{aligned}$$

[6.19] Let the function $\phi(x)$ be defined as $\phi(x) = \tanh^{-1} x$. Prove that $\forall (x, n) \in (-1, 1) \times (\mathbb{N} - 1)$,

$$\begin{aligned}
&\sum_{j=1}^{\infty} \frac{(1-x^2)^j \sinh 2j\phi(x)}{(2j)!(n+2j)!} + \sqrt{1-x^2} \sum_{j=0}^{\infty} \frac{(1-x^2)^j \sinh(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!} \\
&= \left( \frac{(1-x^2)\sinh 2\phi(x)}{2!(n+2)!} + \frac{(1-x^2)^2 \sinh 4\phi(x)}{4!(n+4)!} + \frac{(1-x^2)^3 \sinh 6\phi(x)}{6!(n+6)!} \right. \\
&\quad \left. + \frac{(1-x^2)^4 \sinh 8\phi(x)}{8!(n+8)!} + \cdots \right) \\
&\quad + \sqrt{1-x^2} \left( \frac{\sinh\phi(x)}{1!(n+1)!} + \frac{(1-x^2)\sinh 3\phi(x)}{3!(n+3)!} + \frac{(1-x^2)^2 \sinh 5\phi(x)}{5!(n+5)!} \right. \\
&\quad \left. + \frac{(1-x^2)^3 \sinh 7\phi(x)}{7!(n+7)!} + \cdots \right)
\end{aligned}$$

$$
\begin{aligned}
&= \frac{2}{\pi} \cdot \frac{1 + (1-x^2)^{n/2} \cosh n\phi(x)}{1 + (1-x^2)^n + 2(1-x^2)^{n/2} \cosh n\phi(x)} \\
&\int_0^\pi e^{2\cos\theta} \sinh(x\cos\theta) \cos(x\sin\theta) \cos n\theta \, d\theta \\
&- \frac{2}{\pi} \cdot \frac{(1-x^2)^{n/2} \sinh n\phi(x)}{1 + (1-x^2)^n + 2(1-x^2)^{n/2} \cosh n\phi(x)} \\
&\int_0^\pi e^{2\cos\theta} \cosh(x\cos\theta) \cos(x\sin\theta) \cos n\theta \, d\theta.
\end{aligned}
$$

[6.20] Let the function $\phi(x)$ be defined as $\phi(x) = \tanh^{-1} x$. Prove that $\forall x \in (-1, 1)$,

$$
\begin{aligned}
&\sum_{j=1}^{\infty} \frac{(1-x^2)^j \sinh 2j\phi(x)}{((2j)!)^2} + \sqrt{1-x^2} \sum_{j=0}^{\infty} \frac{(1-x^2)^j \sinh(2j+1)\phi(x)}{((2j+1)!)^2} \\
&= \left( \frac{(1-x^2)\sinh 2\phi(x)}{(2!)^2} + \frac{(1-x^2)^2 \sinh 4\phi(x)}{(4!)^2} + \frac{(1-x^2)^3 \sinh 6\phi(x)}{(6!)^2} \right. \\
&\quad \left. + \frac{(1-x^2)^4 \sinh 8\phi(x)}{(8!)^2} + \cdots \right) \\
&\quad + \sqrt{1-x^2} \left( \frac{\sinh\phi(x)}{(1!)^2} + \frac{(1-x^2)\sinh 3\phi(x)}{(3!)^2} + \frac{(1-x^2)^2 \sinh 5\phi(x)}{(5!)^2} \right. \\
&\quad \left. + \frac{(1-x^2)^3 \sinh 7\phi(x)}{(7!)^2} + \cdots \right) \\
&= \frac{1}{\pi} \int_0^\pi e^{2\cos\theta} \sinh(x\cos\theta) \cos(x\sin\theta) \, d\theta.
\end{aligned}
$$

[6.21] Let the function $C_5(x, \phi, \theta)$ be defined as

$$
\begin{aligned}
&C_5(x, \phi, \theta) \\
&= \cosh(x\sin\phi\sin\theta) \cos(x\sin\phi\cos\theta) \cos((x\cos\phi - 1)\sin\theta) \\
&\quad + \sinh(x\sin\phi\sin\theta) \sin(x\sin\phi\cos\theta) \sin((x\cos\phi - 1)\sin\theta),
\end{aligned} \tag{6.32}
$$

and the function $S_5(x, \phi, \theta)$ be defined as

$$
\begin{aligned}
&S_5(x, \phi, \theta) \\
&= \cosh(x\sin\phi\sin\theta) \sin(x\sin\phi\cos\theta) \cos((x\cos\phi - 1)\sin\theta) \\
&\quad - \sinh(x\sin\phi\sin\theta) \cos(x\sin\phi\cos\theta) \sin((x\cos\phi - 1)\sin\theta).
\end{aligned} \tag{6.33}
$$

Prove that $\forall (x, \phi, n) \in \mathbb{R}^2 \times (\mathbb{N} - 1)$,

$$\sum_{j=0}^{\infty} \frac{x^j \cos j\phi}{j!(n+j)!} = \frac{1}{0!n!} + \frac{x\cos\phi}{1!(n+1)!} + \frac{x^2\cos 2\phi}{2!(n+2)!} + \frac{x^3\cos 3\phi}{3!(n+3)!} + \cdots$$
$$= \frac{2}{\pi}\cdot\frac{1+x^n\cos n\phi}{1+x^{2n}+2x^n\cos n\phi}\int_0^{\pi} C_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta$$
$$+ \frac{2}{\pi}\cdot\frac{x^n\sin n\phi}{1+x^{2n}+2x^n\cos n\phi}\int_0^{\pi} S_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta.$$

[6.22] Let $C_5(x,\phi,\theta)$ be given by (6.32). Prove that $\forall(x,\phi)\in\mathbb{R}^2$,

$$\sum_{j=0}^{\infty} \frac{x^j \cos j\phi}{(j!)^2} = \frac{1}{(0!)^2} + \frac{x\cos\phi}{(1!)^2} + \frac{x^2\cos 2\phi}{(2!)^2} + \frac{x^3\cos 3\phi}{(3!)^2} + \cdots$$
$$= \frac{1}{\pi}\int_0^{\pi} C_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\, d\theta.$$

[6.23] Let $S_5(x,\phi,\theta)$ be given by (6.33) and $C_5(x,\phi,\theta)$ be given by (6.32). Prove that $\forall(x,\phi,n)\in\mathbb{R}^2\times(\mathbb{N}-1)$,

$$\sum_{j=1}^{\infty} \frac{x^j \sin j\phi}{j!(n+j)!} = \frac{x\sin\phi}{1!(n+1)!} + \frac{x^2\sin 2\phi}{2!(n+2)!} + \frac{x^3\sin 3\phi}{3!(n+3)!} + \frac{x^4\sin 4\phi}{4!(n+4)!} + \cdots$$
$$= \frac{2}{\pi}\cdot\frac{1+x^n\cos n\phi}{1+x^{2n}+2x^n\cos n\phi}\int_0^{\pi} S_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta$$
$$- \frac{2}{\pi}\cdot\frac{x^n\sin n\phi}{1+x^{2n}+2x^n\cos n\phi}\int_0^{\pi} C_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta.$$

[6.24] Let $S_5(x,\phi,\theta)$ be given by (6.33). Prove that $\forall(x,\phi)\in\mathbb{R}^2$,

$$\sum_{j=1}^{\infty} \frac{x^j \sin j\phi}{(j!)^2} = \frac{x\sin\phi}{(1!)^2} + \frac{x^2\sin 2\phi}{(2!)^2} + \frac{x^3\sin 3\phi}{(3!)^2} + \frac{x^4\sin 4\phi}{(4!)^2} + \cdots$$
$$= \frac{1}{\pi}\int_0^{\pi} S_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\, d\theta.$$

[6.25] Let the function $C_6(x,\phi,\theta)$ be defined as

$$\begin{aligned} &C_6(x,\phi,\theta)\\ &= \cos(x\sinh\phi\sin\theta)\cosh(x\sinh\phi\cos\theta)\cos((x\cosh\phi-1)\sin\theta)\\ &\quad - \sin(x\sinh\phi\sin\theta)\sinh(x\sinh\phi\cos\theta)\sin((x\cosh\phi-1)\sin\theta), \end{aligned} \tag{6.34}$$

and the function $S_6(x,\phi,\theta)$ be defined as

$$
\begin{aligned}
&S_6(x,\phi,\theta)\\
&=\cos\left(x\sinh\phi\sin\theta\right)\sinh\left(x\sinh\phi\cos\theta\right)\cos\left((x\cosh\phi-1)\sin\theta\right)\\
&\quad-\sin\left(x\sinh\phi\sin\theta\right)\cosh\left(x\sinh\phi\cos\theta\right)\sin\left((x\cosh\phi-1)\sin\theta\right).
\end{aligned}
\tag{6.35}
$$

Prove that $\forall(x,\phi,n)\in\mathbb{R}^2\times(\mathbb{N}-1)$,

$$
\begin{aligned}
&\sum_{j=0}^{\infty}\frac{x^j\cosh j\phi}{j!(n+j)!}=\frac{1}{0!n!}+\frac{x\cosh\phi}{1!(n+1)!}+\frac{x^2\cosh 2\phi}{2!(n+2)!}+\frac{x^3\cosh 3\phi}{3!(n+3)!}+\cdots\\
&=\frac{2}{\pi}\cdot\frac{1+x^n\cosh n\phi}{1+x^{2n}+2x^n\cosh n\phi}\int_0^{\pi}C_6(x,\phi,\theta)e^{(1+x\cosh\phi)\cos\theta}\cos n\theta\,d\theta\\
&-\frac{2}{\pi}\cdot\frac{x^n\sinh n\phi}{1+x^{2n}+2x^n\cosh n\phi}\int_0^{\pi}S_6(x,\phi,\theta)e^{(1+x\cosh\phi)\cos\theta}\cos n\theta\,d\theta.
\end{aligned}
$$

[6.26] Let $C_6(x,\phi,\theta)$ be given by (6.34). Prove that $\forall(x,\phi)\in\mathbb{R}^2$,

$$
\begin{aligned}
\sum_{j=0}^{\infty}\frac{x^j\cosh j\phi}{(j!)^2}&=\frac{1}{(0!)^2}+\frac{x\cosh\phi}{(1!)^2}+\frac{x^2\cosh 2\phi}{(2!)^2}+\frac{x^3\cosh 3\phi}{(3!)^2}\\
&=\frac{1}{\pi}\int_0^{\pi}C_6(x,\phi,\theta)e^{(1+x\cosh\phi)\cos\theta}\,d\theta.
\end{aligned}
$$

[6.27] Let $S_6(x,\phi,\theta)$ be given by (6.35) and $C_6(x,\phi,\theta)$ be given by (6.34). Prove that $\forall(x,\phi,n)\in\mathbb{R}^2\times(\mathbb{N}-1)$,

$$
\begin{aligned}
&\sum_{j=1}^{\infty}\frac{x^j\sinh j\phi}{j!(n+j)!}=\frac{x\sinh\phi}{1!(n+1)!}+\frac{x^2\sinh 2\phi}{2!(n+2)!}+\frac{x^3\sinh 3\phi}{3!(n+3)!}+\frac{x^4\sinh 4\phi}{4!(n+4)!}+\cdots\\
&=\frac{2}{\pi}\cdot\frac{1+x^n\cosh n\phi}{1+x^{2n}+2x^n\cosh n\phi}\int_0^{\pi}S_6(x,\phi,\theta)e^{(1+x\cosh\phi)\cos\theta}\cos n\theta\,d\theta\\
&-\frac{2}{\pi}\cdot\frac{x^n\sinh n\phi}{1+x^{2n}+2x^n\cosh n\phi}\int_0^{\pi}C_6(x,\phi,\theta)e^{(1+x\cosh\phi)\cos\theta}\cos n\theta\,d\theta.
\end{aligned}
$$

[6.28] Let $S_6(x,\phi,\theta)$ is given by (6.35). Prove that $\forall(x,\phi)\in\mathbb{R}^2$,

$$
\begin{aligned}
\sum_{j=1}^{\infty}\frac{x^j\sinh j\phi}{(j!)^2}&=\frac{x\sinh\phi}{(1!)^2}+\frac{x^2\sinh 2\phi}{(2!)^2}+\frac{x^3\sinh 3\phi}{(3!)^2}+\frac{x^4\sinh 4\phi}{(4!)^2}+\cdots\\
&=\frac{1}{\pi}\int_0^{\pi}S_6(x,\phi,\theta)e^{(1+x\cosh\phi)\cos\theta}\,d\theta.
\end{aligned}
$$

[6.29] Prove that $\forall \phi \in \mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\sum_{j=0}^{\infty} \frac{\cos j\phi}{(j!)^2 \cos^j \phi} = \frac{1}{(0!)^2} + \frac{\cos\phi}{(1!)^2\cos\phi} + \frac{\cos 2\phi}{(2!)^2\cos^2\phi} + \frac{\cos 3\phi}{(3!)^2\cos^3\phi} + \cdots$$
$$= \frac{1}{\pi}\int_0^{\pi} \cosh(\tan\phi\sin\theta)\cos(\tan\phi\cos\theta)\, e^{2\cos\theta}\, d\theta.$$

[6.30] Prove that $\forall \phi \in \mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\sum_{j=0}^{\infty}(-1)^j \frac{\cos j\phi}{(j!)^2\cos^j\phi} = \frac{1}{(0!)^2} - \frac{\cos\phi}{(1!)^2\cos\phi} + \frac{\cos 2\phi}{(2!)^2\cos^2\phi}$$
$$- \frac{\cos 3\phi}{(3!)^2\cos^3\phi} + \cdots$$
$$= \frac{1}{\pi}\int_0^{\pi} (\cosh(\tan\phi\sin\theta)\cos(\tan\phi\cos\theta)\cos(2\sin\theta)$$
$$- \sinh(\tan\phi\sin\theta)\sin(\tan\phi\cos\theta)\sin(2\sin\theta))\, d\theta.$$

[6.31] Prove that $\forall \phi \in \mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\sum_{j=1}^{\infty} \frac{\sin j\phi}{(j!)^2\cos^j\phi} = \frac{\sin\phi}{(1!)^2\cos\phi} + \frac{\sin 2\phi}{(2!)^2\cos^2\phi} + \frac{\sin 3\phi}{(3!)^2\cos^3\phi}$$
$$+ \frac{\sin 4\phi}{(4!)^2\cos^4\phi} + \cdots$$
$$= \frac{1}{\pi}\int_0^{\pi} \cosh(\tan\phi\sin\theta)\sin(\tan\phi\cos\theta)\, e^{2\cos\theta}\, d\theta.$$

[6.32] Prove that $\forall \phi \in \mathbb{R} - (2\mathbb{Z}+1)\frac{\pi}{2}$,

$$\sum_{j=1}^{\infty}(-1)^{j-1}\frac{\sin j\phi}{(j!)^2\cos^j\phi}$$
$$= \frac{\sin\phi}{(1!)^2\cos\phi} - \frac{\sin 2\phi}{(2!)^2\cos^2\phi} + \frac{\sin 3\phi}{(3!)^2\cos^3\phi} - \frac{\sin 4\phi}{(4!)^2\cos^4\phi} + \cdots$$
$$= \frac{1}{\pi}\int_0^{\pi} (\cosh(\tan\phi\sin\theta)\sin(\tan\phi\cos\theta)\cos(2\sin\theta)$$
$$+ \sinh(\tan\phi\sin\theta)\cos(\tan\phi\cos\theta)\sin(2\sin\theta))\, d\theta.$$

[6.33] Prove that $\forall \phi \in \mathbb{R}$,

$$\sum_{j=0}^{\infty} \frac{\cosh j\phi}{(j!)^2 \cosh^j \phi} = \frac{1}{(0!)^2} + \frac{\cosh \phi}{(1!)^2 \cosh \phi} + \frac{\cosh 2\phi}{(2!)^2 \cosh^2 \phi} + \frac{\cosh 3\phi}{(3!)^2 \cosh^3 \phi} + \cdots$$
$$= \frac{1}{\pi} \int_0^{\pi} \cos(\tanh \phi \sin \theta) \cosh(\tanh \phi \cos \theta)\, e^{2\cos\theta}\, d\theta.$$

[6.34] Prove that $\forall \phi \in \mathbb{R}$,

$$\sum_{j=0}^{\infty} (-1)^j \frac{\cosh j\phi}{(j!)^2 \cosh^j \phi} = \frac{1}{(0!)^2} - \frac{\cosh \phi}{(1!)^2 \cosh \phi} + \frac{\cosh 2\phi}{(2!)^2 \cosh^2 \phi} - \frac{\cosh 3\phi}{(3!)^2 \cosh^3 \phi} + \cdots$$
$$= \frac{1}{\pi} \int_0^{\pi} (\cos(\tanh \phi \sin \theta) \cosh(\tanh \phi \cos \theta) \cos(2 \sin \theta) + \sin(\tanh \phi \sin \theta) \sinh(\tanh \phi \cos \theta) \sin(2 \sin \theta))\, d\theta.$$

[6.35] Prove that $\forall \phi \in \mathbb{R}$,

$$\sum_{j=1}^{\infty} \frac{\sinh j\phi}{(j!)^2 \cosh^j \phi} = \frac{\sinh \phi}{(1!)^2 \cosh \phi} + \frac{\sinh 2\phi}{(2!)^2 \cosh^2 \phi} + \frac{\sinh 3\phi}{(3!)^2 \cosh^3 \phi} + \frac{\sinh 4\phi}{(4!)^2 \cosh^4 \phi} + \cdots$$
$$= \frac{1}{\pi} \int_0^{\pi} \cos(\tanh \phi \sin \theta) \sinh(\tanh \phi \cos \theta)\, e^{2\cos\theta}\, d\theta.$$

[6.36] Prove that $\forall \phi \in \mathbb{R}$,

$$\sum_{j=1}^{\infty} (-1)^{j-1} \frac{\sinh j\phi}{(j!)^2 \cosh^j \phi}$$
$$= \frac{\sinh \phi}{(1!)^2 \cosh \phi} - \frac{\sinh 2\phi}{(2!)^2 \cosh^2 \phi} + \frac{\sinh 3\phi}{(3!)^2 \cosh^3 \phi} - \frac{\sinh 4\phi}{(4!)^2 \cosh^4 \phi} + \cdots$$
$$= \frac{1}{\pi} \int_0^{\pi} (\cos(\tanh \phi \sin \theta) \sinh(\tanh \phi \cos \theta) \cos(2 \sin \theta) + \sin(\tanh \phi \sin \theta) \cosh(\tanh \phi \cos \theta) \sin(2 \sin \theta))\, d\theta.$$

[6.37] Assuming the convergence conditions of Theorem 6.13, prove that $\forall x \in \mathbb{R}$,

$$\begin{aligned} f(1) \star e^x &= \sum_{j=0}^{\infty} \frac{a_j}{j!} x^j \\ &= a_0 + a_1 x + \frac{a_2}{2!} x^2 + \frac{a_3}{3!} x^3 + \cdots \\ &= -f(0) + \frac{1}{\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \cos(x \sin\theta) e^{x \cos\theta} \, d\theta. \end{aligned}$$

[6.38] Assuming the convergence conditions of Theorem 6.13, prove that $\forall x \in (-1, 1)$,

$$\begin{aligned} f(1) \star \frac{1}{1-x} &= \sum_{j=0}^{\infty} a_j x^j \\ &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots \\ &= f(x) \\ &= -f(0) + \frac{1}{\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta} \, d\theta. \end{aligned}$$

[6.39] Assuming the convergence conditions of Theorem 6.13, prove that $\forall x \in (-1, 1)$,

$$\begin{aligned} f(1) \star (-\log(1-x)) &= \sum_{j=1}^{\infty} \frac{a_j}{j} x^j \\ &= a_1 x + \frac{a_2}{2} x^2 + \frac{a_3}{3} x^3 + \frac{a_4}{4} x^4 + \cdots \\ &= \int_0^x \frac{f(x) - f(0)}{x} \, dx \\ &= -\frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \log(1 + x^2 - 2x\cos\theta) \, d\theta. \end{aligned}$$

[6.40] Assuming the convergence conditions of Theorem 6.13, prove that $\forall x \in (-1, 1)$,

$$\begin{aligned} f(1) \star \frac{x}{(1-x)^2} &= \sum_{j=1}^{\infty} j a_j x^j \\ &= a_1 x + 2a_2 x^2 + 3a_3 x^3 + 4a_4 x^4 + \cdots \\ &= x f'(x) \\ &= \frac{1}{\pi} \int_0^{\pi} x \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \frac{(1 + x^2)\cos\theta - 2x}{(1 + x^2 - 2x\cos\theta)^2} \, d\theta. \end{aligned}$$

[6.41] Prove that $\forall x \in \mathbb{R}$,

$$e(x,0) = -1 + \frac{2}{\pi}\int_0^\pi e^{(x+1)\cos\theta}\cos(\sin\theta)\cos(x\sin\theta)\,d\theta.$$

[6.42] Prove that $\forall x \in \mathbb{R}$,

$$e(x,0) = 1 + \frac{2}{\pi}\int_0^\pi e^{(x+1)\cos\theta}\sin(\sin\theta)\sin(x\sin\theta)\,d\theta.$$

[6.43] Prove that $\forall x \in \mathbb{R}$,

$$\int_0^\pi e^{(x+1)\cos\theta}\cos\left((x+1)\sin\theta\right)\,d\theta = \pi.$$

[6.44] Assuming that $f(z)$ is convergent for $|z| = 1$, prove that

$$\begin{aligned}\sum_{j=0}^{\infty} a_j^2 &= a_0^2 + a_1^2 + a_2^2 + a_3^2 + \cdots \\ &= -f(0)^2 + \frac{1}{2\pi}\int_0^\pi \left(f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)\right)^2\,d\theta.\end{aligned}$$

[6.45] Prove that $\forall (x, y) \in \mathbb{R} \times (-1, 1)$,

$$\int_0^\pi \frac{1 - y\cos\theta}{1 + y^2 - 2y\cos\theta}e^{x\cos\theta}\cos\left(x\sin\theta\right)\,d\theta = \frac{\pi}{2}(e^{xy} + 1).$$

[6.46] Prove that if $f(0) = 0$ and $x \in (-1, 1)$, then

$$\int_0^x \frac{f(x)}{x}\,dx = -\frac{1}{2\pi}\int_0^\pi \left(f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right)\right)\log(1 + x^2 - 2x\cos\theta)\,d\theta.$$

[6.47] Prove that $\forall x \in (-1, 1)$,

$$\int_0^x \frac{\log(1+x)}{x}\,dx = -\frac{1}{\pi}\int_0^\pi \log\left(2\cos\frac{\theta}{2}\right)\log(1 + x^2 - 2x\cos\theta)\,d\theta.$$

[6.48] Prove that $\forall x \in (-1, 1)$,

$$\int_0^x \frac{\sin x}{x}\,dx = -\frac{1}{\pi}\int_0^\pi \sin(\cos\theta)\cosh(\sin\theta)\log(1 + x^2 - 2x\cos\theta)\,d\theta.$$

[6.49] Prove that $\forall x \in (-1, 1)$,

$$\int_0^x \frac{\sinh x}{x}\, dx = -\frac{1}{\pi} \int_0^\pi \sinh(\cos\theta) \cos(\sin\theta) \log(1 + x^2 - 2x\cos\theta)\, d\theta.$$

[6.50] Prove that $\forall (x, y) \in (-1, 1)^2$,

$$\log(1 - xy) = \frac{1}{\pi} \int_0^\pi \frac{1 - y\cos\theta}{1 + y^2 - 2y\cos\theta} \log(1 + x^2 - 2x\cos\theta)\, d\theta.$$

[6.51] Prove that $\forall x \in (-1, 1)$,

$$\log(1 + x) = \frac{2}{\pi} \int_0^\pi \log\left(2\cos\frac{\theta}{2}\right) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta}\, d\theta.$$

[6.52] Prove that $\forall x \in (-1, 1)$,

$$\sin x = \frac{2}{\pi} \int_0^\pi \sin(\cos\theta) \cosh(\sin\theta) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta}\, d\theta.$$

[6.53] Prove that $\forall x \in (-1, 1)$,

$$\sinh x = \frac{2}{\pi} \int_0^\pi \sinh(\cos\theta) \cos(\sin\theta) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta}\, d\theta.$$

[6.54] Prove that $\forall (x, y) \in (-1, 1)^2$,

$$\int_0^\pi \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta} \cdot \frac{1 - y\cos\theta}{1 + y^2 - 2y\cos\theta}\, d\theta = \frac{\pi}{2} \cdot \frac{2 - xy}{1 - xy}.$$

[6.55] Prove that $\forall (x, y) \in (-1, 1)^2$,

$$\int_0^\pi x \log\left(1 + y^2 - 2y\cos\theta\right) \frac{(1 + x^2)\cos\theta - 2x}{(1 + x^2 - 2x\cos\theta)^2}\, d\theta = -\frac{\pi xy}{1 - xy}.$$

[6.56] Prove that $\forall x \in (-1, 1)$,

$$\begin{aligned} 2a_0 + \sum_{j=1}^n a_j x^j &= 2a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \\ &= \frac{2}{\pi} \int_0^\pi \left(\sum_{j=0}^n a_j \cos j\theta\right) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta}\, d\theta. \end{aligned}$$

[6.57] Prove that

$$\sum_{j=0}^{\infty} \frac{1}{2^{8j}} \binom{2j}{j}^2 = 1 + \frac{\binom{2}{1}^2}{2^8} + \frac{\binom{4}{2}^2}{2^{16}} + \frac{\binom{6}{3}^2}{2^{24}} + \cdots$$

$$= -\frac{1}{2} + \frac{1}{2\pi} \int_0^{\pi} \frac{15 + 8\sqrt{9 + 16\sin^2 \dfrac{\theta}{2}}}{9 + 16\sin^2 \dfrac{\theta}{2}}\, d\theta.$$

[6.58] Prove that

$$\int_0^{\pi} \cos^2(\sin\theta) e^{2\cos\theta}\, d\theta = \frac{\pi}{2}(e(1,0) + 1).$$

[6.59] Prove that

$$\int_0^{\pi} \sin^2(\sin\theta) e^{2\cos\theta}\, d\theta = \frac{\pi}{2}(e(1,0) - 1).$$

# Part II
# Solutions

# Chapter 7
# Solutions to Binomial Series




[1.1] Put $\theta = \pi/6$ in [2.14].

[1.2] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.40].

[1.3] Put $\theta = \pi/6$ in [2.9].

[1.4] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.35].

[1.5] Put $\theta = \pi/6$ in [2.10].

[1.6] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.36].

[1.7] Put $\theta = \pi/6$ in [2.12].

[1.8] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.38].

[1.9] Put $\theta = \pi/6$ in [2.19].

[1.10] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.45].

[1.11] Put $\theta = \pi/6$ in [2.18].

[1.12] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.44].

[1.13] Put $\theta = \pi/6$ in [2.13].

[1.14] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.39].

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[1.15] Put $\theta = \pi/6$ in [2.8].

[1.16] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.34].

[1.17] Put $\theta = \pi/6$ in [2.11].

[1.18] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.37].

[1.19] Put $\theta = \pi/6$ in [2.7].

[1.20] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.33].

[1.21] Put $\theta = \pi/6$ in [2.6].

[1.22] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.32].

[1.23] Put $\theta = \pi/6$ in [2.15].

[1.24] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.41].

[1.25] Put $\theta = \pi/6$ in [2.16].

[1.26] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.42].

[1.27] Put $\theta = \pi/6$ in [2.17].

[1.28] Put $\theta = \log\left(\left(\sqrt{5}+1\right)/2\right)$ in [2.43].

[1.29] Put $x = 6$ in [2.5].

[1.30] Put $x = 1$ in [2.31].

[1.31] Put $x = \pi$ in [2.31].

[1.32] Put $x = \pi$ in [2.5].

[1.33] We use the *partial fractions* method to simplify the sum as follows. Let

$$\begin{aligned}
\frac{4j+1}{\binom{2j+2}{4}} &= \frac{4!(4j+1)}{(2j+2)(2j+1)(2j)(2j-1)} \\
&= \frac{A}{2j+2} + \frac{B}{2j+1} + \frac{C}{2j} + \frac{D}{2j-1} \qquad (7.1) \\
\implies \quad 96j+24 &= A(2j+1)(2j)(2j-1) + B(2j+2)(2j)(2j-1)
\end{aligned}$$

$$+C(2j+2)(2j+1)(2j-1)+D(2j+2)(2j+1)(2j).$$

Equating the constant term in (7.1), we get

$$-2C=24 \implies C=-12. \tag{7.2}$$

Equating the coefficient of $j$ in (7.1) and using (7.2), we get

$$\begin{aligned} &-2A-4B-2C+4D=96\\ \implies &-A-2B+2D=36. \end{aligned} \tag{7.3}$$

Equating the coefficient of $j^2$ in (7.1) and using (7.2), we get

$$\begin{aligned} &4B+8C+12D=0\\ \implies &B=24-3D. \end{aligned} \tag{7.4}$$

Equating the coefficient of $j^3$ in (7.1) and using (7.2), we get

$$\begin{aligned} &8A+8B+8C+8D=0\\ \implies &A+B+D=12. \end{aligned} \tag{7.5}$$

Putting the value of $B$ from (7.4) into (7.3), we get

$$\begin{aligned} &-A-48+6D+2D=36\\ \implies &-A+8D=84. \end{aligned} \tag{7.6}$$

Putting the value of $B$ from (7.4) into (7.5), we get

$$\begin{aligned} &A+24-3D+D=12\\ \implies &A-2D=-12. \end{aligned} \tag{7.7}$$

Adding (7.6) and (7.7), we get

$$6D=72 \implies D=12. \tag{7.8}$$

Putting the value of $D$ from (7.8) into (7.7), we get

$$A=2D-12=12. \tag{7.9}$$

Putting the value of $D$ from (7.8) into (7.4), we get

$$B=24-3D=-12. \tag{7.10}$$

Using (7.1), (7.9), (7.10), (7.2), and (7.8), we get

$$\begin{aligned}\sum_{j=1}^{\infty}\frac{4j+1}{\binom{2j+2}{4}} &= \sum_{j=1}^{\infty}\left(\frac{12}{2j+2}-\frac{12}{2j+1}-\frac{12}{2j}+\frac{12}{2j-1}\right)\\ &= 12\left(\sum_{j=1}^{\infty}\frac{1}{2j+2}-\sum_{j=1}^{\infty}\frac{1}{2j}+\sum_{j=1}^{\infty}\frac{1}{2j-1}-\sum_{j=1}^{\infty}\frac{1}{2j+1}\right)\\ &= 12\left(\sum_{j=2}^{\infty}\frac{1}{2j}-\sum_{j=1}^{\infty}\frac{1}{2j}+\sum_{j=1}^{\infty}\frac{1}{2j-1}-\sum_{j=2}^{\infty}\frac{1}{2j-1}\right)\\ &= 12\left(-\frac{1}{2}+1\right)\\ &= 6.\end{aligned}$$

[1.34] Using the binomial theorem, we have

$$(1-x)^n=\sum_{j=0}^{n}(-1)^j\binom{n}{j}x^j. \tag{7.11}$$

Integrating (7.11) with respect to $x$ from 0 to 1, we get

$$\begin{aligned}&-\frac{(1-x)^{n+1}}{n+1}\Big|_0^1=\sum_{j=0}^{n}(-1)^j\binom{n}{j}\frac{x^{j+1}}{j+1}\Big|_0^1\\ \iff &\frac{1}{n+1}=\sum_{j=0}^{n}\frac{(-1)^j}{j+1}\binom{n}{j}.\end{aligned} \tag{7.12}$$

[1.35] Integrating (7.11) with respect to $x$ from 0 to $x$, and using the binomial theorem, we get

$$\begin{aligned}&-\frac{(1-x)^{n+1}}{n+1}\Big|_0^x=\sum_{j=0}^{n}(-1)^j\binom{n}{j}\frac{x^{j+1}}{j+1}\Big|_0^x\\ \iff &\frac{1}{n+1}-\frac{(1-x)^{n+1}}{n+1}=\sum_{j=0}^{n}\frac{(-1)^jx^{j+1}}{j+1}\binom{n}{j}\\ \iff &\frac{1}{n+1}\sum_{j=1}^{n+1}(-1)^{j-1}\binom{n+1}{j}x^{j-1}=\sum_{j=0}^{n}\frac{(-1)^jx^j}{j+1}\binom{n}{j}.\end{aligned} \tag{7.13}$$

Integrating (7.13) with respect to $x$ from 0 to 1, we get

$$\begin{aligned}&\frac{1}{n+1}\sum_{j=1}^{n+1}\frac{(-1)^{j-1}}{j}\binom{n+1}{j}x^j\Big|_0^1=\sum_{j=0}^{n}\frac{(-1)^j}{(j+1)^2}\binom{n}{j}x^{j+1}\Big|_0^1\\ \Longleftrightarrow\ &\frac{1}{n+1}\sum_{j=1}^{n+1}\frac{(-1)^{j-1}}{j}\binom{n+1}{j}=\sum_{j=0}^{n}\frac{(-1)^j}{(j+1)^2}\binom{n}{j}.\end{aligned}\tag{7.14}$$

Using (1.11), (7.12), and (7.14), we get

$$\begin{aligned}&\frac{1}{n+1}\sum_{j=1}^{n+1}\frac{1}{j}=\sum_{j=0}^{n}\frac{(-1)^j}{(j+1)^2}\binom{n}{j}\\ \Longleftrightarrow\ &\frac{1}{n+1}+\frac{1}{n+1}\sum_{j=1}^{n}\frac{1}{j+1}=\sum_{j=0}^{n}\frac{(-1)^j}{(j+1)^2}\binom{n}{j}\\ \Longleftrightarrow\ &\frac{1}{n+1}\sum_{j=1}^{n}\frac{1}{j+1}=\sum_{j=0}^{n}\frac{(-1)^j}{(j+1)^2}\binom{n}{j}-\sum_{j=0}^{n}\frac{(-1)^j}{j+1}\binom{n}{j}\\ &=\sum_{j=1}^{n}(-1)^{j-1}\frac{j}{(j+1)^2}\binom{n}{j}.\end{aligned}$$

[1.36] Differentiating (7.11) with respect to $x$, we get

$$\begin{aligned}-n(1-x)^{n-1}&=\sum_{j=1}^{n}(-1)^j j\binom{n}{j}x^{j-1}\\ \Longleftrightarrow\ n(1-x)^{n-1}&=\sum_{j=1}^{n}(-1)^{j-1}j\binom{n}{j}x^{j-1}.\end{aligned}\tag{7.15}$$

Putting $x=1$ in (7.15), we get

$$0=\sum_{j=1}^{n}(-1)^{j-1}j\binom{n}{j}.$$

[1.37] Putting $x = 2$ in [1.70], we get

$$\begin{aligned}\sum_{j=0}^{n} \frac{(-1)^j}{2j+1}\binom{n}{j} &= \frac{n!2^n}{\prod_{j=0}^{n}(1+2j)} \\ &= \frac{2^n n!}{2n+1}\prod_{j=0}^{n-1}\frac{1}{2j+1} \\ &= \frac{2^{2n}(n!)^2}{2n+1}\prod_{j=0}^{n-1}\frac{1}{(2j+1)(2j+2)} \\ &= \frac{2^{2n}(n!)^2}{(2n+1)(2n)!} \\ &= \frac{2^{2n}}{(2n+1)}\binom{2n}{n}^{-1}.\end{aligned}$$

[1.38] Put $\theta = 0$ in (2.16).

[1.39] Find the limit as $\theta \to 0$ in (2.12).

[1.40] Find the limit as $\theta \to 0$ in [2.1].

[1.41] Put $\theta = 0$ in (2.19).

[1.42] Find the limit as $\theta \to 0$ in (2.15).

[1.43] Find the limit as $\theta \to 0$ in [2.2].

[1.44] Using the geometric series, $\forall x \in \mathbb{R} - [-1, 1]$, we have

$$\begin{aligned}&\frac{1}{1-\frac{1}{x}} = \sum_{j=0}^{\infty}\frac{1}{x^j} \\ \Longleftrightarrow\ &\sum_{j=0}^{\infty}\frac{1}{x^j} = \frac{x}{x-1} \\ &= 1 + \frac{1}{x-1}.\end{aligned} \tag{7.16}$$

Differentiating (7.16) with respect to $x$, we get

$$\begin{aligned} -\frac{1}{(x-1)^2} &= -\sum_{j=1}^{\infty} \frac{j}{x^{j+1}} \\ \iff \sum_{j=1}^{\infty} \frac{j}{x^j} &= \frac{x}{(x-1)^2} \\ &= \frac{1}{x-1} + \frac{1}{(x-1)^2}. \end{aligned} \tag{7.17}$$

Differentiating (7.17) with respect to $x$, we get

$$\begin{aligned} -\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} &= -\sum_{j=1}^{\infty} \frac{j^2}{x^{j+1}} \\ \iff \frac{x(x+1)}{(x-1)^3} &= \sum_{j=1}^{\infty} \frac{j^2}{x^j}. \end{aligned} \tag{7.18}$$

[1.45] Replace $x$ with $-x$ in (7.18).

[1.46] We simplify (7.18) using partial fractions to get

$$\begin{aligned} \frac{((x-1)+1)((x-1)+2)}{(x-1)^3} &= \frac{(x-1)^2+3(x-1)+2}{(x-1)^3} \\ &= \frac{1}{x-1} + \frac{3}{(x-1)^2} + \frac{2}{(x-1)^3} \\ &= \sum_{j=1}^{\infty} \frac{j^2}{x^j}. \end{aligned} \tag{7.19}$$

Differentiating (7.19) with respect to $x$, we get

$$\begin{aligned} -\frac{1}{(x-1)^2} - \frac{6}{(x-1)^3} - \frac{6}{(x-1)^4} &= -\sum_{j=1}^{\infty} \frac{j^3}{x^{j+1}} \\ \iff \sum_{j=1}^{\infty} \frac{j^3}{x^j} &= \frac{x\left((x-1)^2+6(x-1)+6\right)}{(x-1)^4} \\ &= \frac{x\left(x^2+4x+1\right)}{(x-1)^4}. \end{aligned} \tag{7.20}$$

[1.47] Replace $x$ with $-x$ in (7.20).

[1.48] We simplify (7.20) using partial fractions to get

$$\begin{aligned}
&\frac{((x-1)+1)\left(((x-1)+1)^2+4((x-1)+1)+1\right)}{(x-1)^4} \\
&= \frac{((x-1)+1)\left((x-1)^2+6(x-1)+6\right)}{(x-1)^4} \\
&= \frac{(x-1)^3+7(x-1)^2+12(x-1)+6}{(x-1)^4} \\
&= \frac{1}{x-1}+\frac{7}{(x-1)^2}+\frac{12}{(x-1)^3}+\frac{6}{(x-1)^4} \\
&= \sum_{j=1}^{\infty}\frac{j^3}{x^j}.
\end{aligned} \tag{7.21}$$

Differentiating (7.21) with respect to $x$, we get

$$\begin{aligned}
&-\frac{1}{(x-1)^2}-\frac{14}{(x-1)^3}-\frac{36}{(x-1)^4}-\frac{24}{(x-1)^5} = -\sum_{j=1}^{\infty}\frac{j^4}{x^{j+1}} \\
\Longleftrightarrow\ &\sum_{j=1}^{\infty}\frac{j^4}{x^j} = \frac{x\left((x-1)^3+14(x-1)^2+36(x-1)+24\right)}{(x-1)^5} \\
&= \frac{x\left(x^3+11x^2+11x+1\right)}{(x-1)^5} \\
&= \frac{x(x+1)\left(x^2+10x+1\right)}{(x-1)^5}.
\end{aligned} \tag{7.22}$$

[1.49] Replace $x$ with $-x$ in (7.22).

[1.50] Put $\theta = \pi$ in [2.46].

[1.51] Put $\theta = \pi/2$ in [2.47].

[1.52] Put $\theta = \pi$ in [2.48].

[1.53] Put $\theta = \pi/2$ in [2.49].

[1.54] Put $\theta = \pi/6$ in [2.20].

[1.55] Put $\theta = \pi/6$ in [2.21].

[1.56] Put $\theta = \pi/6$ in [2.22].

[1.57] Put $\theta = \pi/6$ in [2.23].

[1.58] Put $\theta = 1$ in [2.46].

[1.59] Put $\theta = 1/2$ in [2.47].

[1.60] Put $\theta = 1$ in [2.48].

[1.61] Put $\theta = 1/2$ in [2.49].

[1.62] Put $\theta = 1$ in [2.20].

[1.63] Put $\theta = 1/2$ in [2.21].

[1.64] Put $\theta = 1$ in [2.22].

[1.65] Put $\theta = 1/2$ in [2.23].

[1.66] Using the infinite series for $e$, we get

$$\begin{aligned}\frac{\lfloor n!e \rfloor}{n!} &= \frac{1}{n!}\left\lfloor n! \sum_{j=0}^{\infty} \frac{1}{j!} \right\rfloor \\ &= \frac{1}{n!}\left\lfloor \sum_{j=0}^{n} \frac{n!}{j!} + \sum_{j=n+1}^{\infty} \frac{n!}{j!} \right\rfloor \\ &= \frac{1}{n!}\sum_{j=0}^{n} \frac{n!}{j!} + \frac{1}{n!}\left\lfloor \sum_{j=n+1}^{\infty} \frac{n!}{j!} \right\rfloor \\ &= \sum_{j=0}^{n} \frac{1}{j!} + \frac{1}{n!}\left\lfloor \sum_{j=n+1}^{\infty} \frac{n!}{j!} \right\rfloor .\end{aligned} \tag{7.23}$$

Now, we will prove that

$$0 < \sum_{j=n+1}^{\infty} \frac{n!}{j!} < 1. \tag{7.24}$$

From (7.23) and (7.24), the result will follow. Using the infinite geometric series, we have

$$0 < \sum_{j=n+1}^{\infty} \frac{n!}{j!} = \sum_{j=1}^{\infty} \frac{1}{\prod_{k=1}^{j}(n+k)} < \sum_{j=1}^{\infty} \frac{1}{2^j} = 1.$$

[1.67] Using the infinite series for $1/e$, we get

$$\begin{aligned} \frac{1}{n!}\left\lfloor \frac{n!}{e} \right\rfloor &= \frac{1}{n!}\left\lfloor n! \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \right\rfloor \\ &= \frac{1}{n!}\left\lfloor \sum_{j=0}^{n} (-1)^j \frac{n!}{j!} + \sum_{j=n+1}^{\infty} (-1)^j \frac{n!}{j!} \right\rfloor \\ &= \frac{1}{n!} \sum_{j=0}^{n} (-1)^j \frac{n!}{j!} + \frac{1}{n!}\left\lfloor \sum_{j=n+1}^{\infty} (-1)^j \frac{n!}{j!} \right\rfloor \\ &= \sum_{j=0}^{n} \frac{(-1)^j}{j!} + \frac{1}{n!}\left\lfloor \sum_{j=n+1}^{\infty} (-1)^j \frac{n!}{j!} \right\rfloor. \end{aligned} \tag{7.25}$$

Now, we will prove that

$$0 < \sum_{j=n+1}^{\infty} (-1)^j \frac{n!}{j!} < 1. \tag{7.26}$$

From (7.25) and (7.26), the result will follow. From (7.24), we have

$$\sum_{j=n+1}^{\infty} (-1)^j \frac{n!}{j!} < \sum_{j=n+1}^{\infty} \frac{n!}{j!} < 1.$$

Let $n = 2m - 1$. We get

$$\begin{aligned} \sum_{j=n+1}^{\infty} (-1)^j \frac{n!}{j!} &= \sum_{k=m}^{\infty} \left( (-1)^{2k} \frac{(2m-1)!}{(2k)!} + (-1)^{2k+1} \frac{(2m-1)!}{(2k+1)!} \right) \\ &= \sum_{k=m}^{\infty} \left( \frac{(2m-1)!}{(2k)!} - \frac{(2m-1)!}{(2k+1)!} \right) > 0. \end{aligned}$$

Once again, using the infinite series for $1/e$, we get

$$\begin{aligned}\frac{1}{(n+1)!}\left\lfloor\frac{(n+1)!}{e}\right\rfloor &= \frac{1}{(n+1)!}\left\lfloor(n+1)!\sum_{j=0}^{\infty}\frac{(-1)^j}{j!}\right\rfloor \\ &= \frac{1}{(n+1)!}\left\lfloor\sum_{j=0}^{n}(-1)^j\frac{(n+1)!}{j!}+\sum_{j=n+1}^{\infty}(-1)^j\frac{(n+1)!}{j!}\right\rfloor \\ &= \frac{1}{(n+1)!}\sum_{j=0}^{n}(-1)^j\frac{(n+1)!}{j!}+\frac{1}{(n+1)!}\left\lfloor\sum_{j=n+1}^{\infty}(-1)^j\frac{(n+1)!}{j!}\right\rfloor \\ &= \sum_{j=0}^{n}\frac{(-1)^j}{j!}+\frac{1}{(n+1)!}\left\lfloor\sum_{j=n+1}^{\infty}(-1)^j\frac{(n+1)!}{j!}\right\rfloor.\end{aligned} \tag{7.27}$$

Now, we will prove that

$$0<\sum_{j=n+1}^{\infty}\frac{(-1)^j(n+1)!}{j!}<1. \tag{7.28}$$

From (7.27) and (7.28), the result will follow. Let $n=2m-1$. We get

$$\begin{aligned}\sum_{j=n+1}^{\infty}(-1)^j\frac{(n+1)!}{j!} &= \sum_{k=m}^{\infty}\left((-1)^{2k}\frac{(2m)!}{(2k)!}+(-1)^{2k+1}\frac{(2m)!}{(2k+1)!}\right) \\ &= \sum_{k=m}^{\infty}\left(\frac{(2m)!}{(2k)!}-\frac{(2m)!}{(2k+1)!}\right)>0.\end{aligned}$$

We also have

$$\begin{aligned}\sum_{j=n+1}^{\infty}(-1)^j\frac{(n+1)!}{j!} &= 1-\sum_{j=n+2}^{\infty}(-1)^{j-1}\frac{(n+1)!}{j!} \\ &= 1-\sum_{k=m}^{\infty}\left((-1)^{2k}\frac{(2m)!}{(2k+1)!}+(-1)^{2k+1}\frac{(2m)!}{(2k+2)!}\right) \\ &= 1-\sum_{k=m}^{\infty}\left(\frac{(2m)!}{(2k+1)!}-\frac{(2m)!}{(2k+2)!}\right)<1.\end{aligned}$$

[1.68] Put $x = 100$ in [1.44].

[1.69] Put $\theta = \pi^2$ in [2.46].

[1.70] We will evaluate the integral $\int_0^a (a^x - y^x)^n \, dy$ in two ways to get this equation. First, using the binomial theorem, we get

$$\begin{aligned}\int_0^a \left(a^x - y^x\right)^n dy &= \sum_{j=0}^{n} (-1)^j \binom{n}{j} a^{(n-j)x} \int_0^a y^{jx} \, dy \\ &= \sum_{j=0}^{n} (-1)^j \binom{n}{j} a^{(n-j)x} \left. \frac{y^{jx+1}}{jx+1} \right|_0^a \\ &= \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{a^{nx+1}}{jx+1}.\end{aligned} \tag{7.29}$$

Now we will evaluate the integral $\int_0^a (a^x - y^x)^n \, dy$ using integration by parts. Let

$$I_j = \int_0^a y^{jx} \left(a^x - y^x\right)^{n-j} dy, \quad \forall j \in [0..n].$$

We have to evaluate $I_0$. Integrating the integral $I_j$ by parts, $\forall j \in [0..n-1]$, we get

$$\begin{aligned}I_j &= \int_0^a y^{jx} \left(a^x - y^x\right)^{n-j} dy \\ &= \left. \frac{y^{jx+1}}{jx+1} \left(a^x - y^x\right)^{n-j} \right|_0^a - \int_0^a \frac{y^{jx+1}}{jx+1} (n-j) \left(a^x - y^x\right)^{n-j-1} \left(-x y^{x-1}\right) dy \\ &= \frac{(n-j)x}{jx+1} \int_0^a y^{(j+1)x} \left(a^x - y^x\right)^{n-j-1} dy \\ &= \frac{(n-j)x}{jx+1} I_{j+1}.\end{aligned} \tag{7.30}$$

Using (7.30), we get

$$I_0 = I_n \prod_{j=0}^{n-1} \frac{(n-j)x}{jx+1} = \frac{n! x^n I_n}{\prod_{j=0}^{n-1} (1+jx)}. \tag{7.31}$$

Now we will evaluate $I_n$.

$$I_n = \int_0^a y^{nx}\,dy = \frac{y^{nx+1}}{nx+1}\bigg|_0^a = \frac{a^{nx+1}}{nx+1}. \tag{7.32}$$

From (7.31) and (7.32), we get

$$I_0 = \frac{n!x^n a^{nx+1}}{\prod_{j=0}^{n}(1+jx)}. \tag{7.33}$$

From (7.29) and (7.33), we get our result.

[1.71] For $m = n$, the series has only one term

$$\sum_{j=0}^{0}(-1)^j\binom{n+j}{n}\binom{n}{n+j} = \binom{n}{n}^2 = 1.$$

[1.72] In the case of $m < n$, the series sum is empty. Therefore, it has the sum of 0. Assuming $m > n$, using the binomial theorem, we have

$$(1+x)^m = \sum_{j=0}^{m}\binom{m}{m-j}x^j, \tag{7.34}$$

and

$$(1+x)^{-(n+1)} = \sum_{j=0}^{\infty}(-1)^j\binom{n+j}{n}x^j. \tag{7.35}$$

Multiplying (7.34) and (7.35), we get

$$(1+x)^{m-n-1} = \sum_{k=0}^{\infty}\left(\sum_{j=0}^{k}(-1)^j\binom{n+j}{n}\binom{m}{m-k+j}\right)x^k. \tag{7.36}$$

Equating the coefficient of $x^{m-n}$ in (7.36), we get

$$0 = \sum_{j=0}^{m-n}(-1)^j\binom{n+j}{n}\binom{m}{n+j}.$$

[1.73] Let

$$f(\theta,k) = \frac{1}{2^k}\cot\frac{1}{2^k\theta} + \sum_{j=k+1}^{\infty}\frac{1}{2^j}\tan\frac{1}{2^j\theta}.$$

We have

$$\begin{aligned} f(\theta,k) &= \frac{1}{2^k}\cot\frac{1}{2^k\theta} + \sum_{j=k+1}^{\infty}\frac{1}{2^j}\tan\frac{1}{2^j\theta} \\ &= \frac{1}{2^k}\cdot\frac{1-\tan^2\dfrac{1}{2^{k+1}\theta}}{2\tan\dfrac{1}{2^{k+1}\theta}} + \frac{1}{2^{k+1}}\tan\frac{1}{2^{k+1}\theta} + \sum_{j=k+2}^{\infty}\frac{1}{2^j}\tan\frac{1}{2^j\theta} \\ &= \frac{1}{2^{k+1}}\cot\frac{1}{2^{k+1}\theta} + \sum_{j=k+2}^{\infty}\frac{1}{2^j}\tan\frac{1}{2^j\theta} \\ &= f(\theta,k+1). \end{aligned} \tag{7.37}$$

Applying the recursion in (7.37) for $k \in [0..n-1]$, we get

$$\begin{aligned} f(\theta,0) &= f(\theta,n) \\ &= \frac{1}{2^n}\cot\frac{1}{2^n\theta} + \sum_{j=n+1}^{\infty}\frac{1}{2^j}\tan\frac{1}{2^j\theta}. \end{aligned} \tag{7.38}$$

Taking the limit as $n \to \infty$ in (7.38), we get

$$\begin{aligned} \cot\frac{1}{\theta} + \sum_{j=1}^{\infty}\frac{1}{2^j}\tan\frac{1}{2^j\theta} &= \lim_{n\to\infty}\left(\frac{1}{2^n}\cot\frac{1}{2^n\theta} + \sum_{j=n+1}^{\infty}\frac{1}{2^j}\tan\frac{1}{2^j\theta}\right) \\ &= \lim_{n\to\infty}\theta\frac{\dfrac{1}{2^n\theta}}{\tan\dfrac{1}{2^n\theta}} + \lim_{n\to\infty}\sum_{j=n+1}^{\infty}\frac{1}{2^j}\cdot\frac{1}{2^j\theta}\cdot\frac{\tan\dfrac{1}{2^j\theta}}{\dfrac{1}{2^j\theta}} \\ &= \theta + \frac{1}{\theta}\lim_{n\to\infty}\sum_{j=n+1}^{\infty}\frac{1}{2^{2j}} \\ &= \theta + \frac{1}{\theta}\lim_{n\to\infty}\frac{\dfrac{1}{2^{2n+2}}}{1-\dfrac{1}{4}} \\ &= \theta. \end{aligned}$$

[1.74] Put $y = 1$ in [1.77].

[1.75] Let

$$f(x,n)=\sum_{j=0}^{n-1}\frac{2^{2j}x^{2^j}}{(x^{2^j}+1)^2}+\frac{2^{2n}x^{2^n}}{(x^{2^n}-1)^2}.$$

We have

$$\begin{aligned}
f(x,n)&=\sum_{j=0}^{n-1}\frac{2^{2j}x^{2^j}}{\left(x^{2^j}+1\right)^2}+\frac{2^{2n}x^{2^n}}{\left(x^{2^n}-1\right)^2}\\
&=\sum_{j=0}^{n-2}\frac{2^{2j}x^{2^j}}{\left(x^{2^j}+1\right)^2}+\frac{2^{2n-2}x^{2^{n-1}}}{\left(x^{2^{n-1}}+1\right)^2}+\frac{2^{2n}x^{2^n}}{\left(x^{2^{n-1}}-1\right)^2\left(x^{2^{n-1}}+1\right)^2}\\
&=\sum_{j=0}^{n-2}\frac{2^{2j}x^{2^j}}{\left(x^{2^j}+1\right)^2}+\frac{2^{2n-2}x^{2^{n-1}}}{\left(x^{2^{n-1}}+1\right)^2}+\frac{2^{2n-2}x^{2^{n-1}}}{\left(x^{2^{n-1}}-1\right)^2}-\frac{2^{2n-2}x^{2^{n-1}}}{\left(x^{2^{n-1}}+1\right)^2}\\
&=\sum_{j=0}^{n-2}\frac{2^{2j}x^{2^j}}{\left(x^{2^j}+1\right)^2}+\frac{2^{2n-2}x^{2^{n-1}}}{\left(x^{2^{n-1}}-1\right)^2}\\
&=f(x,n-1).
\end{aligned}\tag{7.39}$$

Applying the recursion in (7.39) for $n-1$ times, we get

$$\begin{aligned}
f(x,n)&=f(x,1)\\
&=\frac{x}{(x+1)^2}+\frac{4x^2}{\left(x^2-1\right)^2}\\
&=\frac{x}{(x+1)^2}+\frac{4x^2}{(x-1)^2(x+1)^2}\\
&=\frac{x}{(x+1)^2}+\frac{x}{(x-1)^2}-\frac{x}{(x+1)^2}\\
&=\frac{x}{(x-1)^2}.
\end{aligned}$$

[1.76] Let

$$f(x,y,n)=\sum_{j=0}^{n-1}\frac{2^jx^{2^j}\left(y^{2^{j+1}}-1\right)}{\left(x^{2^j}+y^{2^j}\right)\left(x^{2^j}y^{2^j}+1\right)}+\frac{2^nx^{2^n}\left(y^{2^{n+1}}-1\right)}{\left(x^{2^n}-y^{2^n}\right)\left(x^{2^n}y^{2^n}-1\right)}.$$

We have

$$
\begin{aligned}
f(x,y,n) &= \sum_{j=0}^{n-1} \frac{2^j x^{2^j}\left(y^{2^{j+1}}-1\right)}{\left(x^{2^j}+y^{2^j}\right)\left(x^{2^j}y^{2^j}+1\right)} + \frac{2^n x^{2^n}\left(y^{2^{n+1}}-1\right)}{\left(x^{2^n}-y^{2^n}\right)\left(x^{2^n}y^{2^n}-1\right)} \\
&= \sum_{j=0}^{n-2} \frac{2^j x^{2^j}\left(y^{2^{j+1}}-1\right)}{\left(x^{2^j}+y^{2^j}\right)\left(x^{2^j}y^{2^j}+1\right)} + \frac{2^{n-1}x^{2^{n-1}}\left(y^{2^n}-1\right)}{\left(x^{2^{n-1}}+y^{2^{n-1}}\right)\left(x^{2^{n-1}}y^{2^{n-1}}+1\right)} \\
&\quad + \frac{2^n x^{2^n}\left(y^{2^{n+1}}-1\right)}{\left(x^{2^{n-1}}-y^{2^{n-1}}\right)\left(x^{2^{n-1}}y^{2^{n-1}}-1\right)\left(x^{2^{n-1}}+y^{2^{n-1}}\right)\left(x^{2^{n-1}}y^{2^{n-1}}+1\right)} \\
&= \sum_{j=0}^{n-2} \frac{2^j x^{2^j}\left(y^{2^{j+1}}-1\right)}{\left(x^{2^j}+y^{2^j}\right)\left(x^{2^j}y^{2^j}+1\right)} + \frac{2^{n-1}x^{2^{n-1}}\left(y^{2^n}-1\right)}{\left(x^{2^{n-1}}+y^{2^{n-1}}\right)\left(x^{2^{n-1}}y^{2^{n-1}}+1\right)} \\
&\quad + \frac{2^{n-1}x^{2^{n-1}}\left(y^{2^n}-1\right)}{\left(x^{2^{n-1}}-y^{2^{n-1}}\right)\left(x^{2^{n-1}}y^{2^{n-1}}-1\right)} - \frac{2^{n-1}x^{2^{n-1}}\left(y^{2^n}-1\right)}{\left(x^{2^{n-1}}+y^{2^{n-1}}\right)\left(x^{2^{n-1}}y^{2^{n-1}}+1\right)} \\
&= \sum_{j=0}^{n-2} \frac{2^j x^{2^j}\left(y^{2^{j+1}}-1\right)}{\left(x^{2^j}+y^{2^j}\right)\left(x^{2^j}y^{2^j}+1\right)} + \frac{2^{n-1}x^{2^{n-1}}\left(y^{2^n}-1\right)}{\left(x^{2^{n-1}}-y^{2^{n-1}}\right)\left(x^{2^{n-1}}y^{2^{n-1}}-1\right)} \\
&= f(x,y,n-1). \qquad (7.40)
\end{aligned}
$$

Applying the recursion in (7.40) for $n-1$ times, we get

$$
\begin{aligned}
f(x,y,n) &= f(x,y,1) \\
&= \frac{x\left(y^2-1\right)}{(x+y)(xy+1)} + \frac{2x^2\left(y^4-1\right)}{\left(x^2-y^2\right)\left(x^2y^2-1\right)} \\
&= \frac{x\left(y^2-1\right)}{(x+y)(xy+1)} + \frac{2x^2\left(y^4-1\right)}{(x-y)\left(xy-1\right)(x+y)\left(xy+1\right)} \\
&= \frac{x\left(y^2-1\right)}{(x+y)(xy+1)} + \frac{x\left(y^2-1\right)}{(x-y)(xy-1)} - \frac{x\left(y^2-1\right)}{(x+y)(xy+1)} \\
&= \frac{x\left(y^2-1\right)}{(x-y)(xy-1)}.
\end{aligned}
$$

[1.77] Let

$$
f(x,y,n) = \sum_{j=0}^{n-1} \frac{2^j y^{2^j}}{x^{2^j}+y^{2^j}} + \frac{2^n y^{2^n}}{x^{2^n}-y^{2^n}}.
$$

We have

$$
\begin{aligned}
f(x,y,n) &= \sum_{j=0}^{n-1} \frac{2^j y^{2^j}}{x^{2^j}+y^{2^j}} + \frac{2^n y^{2^n}}{x^{2^n}-y^{2^n}} \\
&= \sum_{j=0}^{n-2} \frac{2^j y^{2^j}}{x^{2^j}+y^{2^j}} + \frac{2^{n-1} y^{2^{n-1}}}{x^{2^{n-1}}+y^{2^{n-1}}} + \frac{2^n y^{2^n}}{\left(x^{2^{n-1}}-y^{2^{n-1}}\right)\left(x^{2^{n-1}}+y^{2^{n-1}}\right)} \\
&= \sum_{j=0}^{n-2} \frac{2^j y^{2^j}}{x^{2^j}+y^{2^j}} + \frac{2^{n-1} y^{2^{n-1}}}{x^{2^{n-1}}+y^{2^{n-1}}} + \frac{2^{n-1} y^{2^{n-1}}}{x^{2^{n-1}}-y^{2^{n-1}}} - \frac{2^{n-1} y^{2^{n-1}}}{x^{2^{n-1}}+y^{2^{n-1}}} \\
&= \sum_{j=0}^{n-2} \frac{2^j y^{2^j}}{x^{2^j}+y^{2^j}} + \frac{2^{n-1} y^{2^{n-1}}}{x^{2^{n-1}}-y^{2^{n-1}}} \\
&= f(x,y,n-1). \qquad (7.41)
\end{aligned}
$$

Applying the recursion in (7.41) for $n-1$ times, we get

$$
\begin{aligned}
f(x,y,n) &= f(x,y,1) \\
&= \frac{y}{x+y} + \frac{2y^2}{x^2-y^2} \\
&= \frac{y}{x+y} + \frac{2y^2}{(x-y)(x+y)} \\
&= \frac{y}{x+y} + \frac{y}{x-y} - \frac{y}{x+y} \\
&= \frac{y}{x-y}.
\end{aligned}
$$

[1.78] Replacing $x$ with $xy$ in [1.74], we get

$$
\frac{1}{xy-1} = \sum_{j=0}^{n-1} \frac{2^j}{x^{2^j} y^{2^j}+1} + \frac{2^n}{x^{2^n} y^{2^n}-1}. \qquad (7.42)
$$

Adding [1.77] and (7.42), we get

$$\begin{aligned}
&\frac{y}{x-y}+\frac{1}{xy-1}=\sum_{j=0}^{n-1}2^j\left(\frac{y^{2^j}}{x^{2^j}+y^{2^j}}+\frac{1}{x^{2^j}y^{2^j}+1}\right)\\
&\qquad+2^n\left(\frac{y^{2^n}}{x^{2^n}-y^{2^n}}+\frac{1}{x^{2^n}y^{2^n}-1}\right)\\
&\iff\frac{y(xy-1)+(x-y)}{(x-y)(xy-1)}\\
&=\sum_{j=0}^{n-1}2^j\frac{y^{2^j}(x^{2^j}y^{2^j}+1)+(x^{2^j}+y^{2^j})}{(x^{2^j}+y^{2^j})(x^{2^j}y^{2^j}+1)}+2^n\frac{y^{2^n}(x^{2^n}y^{2^n}-1)+(x^{2^n}-y^{2^n})}{(x^{2^n}-y^{2^n})(x^{2^n}y^{2^n}-1)}.
\end{aligned}$$

[1.79] Let

$$f(x,\theta,n)=\sum_{j=0}^{n-1}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^n\left(x^{2^n}\cos 2^n\theta-1\right)}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}.$$

We have

$$\begin{aligned}
f(x,\theta,n)&=\sum_{j=0}^{n-1}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^n\left(x^{2^n}\cos 2^n\theta-1\right)}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}\\
&=\sum_{j=0}^{n-2}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&\quad+\frac{2^n\left(x^{2^n}\cos 2^n\theta-1\right)}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}\\
&=\sum_{j=0}^{n-2}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&\quad+\frac{2^n\left(2x^{2^n}\cos^2 2^{n-1}\theta-x^{2^n}-1\right)}{x^{2^{n+1}}-4x^{2^n}\cos^2 2^{n-1}\theta+2x^{2^n}+1}\\
&=\sum_{j=0}^{n-2}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&\quad+\frac{2^n\left(2x^{2^n}\cos^2 2^{n-1}\theta-x^{2^n}-1\right)}{\left(x^{2^n}-2x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)\left(x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}\\
&=\sum_{j=0}^{n-2}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}
\end{aligned}$$

$$
\begin{aligned}
&+\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta-1\right)}{x^{2^n}-2x^{2^{n-1}}\cos 2^{n-1}\theta+1}-\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&=\sum_{j=0}^{n-2}\frac{2^j\left(x^{2^j}\cos 2^j\theta+1\right)}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}\left(x^{2^{n-1}}\cos 2^{n-1}\theta-1\right)}{x^{2^n}-2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&=f(x,\theta,n-1). \qquad (7.43)
\end{aligned}
$$

Applying the recursion in (7.43) for $n-1$ times, we get

$$
\begin{aligned}
f(x,\theta,n)&=f(x,\theta,1)=\frac{x\cos\theta+1}{x^2+2x\cos\theta+1}+\frac{2\left(x^2\cos 2\theta-1\right)}{x^4-2x^2\cos 2\theta+1}\\
&=\frac{x\cos\theta+1}{x^2+2x\cos\theta+1}+\frac{2\left(2x^2\cos^2\theta-x^2-1\right)}{x^4-4x^2\cos^2\theta+2x^2+1}\\
&=\frac{x\cos\theta+1}{x^2+2x\cos\theta+1}+\frac{2\left(2x^2\cos^2\theta-x^2-1\right)}{\left(x^2-2x\cos\theta+1\right)\left(x^2+2x\cos\theta+1\right)}\\
&=\frac{x\cos\theta+1}{x^2+2x\cos\theta+1}+\frac{x\cos\theta-1}{x^2-2x\cos\theta+1}-\frac{x\cos\theta+1}{x^2+2x\cos\theta+1}\\
&=\frac{x\cos\theta-1}{x^2-2x\cos\theta+1}.
\end{aligned}
$$

[1.80] Let

$$
f(x,\theta,n)=\sum_{j=0}^{n-1}\frac{2^jx^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^nx^{2^n}\sin 2^n\theta}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}.
$$

We have

$$
\begin{aligned}
f(x,\theta,n)&=\sum_{j=0}^{n-1}\frac{2^jx^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^nx^{2^n}\sin 2^n\theta}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}\\
&=\sum_{j=0}^{n-2}\frac{2^jx^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&\quad+\frac{2^nx^{2^n}\sin 2^n\theta}{x^{2^{n+1}}-2x^{2^n}\cos 2^n\theta+1}\\
&=\sum_{j=0}^{n-2}\frac{2^jx^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{2^{n+1}x^{2^n}\sin 2^{n-1}\theta\cos 2^{n-1}\theta}{x^{2^{n+1}}-4x^{2^n}\cos^2 2^{n-1}\theta+2x^{2^n}+1}\\
&=\sum_{j=0}^{n-2}\frac{2^j x^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&\quad+\frac{2^{n+1}x^{2^n}\sin 2^{n-1}\theta\cos 2^{n-1}\theta}{\left(x^{2^n}-2x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)\left(x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1\right)}\\
&=\sum_{j=0}^{n-2}\frac{2^j x^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&\quad+\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}-2x^{2^{n-1}}\cos 2^{n-1}\theta+1}-\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}+2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&=\sum_{j=0}^{n-2}\frac{2^j x^{2^j}\sin 2^j\theta}{x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1}+\frac{2^{n-1}x^{2^{n-1}}\sin 2^{n-1}\theta}{x^{2^n}-2x^{2^{n-1}}\cos 2^{n-1}\theta+1}\\
&=f(x,\theta,n-1). \qquad (7.44)
\end{aligned}
$$

Applying the recursion in (7.44) for $n-1$ times, we get

$$
\begin{aligned}
f(x,\theta,n)&=f(x,\theta,1)=\frac{x\sin\theta}{x^2+2x\cos\theta+1}+\frac{2x^2\sin 2\theta}{x^4-2x^2\cos 2\theta+1}\\
&=\frac{x\sin\theta}{x^2+2x\cos\theta+1}+\frac{4x^2\sin\theta\cos\theta}{x^4-4x^2\cos^2\theta+2x^2+1}\\
&=\frac{x\sin\theta}{x^2+2x\cos\theta+1}+\frac{4x^2\sin\theta\cos\theta}{\left(x^2-2x\cos\theta+1\right)\left(x^2+2x\cos\theta+1\right)}\\
&=\frac{x\sin\theta}{x^2+2x\cos\theta+1}+\frac{x\sin\theta}{x^2-2x\cos\theta+1}-\frac{x\sin\theta}{x^2+2x\cos\theta+1}\\
&=\frac{x\sin\theta}{x^2-2x\cos\theta+1}.
\end{aligned}
$$

[1.81] We will prove by induction. Let $P(n)$ be the statement that

$$
\frac{x^{2^{n+2}}-2x^{2^{n+1}}\cos 2^{n+1}\theta+1}{x^2-2x\cos\theta+1}=\prod_{j=0}^{n}\left(x^{2^{j+1}}+2x^{2^j}\cos 2^j\theta+1\right). \qquad (7.45)
$$

*Basis Step:* $P(0)$ states that

$$\frac{x^4 - 2x^2\cos 2\theta + 1}{x^2 - 2x\cos\theta + 1} = x^2 + 2x\cos\theta + 1.$$

We have

$$\begin{aligned}\frac{x^4 - 2x^2\cos 2\theta + 1}{x^2 - 2x\cos\theta + 1} &= \frac{x^4 - 2x^2\left(2\cos^2\theta - 1\right) + 1}{x^2 - 2x\cos\theta + 1} \\ &= \frac{x^4 - 4x^2\cos^2\theta + 2x^2 + 1}{x^2 - 2x\cos\theta + 1} \\ &= \frac{\left(x^2 - 2x\cos\theta + 1\right)\left(x^2 + 2x\cos\theta + 1\right)}{x^2 - 2x\cos\theta + 1} \\ &= x^2 + 2x\cos\theta + 1,\end{aligned}$$

proving $P(0)$ to be correct.
*Induction Step:* Assuming $P(n)$ to be true, using (7.45), we have

$$\begin{aligned}&\prod_{j=0}^{n+1}\left(x^{2^{j+1}} + 2x^{2^j}\cos 2^j\theta + 1\right) \\ &= \left(\prod_{j=0}^{n}\left(x^{2^{j+1}} + 2x^{2^j}\cos 2^j\theta + 1\right)\right)\left(x^{2^{n+2}} + 2x^{2^{n+1}}\cos 2^{n+1}\theta + 1\right) \\ &= \frac{\left(x^{2^{n+2}} - 2x^{2^{n+1}}\cos 2^{n+1}\theta + 1\right)\left(x^{2^{n+2}} + 2x^{2^{n+1}}\cos 2^{n+1}\theta + 1\right)}{x^2 - 2x\cos\theta + 1} \\ &= \frac{x^{2^{n+3}} - 2x^{2^{n+2}}\left(2\cos^2 2^{n+1}\theta - 1\right) + 1}{x^2 - 2x\cos\theta + 1} \\ &= \frac{x^{2^{n+3}} - 2x^{2^{n+2}}\cos 2^{n+2}\theta + 1}{x^2 - 2x\cos\theta + 1},\end{aligned}$$

proving $P(n+1)$ to be correct.

[1.82] We have

$$
\begin{aligned}
T(x,\theta,n) &= \sum_{j=0}^{n-1} \tan^{-1} \frac{x^{2^j} + \cos 2^j\theta}{\sin 2^j\theta} + \tan^{-1} \frac{x^{2^n} - \cos 2^n\theta}{\sin 2^n\theta} \\
&= \sum_{j=0}^{n-2} \tan^{-1} \frac{x^{2^j} + \cos 2^j\theta}{\sin 2^j\theta} + \tan^{-1} \frac{x^{2^{n-1}} + \cos 2^{n-1}\theta}{\sin 2^{n-1}\theta} \\
&\qquad + \tan^{-1} \frac{x^{2^n} - \cos 2^n\theta}{\sin 2^n\theta} \\
&= \sum_{j=0}^{n-2} \tan^{-1} \frac{x^{2^j} + \cos 2^j\theta}{\sin 2^j\theta} + \{-1, 0, 1\}\pi \\
&\quad + \tan^{-1} \frac{\left(x^{2^{n-1}} + \cos 2^{n-1}\theta\right) \sin 2^n\theta + \left(x^{2^n} - \cos 2^n\theta\right) \sin 2^{n-1}\theta}{\sin 2^{n-1}\theta \sin 2^n\theta - \left(x^{2^{n-1}} + \cos 2^{n-1}\theta\right)\left(x^{2^n} - \cos 2^n\theta\right)} \\
&= \sum_{j=0}^{n-2} \tan^{-1} \frac{x^{2^j} + \cos 2^j\theta}{\sin 2^j\theta} + \{-1, 0, 1\}\pi \\
&\quad + \tan^{-1} \left( - \frac{x^{2^n} \sin 2^{n-1}\theta + 2x^{2^{n-1}} \sin 2^{n-1}\theta \cos 2^{n-1}\theta + \sin 2^{n-1}\theta}{x^{2^n+2^{n-1}} + x^{2^n} \cos 2^{n-1}\theta - x^{2^{n-1}} \left(2\cos^2 2^{n-1}\theta - 1\right) - \cos 2^{n-1}\theta} \right) \\
&= \sum_{j=0}^{n-2} \tan^{-1} \frac{x^{2^j} + \cos 2^j\theta}{\sin 2^j\theta} + \{-1, 0, 1\}\pi \\
&\quad + \tan^{-1} \left( - \frac{\left(x^{2^n} + 2x^{2^{n-1}} \cos 2^{n-1}\theta + 1\right) \sin 2^{n-1}\theta}{\left(x^{2^{n-1}} - \cos 2^{n-1}\theta\right)\left(x^{2^n} + 2x^{2^{n-1}} \cos 2^{n-1}\theta + 1\right)} \right) \\
&= \sum_{j=0}^{n-2} \tan^{-1} \frac{x^{2^j} + \cos 2^j\theta}{\sin 2^j\theta} + \tan^{-1} \left( - \frac{\sin 2^{n-1}\theta}{x^{2^{n-1}} - \cos 2^{n-1}\theta} \right) + \{-1, 0, 1\}\pi \\
&= \sum_{j=0}^{n-2} \tan^{-1} \frac{x^{2^j} + \cos 2^j\theta}{\sin 2^j\theta} + \tan^{-1} \frac{x^{2^{n-1}} - \cos 2^{n-1}\theta}{\sin 2^{n-1}\theta} - \frac{\pi}{2} + \{-1, 0, 1\}\pi \\
&= T(x,\theta,n-1) + \frac{\pi}{2}\{-3, -1, 1\} \\
\implies & T(x,\theta,n) - T(x,\theta,n-1) \in \frac{\pi}{2}\mathbb{Z}.
\end{aligned}
\tag{7.46}
$$

Applying the recursion in (7.46) for $n-1$ times and adding all the equations, we get

$$
\begin{aligned}
& T(x,\theta,n) - T(x,\theta,1) \in \frac{\pi}{2}\mathbb{Z} \\
\iff & T(x,\theta,n) - \tan^{-1}\frac{x+\cos\theta}{\sin\theta} - \tan^{-1}\frac{x^2-\cos 2\theta}{\sin 2\theta} \in \frac{\pi}{2}\mathbb{Z} \\
\iff & T(x,\theta,n) - \tan^{-1}\frac{(x+\cos\theta)\sin 2\theta + \left(x^2-\cos 2\theta\right)\sin\theta}{\sin\theta\sin 2\theta - (x+\cos\theta)\left(x^2-\cos 2\theta\right)} \\
& + \{-1,0,1\}\pi \in \frac{\pi}{2}\mathbb{Z} \\
\iff & T(x,\theta,n) - \tan^{-1}\left(-\frac{x^2\sin\theta + 2x\sin\theta\cos\theta + \sin\theta}{x^3 + x^2\cos\theta - \left(2\cos^2\theta - 1\right)x + \cos\theta}\right) \in \frac{\pi}{2}\mathbb{Z} \\
\iff & T(x,\theta,n) - \tan^{-1}\left(-\frac{\left(x^2+2x\cos\theta+1\right)\sin\theta}{(x-\cos\theta)\left(x^2+2x\cos\theta+1\right)}\right) \in \frac{\pi}{2}\mathbb{Z} \\
\iff & T(x,\theta,n) - \tan^{-1}\frac{x-\cos\theta}{\sin\theta} + \frac{\pi}{2} \in \frac{\pi}{2}\mathbb{Z} \\
\iff & T(x,\theta,n) - \tan^{-1}\frac{x-\cos\theta}{\sin\theta} \in \frac{\pi}{2}\mathbb{Z}.
\end{aligned}
$$

[1.83] Let

$$
f(x,k) = \frac{1}{2^k\left(x^{1/2^k}-1\right)} + \sum_{j=k+1}^{\infty}\frac{1}{2^j\left(1+x^{1/2^j}\right)}.
$$

We have

$$
\begin{aligned}
f(x,k) &= \frac{1}{2^k\left(x^{1/2^k}-1\right)} + \sum_{j=k+1}^{\infty}\frac{1}{2^j\left(1+x^{1/2^j}\right)} \\
&= \frac{1}{2^k\left(x^{1/2^{k+1}}-1\right)\left(x^{1/2^{k+1}}+1\right)} + \frac{1}{2^{k+1}\left(1+x^{1/2^{k+1}}\right)} \\
&\qquad + \sum_{j=k+2}^{\infty}\frac{1}{2^j\left(1+x^{1/2^j}\right)} \\
&= \frac{1}{2^{k+1}\left(x^{1/2^{k+1}}-1\right)} - \frac{1}{2^{k+1}\left(1+x^{1/2^{k+1}}\right)} + \frac{1}{2^{k+1}\left(1+x^{1/2^{k+1}}\right)} \\
&\quad + \sum_{j=k+2}^{\infty}\frac{1}{2^j\left(1+x^{1/2^j}\right)} \\
&= \frac{1}{2^{k+1}\left(x^{1/2^{k+1}}-1\right)} + \sum_{j=k+2}^{\infty}\frac{1}{2^j\left(1+x^{1/2^j}\right)} \\
&= f(x,k+1).
\end{aligned}
\tag{7.47}
$$

Applying the recursion in (7.47) for $k \in [0..n-1]$, we get

$$\begin{aligned} f(x,0) &= \frac{1}{x-1} + \sum_{j=1}^{\infty} \frac{1}{2^j \left(1+x^{1/2^j}\right)} \\ &= f(x,n) = \frac{1}{2^n \left(x^{1/2^n}-1\right)} + \sum_{j=n+1}^{\infty} \frac{1}{2^j \left(1+x^{1/2^j}\right)}. \end{aligned} \tag{7.48}$$

Taking the limit as $n \to \infty$ in (7.48) and using (1.19), we get

$$\begin{aligned} \frac{1}{x-1} + \sum_{j=1}^{\infty} \frac{1}{2^j \left(1+x^{1/2^j}\right)} &= \lim_{n\to\infty} \frac{1}{2^n \left(x^{1/2^n}-1\right)} + \lim_{n\to\infty} \sum_{j=n+1}^{\infty} \frac{1}{2^j \left(1+x^{1/2^j}\right)} \\ &= \frac{1}{\log x} + \lim_{n\to\infty} \sum_{j=n+1}^{\infty} \frac{1}{2^{j+1}} \\ &= \frac{1}{\log x} + \lim_{n\to\infty} \frac{1}{2^{n+2}\left(1-\frac{1}{2}\right)} \\ &= \frac{1}{\log x}. \end{aligned} \tag{7.49}$$

[1.84] Differentiating (7.49) with respect to $x$, we get

$$\begin{aligned} & \frac{d}{dx}\left(\frac{1}{\log x}\right) = \frac{d}{dx}\left(\frac{1}{x-1}\right) + \sum_{j=1}^{\infty} \frac{d}{dx}\left(\frac{1}{2^j \left(1+x^{1/2^j}\right)}\right) \\ \iff & -\frac{1}{x \log^2 x} = -\frac{1}{(x-1)^2} - \sum_{j=1}^{\infty} \frac{x^{1/2^j}}{2^{2j} x \left(1+x^{1/2^j}\right)^2} \\ \iff & \frac{1}{\log^2 x} = \frac{x}{(x-1)^2} + \sum_{j=1}^{\infty} \frac{x^{1/2^j}}{2^{2j} \left(1+x^{1/2^j}\right)^2}. \end{aligned}$$

[1.85] We will prove by induction. Let $P(n)$ be the statement that $\forall n \in \mathbb{N}$, $\exists (a_k)_{k=1}^n \in \mathbb{R}^n$ and $\exists (b_k)_{k=1}^n \in \mathbb{R}^n$ such that

$$\frac{1}{\log^n x} = \sum_{k=1}^{n} \frac{a_k}{(x-1)^k} + \sum_{k=1}^{n} \sum_{j=1}^{\infty} \frac{b_k}{2^{jn} \left(1+x^{1/2^j}\right)^k}. \tag{7.50}$$

*Basis Step:* $P(1)$ is proved in [1.83].
*Induction Step:* Assuming $P(n)$ to be true, differentiating (7.50) with respect to $x$, we get

$$\begin{aligned}
&\frac{d}{dx}\left(\frac{1}{\log^n x}\right) = \sum_{k=1}^{n} \frac{d}{dx}\left(\frac{a_k}{(x-1)^k}\right) + \sum_{k=1}^{n}\sum_{j=1}^{\infty} \frac{d}{dx}\left(\frac{b_k}{2^{jn}\left(1+x^{1/2^j}\right)^k}\right) \\
\iff & -\frac{n}{x\log^{n+1} x} = -\sum_{k=1}^{n} \frac{ka_k}{(x-1)^{k+1}} - \sum_{k=1}^{n}\sum_{j=1}^{\infty} \frac{kb_k x^{1/2^j}}{2^{j(n+1)} x\left(1+x^{1/2^j}\right)^{k+1}} \\
\iff & \frac{1}{\log^{n+1} x} = \sum_{k=1}^{n} \frac{ka_k x}{n(x-1)^{k+1}} + \sum_{k=1}^{n}\sum_{j=1}^{\infty} \frac{kb_k x^{1/2^j}}{n2^{j(n+1)}\left(1+x^{1/2^j}\right)^{k+1}} \\
&= \sum_{k=1}^{n}\left(\frac{ka_k}{n(x-1)^k} + \frac{ka_k}{n(x-1)^{k+1}}\right) \\
&\quad + \sum_{k=1}^{n}\sum_{j=1}^{\infty}\left(\frac{kb_k}{n2^{j(n+1)}\left(1+x^{1/2^j}\right)^k} - \frac{kb_k}{n2^{j(n+1)}\left(1+x^{1/2^j}\right)^{k+1}}\right) \\
&= \frac{a_1}{n(x-1)} + \sum_{k=2}^{n} \frac{(k-1)a_{k-1} + ka_k}{n(x-1)^k} + \frac{a_n}{(x-1)^{n+1}} \\
&\quad + \sum_{j=1}^{\infty} \frac{b_1}{n2^{j(n+1)}\left(1+x^{1/2^j}\right)} + \sum_{k=2}^{n}\sum_{j=1}^{\infty} \frac{kb_k - (k-1)b_{k-1}}{n2^{j(n+1)}\left(1+x^{1/2^j}\right)^k} \\
&\quad - \sum_{j=1}^{\infty} \frac{b_n}{2^{j(n+1)}\left(1+x^{1/2^j}\right)^{n+1}},
\end{aligned}$$

proving $P(n+1)$ to be correct.

[1.86] Let

$$\begin{aligned}
&f(x,\theta,k) \\
&= \frac{x^{1/2^k}\cos\dfrac{\theta}{2^k} - 1}{2^k\left(x^{1/2^{k-1}} - 2x^{1/2^k}\cos\dfrac{\theta}{2^k} + 1\right)} + \sum_{j=k+1}^{\infty} \frac{x^{1/2^j}\cos\dfrac{\theta}{2^j} + 1}{2^j\left(x^{1/2^{j-1}} + 2x^{1/2^j}\cos\dfrac{\theta}{2^j} + 1\right)}.
\end{aligned}$$

We have

$$
\begin{aligned}
&f(x,\theta,k)\\
&=\frac{x^{1/2^k}\cos\dfrac{\theta}{2^k}-1}{2^k\left(x^{1/2^{k-1}}-2x^{1/2^k}\cos\dfrac{\theta}{2^k}+1\right)}+\sum_{j=k+1}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}\\
&=\frac{2x^{1/2^k}\cos^2\dfrac{\theta}{2^k}-x^{1/2^k}-1}{2^k\left(x^{1/2^{k-1}}-4x^{1/2^k}\cos^2\dfrac{\theta}{2^k}+2x^{1/2^k}+1\right)}\\
&\quad+\frac{x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1}{2^{k+1}\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)}\\
&\quad+\sum_{j=k+2}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}\\
&=\frac{2x^{1/2^k}\cos^2\dfrac{\theta}{2^k}-x^{1/2^k}-1}{2^k\left(x^{1/2^k}-2x^{1/2^{k+1}}\cos^2\dfrac{\theta}{2^{k+1}}+1\right)\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos^2\dfrac{\theta}{2^{k+1}}+1\right)}\\
&\quad+\frac{x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1}{2^{k+1}\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)}\\
&\quad+\sum_{j=k+2}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}\\
&=\frac{x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}-1}{2^{k+1}\left(x^{1/2^k}-2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)}\\
&\quad-\frac{x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1}{2^{k+1}\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)}\\
&\quad+\frac{x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1}{2^{k+1}\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)}
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{j=k+2}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}\\
&=\frac{x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}-1}{2^{k+1}\left(x^{1/2^k}-2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)}\\
&+\sum_{j=k+2}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}\\
&=f(x,\theta,k+1).
\end{aligned}
\tag{7.51}
$$

Applying the recursion in (7.51) for $k\in[0..n-1]$, we get

$$
\begin{aligned}
f(x,\theta,0)&=\frac{x\cos\theta-1}{x^2-2x\cos\theta+1}+\sum_{j=1}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}\\
&=f(x,\theta,n)\\
&=\frac{x^{1/2^n}\cos\dfrac{\theta}{2^n}-1}{2^n\left(x^{1/2^{n-1}}-2x^{1/2^n}\cos\dfrac{\theta}{2^n}+1\right)}\\
&\quad+\sum_{j=n+1}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}.
\end{aligned}
\tag{7.52}
$$

Taking the limit as $n\to\infty$ in (7.52) and using (1.19), we get

$$\frac{x\cos\theta-1}{x^2-2x\cos\theta+1}+\sum_{j=1}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}$$

$$=\lim_{n\to\infty}\frac{x^{1/2^n}\cos\dfrac{\theta}{2^n}-1}{2^n\left(x^{1/2^{n-1}}-2x^{1/2^n}\cos\dfrac{\theta}{2^n}+1\right)}$$

$$+\lim_{n\to\infty}\sum_{j=n+1}^{\infty}\frac{x^{1/2^j}\cos\dfrac{\theta}{2^j}+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}$$

$$=\lim_{n\to\infty}\frac{x^{1/2^n}\left(1-\dfrac{\theta^2}{2^{2n+1}}\right)-1}{2^n\left(x^{1/2^{n-1}}-2x^{1/2^n}\left(1-\dfrac{\theta^2}{2^{2n+1}}\right)+1\right)}$$

$$+\lim_{n\to\infty}\sum_{j=n+1}^{\infty}\frac{x^{1/2^j}\left(1-\dfrac{\theta^2}{2^{2j+1}}\right)+1}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\left(1-\dfrac{\theta^2}{2^{2j+1}}\right)+1\right)}$$

$$=\lim_{n\to\infty}\frac{2^n\left(x^{1/2^n}-1\right)-\dfrac{\theta^2x^{1/2^n}}{2^{n+1}}}{\theta^2x^{1/2^n}+2^{2n}\left(x^{1/2^n}-1\right)^2}+\lim_{n\to\infty}\sum_{j=n+1}^{\infty}\frac{x^{1/2^j}+1-\dfrac{\theta^2x^{1/2^j}}{2^{2j+1}}}{2^j\left(x^{1/2^j}+1\right)^2-\dfrac{\theta^2x^{1/2^j}}{2^j}}$$

$$=\frac{\log x}{\theta^2+\log^2 x}+\lim_{n\to\infty}\sum_{j=n+1}^{\infty}\frac{1}{2^{j+1}}$$

$$=\frac{\log x}{\theta^2+\log^2 x}+\lim_{n\to\infty}\frac{1}{2^{n+2}\left(1-\dfrac{1}{2}\right)}$$

$$=\frac{\log x}{\theta^2+\log^2 x}.$$

[1.87] Let

$$f(x,\theta,k)$$

$$=\frac{x^{1/2^k}\sin\dfrac{\theta}{2^k}}{2^k\left(x^{1/2^{k-1}}-2x^{1/2^k}\cos\dfrac{\theta}{2^k}+1\right)}+\sum_{j=k+1}^{\infty}\frac{x^{1/2^j}\sin\dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)}.$$

We have

$$
\begin{aligned}
f(x,\theta,k) &= \frac{x^{1/2^k}\sin\dfrac{\theta}{2^k}}{2^k\left(x^{1/2^{k-1}}-2x^{1/2^k}\cos\dfrac{\theta}{2^k}+1\right)} \\
&\quad+\sum_{j=k+1}^{\infty}\frac{x^{1/2^j}\sin\dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)} \\
&= \frac{2x^{1/2^k}\sin\dfrac{\theta}{2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}}{2^k\left(x^{1/2^{k-1}}-4x^{1/2^k}\cos^2\dfrac{\theta}{2^k}+2x^{1/2^k}+1\right)} \\
&\quad+\frac{x^{1/2^{k+1}}\sin\dfrac{\theta}{2^{k+1}}}{2^{k+1}\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)} \\
&\quad+\sum_{j=k+2}^{\infty}\frac{x^{1/2^j}\sin\dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)} \\
&= \frac{2x^{1/2^k}\sin\dfrac{\theta}{2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}}{2^k\left(x^{1/2^k}-2x^{1/2^{k+1}}\cos^2\dfrac{\theta}{2^{k+1}}+1\right)\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos^2\dfrac{\theta}{2^{k+1}}+1\right)} \\
&\quad+\frac{x^{1/2^{k+1}}\sin\dfrac{\theta}{2^{k+1}}}{2^{k+1}\left(x^{1/2^k}+2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)} \\
&\quad+\sum_{j=k+2}^{\infty}\frac{x^{1/2^j}\sin\dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}}+2x^{1/2^j}\cos\dfrac{\theta}{2^j}+1\right)} \\
&= \frac{x^{1/2^{k+1}}\sin\dfrac{\theta}{2^{k+1}}}{2^{k+1}\left(x^{1/2^k}-2x^{1/2^{k+1}}\cos\dfrac{\theta}{2^{k+1}}+1\right)}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{x^{1/2^{k+1}} \sin \dfrac{\theta}{2^{k+1}}}{2^{k+1}\left(x^{1/2^k} + 2x^{1/2^{k+1}} \cos \dfrac{\theta}{2^{k+1}} + 1\right)} \\
&+ \frac{x^{1/2^{k+1}} \sin \dfrac{\theta}{2^{k+1}}}{2^{k+1}\left(x^{1/2^k} + 2x^{1/2^{k+1}} \cos \dfrac{\theta}{2^{k+1}} + 1\right)} \\
&+ \sum_{j=k+2}^{\infty} \frac{x^{1/2^j} \sin \dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}} + 2x^{1/2^j} \cos \dfrac{\theta}{2^j} + 1\right)} \\
&= \frac{x^{1/2^{k+1}} \sin \dfrac{\theta}{2^{k+1}}}{2^{k+1}\left(x^{1/2^k} - 2x^{1/2^{k+1}} \cos \dfrac{\theta}{2^{k+1}} + 1\right)} \\
&+ \sum_{j=k+2}^{\infty} \frac{x^{1/2^j} \sin \dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}} + 2x^{1/2^j} \cos \dfrac{\theta}{2^j} + 1\right)} \\
&= f(x, \theta, k+1). \qquad (7.53)
\end{aligned}
$$

Applying the recursion in (7.53) for $k \in [0..n-1]$, we get

$$
\begin{aligned}
f(x, \theta, 0) &= \frac{x \sin \theta}{x^2 - 2x \cos \theta + 1} + \sum_{j=1}^{\infty} \frac{x^{1/2^j} \sin \dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}} + 2x^{1/2^j} \cos \dfrac{\theta}{2^j} + 1\right)} \\
&= f(x, \theta, n) \\
&= \frac{x^{1/2^n} \sin \dfrac{\theta}{2^n}}{2^n\left(x^{1/2^{n-1}} - 2x^{1/2^n} \cos \dfrac{\theta}{2^n} + 1\right)} + \sum_{j=n+1}^{\infty} \frac{x^{1/2^j} \sin \dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}} + 2x^{1/2^j} \cos \dfrac{\theta}{2^j} + 1\right)}. \qquad (7.54)
\end{aligned}
$$

Taking the limit as $n \to \infty$ in (7.54) and using (1.19), we get

$$
\frac{x \sin \theta}{x^2 - 2x \cos \theta + 1} + \sum_{j=1}^{\infty} \frac{x^{1/2^j} \sin \dfrac{\theta}{2^j}}{2^j\left(x^{1/2^{j-1}} + 2x^{1/2^j} \cos \dfrac{\theta}{2^j} + 1\right)}
$$

$$
\begin{aligned}
&= \lim_{n\to\infty} \frac{x^{1/2^n} \sin \dfrac{\theta}{2^n}}{2^n \left( x^{1/2^{n-1}} - 2x^{1/2^n} \cos \dfrac{\theta}{2^n} + 1 \right)} \\
&\quad + \lim_{n\to\infty} \sum_{j=n+1}^{\infty} \frac{x^{1/2^j} \sin \dfrac{\theta}{2^j}}{2^j \left( x^{1/2^{j-1}} + 2x^{1/2^j} \cos \dfrac{\theta}{2^j} + 1 \right)} \\
&= \lim_{n\to\infty} \frac{\dfrac{\theta x^{1/2^n}}{2^n}}{2^n \left( x^{1/2^{n-1}} - 2x^{1/2^n} \left( 1 - \dfrac{\theta^2}{2^{2n+1}} \right) + 1 \right)} \\
&\quad + \lim_{n\to\infty} \sum_{j=n+1}^{\infty} \frac{\dfrac{\theta x^{1/2^j}}{2^j}}{2^j \left( x^{1/2^{j-1}} + 2x^{1/2^j} \left( 1 - \dfrac{\theta^2}{2^{2j+1}} \right) + 1 \right)} \\
&= \lim_{n\to\infty} \frac{\theta x^{1/2^n}}{\theta^2 x^{1/2^n} + 2^{2n} \left( x^{1/2^n} - 1 \right)^2} + \lim_{n\to\infty} \sum_{j=n+1}^{\infty} \frac{\dfrac{\theta x^{1/2^j}}{2^j}}{2^j \left( x^{1/2^j} + 1 \right)^2 - \dfrac{\theta^2 x^{1/2^j}}{2^j}} \\
&= \frac{\theta}{\theta^2 + \log^2 x} + \theta \lim_{n\to\infty} \sum_{j=n+1}^{\infty} \frac{1}{2^{2j+2}} \\
&= \frac{\theta}{\theta^2 + \log^2 x} + \theta \lim_{n\to\infty} \frac{1}{2^{2n+4} \left( 1 - \dfrac{1}{4} \right)} \\
&= \frac{\theta}{\theta^2 + \log^2 x}.
\end{aligned}
$$

[1.88] Replace $\theta$ with $i\theta$ in [1.86].

[1.89] Replace $\theta$ with $i\theta$ in [1.87].

[1.90] Putting $\theta = \log y$ in [1.88], we get

$$\frac{\log x}{\log^2 x - \log^2 y} = \frac{x\left(\dfrac{y+y^{-1}}{2}\right) - 1}{x^2 - x\left(y+y^{-1}\right)+1} + \sum_{j=1}^{\infty} \frac{x^{1/2^j}\left(\dfrac{y^{1/2^j}+y^{-1/2^j}}{2}\right)+1}{2^j\left(x^{1/2^{j-1}} + x^{1/2^j}\left(y^{1/2^j}+y^{-1/2^j}\right)+1\right)}$$

$$\iff \frac{\log x^2}{\log^2 x - \log^2 y} = \frac{x\left(y^2+1\right)-2y}{x^2y - x\left(y^2+1\right)+y} + \sum_{j=1}^{\infty} \frac{x^{1/2^j}\left(y^{1/2^{j-1}}+1\right)+2y^{1/2^j}}{2^j\left(x^{1/2^{j-1}}y^{1/2^j} + x^{1/2^j}\left(y^{1/2^{j-1}}+1\right)+y^{1/2^j}\right)}$$

$$= \frac{y(xy-1)+x-y}{(x-y)(xy-1)} + \sum_{j=1}^{\infty} \frac{y^{1/2^j}\left(x^{1/2^j}y^{1/2^j}+1\right)+x^{1/2^j}+y^{1/2^j}}{2^j\left(x^{1/2^j}+y^{1/2^j}\right)\left(x^{1/2^j}y^{1/2^j}+1\right)}.$$

[1.91] Putting $\theta = \log y$ in [1.89], we get

$$\frac{\log y}{\log^2 x - \log^2 y} = \frac{x\left(\dfrac{y-y^{-1}}{2}\right)}{x^2 - x\left(y+y^{-1}\right)+1} + \sum_{j=1}^{\infty} \frac{x^{1/2^j}\left(\dfrac{y^{1/2^j}-y^{-1/2^j}}{2}\right)}{2^j\left(x^{1/2^{j-1}} + x^{1/2^j}\left(y^{1/2^j}+y^{-1/2^j}\right)+1\right)}$$

$$\iff \frac{\log y^2}{\log^2 x - \log^2 y} = \frac{x\left(y^2-1\right)}{x^2y - x\left(y^2+1\right)+y} + \sum_{j=1}^{\infty} \frac{x^{1/2^j}\left(y^{1/2^{j-1}}-1\right)}{2^j\left(x^{1/2^{j-1}}y^{1/2^j} + x^{1/2^j}\left(y^{1/2^{j-1}}+1\right)+y^{1/2^j}\right)}$$

$$= \frac{x(y^2-1)}{(x-y)(xy-1)} + \sum_{j=1}^{\infty} \frac{x^{1/2^j}\left(y^{1/2^{j-1}}-1\right)}{2^j\left(x^{1/2^j}+y^{1/2^j}\right)\left(x^{1/2^j}y^{1/2^j}+1\right)}.$$

[1.92] From [1.74], we have

$$\begin{aligned}
\frac{1}{x-1} &= \sum_{j=0}^{n-1} \frac{2^j}{x^{2^j}+1} + \frac{2^n}{x^{2^n}-1} \\
&\iff -1 - \frac{2}{x-1} = -1 - \sum_{j=0}^{n-1} \frac{2^{j+1}}{x^{2^j}+1} - \frac{2^{n+1}}{x^{2^n}-1} \\
&\iff \frac{1+x}{1-x} = -1 - \sum_{j=0}^{n-1} 2^j \frac{\left(x^{2^j}+1\right) - \left(x^{2^j}-1\right)}{x^{2^j}+1} \\
&\quad + 2^n \frac{\left(1-x^{2^n}\right) + \left(1+x^{2^n}\right)}{1-x^{2^n}} \\
&\iff \frac{1+x}{1-x} = -1 - \sum_{j=0}^{n-1} 2^j + \sum_{j=0}^{n-1} 2^j \frac{x^{2^j}-1}{x^{2^j}+1} + 2^n + 2^n \frac{1+x^{2^n}}{1-x^{2^n}} \\
&= -1 - \left(2^n - 1\right) + 2^n + \sum_{j=0}^{n-1} 2^j \frac{x^{2^j}-1}{x^{2^j}+1} + 2^n \frac{1+x^{2^j}}{1-x^{2^n}} \\
&= \sum_{j=0}^{n-1} 2^j \frac{x^{2^j}-1}{x^{2^j}+1} + 2^n \frac{1+x^{2^n}}{1-x^{2^n}}.
\end{aligned}$$

# Chapter 8
# Solutions to Trigonometrical Series




[2.1] Differentiating (2.12) with respect to $\theta$, we get

$$\begin{aligned}
&\frac{n\cos n\theta\sin\theta-\sin n\theta\cos\theta}{\sin^2\theta}\\
&=(-1)^{n/2}n\sum_{j=0}^{n/2-1}(-1)^j\frac{\prod_{k=1}^{j}\left(n^2-(2k)^2\right)}{(2j)!}\cos^{2j}\theta\sin\theta\\
\Longleftrightarrow\ &(-1)^{n/2}\frac{n\cos n\theta\sin\theta-\sin n\theta\cos\theta}{n\sin^3\theta}\\
&=\sum_{j=0}^{n/2-1}(-1)^j\frac{\prod_{k=1}^{j}\left(n^2-(2k)^2\right)}{(2j)!}\cos^{2j}\theta.
\end{aligned}$$

[2.2] Differentiating (2.15) with respect to $\theta$, we get

$$\begin{aligned}
&\frac{n\cos n\theta\sin\theta-\sin n\theta\cos\theta}{\sin^2\theta}\\
&=(-1)^{(n-1)/2}\sum_{j=1}^{(n-1)/2}(-1)^{j-1}\frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j-1)!}\cos^{2j-1}\theta\sin\theta
\end{aligned}$$

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A. Mishra, *Special Integrals*, University Texts in the Mathematical Sciences,
https://doi.org/10.1007/978-981-97-7514-9_8

$$\begin{aligned}\iff (-1)^{(n-1)/2}\frac{n\cos n\theta\sin\theta-\sin n\theta\cos\theta}{\sin^3\theta}\\ =\sum_{j=1}^{(n-1)/2}(-1)^{j-1}\frac{\prod_{k=0}^{j-1}\left(n^2-(2k+1)^2\right)}{(2j-1)!}\cos^{2j-1}\theta.\end{aligned}$$

[2.3] Replace $\theta$ with $\pi/2-\theta$ in [2.2].

[2.4] Replace $\theta$ with $\pi/2-\theta$ in [2.1].

[2.5] Replacing $\theta$ with $\theta/2$ in [2.21], we get

$$\begin{aligned}\frac{\tan\frac{\theta}{2}}{4\theta}&=\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2\pi^2-\theta^2}\\ \iff\frac{\theta}{2}\tan\frac{\theta}{2}&=2\theta^2\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2\pi^2-\theta^2}.\end{aligned}\tag{8.1}$$

Replacing $\theta$ with $\theta/2$ in [2.20], we get

$$\begin{aligned}\frac{2-\theta\cot\frac{\theta}{2}}{\theta^2}&=4\sum_{j=0}^{\infty}\frac{1}{(2j)^2\pi^2-\theta^2}\\ \iff\frac{\theta}{2}\cot\frac{\theta}{2}&=1-2\theta^2\sum_{j=0}^{\infty}\frac{1}{(2j)^2\pi^2-\theta^2}.\end{aligned}\tag{8.2}$$

Adding (8.1) and (8.2), we get

$$\frac{\theta}{\sin\theta}=1+2\theta^2\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{j^2\pi^2-\theta^2}.\tag{8.3}$$

Putting $\theta=\pi/x$ in (8.3), we get

$$\frac{\pi}{x\sin\frac{\pi}{x}}=1+2\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{j^2x^2-1}.$$

[2.6] Using the binomial theorem, we have

$$\begin{aligned}
&\left(1-\sin^2\theta\right)^{-3/2} = \sum_{j=0}^{\infty}(-1)^{2j}\frac{\sin^{2j}\theta}{j!}\prod_{k=1}^{j}\left(\frac{3}{2}+k-1\right)\\
\iff &\sec^3\theta = \sum_{j=0}^{\infty}\frac{\sin^{2j}\theta}{2^j\,j!}\prod_{k=1}^{j}(2k+1)\\
&= \sum_{j=0}^{\infty}\frac{\sin^{2j}\theta}{2^{2j}(j!)^2}\prod_{k=1}^{j}(2k)(2k+1)\\
&= \sum_{j=0}^{\infty}\frac{2j+1}{2^{2j}}\binom{2j}{j}\sin^{2j}\theta\\
&= \sum_{j=1}^{\infty}\frac{2j-1}{2^{2j-2}}\binom{2j-2}{j-1}\sin^{2j-2}\theta\\
&= \sum_{j=1}^{\infty}\frac{2j-1}{2^{2j-2}}\cdot\frac{j^2}{(2j-1)(2j)}\binom{2j}{j}\sin^{2j-2}\theta\\
&= \sum_{j=1}^{\infty}\frac{j}{2^{2j-1}}\binom{2j}{j}\sin^{2j-2}\theta.
\end{aligned}$$

[2.7] Using the binomial theorem, we have

$$\begin{aligned}
&\left(1-\sin^2\theta\right)^{-1/2} = \sum_{j=0}^{\infty}(-1)^{2j}\frac{\sin^{2j}\theta}{j!}\prod_{k=1}^{j}\left(\frac{1}{2}+k-1\right)\\
\iff &\sec\theta = \sum_{j=0}^{\infty}\frac{\sin^{2j}\theta}{2^j\,j!}\prod_{k=1}^{j}(2k-1)\\
&= \sum_{j=0}^{\infty}\frac{\sin^{2j}\theta}{2^{2j}(j!)^2}\prod_{k=1}^{j}(2k-1)(2k)\\
&= \sum_{j=0}^{\infty}\binom{2j}{j}\frac{\sin^{2j}\theta}{2^{2j}}.
\end{aligned}$$

[2.8] Multiplying [2.7] by $\cos\theta$, we get

$$1 = \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j}\theta\cos\theta}{2^{2j}}. \tag{8.4}$$

Integrating (8.4) with respect to $\theta$ from 0 to $\theta$, we get

$$\begin{aligned} \theta|_0^\theta &= \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+1}\theta}{2^{2j}(2j+1)}\bigg|_0^\theta \\ \Longleftrightarrow \theta &= \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+1}\theta}{2^{2j}(2j+1)}. \end{aligned}$$

[2.9] Putting $x = \tan\theta$ in (1.13), multiplying it by $2\tan\theta$, and using [1.37], we get

$$\begin{aligned} 2\theta\tan\theta &= \sum_{l=0}^{\infty}(-1)^l \frac{2}{2l+1}\tan^{2l+2}\theta \\ &= \sum_{l=0}^{\infty}(-1)^l \frac{2}{2l+1}\sin^{2l+2}\theta\left(1-\sin^2\theta\right)^{-(l+1)} \\ &= \sum_{l=0}^{\infty}(-1)^l \frac{2}{2l+1}\sin^{2l+2}\theta\sum_{k=0}^{\infty}(-1)^{2k}\frac{(l+k)!}{l!k!}\sin^{2k}\theta \\ &= \sum_{l=0}^{\infty}(-1)^l \frac{2}{2l+1}\sin^{2l+2}\theta\sum_{k=0}^{\infty}\binom{l+k}{l}\sin^{2k}\theta \\ &= \sum_{j=1}^{\infty}\left(\sum_{l=0}^{j-1}(-1)^l\frac{2}{2l+1}\binom{j-1}{l}\right)\sin^{2j}\theta \\ &= \sum_{j=1}^{\infty}2\frac{2^{2j-2}}{2j-1}\binom{2j-2}{j-1}^{-1}\sin^{2j}\theta \\ &= \sum_{j=1}^{\infty}\frac{2^{2j-1}\cdot 2j}{2j(2j-1)}\frac{((j-1)!)^2}{(2j-2)!}\sin^{2j}\theta \\ &= \sum_{j=1}^{\infty}\frac{2^{2j}(j!)^2}{j(2j)!}\sin^{2j}\theta \\ &= \sum_{j=1}^{\infty}\frac{2^{2j}}{j}\binom{2j}{j}^{-1}\sin^{2j}\theta. \end{aligned}$$

[2.10] The identity is true for $\theta = 0$. Assuming $\theta \neq 0$, multiplying [2.9] by $2\cot\theta$, we get

$$4\theta = \sum_{j=1}^{\infty} \frac{2^{2j+1}}{j} \binom{2j}{j}^{-1} \sin^{2j-1}\theta \cos\theta. \tag{8.5}$$

Integrating (8.5) with respect to $\theta$ from 0 to $\theta$, we get

$$2\theta^2\Big|_0^\theta = \sum_{j=1}^{\infty} \frac{2^{2j+1}}{2j^2} \binom{2j}{j}^{-1} \sin^{2j}\theta \Bigg|_0^\theta$$
$$\Longleftrightarrow 2\theta^2 = \sum_{j=1}^{\infty} \frac{2^{2j}}{j^2} \binom{2j}{j}^{-1} \sin^{2j}\theta.$$

[2.11] Multiplying [2.8] by $\cos\theta$, we get

$$\theta\cos\theta = \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+1}\theta\cos\theta}{2^{2j}(2j+1)}. \tag{8.6}$$

Integrating (8.6) with respect to $\theta$ from 0 to $\theta$, we get

$$\theta\sin\theta + \cos\theta\Big|_0^\theta = \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)} \Bigg|_0^\theta$$
$$\Longleftrightarrow \theta\sin\theta + \cos\theta = 1 + \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)}.$$

[2.12] Multiplying [2.9] by $(1/4)\cos\theta$, we get

$$\frac{\theta}{2}\sin\theta = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j} \binom{2j}{j}^{-1} \sin^{2j}\theta\cos\theta. \tag{8.7}$$

Integrating (8.7) with respect to $\theta$ from 0 to $\theta$, we get

$$\frac{\sin\theta - \theta\cos\theta}{2}\Bigg|_0^\theta = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j(2j+1)} \binom{2j}{j}^{-1} \sin^{2j+1}\theta \Bigg|_0^\theta$$
$$\Longleftrightarrow \frac{\sin\theta - \theta\cos\theta}{2} = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j(2j+1)} \binom{2j}{j}^{-1} \sin^{2j+1}\theta.$$

[2.13] Multiplying [2.12] by $\cos\theta$, we get

$$\sum_{j=1}^{\infty} \frac{2^{2j-2}}{j(2j+1)} \binom{2j}{j}^{-1} \sin^{2j+1}\theta\cos\theta = \frac{\sin\theta\cos\theta - \theta\cos^2\theta}{2} = \frac{\sin 2\theta - \theta - \theta\cos 2\theta}{4}. \tag{8.8}$$

Integrating (8.8) with respect to $\theta$ from 0 to $\theta$, we get

$$\frac{-\frac{3}{2}\cos 2\theta - \theta^2 - \theta\sin 2\theta}{8}\bigg|_0^{\theta} = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j(2j+1)(2j+2)} \binom{2j}{j}^{-1} \sin^{2j+2}\theta\bigg|_0^{\theta}$$

$$\iff \frac{3\sin^2\theta - \theta^2 - \theta\sin 2\theta}{8} = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j(2j+1)(2j+2)} \binom{2j}{j}^{-1} \sin^{2j+2}\theta.$$

[2.14] Differentiating [2.9] with respect to $\theta$, we get

$$2\theta\sec^2\theta + 2\tan\theta = \sum_{j=1}^{\infty} 2^{2j+1} \binom{2j}{j}^{-1} \sin^{2j-1}\theta\cos\theta$$

$$\iff \theta\tan\theta\sec^2\theta + \tan^2\theta = \sum_{j=1}^{\infty} 2^{2j} \binom{2j}{j}^{-1} \sin^{2j}\theta$$

$$\iff \theta\tan\theta\sec^2\theta + \sec^2\theta - 1 = \sum_{j=1}^{\infty} 2^{2j} \binom{2j}{j}^{-1} \sin^{2j}\theta$$

$$\iff \sec^2\theta(1 + \theta\tan\theta) = \sum_{j=0}^{\infty} 2^{2j} \binom{2j}{j}^{-1} \sin^{2j}\theta.$$

[2.15] Multiplying the second part of [2.6] by $\sin\theta$, we get

$$\frac{\sin\theta}{\cos^3\theta} = \sum_{j=0}^{\infty} \frac{2j+1}{2^{2j}} \binom{2j}{j} \sin^{2j+1}\theta. \tag{8.9}$$

Differentiating (8.9) with respect to $\theta$, we get

$$\frac{\cos^4\theta + 3\cos^2\theta\sin^2\theta}{\cos^6\theta} = \sum_{j=0}^{\infty} \frac{(2j+1)^2}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta\cos\theta$$

$$\Longleftrightarrow \sum_{j=0}^{\infty} \frac{(2j+1)^2}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta = \frac{\cos^2\theta + 3\sin^2\theta}{\cos^5\theta}$$

$$= \frac{1 + 2\sin^2\theta}{\cos^5\theta}.$$

[2.16] Multiplying [2.15] by $\sin\theta$, we get

$$\frac{\sin\theta + 2\sin^3\theta}{\cos^5\theta} = \sum_{j=0}^{\infty} \frac{(2j+1)^2}{2^{2j}} \binom{2j}{j} \sin^{2j+1}\theta. \tag{8.10}$$

Differentiating (8.10) with respect to $\theta$, we get

$$\frac{\left(\cos\theta + 6\sin^2\theta\cos\theta\right)\cos^5\theta + 5\cos^4\theta\sin\theta\left(\sin\theta + 2\sin^3\theta\right)}{\cos^{10}\theta}$$

$$= \sum_{j=0}^{\infty} \frac{(2j+1)^3}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta\cos\theta$$

$$\Longleftrightarrow \sum_{j=0}^{\infty} \frac{(2j+1)^3}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta$$

$$= \frac{\cos^2\theta + 6\sin^2\theta\cos^2\theta + 5\sin^2\theta + 10\sin^4\theta}{\cos^7\theta}$$

$$= \frac{1 + 2\sin^2\theta\left(5 + 2\sin^2\theta\right)}{\cos^7\theta}.$$

[2.17] Multiplying [2.16] by $\sin\theta$, we get

$$\frac{\sin\theta + 10\sin^3\theta + 4\sin^5\theta}{\cos^7\theta} = \sum_{j=0}^{\infty} \frac{(2j+1)^3}{2^{2j}} \binom{2j}{j} \sin^{2j+1}\theta. \tag{8.11}$$

Differentiating (8.11) with respect to $\theta$, we get

$$\begin{aligned}
&\sum_{j=0}^{\infty} \frac{(2j+1)^4}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta \cos\theta \\
&= \left(\left(\cos\theta + 30\sin^2\theta\cos\theta + 20\sin^4\theta\cos\theta\right)\cos^7\theta \right. \\
&\quad \left. +7\cos^6\theta\sin\theta\left(\sin\theta + 10\sin^3\theta + 4\sin^5\theta\right)\right) / \cos^{14}\theta \\
\Longleftrightarrow &\sum_{j=0}^{\infty} \frac{(2j+1)^4}{2^{2j}} \binom{2j}{j} \sin^{2j}\theta \\
&= \frac{\cos^2\theta\left(1 + 30\sin^2\theta + 20\sin^4\theta\right) + 7\sin^2\theta\left(1 + 10\sin^2\theta + 4\sin^4\theta\right)}{\cos^9\theta} \\
&= \frac{1 + 4\sin^2\theta\left(9 + 15\sin^2\theta + 2\sin^4\theta\right)}{\cos^9\theta}.
\end{aligned}$$

[2.18] Multiplying [2.10] by $\cos\theta$, we get

$$2\theta^2\cos\theta = \sum_{j=1}^{\infty} \frac{2^{2j}}{j^2} \binom{2j}{j}^{-1} \sin^{2j}\theta\cos\theta. \tag{8.12}$$

Integrating (8.12) with respect to $\theta$ from 0 to $\theta$, we get

$$\begin{aligned}
&2\theta^2\sin\theta + 4\theta\cos\theta - 4\sin\theta\Big|_0^{\theta} = \sum_{j=1}^{\infty} \frac{2^{2j}}{j^2(2j+1)} \binom{2j}{j}^{-1} \sin^{2j+1}\theta\Big|_0^{\theta} \\
\Longleftrightarrow &\frac{\theta^2}{2}\sin\theta + \theta\cos\theta - \sin\theta = \sum_{j=1}^{\infty} \frac{2^{2j-2}}{j^2(2j+1)} \binom{2j}{j}^{-1} \sin^{2j+1}\theta.
\end{aligned}$$

[2.19] Multiplying [2.11] by $\cot\theta$, we get

$$\begin{aligned}
&\theta\cos\theta + \frac{\cos^2\theta}{\sin\theta} = \frac{\cos\theta}{\sin\theta} + \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+1}\theta\cos\theta}{2^{2j}(2j+1)(2j+2)} \\
\Longleftrightarrow &\sum_{j=0}^{\infty} \binom{2j}{j} \frac{\sin^{2j+1}\theta\cos\theta}{2^{2j}(2j+1)(2j+2)} = \theta\cos\theta - \sin\theta + \frac{1-\cos\theta}{\sin\theta} \\
&= \theta\cos\theta - \sin\theta + \tan\frac{\theta}{2}.
\end{aligned} \tag{8.13}$$

Integrating (8.13) with respect to $\theta$ from 0 to $\theta$, we get

$$\theta \sin\theta + 2\cos\theta - 2\log\cos\frac{\theta}{2}\bigg|_0^\theta = \sum_{j=0}^{\infty}\binom{2j}{j}\frac{\sin^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)^2}\bigg|_0^\theta$$

$$\Longleftrightarrow \sum_{j=0}^{\infty}\binom{2j}{j}\frac{\sin^{2j+2}\theta}{2^{2j}(2j+1)(2j+2)^2} = \theta\sin\theta + 2\cos\theta - 2 - 2\log\cos\frac{\theta}{2}$$

$$= \theta\sin\theta + 2\cos\theta - \log\left(\frac{e^2}{2}(1+\cos\theta)\right).$$

[2.20] Taking log of (2.22), we get

$$\log\sin\theta = \log\theta + \sum_{j=1}^{\infty}\log\left(1 - \frac{\theta^2}{j^2\pi^2}\right). \tag{8.14}$$

Differentiating (8.14) with respect to $\theta$, we get

$$\cot\theta = \frac{1}{\theta} - \sum_{j=1}^{\infty}\frac{2\theta}{j^2\pi^2 - \theta^2}$$

$$\Longleftrightarrow \frac{1-\theta\cot\theta}{2\theta^2} = \sum_{j=1}^{\infty}\frac{1}{j^2\pi^2 - \theta^2}.$$

[2.21] Taking log of (2.29), we get

$$\log\cos\theta = \sum_{j=0}^{\infty}\log\left(1 - \frac{4\theta^2}{(2j+1)^2\pi^2}\right). \tag{8.15}$$

Differentiating (8.15) with respect to $\theta$, we get

$$-\tan\theta = -\sum_{j=0}^{\infty}\frac{8\theta}{(2j+1)^2\pi^2 - 4\theta^2}$$

$$\Longleftrightarrow \frac{\tan\theta}{8\theta} = \sum_{j=0}^{\infty}\frac{1}{(2j+1)^2\pi^2 - 4\theta^2}.$$

[2.22] Differentiating [2.20] with respect to $\theta$, we get

$$\begin{aligned}
&\frac{\left(\theta \csc^2\theta - \cot\theta\right)\theta^2 - 2\theta(1-\theta\cot\theta)}{2\theta^4} = \sum_{j=1}^{\infty} \frac{2\theta}{\left(j^2\pi^2 - \theta^2\right)^2} \\
\iff &\sum_{j=1}^{\infty} \frac{1}{\left(j^2\pi^2 - \theta^2\right)^2} = \frac{\theta^2 - \theta\cos\theta\sin\theta - 2\sin^2\theta + 2\theta\cos\theta\sin\theta}{4\theta^4\sin^2\theta} \\
&= \frac{2\theta^2 + \theta\sin 2\theta - 4\sin^2\theta}{8\theta^4\sin^2\theta}.
\end{aligned}$$

[2.23] Differentiating [2.21] with respect to $\theta$, we get

$$\begin{aligned}
&\frac{\theta\sec^2\theta - \tan\theta}{8\theta^2} = \sum_{j=0}^{\infty} \frac{8\theta}{\left((2j+1)^2\pi^2 - 4\theta^2\right)^2} \\
\iff &\sum_{j=0}^{\infty} \frac{1}{\left((2j+1)^2\pi^2 - 4\theta^2\right)^2} = \frac{\theta - \sin\theta\cos\theta}{64\theta^3\cos^2\theta} \\
&= \frac{2\theta - \sin 2\theta}{128\theta^3\cos^2\theta}.
\end{aligned}$$

[2.24] The sum of the GP $\left((-1)^{j-1}\tan^{2j-1}\theta\right)_{j=1}^{\infty}$ is given by

$$\sum_{j=1}^{\infty}(-1)^{j-1}\tan^{2j-1}\theta = \frac{\tan\theta}{1+\tan^2\theta} = \sin\theta\cos\theta. \tag{8.16}$$

Multiplying (8.16) by $\tan\theta$, we get

$$\sin^2\theta = \sum_{j=1}^{\infty}(-1)^{j-1}\tan^{2j}\theta. \tag{8.17}$$

Differentiating (8.17) with respect to $\theta$, we get

$$\begin{aligned}
&2\sin\theta\cos\theta = \sum_{j=1}^{\infty}(-1)^{j-1}2j\tan^{2j-1}\theta\sec^2\theta \\
\iff &\sin\theta\cos^3\theta = \sum_{j=1}^{\infty}(-1)^{j-1}j\tan^{2j-1}\theta.
\end{aligned} \tag{8.18}$$

Multiplying (8.18) by $\tan\theta$, we get

$$\sin^2\theta\cos^2\theta = \sum_{j=1}^{\infty}(-1)^{j-1} j \tan^{2j}\theta. \tag{8.19}$$

Differentiating (8.19) with respect to $\theta$, we get

$$\begin{aligned} & 2\left(\sin\theta\cos^3\theta - \sin^3\theta\cos\theta\right) = \sum_{j=1}^{\infty}(-1)^{j-1} 2j^2 \tan^{2j-1}\theta\sec^2\theta \\ \Longleftrightarrow & \sum_{j=1}^{\infty}(-1)^{j-1} j^2 \tan^{2j-1}\theta = \sin\theta\cos^3\theta\cos 2\theta \\ & = \frac{1}{2}\cos^2\theta\sin 2\theta\cos 2\theta \\ & = \frac{1}{4}\sin 4\theta\cos^2\theta. \end{aligned}$$

[2.25] Multiplying [2.24] by $\tan\theta$, we get

$$\begin{aligned} \sum_{j=1}^{\infty}(-1)^{j-1} j^2 \tan^{2j}\theta &= \frac{1}{4}\sin 4\theta\sin\theta\cos\theta \\ &= \frac{1}{8}\sin 4\theta\sin 2\theta. \end{aligned} \tag{8.20}$$

Differentiating (8.20) with respect to $\theta$, we get

$$\begin{aligned} & \sum_{j=1}^{\infty}(-1)^{j-1} 2j^3 \tan^{2j-1}\theta\sec^2\theta = \frac{1}{4}\left(2\cos 4\theta\sin 2\theta + \sin 4\theta\cos 2\theta\right) \\ \Longleftrightarrow & \sum_{j=1}^{\infty}(-1)^{j-1} j^3 \tan^{2j-1}\theta = \frac{1}{8}\cos^2\theta\left(\sin 6\theta + \sin 2\theta\cos 4\theta\right). \end{aligned}$$

[2.26] Multiplying [2.25] by $\tan\theta$, we get

$$\begin{aligned}\sum_{j=1}^{\infty}(-1)^{j-1}j^3\tan^{2j}\theta &= \frac{1}{8}\sin\theta\cos\theta\,(\sin 6\theta+\sin 2\theta\cos 4\theta)\\ &= \frac{1}{16}\sin 2\theta\,(\sin 6\theta+\sin 2\theta\cos 4\theta)\\ &= \frac{1}{32}(\cos 4\theta-\cos 8\theta+(1-\cos 4\theta)\cos 4\theta)\\ &= \frac{1}{32}\left(2\cos 4\theta-\cos 8\theta-\cos^2 4\theta\right)\\ &= \frac{1}{32}\left(2\cos 4\theta-\frac{3}{2}\cos 8\theta-\frac{1}{2}\right).\end{aligned} \tag{8.21}$$

Differentiating (8.21) with respect to $\theta$, we get

$$\begin{aligned}&\sum_{j=1}^{\infty}(-1)^{j-1}2j^4\tan^{2j-1}\theta\sec^2\theta = \frac{1}{8}\,(3\sin 8\theta-2\sin 4\theta)\\ \Longleftrightarrow\ &\sum_{j=1}^{\infty}(-1)^{j-1}j^4\tan^{2j-1}\theta = \frac{1}{16}\cos^2\theta\,(3\sin 8\theta-2\sin 4\theta)\,.\end{aligned}$$

[2.27] Replace $\theta$ with $i\theta$ in [2.1].

[2.28] Replace $\theta$ with $i\theta$ in [2.2].

[2.29] Replace $\theta$ with $i\theta$ in [2.3].

[2.30] Replace $\theta$ with $i\theta$ in [2.4].

[2.31] Replace $x$ with $ix$ in [2.5].

[2.32] Replace $\theta$ with $i\theta$ in [2.6].

[2.33] Replace $\theta$ with $i\theta$ in [2.7].

[2.34] Replace $\theta$ with $i\theta$ in [2.8].

[2.35] Replace $\theta$ with $i\theta$ in [2.9].

[2.36] Replace $\theta$ with $i\theta$ in [2.10].

[2.37] Replace $\theta$ with $i\theta$ in [2.11].

[2.38] Replace $\theta$ with $i\theta$ in [2.12].

[2.39] Replace $\theta$ with $i\theta$ in [2.13].

[2.40] Replace $\theta$ with $i\theta$ in [2.14].

[2.41] Replace $\theta$ with $i\theta$ in [2.15].

[2.42] Replace $\theta$ with $i\theta$ in [2.16].

[2.43] Replace $\theta$ with $i\theta$ in [2.17].

[2.44] Replace $\theta$ with $i\theta$ in [2.18].

[2.45] Replace $\theta$ with $i\theta$ in [2.19].

[2.46] Replace $\theta$ with $i\theta$ in [2.20].

[2.47] Replace $\theta$ with $i\theta$ in [2.21].

[2.48] Replace $\theta$ with $i\theta$ in [2.22].

[2.49] Replace $\theta$ with $i\theta$ in [2.23].

[2.50] Replace $\theta$ with $i\theta$ in [2.24].

[2.51] Replace $\theta$ with $i\theta$ in [2.25].

[2.52] Replace $\theta$ with $i\theta$ in [2.26].

# Chapter 9
# Solutions to Bessel Functions




[3.1] Using the binomial theorem and (1.11), we have

$$\begin{aligned}
\int_0^1 \frac{1-(1-t)^n}{t}\,dt &= \int_0^1 \left(1-\sum_{j=0}^{n}(-1)^j t^j \binom{n}{j}\right)\frac{dt}{t} \\
&= \int_0^1 \left(\sum_{j=1}^{n}(-1)^{j-1} t^{j-1}\binom{n}{j}\right) dt \\
&= \sum_{j=1}^{n}\frac{(-1)^{j-1}t^j}{j}\binom{n}{j}\Bigg|_0^1 \\
&= \sum_{j=1}^{n}\frac{(-1)^{j-1}}{j}\binom{n}{j} \\
&= \sum_{j=1}^{n}\frac{1}{j} \\
&= H(n).
\end{aligned}$$

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[3.2] From [3.1], we have

$$\begin{aligned}
H(n) &= \int_0^1 \frac{1-(1-t)^n}{t}\,dt \\
&= \int_0^n \frac{1-(1-t/n)^n}{t/n}\,d\left(\frac{t}{n}\right) \\
&= \int_0^n \frac{1-(1-t/n)^n}{t}\,dt \\
&= \int_0^1 \frac{1-(1-t/n)^n}{t}\,dt + \int_1^n \frac{1-(1-t/n)^n}{t}\,dt \qquad (9.1) \\
&= \int_0^1 \frac{1-(1-t/n)^n}{t}\,dt + \log x|_1^n - \int_1^n \frac{(1-t/n)^n}{t}\,dt \\
&= \int_0^1 \frac{1-(1-t/n)^n}{t}\,dt + \log n - \int_1^n \frac{(1-t/n)^n}{t}\,dt \\
\Longrightarrow H(n) - \log n &= \int_0^1 \frac{1-(1-t/n)^n}{t}\,dt - \int_1^n \frac{(1-t/n)^n}{t}\,dt.
\end{aligned}$$

Taking the limit as $n \to \infty$ in (9.1) and using (3.3), we get the result.

[3.3] Take the limit as $n \to \infty$ in [3.2].

[3.4] From (3.3) and (3.4), we have

$$\begin{aligned}
\Gamma(1) &= \lim_{n\to\infty} \frac{e^{\sum_{j=1}^n 1/j}}{e^{\sum_{j=1}^n 1/j - \log n} \prod_{j=1}^n \left(1+\frac{1}{j}\right)} \\
&= \lim_{n\to\infty} \frac{n}{\prod_{j=1}^n \frac{j+1}{j}} \\
&= \lim_{n\to\infty} \frac{n}{n+1} \\
&= 1.
\end{aligned}$$

[3.5] Taking log of (3.4), we get

$$\log\Gamma(z) = -\log z - \gamma z + \sum_{j=1}^{\infty} \left(\frac{z}{j} - \log\left(1+\frac{z}{j}\right)\right). \qquad (9.2)$$

Differentiating (9.2) with respect to $z$, we get

$$\begin{aligned}\frac{\Gamma'(z)}{\Gamma(z)} &= -\frac{1}{z} - \gamma + \sum_{j=1}^{\infty}\left(\frac{1}{j} - \frac{1/j}{1+z/j}\right) \\ &= -\frac{1}{z} - \gamma + \sum_{j=1}^{\infty}\left(\frac{1}{j} - \frac{1}{j+z}\right).\end{aligned} \tag{9.3}$$

Putting $z = 1$ in (9.3) and using [3.4], we get

$$\begin{aligned}\Gamma'(1) &= -1 - \gamma + \sum_{j=1}^{\infty}\left(\frac{1}{j} - \frac{1}{j+1}\right) \\ &= -1 - \gamma + 1 \\ &= -\gamma.\end{aligned}$$

[3.6] The result follows from [3.4] and (3.6).

[3.7] From (3.4), (3.6), and (2.22), we have

$$\begin{aligned}&\Gamma^2\left(\frac{1}{2}\right) = \Gamma\left(1-\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right) \\ &= -\frac{1}{2}\Gamma\left(\frac{1}{2}\right)\Gamma\left(-\frac{1}{2}\right) \\ &= -\frac{1}{2}\left(2e^{-\gamma/2}\prod_{j=1}^{\infty}e^{1/(2j)}\left(1+\frac{1}{2j}\right)^{-1}\right)\left(-2e^{\gamma/2}\prod_{j=1}^{\infty}e^{-1/(2j)}\left(1-\frac{1}{2j}\right)^{-1}\right) \\ &= 2\prod_{j=1}^{\infty}\left(1-\frac{1}{4j^2}\right)^{-1} \\ &= 2\prod_{j=1}^{\infty}\left(1-\frac{(\pi/2)^2}{j^2\pi^2}\right)^{-1} \\ &= 2\frac{\pi/2}{\sin(\pi/2)} \\ &= \pi \\ &\implies \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.\end{aligned}$$

[3.8] From (3.5), we have

$$\begin{aligned}\Gamma(z) &= \frac{1}{z}\lim_{n\to\infty}\prod_{j=1}^{n-1}\left(1+\frac{1}{j}\right)^{z}\left(1+\frac{z}{j}\right)^{-1}\\ &= \frac{1}{z}\lim_{n\to\infty}\prod_{j=1}^{n-1}\left(\frac{j+1}{j}\right)^{z}\left(\frac{j}{z+j}\right)\\ &= \frac{1}{z}\lim_{n\to\infty} n^{z}\prod_{j=1}^{n-1}\frac{j}{z+j}.\end{aligned}$$

[3.9] Replacing $t$ with $-1/t$ in (3.7), we get

$$\begin{aligned}e^{z(-1/t+t)/2} &= \sum_{n=-\infty}^{\infty}(-1)^n J_n(z)t^{-n}\\ &= \sum_{n=-\infty}^{\infty}(-1)^{-n} J_{-n}(z)t^{n} \qquad (9.4)\\ &= \sum_{n=-\infty}^{\infty}(-1)^{n} J_{-n}(z)t^{n}.\end{aligned}$$

We get [3.9] by equating the coefficients of $t^n$ in (3.7) and (9.4).

[3.10] Put $n = 0$ in (3.12) and use [3.9].

[3.11] The case of $n = 0$ is [3.10]. Assuming $n \neq 0$, using (3.10) and (3.12), we have

$$\begin{aligned}\frac{d}{dz}\left(z^n J_n(z)\right) &= nz^{n-1}J_n(z) + z^n J_n'(z)\\ &= nz^{n-1}\frac{z}{2n}\left(J_{n-1}(z)+J_{n+1}(z)\right) + \frac{z^n}{2}\left(J_{n-1}(z)-J_{n+1}(z)\right)\\ &= \frac{z^n}{2}\left(2J_{n-1}(z)\right)\\ &= z^n J_{n-1}(z).\end{aligned}$$

[3.12] The case of $n = 0$ is [3.10]. Assuming $n \neq 0$, using (3.10) and (3.12), we have

$$\begin{aligned}\frac{d}{dz}\left(z^{-n}J_n(z)\right) &= -nz^{-n-1}J_n(z) + z^{-n}J_n'(z)\\ &= -nz^{-n-1}\frac{z}{2n}\left(J_{n-1}(z) + J_{n+1}(z)\right) + \frac{z^{-n}}{2}\left(J_{n-1}(z) - J_{n+1}(z)\right)\\ &= \frac{z^{-n}}{2}\left(2J_{n+1}(z)\right)\\ &= z^{-n}J_{n+1}(z).\end{aligned}$$

[3.13] Putting $t = e^{i\theta}$ in (3.7), we get

$$\begin{aligned}e^{z(e^{i\theta}-e^{-i\theta})/2} &= \sum_{j=-\infty}^{\infty} J_j(z)e^{ij\theta}\\ \iff e^{iz\sin\theta} &= \sum_{j=-\infty}^{\infty} J_j(z)e^{ij\theta}.\end{aligned} \tag{9.5}$$

Putting $t = e^{-i\theta}$ in (3.7), we get

$$\begin{aligned}e^{z(e^{-i\theta}-e^{i\theta})/2} &= \sum_{j=-\infty}^{\infty} J_j(z)e^{-ij\theta}\\ \iff e^{-iz\sin\theta} &= \sum_{j=-\infty}^{\infty} J_j(z)e^{-ij\theta}.\end{aligned} \tag{9.6}$$

Taking the average of (9.5) and (9.6), and using [3.9], we get

$$\begin{aligned}\frac{e^{iz\sin\theta} + e^{-iz\sin\theta}}{2} &= \sum_{j=-\infty}^{\infty} J_j(z)\frac{e^{ij\theta} + e^{ij\theta}}{2}\\ \iff \cos(z\sin\theta) &= \sum_{j=-\infty}^{\infty} J_j(z)\cos j\theta\\ &= J_0(z) + \sum_{j=1}^{\infty}\left(J_j(z) + J_{-j}(z)\right)\cos j\theta\\ &= J_0(z) + \sum_{j=1}^{\infty}\left(1 + (-1)^j\right)J_j(z)\cos j\theta\\ &= J_0(z) + 2\sum_{j=1}^{\infty} J_{2j}(z)\cos 2j\theta.\end{aligned}$$

[3.14] Subtracting (9.6) from (9.5), dividing the result by $2i$, and using [3.9], we get

$$\begin{aligned}
&\frac{e^{iz\sin\theta} - e^{-iz\sin\theta}}{2i} = \sum_{j=-\infty}^{\infty} J_j(z)\frac{e^{ij\theta} - e^{ij\theta}}{2i} \\
\iff &\sin(z\sin\theta) = \sum_{j=-\infty}^{\infty} J_j(z)\sin j\theta \\
&= \sum_{j=1}^{\infty} \big(J_j(z) - J_{-j}(z)\big)\sin j\theta \\
&= \sum_{j=1}^{\infty} \big(1 - (-1)^j\big) J_j(z)\sin j\theta \\
&= 2\sum_{j=0}^{\infty} J_{2j+1}(z)\sin(2j+1)\theta.
\end{aligned}$$

[3.15] Replace $\theta$ with $\pi/2 - \theta$ in [3.13].

[3.16] Replace $\theta$ with $\pi/2 - \theta$ in [3.14].

[3.17] Differentiating [3.15] with respect to $\theta$, we get

$$\begin{aligned}
&(-z\sin\theta)(-\sin(z\cos\theta)) = 4\sum_{j=1}^{\infty}(-1)^{j-1} j J_{2j}(z)\sin 2j\theta \\
\iff &z\sin\theta\sin(z\cos\theta) = 4\sum_{j=1}^{\infty}(-1)^{j-1} j J_{2j}(z)\sin 2j\theta.
\end{aligned} \tag{9.7}$$

Differentiating (9.7) with respect to $\theta$, we get

$$\begin{aligned}
&z\cos\theta\sin(z\cos\theta) + (z\sin\theta)(-z\sin\theta)\cos(z\cos\theta) \\
&\quad = 8\sum_{j=1}^{\infty}(-1)^{j-1} j^2 J_{2j}(z)\cos 2j\theta \\
\iff &z\cos\theta\sin(z\cos\theta) - z^2\sin^2\theta\cos(z\cos\theta) \\
&\quad = 2\sum_{j=1}^{\infty}(-1)^{j-1}(2j)^2 J_{2j}(z)\cos 2j\theta.
\end{aligned} \tag{9.8}$$

We get the result by putting $\theta = 0$ in (9.8).

[3.18] Differentiating [3.16] with respect to $\theta$, we get

$$\begin{aligned}(-z\sin\theta)\cos(z\cos\theta) &= 2\sum_{j=0}^{\infty}(-1)^{j-1}(2j+1)J_{2j+1}(z)\sin(2j+1)\theta \\ \Longleftrightarrow z\sin\theta\cos(z\cos\theta) &= 2\sum_{j=0}^{\infty}(-1)^{j}(2j+1)J_{2j+1}(z)\sin(2j+1)\theta.\end{aligned} \tag{9.9}$$

Differentiating (9.9) with respect to $\theta$, we get

$$\begin{aligned}&z\cos\theta\cos(z\cos\theta) + (z\sin\theta)(-z\sin\theta)(-\sin(z\cos\theta)) \\ &\quad = 2\sum_{j=0}^{\infty}(-1)^{j}(2j+1)^2 J_{2j+1}(z)\cos(2j+1)\theta \\ \Longleftrightarrow\ &z\cos\theta\cos(z\cos\theta) + z^2\sin^2\theta\sin(z\cos\theta) \\ &\quad = 2\sum_{j=0}^{\infty}(-1)^{j}(2j+1)^2 J_{2j+1}(z)\cos(2j+1)\theta.\end{aligned} \tag{9.10}$$

We get the result by putting $\theta = 0$ in (9.10).

[3.19] Using (3.17) and [3.9], we have

$$\begin{aligned}I_{-n}(z) &= i^n J_{-n}(iz) \\ &= i^n\left((-1)^n J_n(iz)\right) \\ &= (-i)^n\left(i^n I_n(z)\right) \\ &= I_n(z).\end{aligned}$$

[3.20] Using (3.17) and [3.10], we have

$$\begin{aligned}I_0'(z) &= \frac{dJ_0(iz)}{dz} \\ &= iJ_0'(iz) \\ &= i\left(-J_1(iz)\right) \\ &= i\left(-iI_1(z)\right) \\ &= I_1(z).\end{aligned}$$

[3.21] Replace $z$ with $iz$ in [3.11] and use (3.17).

[3.22] Replace $z$ with $iz$ in [3.12] and use (3.17).

[3.23] Replace $z$ with $iz$ in [3.13] and use (3.17).

[3.24] Replace $z$ with $iz$ in [3.14] and use (3.17).

[3.25] Replace $z$ with $iz$ in [3.15] and use (3.17).

[3.26] Replace $z$ with $iz$ in [3.16] and use (3.17).

[3.27] Replace $z$ with $iz$ in [3.17] and use (3.17).

[3.28] Replace $z$ with $iz$ in [3.18] and use (3.17).

[3.29] Using (3.27) and (3.6), we have

$$\begin{aligned}\frac{d}{dz}\left(z^v J_v(z)\right) &= \frac{d}{dz}\left(\sum_{j=0}^{\infty}(-1)^j \frac{z^{2v+2j}}{j!\Gamma(v+j+1)2^{v+2j}}\right)\\ &= \sum_{j=0}^{\infty}(-1)^j \frac{(2v+2j)z^{2v+2j-1}}{j!(v+j)\Gamma(v+j)2^{v+2j}}\\ &= z^v \sum_{j=0}^{\infty}(-1)^j \frac{z^{v-1+2j}}{j!\Gamma(v-1+j+1)2^{v-1+2j}}\\ &= z^v J_{v-1}(z).\end{aligned}$$

[3.30] Using (3.27), we have

$$\begin{aligned}\frac{d}{dz}\left(z^{-v} J_v(z)\right) &= \frac{d}{dz}\left(\sum_{j=0}^{\infty}(-1)^j \frac{z^{2j}}{j!\Gamma(v+j+1)2^{v+2j}}\right)\\ &= \sum_{j=1}^{\infty}(-1)^j \frac{(2j)z^{2j-1}}{j!\Gamma(v+j+1)2^{v+2j}}\\ &= \sum_{j=1}^{\infty}(-1)^j \frac{z^{2j-1}}{(j-1)!\Gamma(v+j+1)2^{v+2j-1}}\\ &= z^{-v} \sum_{j=0}^{\infty}(-1)^{j+1} \frac{z^{v+1+2j}}{j!\Gamma(v+1+j+1)2^{v+1+2j}}\\ &= -z^{-v} J_{v+1}(z).\end{aligned}$$

[3.31] Using (3.32) and (3.6), we have

$$\begin{aligned}\frac{d}{dz}\left(z^{v}I_{v}(z)\right) &= \frac{d}{dz}\left(\sum_{j=0}^{\infty}\frac{z^{2v+2j}}{j!\Gamma(v+j+1)2^{v+2j}}\right)\\ &= \sum_{j=0}^{\infty}\frac{(2v+2j)z^{2v+2j-1}}{j!(v+j)\Gamma(v+j)2^{v+2j}}\\ &= z^{v}\sum_{j=0}^{\infty}\frac{z^{v-1+2j}}{j!\Gamma(v-1+j+1)2^{v-1+2j}}\\ &= z^{v}I_{v-1}(z).\end{aligned}$$

[3.32] Using (3.32), we have

$$\begin{aligned}\frac{d}{dz}\left(z^{-v}I_{v}(z)\right) &= \frac{d}{dz}\left(\sum_{j=0}^{\infty}\frac{z^{2j}}{j!\Gamma(v+j+1)2^{v+2j}}\right)\\ &= \sum_{j=1}^{\infty}\frac{(2j)z^{2j-1}}{j!\Gamma(v+j+1)2^{v+2j}}\\ &= \sum_{j=1}^{\infty}\frac{z^{2j-1}}{(j-1)!\Gamma(v+j+1)2^{v+2j-1}}\\ &= z^{-v}\sum_{j=0}^{\infty}\frac{z^{v+1+2j}}{j!\Gamma(v+1+j+1)2^{v+1+2j}}\\ &= z^{-v}J_{v+1}(z).\end{aligned}$$

# Chapter 10
# Solutions to Special Integrals




[4.1] Putting $n = 0$ in [4.2], we get

$$\int_0^{\pi} e^{\cos\theta} \cos(-\sin\theta)\, d\theta = \int_0^{\pi} e^{\cos\theta} \cos(\sin\theta)\, d\theta = \pi. \tag{10.1}$$

Putting $n = 0$ in [4.4], we get

$$\int_0^{\pi} e^{-\cos\theta} \cos(\sin\theta)\, d\theta = \pi. \tag{10.2}$$

Taking the average of (10.1) and (10.2), we get

$$\int_0^{\pi} \frac{e^{\cos\theta} + e^{-\cos\theta}}{2} \cos(\sin\theta)\, d\theta = \int_0^{\pi} \cosh(\cos\theta) \cos(\sin\theta)\, d\theta = \pi.$$

[4.2] Put $x = 1$ in [4.6].

[4.3] Put $x = 1$ in [4.7].

[4.4] Put $x = -1$ in [4.6].

[4.5] Put $x = -1$ in [4.7].

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[4.6] Applying Theorem 4.2 to $f(z) = e^{xz}$ for $(x, n) \in \mathbb{R} \times (\mathbb{N} - 1)$, we get

$$\begin{aligned}
\pi a_n &= \frac{\pi}{n!} x^n \\
&= \int_0^\pi \frac{e^{-in\theta} e^{xe^{i\theta}} + e^{in\theta} e^{xe^{-i\theta}}}{2}\, d\theta \\
&= \int_0^\pi \frac{e^{-in\theta} e^{x(\cos\theta + i\sin\theta)} + e^{in\theta} e^{x(\cos\theta - i\sin\theta)}}{2}\, d\theta \\
&= \int_0^\pi e^{x\cos\theta} \frac{e^{-i(n\theta - x\sin\theta)} + e^{i(n\theta - x\sin\theta)}}{2}\, d\theta \\
&= \int_0^\pi e^{x\cos\theta} \cos(n\theta - x\sin\theta)\, d\theta.
\end{aligned} \tag{10.3}$$

[4.7] Applying Theorem 4.2 to $f(z) = e^{xz}$ for $(x, n) \in \mathbb{R} \times (-\mathbb{N})$ and using (10.3), we get

$$\pi a_n = 0 = \int_0^\pi e^{x\cos\theta} \cos(n\theta - x\sin\theta)\, d\theta.$$

[4.8] Applying Theorem 4.2 to $f(z) = \sin(xz)$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, we get

$$\begin{aligned}
&\pi a_n = (-1)^{(n-1)/2} \frac{\pi}{n!} x^n \\
&= \int_0^\pi \frac{e^{-in\theta} \sin\left(xe^{i\theta}\right) + e^{in\theta} \sin\left(xe^{-i\theta}\right)}{2}\, d\theta \\
&= \int_0^\pi \frac{e^{-in\theta} \sin\left(x(\cos\theta + i\sin\theta)\right) + e^{in\theta} \sin\left(x(\cos\theta - i\sin\theta)\right)}{2}\, d\theta \\
&= \frac{1}{2} \int_0^\pi \Big( e^{-in\theta} \left(\sin(x\cos\theta)\cos(ix\sin\theta) + \cos(x\cos\theta)\sin(ix\sin\theta)\right) \\
&\quad + e^{in\theta} \left(\sin(x\cos\theta)\cos(ix\sin\theta) - \cos(x\cos\theta)\sin(ix\sin\theta)\right) \Big)\, d\theta \\
&= \int_0^\pi \Bigg( \frac{e^{-in\theta} + e^{in\theta}}{2} \sin(x\cos\theta)\cosh(x\sin\theta) \\
&\quad - \frac{e^{-in\theta} - e^{in\theta}}{2i} \cos(x\cos\theta)\sinh(x\sin\theta) \Bigg)\, d\theta \\
&= \int_0^\pi \left(\cos(n\theta)\sin\left(x\cos\theta\right)\cosh\left(x\sin\theta\right)\right. \\
&\quad \left. + \sin(n\theta)\cos\left(x\cos\theta\right)\sinh\left(x\sin\theta\right)\right)\, d\theta.
\end{aligned} \tag{10.4}$$

[4.9] Applying Theorem 4.2 to $f(z) = \sin(xz)$ for $(x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 2))$ and using (10.4), we get

$$\begin{aligned}
&\pi a_n = 0 \\
&= \int_0^\pi (\cos(n\theta) \sin(x \cos\theta) \cosh(x \sin\theta) + \sin(n\theta) \cos(x \cos\theta) \\
&\quad \sinh(x \sin\theta))\, d\theta.
\end{aligned}$$

[4.10] Applying Theorem 4.2 to $f(z) = \cos(xz)$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, we get

$$\begin{aligned}
&\pi a_n = (-1)^{n/2} \frac{\pi}{n!} x^n \\
&= \int_0^\pi \frac{e^{-in\theta} \cos\left(xe^{i\theta}\right) + e^{in\theta} \cos\left(xe^{-i\theta}\right)}{2}\, d\theta \\
&= \int_0^\pi \frac{e^{-in\theta} \cos(x(\cos\theta + i \sin\theta)) + e^{in\theta} \cos(x(\cos\theta - i \sin\theta))}{2}\, d\theta \\
&= \frac{1}{2} \int_0^\pi \left(e^{-in\theta} (\cos(x \cos\theta) \cos(ix \sin\theta) - \sin(x \cos\theta) \sin(ix \sin\theta))\right. \\
&\quad \left. + e^{in\theta} (\cos(x \cos\theta) \cos(ix \sin\theta) + \sin(x \cos\theta) \sin(ix \sin\theta))\right) d\theta \\
&= \int_0^\pi \left( \frac{e^{-in\theta} + e^{in\theta}}{2} \cos(x \cos\theta) \cosh(x \sin\theta) \right. \\
&\quad \left. - \frac{e^{in\theta} - e^{-in\theta}}{2i} \sin(x \cos\theta) \sinh(x \sin\theta) \right) d\theta \\
&= \int_0^\pi (\cos(n\theta) \cos(x \cos\theta) \cosh(x \sin\theta) - \sin(n\theta) \sin(x \cos\theta) \\
&\quad \sinh(x \sin\theta))\, d\theta.
\end{aligned} \tag{10.5}$$

[4.11] Applying Theorem 4.2 to $f(z) = \cos(xz)$ for $(x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 1))$, and using (10.5), we get

$$\begin{aligned}
&\pi a_n = 0 \\
&= \int_0^\pi (\cos(n\theta) \cos(x \cos\theta) \cosh(x \sin\theta) - \sin(n\theta) \sin(x \cos\theta) \\
&\quad \sinh(x \sin\theta))\, d\theta.
\end{aligned}$$

[4.12] Applying Theorem 4.2 to $f(z) = \sinh(xz)$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, we get

$$\begin{aligned}
\pi a_n &= \frac{\pi}{n!} x^n \\
&= \int_0^\pi \frac{e^{-in\theta} \sinh\left(xe^{i\theta}\right) + e^{in\theta} \sinh\left(xe^{-i\theta}\right)}{2}\, d\theta \\
&= \int_0^\pi \frac{e^{-in\theta} \sinh\left(x(\cos\theta + i\sin\theta)\right) + e^{in\theta} \sinh\left(x(\cos\theta - i\sin\theta)\right)}{2}\, d\theta \\
&= \frac{1}{2} \int_0^\pi \Big(e^{-in\theta} \left(\sinh(x\cos\theta)\cosh(ix\sin\theta) + \cosh(x\cos\theta)\sinh(ix\sin\theta)\right) \\
&\quad + e^{in\theta} \left(\sinh(x\cos\theta)\cosh(ix\sin\theta) - \cosh(x\cos\theta)\sinh(ix\sin\theta)\right)\Big)\, d\theta \\
&= \int_0^\pi \left(\frac{e^{-in\theta} + e^{in\theta}}{2} \sinh(x\cos\theta)\cos(x\sin\theta)\right. \\
&\quad \left. + \frac{e^{in\theta} - e^{-in\theta}}{2i} \cosh(x\cos\theta)\sin(x\sin\theta)\right) d\theta \\
&= \int_0^\pi \left(\cos(n\theta)\sinh(x\cos\theta)\cos(x\sin\theta) + \sin(n\theta)\cosh(x\cos\theta)\right. \\
&\quad \left. \sin(x\sin\theta)\right)\, d\theta.
\end{aligned} \tag{10.6}$$

[4.13] Applying Theorem 4.2 to $f(z) = \sinh(xz)$ for $(x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 2))$ and using (10.6), we get

$$\begin{aligned}
\pi a_n &= 0 \\
&= \int_0^\pi \left(\cos(n\theta)\sinh(x\cos\theta)\cos(x\sin\theta) + \sin(n\theta)\cosh(x\cos\theta)\right. \\
&\quad \left. \sin(x\sin\theta)\right)\, d\theta.
\end{aligned}$$

[4.14] Applying Theorem 4.2 to $f(z) = \cosh(xz)$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, we get

$$\begin{aligned}
\pi a_n &= \frac{\pi}{n!} x^n \\
&= \int_0^\pi \frac{e^{-in\theta} \cosh\left(xe^{i\theta}\right) + e^{in\theta} \cosh\left(xe^{-i\theta}\right)}{2}\, d\theta \\
&= \int_0^\pi \frac{e^{-in\theta} \cosh\left(x(\cos\theta + i\sin\theta)\right) + e^{in\theta} \cosh\left(x(\cos\theta - i\sin\theta)\right)}{2}\, d\theta
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2}\int_0^{\pi} \left(e^{-in\theta}\left(\cosh(x\cos\theta)\cosh(ix\sin\theta) + \sinh(x\cos\theta)\sinh(ix\sin\theta)\right)\right. \\
&\quad \left. + e^{in\theta}\left(\cosh(x\cos\theta)\cosh(ix\sin\theta) - \sinh(x\cos\theta)\sinh(ix\sin\theta)\right)\right) d\theta \\
&= \int_0^{\pi} \left(\frac{e^{-in\theta} + e^{in\theta}}{2}\cosh(x\cos\theta)\cos(x\sin\theta)\right. \\
&\quad \left. + \frac{e^{in\theta} - e^{-in\theta}}{2i}\sinh(x\cos\theta)\sin(x\sin\theta)\right) d\theta \\
&= \int_0^{\pi} \left(\cos(n\theta)\cosh(x\cos\theta)\cos(x\sin\theta) + \sin(n\theta)\sinh(x\cos\theta)\right. \\
&\quad \left. \sin(x\sin\theta)\right) d\theta.
\end{aligned}
\tag{10.7}
$$

[4.15] Applying Theorem 4.2 to $f(z) = \cosh(xz)$ for $(x, n) \in \mathbb{R} \times (-\mathbb{N} \cup (2\mathbb{N} - 1))$ and using (10.7), we get

$$
\begin{aligned}
&\pi a_n = 0 \\
&= \int_0^{\pi} \left(\cos(n\theta)\cosh(x\cos\theta)\cos(x\sin\theta) + \sin(n\theta)\sinh(x\cos\theta)\right. \\
&\quad \left. \sin(x\sin\theta)\right) d\theta.
\end{aligned}
$$

[4.16] Applying Theorem 4.2 to $f(z) = (1+z)^m$ for $(m, n) \in (\mathbb{N} - 1) \times [0\,..\,m]$, we get

$$
\begin{aligned}
\frac{\pi}{2^m} a_n &= \frac{\pi}{2^m}\binom{m}{n} \\
&= \frac{1}{2^m}\int_0^{\pi} \frac{e^{-in\theta}\left(1 + e^{i\theta}\right)^m + e^{in\theta}\left(1 + e^{-i\theta}\right)^m}{2} d\theta \\
&= \frac{1}{2^m}\int_0^{\pi} \left(\frac{e^{-in\theta}\left((1+\cos\theta) + i\sin\theta\right)^m}{2}\right. \\
&\quad \left. + \frac{e^{in\theta}\left((1+\cos\theta) - i\sin\theta\right)^m}{2}\right) d\theta \\
&= \frac{1}{2^m}\int_0^{\pi} \left(\frac{e^{-in\theta}\left(2\cos^2\frac{\theta}{2} + 2i\cos\frac{\theta}{2}\sin\frac{\theta}{2}\right)^m}{2}\right.
\end{aligned}
$$

$$
\begin{aligned}
&\quad + \frac{e^{in\theta}\left(2\cos^2\frac{\theta}{2} - 2i\cos\frac{\theta}{2}\sin\frac{\theta}{2}\right)^m}{2}\Bigg)\, d\theta \\
&= \frac{1}{2^m}\int_0^\pi \Bigg(\frac{2^m e^{-in\theta}\cos^m\frac{\theta}{2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)^m}{2} \\
&\quad + \frac{2^m e^{in\theta}\cos^m\frac{\theta}{2}\left(\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\right)^m}{2}\Bigg)\, d\theta \\
&= \int_0^\pi \frac{e^{-in\theta}e^{im\theta/2}\cos^m\frac{\theta}{2} + e^{in\theta}e^{-im\theta/2}\cos^m\frac{\theta}{2}}{2}\, d\theta \\
&= \int_0^\pi \frac{e^{-i(2n-m)\theta/2} + e^{i(2n-m)\theta/2}}{2}\cos^m\frac{\theta}{2}\, d\theta \\
&= \int_0^\pi \cos\frac{(2n-m)\theta}{2}\cos^m\frac{\theta}{2}\, d\theta. \qquad (10.8)
\end{aligned}
$$

[4.17] Applying Theorem 4.2 to $f(z) = (1+z)^m$ for $(m,n) \in (\mathbb{N}-1)\times(\mathbb{Z} - [0\,..\,m])$ and using (10.8), we get

$$
\frac{\pi}{2^m}a_n = 0 = \int_0^\pi \cos\frac{(2n-m)\theta}{2}\cos^m\frac{\theta}{2}\, d\theta.
$$

[4.18] Applying Theorem 4.2 to $f(z) = (1-z)^m$ for $(m,n) \in (\mathbb{N}-1)\times[0\,..\,m]$, we get

$$
\begin{aligned}
\frac{\pi}{2^m}a_n &= (-1)^n\frac{\pi}{2^m}\binom{m}{n} \\
&= \frac{1}{2^m}\int_0^\pi \frac{e^{-in\theta}\left(1-e^{i\theta}\right)^m + e^{in\theta}\left(1-e^{-i\theta}\right)^m}{2}\, d\theta \\
&= \frac{1}{2^m}\int_0^\pi \Bigg(\frac{e^{-in\theta}\left((1-\cos\theta) - i\sin\theta\right)^m}{2} \\
&\quad + \frac{e^{in\theta}\left((1-\cos\theta) + i\sin\theta\right)^m}{2}\Bigg)\, d\theta
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{1}{2^m}\int_0^\pi \left( \frac{e^{-in\theta}\left(2\sin^2\frac{\theta}{2} - 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)^m}{2} \right. \\
&\quad \left. + \frac{e^{in\theta}\left(2\sin^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)^m}{2} \right) d\theta \\
&= \frac{1}{2^m}\int_0^\pi \left( \frac{2^m e^{-in\theta}\sin^m\frac{\theta}{2}\left(\sin\frac{\theta}{2} - i\cos\frac{\theta}{2}\right)^m}{2} \right. \\
&\quad \left. + \frac{2^m e^{in\theta}\sin^m\frac{\theta}{2}\left(\sin\frac{\theta}{2} + i\cos\frac{\theta}{2}\right)^m}{2} \right) d\theta \\
&= \int_0^\pi \frac{e^{-in\theta}e^{-i(\pi-\theta)m/2}\sin^m\frac{\theta}{2} + e^{in\theta}e^{i(\pi-\theta)m/2}\sin^m\frac{\theta}{2}}{2}\, d\theta \\
&= \int_0^\pi \frac{e^{-i((2n-m)\theta+m\pi)/2} + e^{i((2n-m)\theta+m\pi)/2}}{2}\sin^m\frac{\theta}{2}\, d\theta \\
&= \int_0^\pi \cos\frac{(2n-m)\theta + m\pi}{2}\sin^m\frac{\theta}{2}\, d\theta. \qquad (10.9)
\end{aligned}
$$

[4.19] Applying Theorem 4.2 to $f(z) = (1-z)^m$ for $(m,n) \in (\mathbb{N}-1)\times(\mathbb{Z}-[0\,..\,m])$ and using (10.9), we get

$$
\frac{\pi}{2^m}a_n = 0 = \int_0^\pi \cos\frac{(2n-m)\theta + m\pi}{2}\sin^m\frac{\theta}{2}\, d\theta.
$$

[4.20] Applying Theorem 4.2 to $f(z) = (1 + xz)^m$ for $(x, m, n) \in [-1, 1] \times (\mathbb{N} - 1) \times [0..m]$, we get

$$\begin{aligned}
\pi a_n &= \pi x^n \binom{m}{n} \\
&= \int_0^\pi \frac{e^{-in\theta}\left(1 + xe^{i\theta}\right)^m + e^{in\theta}\left(1 + xe^{-i\theta}\right)^m}{2}\, d\theta \\
&= \int_0^\pi \left(\frac{e^{-in\theta}\left((1 + x\cos\theta) + ix\sin\theta\right)^m}{2} + \frac{e^{in\theta}\left((1 + x\cos\theta) - ix\sin\theta\right)^m}{2}\right) \\
&\quad d\theta \\
&= \int_0^\pi \frac{1}{2}\left(e^{-in\theta}\left(\frac{1 + x\cos\theta}{\sqrt{1 + x^2 + 2x\cos\theta}} + i\frac{x\sin\theta}{\sqrt{1 + x^2 + 2x\cos\theta}}\right)^m\right. \\
&\quad \left. + e^{in\theta}\left(\frac{1 + x\cos\theta}{\sqrt{1 + x^2 + 2x\cos\theta}} - i\frac{x\sin\theta}{\sqrt{1 + x^2 + 2x\cos\theta}}\right)^m\right) \\
&\quad \left(1 + x^2 + 2x\cos\theta\right)^{m/2}\, d\theta \\
&= \int_0^\pi \frac{1}{2}\left(e^{-in\theta}e^{im\tan^{-1}(x\sin\theta/(1+x\cos\theta))}\right. \\
&\quad \left. + e^{in\theta}e^{-im\tan^{-1}(x\sin\theta/(1+x\cos\theta))}\right)\left(1 + x^2 + 2x\cos\theta\right)^{m/2}\, d\theta \\
&= \int_0^\pi \left(1 + x^2 + 2x\cos\theta\right)^{m/2} \cos\left(n\theta - m\tan^{-1}\left(\frac{x\sin\theta}{1 + x\cos\theta}\right)\right)\, d\theta.
\end{aligned} \tag{10.10}$$

[4.21] Applying Theorem 4.2 to $f(z) = (1 + xz)^m$ for $(x, m, n) \in [-1, 1] \times (\mathbb{N} - 1) \times (\mathbb{Z} - [0..m])$ and using (10.10), we get

$$\begin{aligned}
\pi a_n &= 0 \\
&= \int_0^\pi \left(1 + x^2 + 2x\cos\theta\right)^{m/2} \cos\left(n\theta - m\tan^{-1}\left(\frac{x\sin\theta}{1 + x\cos\theta}\right)\right)\, d\theta.
\end{aligned}$$

[4.22] Put $x = 1$ in [4.24].

[4.23] Put $x = 1$ in [4.25].

[4.24] Applying Theorem 4.5 to $f(z) = e^{xz}$, and using (3.19), we get

$$\begin{aligned}\pi \sum_{j=0}^{\infty} a_j^2 &= \pi \sum_{j=0}^{\infty} \frac{x^{2j}}{(j!)^2} \\ &= \pi I_0(2x) \\ &= \int_0^{\pi} e^{xe^{i\theta}} e^{xe^{-i\theta}}\, d\theta \\ &= \int_0^{\pi} e^{x(e^{i\theta}+e^{-i\theta})}\, d\theta \\ &= \int_0^{\pi} e^{2x\cos\theta}\, d\theta.\end{aligned}$$

[4.25] Applying Theorem 4.6 to $f(z) = e^{xz}$, and using (3.19), we get

$$\begin{aligned}\pi \sum_{j=0}^{\infty} a_j a_{n+j} &= \pi \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!} \\ &= \pi I_n(2x) \\ &= \int_0^{\pi} e^{xe^{i\theta}} e^{xe^{-i\theta}} \cos n\theta\, d\theta \\ &= \int_0^{\pi} e^{x(e^{i\theta}+e^{-i\theta})} \cos n\theta\, d\theta \\ &= \int_0^{\pi} e^{2x\cos\theta} \cos n\theta\, d\theta.\end{aligned}$$

[4.26] Applying Theorem 4.5 to $f(z) = \sum_{j=0}^{n} a_j z^j$, we get

$$\begin{aligned}\pi \sum_{j=0}^{n} a_j^2 &= \int_0^{\pi} \left(\sum_{j=0}^{n} a_j e^{ji\theta}\right)\left(\sum_{j=0}^{n} a_j e^{-ji\theta}\right) d\theta \\ &= \int_0^{\pi} \left(\sum_{j=0}^{n} a_j \cos j\theta + i \sum_{j=1}^{n} a_j \sin j\theta\right)\left(\sum_{j=0}^{n} a_j \cos j\theta - i \sum_{j=1}^{n} a_j \sin j\theta\right) \\ &\qquad d\theta \\ &= \int_0^{\pi} \left(\left(\sum_{j=0}^{n} a_j \cos j\theta\right)^2 + \left(\sum_{j=1}^{n} a_j \sin j\theta\right)^2\right) d\theta \\ &= \int_0^{\pi} \left(C^2(\theta) + S^2(\theta)\right) d\theta,\end{aligned} \tag{10.11}$$

where $C(\theta)$ and $S(\theta)$ are as defined in (4.46) and (4.47), respectively.

[4.27] Applying Theorem 4.6 to $f(z) = \sum_{j=0}^{n} a_j z^j$ and using (10.11), we get

$$\begin{aligned}\pi \sum_{j=0}^{\infty} a_j a_{m+j} &= \pi \sum_{j=0}^{n-m} a_j a_{m+j} \\ &= \int_0^{\pi} \left( \sum_{j=0}^{n} a_j e^{ji\theta} \right) \left( \sum_{j=0}^{n} a_j e^{-ji\theta} \right) \cos m\theta \, d\theta \\ &= \int_0^{\pi} \left( C^2(\theta) + S^2(\theta) \right) \cos m\theta \, d\theta,\end{aligned}$$

where $C(\theta)$ and $S(\theta)$ are as defined in (4.46) and (4.47), respectively.

[4.28] Applying Theorem 4.5 to $f(z) = (1-z)^n$, we get

$$\begin{aligned}&\frac{\pi}{2^{2n}} \sum_{j=0}^{\infty} a_j^2 = \frac{\pi}{2^{2n}} \sum_{j=0}^{n} \binom{n}{j}^2 \\ &= \frac{1}{2^{2n}} \int_0^{\pi} \left(1 - e^{i\theta}\right)^n \left(1 - e^{-i\theta}\right)^n \, d\theta \\ &= \frac{1}{2^{2n}} \int_0^{\pi} \left((1 - \cos\theta) - i \sin\theta\right)^n \left((1 - \cos\theta) + i \sin\theta\right)^n \, d\theta \\ &= \frac{1}{2^{2n}} \int_0^{\pi} \left(2 \sin^2 \frac{\theta}{2} - 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2}\right)^n \left(2 \sin^2 \frac{\theta}{2} + 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2}\right)^n \, d\theta \\ &= \frac{1}{2^{2n}} \int_0^{\pi} 2^{2n} \sin^{2n}\frac{\theta}{2} e^{-in(\pi-\theta)/2} e^{in(\pi-\theta)/2} \, d\theta \\ &= \int_0^{\pi} \sin^{2n}\frac{\theta}{2} \, d\theta.\end{aligned} \tag{10.12}$$

[4.29] Applying Theorem 4.6 to $f(z) = (1-z)^n$ for $(n,m) \in \mathbb{N} \times [0..n]$ and using (10.12), we get

$$\begin{aligned}\frac{\pi}{2^{2n}} \sum_{j=0}^{\infty} a_j a_{m+j} &= \frac{\pi}{2^{2n}} \sum_{j=0}^{n-m} a_j a_{m+j} \\ &= (-1)^m \frac{\pi}{2^{2n}} \sum_{j=0}^{n-m} \binom{n}{j}\binom{n}{m+j} \\ &= \frac{1}{2^{2n}} \int_0^{\pi} \left(1-e^{i\theta}\right)^n \left(1-e^{-i\theta}\right)^n \cos m\theta \, d\theta \\ &= \int_0^{\pi} \sin^{2n}\frac{\theta}{2} \cos m\theta \, d\theta.\end{aligned}$$

[4.30] Applying Theorem 4.6 to $f(z) = (1-z)^n$ for $(n,m) \in \mathbb{N} \times ((\mathbb{N}-1) - [0..n])$ and using (10.12), we get

$$\begin{aligned}\frac{\pi}{2^{2n}} \sum_{j=0}^{\infty} a_j a_{m+j} &= \frac{\pi}{2^{2n}} \sum_{j=0}^{n-m} a_j a_{m+j} \\ &= 0 \\ &= \frac{1}{2^{2n}} \int_0^{\pi} \left(1-e^{i\theta}\right)^n \left(1-e^{-i\theta}\right)^n \cos m\theta \, d\theta \\ &= \int_0^{\pi} \sin^{2n}\frac{\theta}{2} \cos m\theta \, d\theta.\end{aligned}$$

[4.31] Applying Theorem 4.5 to $f(z) = \sin xz$, we get

$$\begin{aligned}\pi \sum_{j=0}^{\infty} a_j^2 &= \pi \sum_{j=0}^{\infty} \frac{x^{4j+2}}{((2j+1)!)^2} \\ &= \int_0^{\pi} \sin\left(xe^{i\theta}\right) \sin\left(xe^{-i\theta}\right) d\theta \\ &= \int_0^{\pi} \sin(x\cos\theta + ix\sin\theta) \sin(x\cos\theta - ix\sin\theta) \, d\theta \\ &= \int_0^{\pi} (\sin(x\cos\theta)\cos(ix\sin\theta) + \cos(x\cos\theta)\sin(ix\sin\theta)) \\ &\quad (\sin(x\cos\theta)\cos(ix\sin\theta) - \cos(x\cos\theta)\sin(ix\sin\theta)) \, d\theta \\ &= \int_0^{\pi} (\sin(x\cos\theta)\cosh(x\sin\theta) + i\cos(x\cos\theta)\sinh(x\sin\theta)) \\ &\quad (\sin(x\cos\theta)\cosh(x\sin\theta) - i\cos(x\cos\theta)\sinh(x\sin\theta)) \, d\theta\end{aligned}$$

$$= \int_0^\pi \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta) + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right) d\theta. \tag{10.13}$$

[4.32] Applying Theorem 4.5 to $f(z) = \sinh xz$, we get

$$\begin{aligned}
\pi\sum_{j=0}^{\infty} a_j^2 &= \pi\sum_{j=0}^{\infty} \frac{x^{4j+2}}{((2j+1)!)^2} \\
&= \int_0^\pi \sinh\left(xe^{i\theta}\right)\sinh\left(xe^{-i\theta}\right) d\theta \\
&= \int_0^\pi \sinh(x\cos\theta + ix\sin\theta)\sinh(x\cos\theta - ix\sin\theta)\, d\theta \\
&= \int_0^\pi (\sinh(x\cos\theta)\cosh(ix\sin\theta) + \cosh(x\cos\theta)\sinh(ix\sin\theta)) \\
&\quad (\sinh(x\cos\theta)\cosh(ix\sin\theta) - \cosh(x\cos\theta)\sinh(ix\sin\theta))\, d\theta \\
&= \int_0^\pi (\sinh(x\cos\theta)\cos(x\sin\theta) + i\cosh(x\cos\theta)\sin(x\sin\theta)) \\
&\quad (\sinh(x\cos\theta)\cos(x\sin\theta) - i\cosh(x\cos\theta)\sin(x\sin\theta))\, d\theta \\
&= \int_0^\pi \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta) + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right) d\theta.
\end{aligned} \tag{10.14}$$

[4.33] Applying Theorem 4.6 to $f(z) = \sin xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$ and using (10.13), we get

$$\begin{aligned}
\pi\sum_{j=0}^{\infty} a_j a_{n+j} &= (-1)^{n/2}\pi x^n \sum_{j=0}^{\infty} \frac{x^{4j+2}}{(2j+1)!(n+2j+1)!} \\
&= \int_0^\pi \sin\left(xe^{i\theta}\right)\sin\left(xe^{-i\theta}\right)\cos n\theta\, d\theta \\
&= \int_0^\pi \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta) + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos n\theta\, d\theta.
\end{aligned}$$

[4.34] Applying Theorem 4.6 to $f(z) = \sin xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$ and using (10.13), we get

$$\begin{aligned}
&\pi \sum_{j=0}^{\infty} a_j a_{n+j} = 0 \\
&= \int_0^{\pi} \sin\left(xe^{i\theta}\right) \sin\left(xe^{-i\theta}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta) + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right) \cos n\theta \, d\theta.
\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.35] Applying Theorem 4.6 to $f(z) = \sinh xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$ and using (10.14), we get

$$\begin{aligned}
&\pi \sum_{j=0}^{\infty} a_j a_{n+j} = \pi x^n \sum_{j=0}^{\infty} \frac{x^{4j+2}}{(2j+1)!(n+2j+1)!} \\
&= \int_0^{\pi} \sinh\left(xe^{i\theta}\right) \sinh\left(xe^{-i\theta}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta) + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right) \cos n\theta \, d\theta.
\end{aligned}$$

[4.36] Applying Theorem 4.6 to $f(z) = \sinh xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$ and using (10.14), we get

$$\begin{aligned}
&\pi \sum_{j=0}^{\infty} a_j a_{n+j} = 0 \\
&= \int_0^{\pi} \sinh\left(xe^{i\theta}\right) \sinh\left(xe^{-i\theta}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta) + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right) \cos n\theta \, d\theta.
\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.37] Applying Theorem 4.5 to $f(z) = \cos xz$, we get

$$\begin{aligned}
\pi \sum_{j=0}^{\infty} a_j^2 &= \pi \sum_{j=0}^{\infty} \frac{x^{4j}}{((2j)!)^2} \\
&= \int_0^{\pi} \cos\left(xe^{i\theta}\right) \cos\left(xe^{-i\theta}\right)\, d\theta \\
&= \int_0^{\pi} \cos(x\cos\theta + ix\sin\theta)\cos(x\cos\theta - ix\sin\theta)\, d\theta \\
&= \int_0^{\pi} (\cos(x\cos\theta)\cos(ix\sin\theta) - \sin(x\cos\theta)\sin(ix\sin\theta)) \\
&\quad (\cos(x\cos\theta)\cos(ix\sin\theta) + \sin(x\cos\theta)\sin(ix\sin\theta))\, d\theta \\
&= \int_0^{\pi} (\cos(x\cos\theta)\cosh(x\sin\theta) - i\sin(x\cos\theta)\sinh(x\sin\theta)) \\
&\quad (\cos(x\cos\theta)\cosh(x\sin\theta) + i\sin(x\cos\theta)\sinh(x\sin\theta))\, d\theta \\
&= \int_0^{\pi} \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\, d\theta.
\end{aligned} \tag{10.15}$$

[4.38] Applying Theorem 4.5 to $f(z) = \cosh xz$, we get

$$\begin{aligned}
\pi \sum_{j=0}^{\infty} a_j^2 &= \pi \sum_{j=0}^{\infty} \frac{x^{4j}}{((2j)!)^2} \\
&= \int_0^{\pi} \cosh\left(xe^{i\theta}\right) \cosh\left(xe^{-i\theta}\right)\, d\theta \\
&= \int_0^{\pi} \cosh(x\cos\theta + ix\sin\theta)\cosh(x\cos\theta - ix\sin\theta)\, d\theta \\
&= \int_0^{\pi} (\cosh(x\cos\theta)\cosh(ix\sin\theta) + \sinh(x\cos\theta)\sinh(ix\sin\theta)) \\
&\quad (\cosh(x\cos\theta)\cosh(ix\sin\theta) - \sinh(x\cos\theta)\sinh(ix\sin\theta))\, d\theta \\
&= \int_0^{\pi} (\cosh(x\cos\theta)\cos(x\sin\theta) + i\sinh(x\cos\theta)\sin(x\sin\theta)) \\
&\quad (\cosh(x\cos\theta)\cos(x\sin\theta) - i\sinh(x\cos\theta)\sin(x\sin\theta))\, d\theta \\
&= \int_0^{\pi} \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\, d\theta.
\end{aligned} \tag{10.16}$$

[4.39] Applying Theorem 4.6 to $f(z) = \cos xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$ and using (10.15), we get

$$\pi \sum_{j=0}^{\infty} a_j a_{n+j} = (-1)^{n/2} \pi x^n \sum_{j=0}^{\infty} \frac{x^{4j}}{(2j)!(n+2j)!}$$

$$= \int_0^{\pi} \cos\left(xe^{i\theta}\right) \cos\left(xe^{-i\theta}\right) \cos n\theta \, d\theta$$

$$= \int_0^{\pi} \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right) \cos n\theta \, d\theta.$$

[4.40] Applying Theorem 4.6 to $f(z) = \cos xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$ and using (10.15), we get

$$\pi \sum_{j=0}^{\infty} a_j a_{n+j} = 0$$

$$= \int_0^{\pi} \cos\left(xe^{i\theta}\right) \cos\left(xe^{-i\theta}\right) \cos n\theta \, d\theta$$

$$= \int_0^{\pi} \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta) + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right) \cos n\theta \, d\theta.$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.41] Applying Theorem 4.6 to $f(z) = \cosh xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$ and using (10.16), we get

$$\pi \sum_{j=0}^{\infty} a_j a_{n+j} = \pi x^n \sum_{j=0}^{\infty} \frac{x^{4j}}{(2j)!(n+2j)!}$$

$$= \int_0^{\pi} \cosh\left(xe^{i\theta}\right) \cosh\left(xe^{-i\theta}\right) \cos n\theta \, d\theta$$

$$= \int_0^{\pi} \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right) \cos n\theta \, d\theta.$$

[4.42] Applying Theorem 4.6 to $f(z) = \cosh xz$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$ and using (10.16), we get

$$\begin{aligned}
&\pi \sum_{j=0}^{\infty} a_j a_{n+j} = 0 \\
&= \int_0^{\pi} \cosh\left(xe^{i\theta}\right) \cosh\left(xe^{-i\theta}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta) + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right) \cos n\theta \, d\theta.
\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.43] Applying Theorem 4.2 to $f(z) = \log(1 + z)$ for $n \in \mathbb{N}$, we get

$$\begin{aligned}
&\pi a_n = (-1)^{n+1} \frac{\pi}{n} \\
&= \frac{1}{2} \int_0^{\pi} \left(e^{-in\theta} \log\left(1 + e^{i\theta}\right) + e^{in\theta} \log\left(1 + e^{-i\theta}\right)\right) d\theta \\
&= \frac{1}{2} \int_0^{\pi} \left(e^{-in\theta} \log\left((1 + \cos\theta) + i \sin\theta\right)\right. \\
&\quad \left. + e^{in\theta} \log\left((1 + \cos\theta) - i \sin\theta\right)\right) d\theta \\
&= \frac{1}{2} \int_0^{\pi} \left(e^{-in\theta} \log\left(2\cos^2\frac{\theta}{2} + 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2}\right)\right. \\
&\quad \left. + e^{in\theta} \log\left(2\cos^2\frac{\theta}{2} - 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2}\right)\right) d\theta
\end{aligned} \tag{10.17}$$

$$
\begin{aligned}
&= \frac{1}{2}\int_0^{\pi}\left(e^{-in\theta}\left(\log\left(2\cos\frac{\theta}{2}\right)+\log\left(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right)\right)\right.\\
&\quad\left.+e^{in\theta}\left(\log\left(2\cos\frac{\theta}{2}\right)+\log\left(\cos\frac{\theta}{2}-i\sin\frac{\theta}{2}\right)\right)\right)d\theta\\
&= \int_0^{\pi}\left(\frac{e^{-in\theta}+e^{in\theta}}{2}\log\left(2\cos\frac{\theta}{2}\right)+\frac{e^{-in\theta}\log e^{i\theta/2}+e^{in\theta}\log e^{-i\theta/2}}{2}\right)d\theta\\
&= \int_0^{\pi}\left(\cos n\theta\log\left(2\cos\frac{\theta}{2}\right)\right.\\
&\quad\left.+\frac{i\frac{\theta}{2}(\cos n\theta-i\sin n\theta)-i\frac{\theta}{2}(\cos n\theta+i\sin n\theta)}{2}\right)d\theta\\
&= \int_0^{\pi}\left(\cos n\theta\log\left(2\cos\frac{\theta}{2}\right)+\frac{\theta}{2}\sin n\theta\right)d\theta.
\end{aligned}
$$

[4.44] Applying Theorem 4.2 to $f(z)=\log(1+z)$ for $n\in-\mathbb{N}\cup\{0\}$ and using (10.17), we get

$$
\pi a_n = 0 = \int_0^{\pi}\left(\cos n\theta\log\left(2\cos\frac{\theta}{2}\right)+\frac{\theta}{2}\sin n\theta\right)d\theta.
$$

[4.45] Applying Theorem 4.2 to $f(z)=\log(1+z/x)$ for $(x,n)\in(\mathbb{R}-(-1,1))\times\mathbb{N}$, we get

$$
\begin{aligned}
\pi a_n &= (-1)^{n+1}\frac{\pi}{nx^n}\\
&= \frac{1}{2}\int_0^{\pi}\left(e^{-in\theta}\log\left(1+\frac{e^{i\theta}}{x}\right)+e^{in\theta}\log\left(1+\frac{e^{-i\theta}}{x}\right)\right)d\theta\\
&= \frac{1}{2}\int_0^{\pi}\left(e^{-in\theta}\log\left(1+\frac{\cos\theta}{x}+\frac{i\sin\theta}{x}\right)\right.\\
&\quad\left.+e^{in\theta}\log\left(1+\frac{\cos\theta}{x}-\frac{i\sin\theta}{x}\right)\right)d\theta
\end{aligned}
\tag{10.18}
$$

$$
\begin{aligned}
&= \frac{1}{2}\int_0^{\pi}\left(e^{-in\theta}\left(\log\sqrt{\left(1+\frac{\cos\theta}{x}\right)^2+\left(\frac{\sin\theta}{x}\right)^2}+\right.\right.\\
&\quad\left.\log\left(\frac{1+\frac{\cos\theta}{x}+\frac{i\sin\theta}{x}}{\sqrt{\left(1+\frac{\cos\theta}{x}\right)^2+\left(\frac{\sin\theta}{x}\right)^2}}\right)\right)\\
&\quad+e^{in\theta}\left(\log\sqrt{\left(1+\frac{\cos\theta}{x}\right)^2+\left(\frac{\sin\theta}{x}\right)^2}\right.\\
&\quad\left.\left.\left.+\log\left(\frac{1+\frac{\cos\theta}{x}-\frac{i\sin\theta}{x}}{\sqrt{\left(1+\frac{\cos\theta}{x}\right)^2+\left(\frac{\sin\theta}{x}\right)^2}}\right)\right)\right)d\theta\\
&= \int_0^{\pi}\left(\left(\frac{e^{-in\theta}+e^{in\theta}}{2}\right)\log\sqrt{1+\frac{1}{x^2}+\frac{2}{x}\cos\theta}\right.\\
&\quad\left.+\frac{1}{2}\left(e^{-in\theta}\log e^{i\tan^{-1}(\sin\theta/(x+\cos\theta))}+e^{in\theta}\log e^{-i\tan^{-1}(\sin\theta/(x+\cos\theta))}\right)\right)d\theta\\
&= \int_0^{\pi}\left(\cos n\theta\log\sqrt{1+\frac{1}{x^2}+\frac{2}{x}\cos\theta}+\left(\frac{e^{in\theta}-e^{-in\theta}}{2i}\right)\right.\\
&\quad\left.\tan^{-1}\left(\frac{\sin\theta}{x+\cos\theta}\right)\right)d\theta\\
&= \int_0^{\pi}\left(\cos n\theta\log\sqrt{1+\frac{1}{x^2}+\frac{2}{x}\cos\theta}+\sin n\theta\tan^{-1}\left(\frac{\sin\theta}{x+\cos\theta}\right)\right)d\theta.
\end{aligned}
$$

[4.46] Applying Theorem 4.2 to $f(z)=\log(1+z/x)$ for $(x,n)\in(\mathbb{R}-(-1,1))\times(-\mathbb{N}\cup\{0\})$ and using (10.18), we get

$$
\begin{aligned}
&\pi a_n = 0\\
&= \frac{1}{2}\int_0^{\pi}\left(e^{-in\theta}\log\left(1+\frac{e^{i\theta}}{x}\right)+e^{in\theta}\log\left(1+\frac{e^{-i\theta}}{x}\right)\right)d\theta\\
&= \int_0^{\pi}\left(\cos n\theta\log\sqrt{1+\frac{1}{x^2}+\frac{2}{x}\cos\theta}+\sin n\theta\tan^{-1}\left(\frac{\sin\theta}{x+\cos\theta}\right)\right)d\theta.
\end{aligned}
$$

[4.47] Applying Theorem 4.2 to $f(z) = (1+z)^y$ for $(y, n) \in \mathbb{R} \times (\mathbb{N} - 1)$, we get

$$\begin{aligned}
\frac{\pi}{2^y} a_n &= \frac{\pi}{2^y n!} \prod_{j=0}^{n-1} (y - j) \\
&= \frac{1}{2^{y+1}} \int_0^{\pi} \left( e^{-in\theta} \left(1 + e^{i\theta}\right)^y + e^{in\theta} \left(1 + e^{-i\theta}\right)^y \right) d\theta \\
&= \frac{1}{2^{y+1}} \int_0^{\pi} \left( e^{-in\theta} \left((1 + \cos\theta) + i \sin\theta\right)^y + e^{in\theta} \left((1 + \cos\theta) - i \sin\theta\right)^y \right) d\theta \\
&= \frac{1}{2^{y+1}} \int_0^{\pi} \left( e^{-in\theta} \left( 2\cos^2 \frac{\theta}{2} + 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2} \right)^y \right. \\
&\quad \left. + e^{in\theta} \left( 2\cos^2 \frac{\theta}{2} - 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2} \right)^y \right) d\theta \\
&= \frac{1}{2^{y+1}} \int_0^{\pi} 2^y \cos^y \frac{\theta}{2} \left( e^{-in\theta} \left( \cos\frac{\theta}{2} + i \sin\frac{\theta}{2} \right)^y \right. \\
&\quad \left. + e^{-in\theta} \left( \cos\frac{\theta}{2} - i \sin\frac{\theta}{2} \right)^y \right) d\theta \\
&= \frac{1}{2} \int_0^{\pi} \left( e^{-in\theta} e^{iy\theta/2} + e^{in\theta} e^{-iy\theta/2} \right) \cos^y \frac{\theta}{2}\, d\theta \\
&= \frac{1}{2} \int_0^{\pi} \left( e^{-i(2n-y)\theta/2} + e^{i(2n-y)\theta/2} \right) \cos^y \frac{\theta}{2}\, d\theta \\
&= \int_0^{\pi} \cos\frac{(2n-y)\theta}{2} \cos^y \frac{\theta}{2}\, d\theta.
\end{aligned} \tag{10.19}$$

[4.48] Applying Theorem 4.2 to $f(z) = (1+z)^y$ for $(y, n) \in \mathbb{R} \times (-\mathbb{N})$ and using (10.19), we get

$$\begin{aligned}
\frac{\pi}{2^y} a_n &= 0 \\
&= \frac{1}{2^{y+1}} \int_0^{\pi} \left( e^{-in\theta} \left(1 + e^{i\theta}\right)^y + e^{in\theta} \left(1 + e^{-i\theta}\right)^y \right) d\theta \\
&= \int_0^{\pi} \cos\frac{(2n-y)\theta}{2} \cos^y \frac{\theta}{2}\, d\theta.
\end{aligned}$$

[4.49] Applying Theorem 4.2 to $f(z) = (1 + z/x)^y$ for $(x, y, n) \in (\mathbb{R} - [-1, 1]) \times \mathbb{R} \times \mathbb{N}$, we get

$$\begin{aligned}
\pi a_n &= \frac{\pi}{n! x^n} \prod_{j=0}^{n-1} (y - j) \\
&= \frac{1}{2} \int_0^\pi \left( e^{-in\theta} \left( 1 + \frac{e^{i\theta}}{x} \right)^y + e^{in\theta} \left( 1 + \frac{e^{-i\theta}}{x} \right)^y \right) d\theta \\
&= \frac{1}{2} \int_0^\pi \left( e^{-in\theta} \left( 1 + \frac{\cos\theta}{x} + \frac{i \sin\theta}{x} \right)^y + e^{in\theta} \left( 1 + \frac{\cos\theta}{x} - \frac{i \sin\theta}{x} \right)^y \right) d\theta \\
&= \frac{1}{2} \int_0^\pi \left( 1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta \right)^{y/2} \left( e^{-in\theta} \left( \frac{1 + \frac{\cos\theta}{x} + \frac{i \sin\theta}{x}}{\sqrt{1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta}} \right)^y \right. \\
&\quad \left. + e^{in\theta} \left( \frac{1 + \frac{\cos\theta}{x} - \frac{i \sin\theta}{x}}{\sqrt{1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta}} \right)^y \right) d\theta \\
&= \frac{1}{2} \int_0^\pi \left( 1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta \right)^{y/2} \left( e^{-in\theta} e^{iy \tan^{-1}(\sin\theta/(x + \cos\theta))} \right. \\
&\quad \left. + e^{in\theta} e^{-iy \tan^{-1}(\sin\theta(x + \cos\theta))} \right) d\theta \\
&= \frac{1}{2} \int_0^\pi \left( 1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta \right)^{y/2} \left( e^{-i\left(n\theta - y \tan^{-1}(\sin\theta/(x + \cos\theta))\right)} \right. \\
&\quad \left. + e^{i\left(n\theta - y \tan^{-1}(\sin\theta/(x + \cos\theta))\right)} \right) d\theta \\
&= \int_0^\pi \left( 1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta \right)^{y/2} \cos\left( n\theta - \tan^{-1}\left( \frac{\sin\theta}{x + \cos\theta} \right) \right) d\theta.
\end{aligned} \tag{10.20}$$

[4.50] Applying Theorem 4.2 to $f(z) = (1 + z/x)^y$ for $(x, y, n) \in (\mathbb{R} - [-1, 1]) \times \mathbb{R} \times (-\mathbb{N} \cup \{0\})$ and using (10.20), we get

$$\begin{aligned}
\pi a_n &= 0 \\
&= \frac{1}{2} \int_0^\pi \left( e^{-in\theta} \left( 1 + \frac{e^{i\theta}}{x} \right)^y + e^{in\theta} \left( 1 + \frac{e^{-i\theta}}{x} \right)^y \right) d\theta \\
&= \int_0^\pi \left( 1 + \frac{1}{x^2} + \frac{2}{x} \cos\theta \right)^{y/2} \cos\left( n\theta - \tan^{-1}\left( \frac{\sin\theta}{x + \cos\theta} \right) \right) d\theta.
\end{aligned}$$

[4.51] Putting $n = 0$ in [4.52] and using (2.30), we get

$$\begin{aligned}
&\int_0^\pi \left(\log^2\left(2\cos\frac{\theta}{2}\right) + \frac{\theta^2}{4}\right) d\theta = \pi \sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^3}{6} \\
\iff &\int_0^\pi \log^2\left(2\cos\frac{\theta}{2}\right) d\theta = \frac{\pi^3}{6} - \int_0^\pi \frac{\theta^2}{4}\, d\theta \\
&= \frac{\pi^3}{6} - \frac{\theta^3}{12}\Big|_0^\pi \\
&= \frac{\pi^3}{6} - \frac{\pi^3}{12} = \frac{\pi^3}{12}.
\end{aligned} \tag{10.21}$$

Using (10.21), we also have

$$\begin{aligned}
\int_0^\pi \log^2\left(2\cos\frac{\theta}{2}\right) d\theta &= \int_0^\pi \log^2\left(2\cos\left(\frac{\pi-\theta}{2}\right)\right) d\theta \\
&= \int_0^\pi \log^2\left(2\sin\frac{\theta}{2}\right) d\theta \\
&= \frac{\pi^3}{12}.
\end{aligned}$$

[4.52] Applying Theorem 4.6 to $f(z) = \log(1+z)$ for $n \in 2\mathbb{N} - 2$, we get

$$\begin{aligned}
\pi \sum_{j=0}^{\infty} a_j a_{n+j} &= \pi \sum_{j=1}^{\infty} \frac{1}{j(n+j)} \\
&= \int_0^{\pi} \log\left(1 + e^{i\theta}\right) \log\left(1 + e^{-i\theta}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \log((1 + \cos\theta) + i \sin\theta) \log((1 + \cos\theta) - i \sin\theta) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \log\left(2\cos^2\frac{\theta}{2} + 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2}\right) \\
&\qquad \log\left(2\cos^2\frac{\theta}{2} - 2i \cos\frac{\theta}{2} \sin\frac{\theta}{2}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log\left(2\cos\frac{\theta}{2}\right) + \log\left(\cos\frac{\theta}{2} + i \sin\frac{\theta}{2}\right)\right) \\
&\quad \left(\log\left(2\cos\frac{\theta}{2}\right) + \log\left(\cos\frac{\theta}{2} - i \sin\frac{\theta}{2}\right)\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log\left(2\cos\frac{\theta}{2}\right) + \log e^{i\theta/2}\right) \left(\log\left(2\cos\frac{\theta}{2}\right) + \log e^{-i\theta/2}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log\left(2\cos\frac{\theta}{2}\right) + i\frac{\theta}{2}\right) \left(\log\left(2\cos\frac{\theta}{2}\right) - i\frac{\theta}{2}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log^2\left(2\cos\frac{\theta}{2}\right) + \frac{\theta^2}{4}\right) \cos n\theta \, d\theta.
\end{aligned} \tag{10.22}$$

[4.53] Applying Theorem 4.6 to $f(z) = \log(1+z)$ for $n \in 2\mathbb{N} - 1$ and using (10.22), we get

$$\begin{aligned}
\pi \sum_{j=0}^{\infty} a_j a_{n+j} &= -\pi \sum_{j=1}^{\infty} \frac{1}{j(n+j)} \\
&= \int_0^{\pi} \log\left(1 + e^{i\theta}\right) \log\left(1 + e^{-i\theta}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log^2\left(2\cos\frac{\theta}{2}\right) + \frac{\theta^2}{4}\right) \cos n\theta \, d\theta.
\end{aligned}$$

[4.54] Put $n = 0$ in [4.55].

[4.55] Applying Theorem 4.6 to $f(z) = \log(1 + z/x)$ for $(x, n) \in [1, \infty) \times (2\mathbb{N} - 2)$, we get

$$
\begin{aligned}
\pi \sum_{j=0}^{\infty} a_j a_{n+j} &= \frac{\pi}{x^n} \sum_{j=1}^{\infty} \frac{1}{j(n+j)x^{2j}} \\
&= \int_0^{\pi} \log\left(1 + \frac{e^{i\theta}}{x}\right) \log\left(1 + \frac{e^{-i\theta}}{x}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \log\left(1 + \frac{\cos\theta}{x} + \frac{i\sin\theta}{x}\right) \log\left(1 + \frac{\cos\theta}{x} - \frac{i\sin\theta}{x}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log\sqrt{1 + \frac{1}{x^2} + \frac{2\cos\theta}{x}} + \log\left(\frac{1 + \dfrac{\cos\theta}{x} + \dfrac{i\sin\theta}{x}}{\sqrt{1 + \dfrac{1}{x^2} + \dfrac{2\cos\theta}{x}}}\right)\right) \\
&\quad \left(\log\sqrt{1 + \frac{1}{x^2} + \frac{2\cos\theta}{x}} + \log\left(\frac{1 + \dfrac{\cos\theta}{x} - \dfrac{i\sin\theta}{x}}{\sqrt{1 + \dfrac{1}{x^2} + \dfrac{2\cos\theta}{x}}}\right)\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log\sqrt{1 + \frac{1}{x^2} + \frac{2\cos\theta}{x}} + \log e^{i\tan^{-1}(\sin\theta/(x+\cos\theta))}\right) \\
&\quad \left(\log\sqrt{1 + \frac{1}{x^2} + \frac{2\cos\theta}{x}} + \log e^{-i\tan^{-1}(\sin\theta/(x+\cos\theta))}\right) \cos n\theta \, d\theta \qquad (10.23) \\
&= \int_0^{\pi} \left(\log\sqrt{1 + \frac{1}{x^2} + \frac{2\cos\theta}{x}} + i\tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right)\right) \\
&\quad \left(\log\sqrt{1 + \frac{1}{x^2} + \frac{2\cos\theta}{x}} - i\tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right)\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(\log^2\sqrt{1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta} + \left(\tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right)\right)^2\right) \cos n\theta \, d\theta.
\end{aligned}
$$

[4.56] Applying Theorem 4.6 to $f(z) = \log(1 + z/x)$ for $(x, n) \in [1, \infty) \times (2\mathbb{N} - 1)$ and using (10.23), we get

$$\begin{aligned}
\pi \sum_{j=0}^{\infty} a_j a_{n+j} &= -\frac{\pi}{x^n} \sum_{j=1}^{\infty} \frac{1}{j(n+j)x^{2j}} \\
&= \int_0^{\pi} \log\left(1 + \frac{e^{i\theta}}{x}\right) \log\left(1 + \frac{e^{-i\theta}}{x}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left( \log^2 \sqrt{1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta} + \left( \tan^{-1}\left( \frac{\sin\theta}{x + \cos\theta} \right) \right)^2 \right) \cos n\theta \, d\theta.
\end{aligned}$$

[4.57] Put $n = 0$ in [4.58].

[4.58] Applying Theorem 4.6 to $f(z) = (1 + z/x)^y$ for $(x, y, n) \in (1, \infty) \times \mathbb{R} \times (\mathbb{N} - 1)$, we get

$$\begin{aligned}
\pi \sum_{j=0}^{\infty} a_j a_{n+j} &= \pi \sum_{j=0}^{\infty} \frac{\prod_{k=0}^{j-1}(y-k)}{j! x^j} \cdot \frac{\prod_{k=0}^{n+j-1}(y-k)}{(n+j)! x^{n+j}} \\
&= \int_0^{\pi} \left(1 + \frac{e^{i\theta}}{x}\right)^y \left(1 + \frac{e^{-i\theta}}{x}\right)^y \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(1 + \frac{\cos\theta}{x} + \frac{i\sin\theta}{x}\right)^y \left(1 + \frac{\cos\theta}{x} - \frac{i\sin\theta}{x}\right)^y \cos n\theta \, d\theta \\
&= \int_0^{\pi} \left(1 + \frac{1}{x^2} + \frac{2\cos\theta}{x}\right)^y \cos n\theta \, d\theta.
\end{aligned}$$

[4.59] Replacing $n$ with $(m - n)/2$ in [4.16], we get

$$\int_0^{\pi} \cos\frac{n\theta}{2} \cos^m\frac{\theta}{2} \, d\theta = \frac{\pi}{2^m} \binom{m}{(m-n)/2}. \tag{10.24}$$

Substituting $\theta = 2x\phi$ in (10.24), we get

$$\begin{aligned}
&\int_0^{\pi/2x} 2x \cos(nx\phi) \cos^m(x\phi) \, d\phi = \frac{\pi}{2^m} \binom{m}{(m-n)/2} \\
\iff &\int_0^{\pi/2x} \cos(nx\phi) \cos^m(x\phi) \, d\phi = \frac{\pi}{2^{m+1}x} \binom{m}{(m-n)/2}.
\end{aligned} \tag{10.25}$$

Consider the integral

$$\int_{\pi/2x}^{\pi/x} \cos(nx\phi)\cos^m(x\phi)\,d\phi.$$

Substituting $\theta = \phi - \pi/2x$ in this integral, and using (10.25), we get

$$\begin{aligned}
&\int_{\pi/2x}^{\pi/x} \cos(nx\phi)\cos^m(x\phi)\,d\phi \\
&= \int_0^{\pi/2x} \cos\left(nx\left(\theta+\frac{\pi}{2x}\right)\right)\cos^m\left(x\left(\theta+\frac{\pi}{2x}\right)\right)\,d\theta \\
&= \int_0^{\pi/2x} \cos\left(nx\theta+\frac{n\pi}{2}\right)\cos^m\left(x\theta+\frac{\pi}{2}\right)\,d\theta \\
&= \int_0^{\pi/2x} \cos\left(nx\left(\frac{\pi}{2x}-\theta\right)+\frac{n\pi}{2}\right)\cos^m\left(x\left(\frac{\pi}{2x}-\theta\right)+\frac{\pi}{2}\right)\,d\theta \qquad (10.26)\\
&= \int_0^{\pi/2x} \cos(n\pi - nx\theta)\cos^m(\pi - x\theta)\,d\theta \\
&= (-1)^{m+n}\int_0^{\pi/2x} \cos(nx\theta)\cos^m(x\theta)\,d\theta \\
&= (-1)^{m+n}\frac{\pi}{2^{m+1}x}\binom{m}{(m-n)/2}.
\end{aligned}$$

Adding (10.25) and (10.26), we get

$$\begin{aligned}
\int_0^{\pi/x} \cos(nx\theta)\cos^m(x\theta)\,d\theta &= \left(1+(-1)^{m+n}\right)\int_0^{\pi/2x} \cos(nx\theta)\cos^m(x\theta)\,d\theta \\
&= \left(1+(-1)^{m+n}\right)\frac{\pi}{2^{m+1}x}\binom{m}{(m-n)/2} \\
&= \frac{\pi}{2^m x}\binom{m}{(m-n)/2},
\end{aligned} \qquad (10.27)$$

for $(x, m, n) \in (\mathbb{R}-\{0\})\times(2\mathbb{N}-2)\times((2\mathbb{N}-2)\cap[0..m]) \cup (\mathbb{R}-\{0\})\times(2\mathbb{N}-1)\times((2\mathbb{N}-1)\cap[0..m])$.

[4.60] Using (10.27) for $(x, m, n) \in (\mathbb{R}-\{0\})\times(2\mathbb{N}-2)\times(2\mathbb{N}-1) \cup (\mathbb{R}-\{0\})\times(2\mathbb{N}-1)\times(2\mathbb{N}-2)$, we get

$$\int_0^{\pi/x} \cos(nx\theta)\cos^m(x\theta)\,d\theta = 0.$$

[4.61] Applying Theorem 4.8 to $f(z) = \sin^{-1} z$ and using [2.8] for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, we get

$$\begin{aligned}\int_0^\pi \sin^{-1}(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2} \frac{x^{n+2j}}{(n+2j)2^{n+2j-1}}.\end{aligned}$$

[4.62] Applying Theorem 4.8 to $f(z) = \sin^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 2)$, we get

$$\begin{aligned}\int_0^\pi \sin^{-1}(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\cos n\theta$ is an even function.

[4.63] Applying Theorem 4.13 to $f(z) = \sin^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, and using [2.8], we get

$$\begin{aligned}&\int_0^\pi -i \sin^{-1}(2ix\sin\theta)\sin n\theta\, d\theta = \int_0^\pi \sinh^{-1}(2x\sin\theta)\sin n\theta\, d\theta\\ &= \pi B_n(x)\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2} \frac{x^{n+2j}}{(n+2j)2^{n+2j-1}}. \qquad (10.28)\end{aligned}$$

[4.64] Applying Theorem 4.13 to $f(z) = \sin^{-1} z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using [2.8] and (10.28), we get

$$\begin{aligned}\int_0^\pi \sinh^{-1}(2x\sin\theta)\sin n\theta\, d\theta &= \pi B_n(x)\\ &= \pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\sin n\theta$ is an odd function.

[4.65] Applying Theorem 4.8 to $f(z) = \sinh^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, and using [2.34], we get

$$\begin{aligned}&\int_0^\pi \sinh^{-1}(2x\cos\theta)\cos n\theta\, d\theta = \pi A_n(x)\\ &= \pi\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= \pi\sum_{j=0}^{\infty}(-1)^{(n+2j-1)/2}\binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2}\frac{x^{n+2j}}{(n+2j)2^{n+2j-1}}\\ &= (-1)^{(n-1)/2}\pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}\binom{n+2j-1}{(n+2j-1)/2}\frac{x^{n+2j}}{(n+2j)2^{n+2j-1}}.\end{aligned}$$

[4.66] Applying Theorem 4.8 to $f(z) = \sinh^{-1} z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using [2.34], we get

$$\begin{aligned}\int_0^\pi \sinh^{-1}(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\cos n\theta$ is an even function.

[4.67] Applying Theorem 4.13 to $f(z) = \sinh^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, and using [2.34], we get

$$\begin{aligned}
&\int_0^\pi -i \sinh^{-1}(2ix \sin\theta) \sin n\theta \, d\theta = \int_0^\pi \sin^{-1}(2x \sin\theta) \sin n\theta \, d\theta \\
&= \pi B_n(x) \\
&= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\
&= \pi \sum_{j=0}^{\infty} (-1)^j (-1)^{(n+2j-1)/2} \binom{n+2j}{j} \binom{n+2j-1}{(n+2j-1)/2} \frac{x^{n+2j}}{(n+2j)2^{n+2j-1}} \\
&= (-1)^{(n-1)/2} \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \binom{n+2j-1}{(n+2j-1)/2} \frac{x^{n+2j}}{(n+2j)2^{n+2j-1}}.
\end{aligned} \tag{10.29}$$

[4.68] Applying Theorem 4.13 to $f(z) = \sinh^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 2)$ and using [2.34] and (10.29), we get

$$\begin{aligned}
\int_0^\pi \sin^{-1}(2x \sin\theta) \sin n\theta \, d\theta &= \pi B_n(x) \\
&= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\
&= 0.
\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\sin n\theta$ is an odd function.

[4.69] Applying Theorem 4.8 to $f(z) = \tan^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, and using (1.13), we get

$$\begin{aligned}
\int_0^\pi \tan^{-1}(2x \cos\theta) \cos n\theta \, d\theta &= \pi A_n(x) \\
&= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\
&= \pi \sum_{j=0}^{\infty} (-1)^{(n+2j-1)/2} \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j} \\
&= (-1)^{(n-1)/2} \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j}.
\end{aligned}$$

[4.70] Applying Theorem 4.8 to $f(z) = \tan^{-1} z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (1.13), we get

$$\begin{aligned}\int_0^\pi \tan^{-1}(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\cos n\theta$ is an even function.

[4.71] Applying Theorem 4.13 to $f(z) = \tan^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, and using (1.13), we get

$$\begin{aligned}\int_0^\pi -i\tan^{-1}(2ix\sin\theta)\sin n\theta d\theta &= \int_0^\pi \tanh^{-1}(2x\sin\theta)\sin n\theta\, d\theta\\ &= \pi B_n(x)\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= \pi \sum_{j=0}^{\infty} (-1)^j (-1)^{(n+2j-1)/2} \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j}\\ &= (-1)^{(n-1)/2}\pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{x^{n+2j}}{n+2j}.\end{aligned} \tag{10.30}$$

[4.72] Applying Theorem 4.13 to $f(z) = \tan^{-1} z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (1.13) and (10.30), we get

$$\begin{aligned}\int_0^\pi \tanh^{-1}(2x\sin\theta)\sin n\theta\, d\theta &= \pi B_n(x)\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\sin n\theta$ is an odd function.

[4.73] Applying Theorem 4.8 to $f(z) = \tanh^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, and using (1.15), we get

$$\begin{aligned}\int_0^\pi \tanh^{-1}(2x\cos\theta)\cos n\theta d\theta &= \pi A_n(x)\\ &= \pi\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= \pi\sum_{j=0}^{\infty}\binom{n+2j}{j}\frac{x^{n+2j}}{n+2j}.\end{aligned}$$

[4.74] Applying Theorem 4.8 to $f(z) = \tanh^{-1} z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (1.15), we get

$$\begin{aligned}\int_0^\pi \tanh^{-1}(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\cos n\theta$ is an even function.

[4.75] Applying Theorem 4.13 to $f(z) = \tanh^{-1} z$ for $(x, n) \in \left[-\frac{1}{2}, \frac{1}{2}\right] \times (2\mathbb{N} - 1)$, and using (1.15), we get

$$\begin{aligned}\int_0^\pi -i\tanh^{-1}(2ix\sin\theta)\sin n\theta d\theta &= \int_0^\pi \tan^{-1}(2x\sin\theta)\sin n\theta\, d\theta\\ &= \pi B_n(x)\\ &= \pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= \pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}\frac{x^{n+2j}}{n+2j}.\end{aligned} \tag{10.31}$$

[4.76] Applying Theorem 4.13 to $f(z) = \tanh^{-1} z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (1.15) and (10.31), we get

$$\begin{aligned}\int_0^\pi \tan^{-1}(2x \sin\theta) \sin n\theta \, d\theta &= \pi B_n(x) \\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\sin n\theta$ is an odd function.

[4.77] Applying Theorem 4.8 to $f(z) = \sin z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (5.2) and (3.8), we get

$$\begin{aligned}\int_0^\pi \sin(2x \cos\theta) \cos n\theta \, d\theta &= \pi A_n(x) \\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\ &= \pi \sum_{j=0}^{\infty} (-1)^{(n+2j-1)/2} \binom{n+2j}{j} \frac{x^{n+2j}}{(n+2j)!} \\ &= (-1)^{(n-1)/2} \pi x^n \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j}}{j!(n+j)!} \\ &= (-1)^{(n-1)/2} \pi J_n(2x).\end{aligned}$$

[4.78] Applying Theorem 4.8 to $f(z) = \sin z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (5.2), we get

$$\begin{aligned}\int_0^\pi \sin(2x \cos\theta) \cos n\theta \, d\theta &= \pi A_n(x) \\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\cos n\theta$ is an even function.

[4.79] Applying Theorem 4.13 to $f(z) = \sin z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (5.2) and (3.19), we get

$$\begin{aligned}
&\int_0^\pi -i \sin(2ix \sin\theta) \sin n\theta \, d\theta = \int_0^\pi \sinh(2x \sin\theta) \sin n\theta \, d\theta \\
&= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\
&= \pi B_n(x) \\
&= \pi \sum_{j=0}^{\infty} (-1)^j (-1)^{(n+2j-1)/2} \binom{n+2j}{j} \frac{x^{n+2j}}{(n+2j)!} \\
&= (-1)^{(n-1)/2} x^n \pi \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!} \\
&= (-1)^{(n-1)/2} \pi I_n(2x).
\end{aligned} \tag{10.32}$$

[4.80] Applying Theorem 4.13 to $f(z) = \sin z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (10.32) and (5.2), we get

$$\begin{aligned}
\int_0^\pi \sinh(2x \sin\theta) \sin n\theta \, d\theta &= \pi B_n(x) \\
&= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\
&= 0.
\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\sin n\theta$ is an odd function.

[4.81] Applying Theorem 4.8 to $f(z) = \sinh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (5.4) and (3.19), we get

$$\begin{aligned}
\int_0^\pi \sinh(2x \cos\theta) \cos n\theta \, d\theta &= \pi A_n(x) \\
&= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\
&= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{x^{n+2j}}{(n+2j)!} \\
&= \pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!} \\
&= \pi I_n(2x).
\end{aligned}$$

[4.82] Applying Theorem 4.8 to $f(z) = \sinh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (5.4), we get

$$\begin{aligned}\int_0^\pi \sinh(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\cos n\theta$ is an even function.

[4.83] Applying Theorem 4.13 to $f(z) = \sinh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (5.4) and (3.8), we get

$$\begin{aligned}\int_0^\pi -i\sinh(2ix\sin\theta)\sin n\theta\, d\theta &= \int_0^\pi \sin(2x\sin\theta)\sin n\theta\, d\theta\\ &= \pi B_n(x)\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} \frac{x^{n+2j}}{(n+2j)!}\\ &= \pi x^n \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j}}{j!(n+j)!}\\ &= \pi J_n(2x).\end{aligned} \tag{10.33}$$

[4.84] Applying Theorem 4.13 to $f(z) = \sinh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (10.33) and (5.4), we get

$$\begin{aligned}\int_0^\pi \sin(2x\sin\theta)\sin n\theta\, d\theta &= \pi B_n(x)\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}$ because $\sin n\theta$ is an odd function.

[4.85] Applying Theorem 4.8 to $f(z) = \cos z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (5.3) and (3.8), we get

$$\begin{aligned}\int_0^\pi \cos(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= \pi \sum_{j=0}^{\infty} (-1)^{(n+2j)/2} \binom{n+2j}{j} \frac{x^{n+2j}}{(n+2j)!}\\ &= (-1)^{n/2}\pi x^n \sum_{j=0}^{\infty} \frac{(-1)^j x^{2j}}{j!(n+j)!}\\ &= (-1)^{n/2}\pi J_n(2x).\end{aligned}$$

[4.86] Applying Theorem 4.8 to $f(z) = \cos z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (5.3), we get

$$\begin{aligned}\int_0^\pi \cos(2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.87] Applying Theorem 4.11 to $f(z) = \cos z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (5.3) and (3.19), we get

$$\begin{aligned}\int_0^\pi \cos(2ix\sin\theta)\cos n\theta\, d\theta &= \int_0^\pi \cosh(2x\sin\theta)\cos n\theta\, d\theta\\ &= \pi B_n(x)\\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j}\\ &= \pi \sum_{j=0}^{\infty} (-1)^j (-1)^{(n+2j)/2} \binom{n+2j}{j} \frac{x^{n+2j}}{(n+2j)!}\\ &= (-1)^{n/2}\pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!}\\ &= (-1)^{n/2}\pi I_n(2x).\end{aligned} \tag{10.34}$$

[4.88] Applying Theorem 4.11 to $f(z) = \cos z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (10.34) and (5.3), we get

$$\begin{aligned}\int_0^{\pi} \cosh(2x \sin\theta) \cos n\theta \, d\theta &= \pi B_n(x) \\ &= \pi \sum_{j=0}^{\infty} (-1)^j \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.89] Applying Theorem 4.8 to $f(z) = \cosh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (5.5) and (3.19), we get

$$\begin{aligned}\int_0^{\pi} \cosh(2x \cos\theta) \cos n\theta \, d\theta &= \pi A_n(x) \\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{x^{n+2j}}{(n+2j)!} \\ &= \pi x^n \sum_{j=0}^{\infty} \frac{x^{2j}}{j!(n+j)!} \\ &= \pi I_n(2x).\end{aligned}$$

[4.90] Applying Theorem 4.8 to $f(z) = \cosh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (5.5), we get

$$\begin{aligned}\int_0^{\pi} \cosh(2x \cos\theta) \cos n\theta \, d\theta &= \pi A_n(x) \\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.91] Applying Theorem 4.11 to $f(z) = \cosh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 2)$, and using (5.5) and (3.8), we get

$$\begin{aligned}\int_0^\pi \cosh(2ix\sin\theta)\cos n\theta\, d\theta &= \int_0^\pi \cos(2x\sin\theta)\cos n\theta\, d\theta \\ &= \pi B_n(x) \\ &= \pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j} \\ &= \pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}\frac{x^{n+2j}}{(n+2j)!} \\ &= \pi x^n\sum_{j=0}^{\infty}(-1)^j\frac{x^{2j}}{j!(n+j)!} \\ &= \pi J_n(2x).\end{aligned} \tag{10.35}$$

[4.92] Applying Theorem 4.11 to $f(z) = \cosh z$ for $(x, n) \in \mathbb{R} \times (2\mathbb{N} - 1)$, and using (10.35) and (5.5), we get

$$\begin{aligned}\int_0^\pi \cos(2x\sin\theta)\cos n\theta\, d\theta &= \pi B_n(x) \\ &= \pi\sum_{j=0}^{\infty}(-1)^j\binom{n+2j}{j}a_{n+2j}x^{n+2j} \\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z} + 1$ because $\cos n\theta$ is an even function.

[4.93] Put $n = 0$ in [4.94].

[4.94] Applying Theorem 4.8 to $f(z) = (y + z)^m$ for $(x, y, m, n) \in \mathbb{R}^2 \times (\mathbb{N} - 1) \times [0\,..\,m]$, and using the binomial theorem, we get

$$\begin{aligned}\int_0^\pi (y + 2x\cos\theta)^m \cos n\theta\, d\theta &= \pi A_n(x) \\ &= \pi\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j} \\ &= \pi\sum_{j=0}^{\lfloor (m-n)/2\rfloor}\binom{n+2j}{j}\binom{m}{n+2j}y^{m-n-2j}x^{n+2j}.\end{aligned}$$

[4.95] Applying Theorem 4.8 to $f(z) = (y+z)^m$ for $(x, y, m, n) \in \mathbb{R}^2 \times (\mathbb{N}-1) \times ((\mathbb{N}-1) - [0..m])$, and using the binomial theorem, we get

$$\begin{aligned}\int_0^\pi (y + 2x\cos\theta)^m \cos n\theta \, d\theta &= \pi A_n(x) \\ &= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j} \\ &= 0.\end{aligned}$$

The above identity is also true for $n \in 2\mathbb{Z}+1$ because $\cos n\theta$ is an even function.

[4.96] *First Method:* Put $n = 0$ in [4.6].
*Second Method:* Applying Theorem 4.1 to $f(z) = e^{xz}$, and using (5.1), we get

$$\begin{aligned}\pi a_0 &= \pi \\ &= \int_0^\pi \frac{e^{xe^{i\theta}} + e^{xe^{-i\theta}}}{2} \, d\theta \\ &= \int_0^\pi \frac{e^{x(\cos\theta + i\sin\theta)} + e^{x(\cos\theta - i\sin\theta)}}{2} \, d\theta \\ &= \int_0^\pi e^{x\cos\theta} \frac{e^{ix\sin\theta} + e^{-ix\sin\theta}}{2} \, d\theta \\ &= \int_0^\pi e^{x\cos\theta} \cos(x\sin\theta) \, d\theta.\end{aligned} \tag{10.36}$$

[4.97] Applying Theorem 4.4 to $f(z) = e^{xz}$ for $(x, n) \in \mathbb{R} \times \mathbb{N}$, and using (10.36) and (5.1), we get

$$\begin{aligned}\frac{\pi}{2} a_n &= \frac{\pi}{2n!} x^n \\ &= \int_0^\pi \frac{e^{xe^{i\theta}} + e^{xe^{-i\theta}}}{2} \cos n\theta \, d\theta \\ &= \int_0^\pi e^{x\cos\theta} \cos(x\sin\theta) \cos n\theta \, d\theta.\end{aligned} \tag{10.37}$$

[4.98] Using [4.6] for $(x, n) \in \mathbb{R} \times \mathbb{N}$, and (10.37), we get

$$\begin{aligned}\frac{\pi}{n!}x^n &= \int_0^\pi e^{x\cos\theta}\cos(n\theta - x\sin\theta)\,d\theta \\ &= \int_0^\pi e^{x\cos\theta}\cos(x\sin\theta)\cos n\theta\,d\theta + \int_0^\pi e^{x\cos\theta}\sin(x\sin\theta)\sin n\theta\,d\theta \\ &= \frac{\pi}{2n!}x^n + \int_0^\pi e^{x\cos\theta}\sin(x\sin\theta)\sin n\theta\,d\theta \\ \Longleftrightarrow \int_0^\pi e^{x\cos\theta}&\sin(x\sin\theta)\sin n\theta\,d\theta = \frac{\pi}{2n!}x^n.\end{aligned}$$

[4.99] Replace $n$ with $n - m$ in [4.6].

[4.100] Replace $n$ with $n - m$ in [4.7].

[4.101] Put $x = 1$ in [4.102].

[4.102] Put $n = 0$ in [4.46].

[4.103] Put $x = 1$ in [4.104].

[4.104] Using [4.45] for $(x, n) \in (\mathbb{R} - (-1, 1)) \times \mathbb{N}$, we have

$$\begin{aligned}&\int_0^\pi \left(\cos n\theta \log\left(\sqrt{1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta}\right) + \sin n\theta \tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right)\right) d\theta \\ &= (-1)^{n+1}\frac{\pi}{nx^n}.\end{aligned} \tag{10.38}$$

Using [4.46] for $(x, n) \in (\mathbb{R} - (-1, 1)) \times \mathbb{N}$, we have

$$\int_0^\pi \left(\cos n\theta \log\left(\sqrt{1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta}\right) - \sin n\theta \tan^{-1}\left(\frac{\sin\theta}{x + \cos\theta}\right)\right) d\theta = 0. \tag{10.39}$$

Taking the average of (10.38) and (10.39), we get

$$\int_0^\pi \cos n\theta \log\sqrt{1 + \frac{1}{x^2} + \frac{2}{x}\cos\theta}\,d\theta = (-1)^{n+1}\frac{\pi}{2nx^n}.$$

[4.105] Subtracting (10.39) from (10.38), and dividing the result by 2, we get

$$\int_0^\pi \sin n\theta \tan^{-1}\left(\frac{\sin\theta}{x+\cos\theta}\right) d\theta = (-1)^{n+1}\frac{\pi}{2nx^n}.$$

[4.106] Applying Theorem 4.7 to $f(z) = \log(1+z)$ for $x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$, and using the logarithmic series, we get

$$\begin{aligned}\int_0^\pi \log(1+2x\cos\theta)\, d\theta &= \pi A_0(x)\\ &= \pi\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}x^{2j}\\ &= -\pi\sum_{j=1}^{\infty}\binom{2j}{j}\frac{x^{2j}}{2j}.\end{aligned}$$

[4.107] Applying Theorem 4.8 to $f(z) = \log(1+z)$ for $(x, n) \in \left(-\frac{1}{2}, \frac{1}{2}\right) \times \mathbb{N}$, and using the logarithmic series, we get

$$\begin{aligned}\int_0^\pi \log(1+2x\cos\theta)\cos n\theta\, d\theta &= \pi A_n(x)\\ &= \pi\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}x^{n+2j}\\ &= \pi\sum_{j=0}^{\infty}(-1)^{n+1+2j}\binom{n+2j}{j}\frac{x^{n+2j}}{n+2j}\\ &= (-1)^{n+1}\pi\sum_{j=0}^{\infty}\binom{n+2j}{j}\frac{x^{n+2j}}{n+2j}.\end{aligned}$$

[4.108] Applying Theorem 4.7 to $f(z) = (1+z)^y$ for $(x, y) \in \left(-\frac{1}{2}, \frac{1}{2}\right) \times \mathbb{R}$, and using the binomial theorem, we get

$$\int_0^{\pi} (1+2x\cos\theta)^y \, d\theta = \pi A_0(x)$$
$$= \pi \sum_{j=0}^{\infty} \binom{2j}{j} a_{2j} x^{2j}$$
$$= \pi \sum_{j=0}^{\infty} \binom{2j}{j} \frac{\prod_{k=0}^{2j-1} (y-k)}{(2j)!} x^{2j}.$$

[4.109] Applying Theorem 4.8 to $f(z) = (1+z)^y$ for $(x, y, n) \in \left(-\frac{1}{2}, \frac{1}{2}\right) \times \mathbb{R} \times \mathbb{N}$, and using the binomial theorem, we get

$$\int_0^{\pi} (1+2x\cos\theta)^y \cos n\theta \, d\theta = \pi A_n(x)$$
$$= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} x^{n+2j}$$
$$= \pi \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{\prod_{k=0}^{n+2j-1} (y-k)}{(n+2j)!} x^{n+2j}.$$

[4.110] Applying Theorem 4.17 to $f(z) = z^m$ for $(x, y, m) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (2\mathbb{N} - 2)$ and $n = 0$, we get

$$\frac{\pi}{2} T_0(x, y) = \pi \sum_{j=0}^{\infty} \binom{2j}{j} a_{2j} (xy)^j$$
$$= \pi \binom{m}{m/2} (xy)^{m/2}$$
$$= \int_0^{\pi} \frac{\left(xe^{i\theta} + ye^{-i\theta}\right)^m + \left(xe^{-i\theta} + ye^{i\theta}\right)^m}{2} \, d\theta$$
$$= \int_0^{\pi} \frac{1}{2} \left(((x+y)\cos\theta + i(x-y)\sin\theta)^m + ((x+y)\cos\theta \right.$$
$$\left. -i(x-y)\sin\theta)^m\right) d\theta$$
$$= \int_0^{\pi} \frac{1}{2} \left(x^2 + y^2 + 2xy\cos 2\theta\right)^{m/2} \left(\left(\frac{(x+y)\cos\theta + i(x-y)\sin\theta}{\sqrt{x^2+y^2+2xy\cos 2\theta}}\right)^m \right.$$

$$+\left(\frac{(x+y)\cos\theta - i(x-y)\sin\theta}{\sqrt{x^2+y^2+2xy\cos 2\theta}}\right)^m\Bigg)\,d\theta$$

$$= \int_0^\pi \left(x^2+y^2+2xy\cos 2\theta\right)^{m/2} \frac{e^{im\phi(x,y,\theta)} + e^{-im\phi(x,y,\theta)}}{2}\,d\theta$$

$$= \int_0^\pi \left(x^2+y^2+2xy\cos 2\theta\right)^{m/2} \cos m\phi(x,y,\theta)\,d\theta, \tag{10.40}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.48).

[4.111] Applying Theorem 4.17 to $f(z) = z^m$ for $(x, y, m) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (2\mathbb{N} - 1)$, $n = 0$, and using (10.40), we get

$$\begin{aligned}\frac{\pi}{2}T_0(x,y) &= \pi\sum_{j=0}^{\infty}\binom{2j}{j}a_{2j}(xy)^j\\ &= 0\\ &= \int_0^\pi \left(x^2+y^2+2xy\cos 2\theta\right)^{m/2}\cos m\phi(x,y,\theta)\,d\theta,\end{aligned}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.48).

[4.112] Applying Theorem 4.17 to $f(z) = z^m$ for $(x, y, m, n) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (\mathbb{N} - 1)^2$ such that $n \le m$ and $(m - n)/2$ is an integer, and using (10.40), we get

$$\begin{aligned}\frac{\pi}{2}T_n(x,y) &= \frac{\pi}{2}\left(x^n+y^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}(xy)^j\\ &= \frac{\pi\left(x^n+y^n\right)}{2}\binom{m}{(m-n)/2}(xy)^{(m-n)/2}\\ &= \int_0^\pi \left(x^2+y^2+2xy\cos 2\theta\right)^{m/2}\cos m\phi(x,y,\theta)\cos n\theta\,d\theta,\end{aligned}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.48).

[4.113] Applying Theorem 4.17 to $f(z) = z^m$ for $(x, y, m, n) \in \mathbb{R} \times (\mathbb{R} - \{-x\}) \times (\mathbb{N} - 1)^2$ such that $n > m$ or $(m - n)/2$ is not an integer, and using (10.40), we get

$$\begin{aligned}\frac{\pi}{2}T_n(x,y) &= \frac{\pi}{2}\left(x^n+y^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}(xy)^j\\ &= 0\\ &= \int_0^{\pi}\left(x^2+y^2+2xy\cos 2\theta\right)^{m/2}\cos m\phi(x,y,\theta)\cos n\theta\,d\theta,\end{aligned}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.48).

[4.114] Applying Theorem 4.28 to $f(z) = z^m$ for $(x, y, m) \in \left(\mathbb{R}^2 - \{(0,0)\}\right) \times (\mathbb{N} - 1)$, we get

$$\begin{aligned}\pi f(x) = \pi x^m &= \int_0^{\pi}\frac{\left(x+ye^{-i\theta}\right)^m+\left(x+ye^{i\theta}\right)^m}{2}\,d\theta\\ &= \int_0^{\pi}\frac{1}{2}\left((x+y\cos\theta+iy\sin\theta)^m+(x+y\cos\theta-iy\sin\theta)^m\right)\,d\theta\\ &= \int_0^{\pi}\frac{1}{2}\left(x^2+y^2+2xy\cos\theta\right)^{m/2}\left(\left(\frac{x+y\cos\theta+iy\sin\theta}{\sqrt{x^2+y^2+2xy\cos\theta}}\right)^m\right.\\ &\quad\left.+\left(\frac{x+y\cos\theta-iy\sin\theta}{\sqrt{x^2+y^2+2xy\cos\theta}}\right)^m\right)\,d\theta\\ &= \int_0^{\pi}\left(x^2+y^2+2xy\cos\theta\right)^{m/2}\frac{e^{im\phi(x,y,\theta)}+e^{-im\phi(x,y,\theta)}}{2}\,d\theta\\ &= \int_0^{\pi}\left(x^2+y^2+2xy\cos\theta\right)^{m/2}\cos m\phi(x,y,\theta)\,d\theta,\end{aligned}\tag{10.41}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.49).

[4.115] Applying Theorem 4.41 to $f(z) = z^m$ for $(x, y, m, n) \in \left(\mathbb{R}^2 - \{(0,0)\}\right) \times \mathbb{N} \times [1..m]$ and using (10.41), we get

$$\begin{aligned}\frac{\pi}{2}S_n(x,y) &= \frac{\pi y^n}{2}\sum_{j=0}^{\infty}\binom{n+j}{j}a_{n+j}x^j\\ &= \frac{\pi}{2}\binom{m}{n}x^{m-n}y^n\\ &= \int_0^{\pi}\left(x^2+y^2+2xy\cos\theta\right)^{m/2}\cos m\phi(x,y,\theta)\cos n\theta\,d\theta,\end{aligned}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.49).

[4.116] Applying Theorem 4.41 to $f(z) = z^m$ for $(x, y, m, n) \in \left(\mathbb{R}^2 - \{(0,0)\}\right) \times \mathbb{N} \times (\mathbb{N} - [1..m])$ and using (10.41), we get

$$\begin{aligned}\frac{\pi}{2} S_n(x, y) &= \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j \\ &= 0 \\ &= \int_0^{\pi} \left(x^2 + y^2 + 2xy\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta \, d\theta,\end{aligned}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.49).

[4.117] Put $n = 0$ in [4.118].

[4.118] Applying Theorem 4.17 to $f(z) = (1+z)^m$ for $(x, y, m, n) \in \mathbb{R}^2 \times (\mathbb{N} - 1)^2$, we get

$$\begin{aligned}\frac{\pi}{2} T_n(x, y) &= \frac{\pi\left(x^n + y^n\right)}{2} \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\ &= \frac{\pi\left(x^n + y^n\right)}{2} \sum_{j=0}^{\lfloor (m-n)/2 \rfloor} \binom{n+2j}{j} \binom{m}{n+2j} (xy)^j \\ &= \int_0^{\pi} \frac{\left(1 + xe^{i\theta} + ye^{-i\theta}\right)^m + \left(1 + xe^{-i\theta} + ye^{i\theta}\right)^m}{2} \cos n\theta \, d\theta \\ &= \int_0^{\pi} \frac{1}{2} \left((1 + (x+y)\cos\theta + i(x-y)\sin\theta)^m \right. \\ &\quad \left. + (1 + (x+y)\cos\theta - i(x-y)\sin\theta)^m\right) \cos n\theta \, d\theta \\ &= \int_0^{\pi} \frac{1}{2} \left(1 + x^2 + y^2 + 2xy\cos 2\theta + 2(x+y)\cos\theta\right)^{m/2} \\ &\quad \left(\left(\frac{1 + (x+y)\cos\theta + i(x-y)\sin\theta}{\sqrt{1 + x^2 + y^2 + 2xy\cos 2\theta + 2(x+y)\cos\theta}}\right)^m \right. \\ &\quad \left. + \left(\frac{1 + (x+y)\cos\theta - i(x-y)\sin\theta}{\sqrt{1 + x^2 + y^2 + 2xy\cos 2\theta + 2(x+y)\cos\theta}}\right)^m\right) \cos n\theta \, d\theta \\ &= \int_0^{\pi} \left(1 + x^2 + y^2 + 2xy\cos 2\theta + 2(x+y)\cos\theta\right)^{m/2} \\ &\quad \frac{e^{im\phi(x,y,\theta)} + e^{-im\phi(x,y,\theta)}}{2} \cos n\theta \, d\theta \\ &= \int_0^{\pi} \left(1 + x^2 + y^2 + 2xy\cos 2\theta + 2(x+y)\cos\theta\right)^{m/2} \cos m\phi(x, y, \theta) \cos n\theta \\ &\quad d\theta,\end{aligned}$$

where the function $\phi(x, y, \theta)$ is as defined in (4.50).

[4.119] Replace $x$ with $1 + x$ in [4.114].

[4.120] Replace $x$ with $1 + x$ in [1.115] and use the binomial theorem.

[4.121] Applying Theorem 4.17 to $f(z) = \log(1 + z)$ for $(x, y) \in \mathbb{R}^2$ such that $|x| + |y| \leq 1$ and $n = 0$, we get

$$\begin{aligned}
\pi T_0(x, y) &= 2\pi \sum_{j=0}^{\infty} \binom{2j}{j} a_{2j} (xy)^j \\
&= -\pi \sum_{j=1}^{\infty} \binom{2j}{j} \frac{(xy)^j}{j} \\
&= 2 \int_0^{\pi} \frac{\log\left(1 + xe^{i\theta} + ye^{-i\theta}\right) + \log\left(1 + xe^{-i\theta} + ye^{i\theta}\right)}{2} \, d\theta \\
&= \int_0^{\pi} (\log(1 + (x + y)\cos\theta + i(x - y)\sin\theta) \\
&\quad + \log(1 + (x + y)\cos\theta - i(x - y)\sin\theta)) \, d\theta \\
&= \int_0^{\pi} \log\left(1 + x^2 + y^2 + 2(x + y)\cos\theta + 2xy\cos 2\theta\right) \, d\theta.
\end{aligned} \tag{10.42}$$

[4.122] Applying Theorem 4.17 to $f(z) = \log(1 + z)$ for $(x, y, n) \in \mathbb{R}^2 \times \mathbb{N}$ such that $|x| + |y| \leq 1$ and using (10.42), we get

$$\begin{aligned}
\pi T_n(x, y) &= \pi \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n + 2j}{j} a_{n+2j} (xy)^j \\
&= (-1)^{n+1} \pi (x^n + y^n) \sum_{j=0}^{\infty} \binom{n + 2j}{j} \frac{(xy)^j}{n + 2j} \\
&= 2 \int_0^{\pi} \frac{\log\left(1 + xe^{i\theta} + ye^{-i\theta}\right) + \log\left(1 + xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^{\pi} \log\left(1 + x^2 + y^2 + 2(x + y)\cos\theta + 2xy\cos 2\theta\right) \cos n\theta \, d\theta.
\end{aligned}$$

[4.123] Put $y = 0$ in [4.121].

[4.124] Put $y = 0$ in [4.122].

[4.125] Applying Theorem 4.28 to $f(z) = \sin z$, we get

$$\begin{aligned}\pi f(x) = \pi \sin x &= \int_0^\pi \frac{\sin\left(x + ye^{-i\theta}\right) + \sin\left(x + ye^{i\theta}\right)}{2}\, d\theta \\ &= \int_0^\pi \sin\left(x + y\frac{e^{i\theta} + e^{-i\theta}}{2}\right)\cos\left(y\frac{e^{i\theta} - e^{-i\theta}}{2}\right)\, d\theta \\ &= \int_0^\pi \sin\left(x + y\cos\theta\right)\cos\left(iy\sin\theta\right)\, d\theta \\ &= \int_0^\pi \sin(x + y\cos\theta)\cosh(y\sin\theta)\, d\theta.\end{aligned} \tag{10.43}$$

[4.126] Applying Theorem 4.28 to $f(z) = \sinh z$, we get

$$\begin{aligned}\pi f(x) = \pi \sinh x &= \int_0^\pi \frac{\sinh\left(x + ye^{-i\theta}\right) + \sinh\left(x + ye^{i\theta}\right)}{2}\, d\theta \\ &= \int_0^\pi \sinh\left(x + y\frac{e^{i\theta} + e^{-i\theta}}{2}\right)\cosh\left(y\frac{e^{i\theta} - e^{-i\theta}}{2}\right)\, d\theta \\ &= \int_0^\pi \sinh\left(x + y\cos\theta\right)\cosh\left(iy\sin\theta\right)\, d\theta \\ &= \int_0^\pi \sinh(x + y\cos\theta)\cos(y\sin\theta)\, d\theta.\end{aligned} \tag{10.44}$$

[4.127] Applying Theorem 4.28 to $f(z) = \cos z$, we get

$$\begin{aligned}\pi f(x) = \pi \cos x &= \int_0^\pi \frac{\cos\left(x + ye^{-i\theta}\right) + \cos\left(x + ye^{i\theta}\right)}{2}\, d\theta \\ &= \int_0^\pi \cos\left(x + y\frac{e^{i\theta} + e^{-i\theta}}{2}\right)\cos\left(y\frac{e^{i\theta} - e^{-i\theta}}{2}\right)\, d\theta \\ &= \int_0^\pi \cos\left(x + y\cos\theta\right)\cos\left(iy\sin\theta\right)\, d\theta \\ &= \int_0^\pi \cos(x + y\cos\theta)\cosh(y\sin\theta)\, d\theta.\end{aligned} \tag{10.45}$$

[4.128] Applying Theorem 4.28 to $f(z) = \cosh z$, we get

$$\begin{aligned}\pi f(x) = \pi \cosh x &= \int_0^\pi \frac{\cosh\left(x + ye^{-i\theta}\right) + \cosh\left(x + ye^{i\theta}\right)}{2}\, d\theta \\ &= \int_0^\pi \cosh\left(x + y\frac{e^{i\theta} + e^{-i\theta}}{2}\right) \cosh\left(y\frac{e^{i\theta} - e^{-i\theta}}{2}\right) d\theta \\ &= \int_0^\pi \cosh\left(x + y\cos\theta\right) \cosh\left(iy\sin\theta\right)\, d\theta \\ &= \int_0^\pi \cosh(x + y\cos\theta)\cos(y\sin\theta)\, d\theta.\end{aligned} \tag{10.46}$$

[4.129] Applying Theorem 4.41 to $f(z) = \sin z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$ and using (10.43), (5.3), and (5.2), we get

$$\begin{aligned}\frac{\pi}{2} S_n(x, y) &= \frac{\pi y^n}{2} \sum_{j=0}^\infty \binom{n+j}{j} a_{n+j} x^j \\ &= \frac{\pi y^n}{2} \sum_{j=0}^\infty \binom{n+2j}{2j} (-1)^{(n+2j-1)/2} \frac{x^{2j}}{(n+2j)!} \\ &= (-1)^{(n-1)/2} \frac{\pi y^n}{2n!} \sum_{j=0}^\infty (-1)^j \frac{x^{2j}}{(2j)!} \\ &= (-1)^{(n-1)/2} \frac{\pi y^n}{2n!} \cos x \\ &= \int_0^\pi \frac{\sin\left(x + ye^{-i\theta}\right) + \sin\left(x + ye^{i\theta}\right)}{2} \cos n\theta\, d\theta \\ &= \int_0^\pi \sin(x + y\cos\theta)\cosh(y\sin\theta)\cos n\theta\, d\theta.\end{aligned}$$

[4.130] Applying Theorem 4.41 to $f(z) = \sin z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 2)$ and using (10.43) and (5.2), we get

$$\begin{aligned}\frac{\pi}{2} S_n(x, y) &= \frac{\pi y^n}{2} \sum_{j=0}^\infty \binom{n+j}{j} a_{n+j} x^j \\ &= \frac{\pi y^n}{2} \sum_{j=0}^\infty \binom{n+2j+1}{2j+1} (-1)^{(n+2j)/2} \frac{x^{2j+1}}{(n+2j+1)!}\end{aligned}$$

$$
\begin{aligned}
&= (-1)^{n/2}\frac{\pi y^n}{2n!}\sum_{j=0}^{\infty}(-1)^j\frac{x^{2j+1}}{(2j+1)!}\\
&= (-1)^{n/2}\frac{\pi y^n}{2n!}\sin x\\
&= \int_0^{\pi}\frac{\sin\left(x+ye^{-i\theta}\right)+\sin\left(x+ye^{i\theta}\right)}{2}\cos n\theta\,d\theta\\
&= \int_0^{\pi}\sin(x+y\cos\theta)\cosh(y\sin\theta)\cos n\theta\,d\theta.
\end{aligned}
$$

[4.131] Applying Theorem 4.41 to $f(z)=\sinh z$ for $(x,y,n)\in\mathbb{R}^2\times(2\mathbb{N}-1)$ and using (10.44), (5.4), and (5.5), we get

$$
\begin{aligned}
\frac{\pi}{2}S_n(x,y) &= \frac{\pi y^n}{2}\sum_{j=0}^{\infty}\binom{n+j}{j}a_{n+j}x^j\\
&= \frac{\pi y^n}{2}\sum_{j=0}^{\infty}\binom{n+2j}{2j}\frac{x^{2j}}{(n+2j)!}\\
&= \frac{\pi y^n}{2n!}\sum_{j=0}^{\infty}\frac{x^{2j}}{(2j)!}\\
&= \frac{\pi y^n}{2n!}\cosh x\\
&= \int_0^{\pi}\frac{\sinh\left(x+ye^{-i\theta}\right)+\sinh\left(x+ye^{i\theta}\right)}{2}\cos n\theta\,d\theta\\
&= \int_0^{\pi}\sinh(x+y\cos\theta)\cos(y\sin\theta)\cos n\theta\,d\theta.
\end{aligned}
$$

[4.132] Applying Theorem 4.41 to $f(z)=\sinh z$ for $(x,y,n)\in\mathbb{R}^2\times(2\mathbb{N}-2)$ and using (10.44) and (5.4), we get

$$
\begin{aligned}
\frac{\pi}{2}S_n(x,y) &= \frac{\pi y^n}{2}\sum_{j=0}^{\infty}\binom{n+j}{j}a_{n+j}x^j\\
&= \frac{\pi y^n}{2}\sum_{j=0}^{\infty}\binom{n+2j+1}{2j+1}\frac{x^{2j+1}}{(n+2j+1)!}\\
&= \frac{\pi y^n}{2n!}\sum_{j=0}^{\infty}\frac{x^{2j+1}}{(2j+1)!}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{\pi y^n}{2n!} \sinh x \\
&= \int_0^\pi \frac{\sinh\left(x + ye^{-i\theta}\right) + \sinh\left(x + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^\pi \sinh(x + y\cos\theta) \cos(y \sin\theta) \cos n\theta \, d\theta.
\end{aligned}
$$

[4.133] Applying Theorem 4.41 to $f(z) = \cos z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$ and using (10.45), (5.3), and (5.2), we get

$$
\begin{aligned}
\frac{\pi}{2} S_n(x, y) &= \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j \\
&= \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+2j+1}{2j+1} (-1)^{(n+2j+1)/2} \frac{x^{2j+1}}{(n+2j+1)!} \\
&= (-1)^{(n+1)/2} \frac{\pi y^n}{2n!} \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j+1}}{(2j+1)!} \\
&= (-1)^{(n+1)/2} \frac{\pi y^n}{2n!} \sin x \\
&= \int_0^\pi \frac{\cos\left(x + ye^{-i\theta}\right) + \cos\left(x + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^\pi \cos(x + y\cos\theta) \cosh(y \sin\theta) \cos n\theta \, d\theta.
\end{aligned}
$$

[4.134] Applying Theorem 4.41 to $f(z) = \cos z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 2)$ and using (10.45) and (5.3), we get

$$
\begin{aligned}
\frac{\pi}{2} S_n(x, y) &= \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j \\
&= \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+2j}{2j} (-1)^{(n+2j)/2} \frac{x^{2j}}{(n+2j)!} \\
&= (-1)^{n/2} \frac{\pi y^n}{2n!} \sum_{j=0}^{\infty} (-1)^j \frac{x^{2j}}{(2j)!} \\
&= (-1)^{n/2} \frac{\pi y^n}{2n!} \cos x
\end{aligned}
$$

$$= \int_0^\pi \frac{\cos\left(x + ye^{-i\theta}\right) + \cos\left(x + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta$$
$$= \int_0^\pi \cos(x + y\cos\theta)\cosh(y\sin\theta)\cos n\theta \, d\theta.$$

[4.135] Applying Theorem 4.41 to $f(z) = \cosh z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$ and using (10.46), (5.5), and (5.4), we get

$$\frac{\pi}{2} S_n(x, y) = \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j$$
$$= \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+2j+1}{2j+1} \frac{x^{2j+1}}{(n+2j+1)!}$$
$$= \frac{\pi y^n}{2n!} \sum_{j=0}^{\infty} \frac{x^{2j+1}}{(2j+1)!}$$
$$= \frac{\pi y^n}{2n!} \sinh x$$
$$= \int_0^\pi \frac{\cosh\left(x + ye^{-i\theta}\right) + \cosh\left(x + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta$$
$$= \int_0^\pi \cosh(x + y\cos\theta)\cos(y\sin\theta)\cos n\theta \, d\theta.$$

[4.136] Applying Theorem 4.41 to $f(z) = \cosh z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 2)$ and using (10.46) and (5.5), we get

$$\frac{\pi}{2} S_n(x, y) = \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+j}{j} a_{n+j} x^j$$
$$= \frac{\pi y^n}{2} \sum_{j=0}^{\infty} \binom{n+2j}{2j} \frac{x^{2j}}{(n+2j)!}$$
$$= \frac{\pi y^n}{2n!} \sum_{j=0}^{\infty} \frac{x^{2j}}{(2j)!}$$
$$= \frac{\pi y^n}{2n!} \cosh x$$
$$= \int_0^\pi \frac{\cosh\left(x + ye^{-i\theta}\right) + \cosh\left(x + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta$$
$$= \int_0^\pi \cosh(x + y\cos\theta)\cos(y\sin\theta)\cos n\theta \, d\theta.$$

[4.137] Put $y = 1/x$ in [4.110].

[4.138] Put $y = 1/x$ in [4.111].

[4.139] Put $x = 1$ and replace $y$ with $x$ in [4.114].

[4.140] Put $y = 1/x$ in [4.114].

[4.141] Put $x = y = 1$ in [4.119].

[4.142] Put $x = y = 1$ in [4.120].

[4.143] Put $x = y = 1$ in [4.110].

[4.144] Put $x = y = 1$ in [4.111].

[4.145] Put $x = y = 1$ and $n = m$ in [4.112].

[4.146] Putting $x = 1/2$ in [4.123], we get

$$\begin{aligned} & \int_0^\pi \log\left(1 + \frac{1}{4} + \cos\theta\right) d\theta = 0 \\ \iff & \int_0^\pi \log\left(\frac{5 + 4\cos\theta}{4}\right) d\theta = 0 \\ \iff & \int_0^\pi \log\left(5 + 4\cos\theta\right)\, d\theta = \int_0^\pi (\log 4)\, d\theta = 2\pi \log 2. \end{aligned}$$

[4.147] Putting $x = -1$ in [4.123], we get

$$\begin{aligned} & \int_0^\pi \log(2 - 2\cos\theta)\, d\theta = 0 \\ \iff & \int_0^\pi \log\left(4\sin^2\frac{\theta}{2}\right) d\theta = 0 \\ \iff & \int_0^\pi \log\left(2\sin\frac{\theta}{2}\right) d\theta = 0. \end{aligned} \tag{10.47}$$

We also have

$$\begin{aligned}\int_0^\pi \log\left(2\sin\frac{\theta}{2}\right)d\theta &= \int_0^\pi \log\left(2\sin\frac{\pi-\theta}{2}\right)d\theta \\ &= \int_0^\pi \log\left(2\cos\frac{\theta}{2}\right)d\theta \\ &= 0.\end{aligned} \tag{10.48}$$

Adding (10.47) and (10.48), we get

$$\begin{aligned}&\int_0^\pi \log\left(4\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)d\theta = 0 \\ \iff &\int_0^\pi \log(2\sin\theta)\,d\theta = 0 \\ \iff &\int_0^\pi \log\sin\theta\,d\theta = -\int_0^\pi (\log 2)d\theta = -\pi\log 2.\end{aligned}$$

[4.148] Putting $x = 1$ in [4.124], we get

$$\begin{aligned}&\int_0^\pi \log(2+2\cos\theta)\cos n\theta\,d\theta = (-1)^{n+1}\frac{\pi}{n} \\ \iff &\int_0^\pi \log\left(4\cos^2\frac{\theta}{2}\right)\cos n\theta\,d\theta = (-1)^{n+1}\frac{\pi}{n} \\ \iff &\int_0^\pi \log\left(2\cos\frac{\theta}{2}\right)\cos n\theta\,d\theta = (-1)^{n+1}\frac{\pi}{2n} \\ \iff &\int_0^\pi \log\left(\cos\frac{\theta}{2}\right)\cos n\theta\,d\theta = (-1)^{n+1}\frac{\pi}{2n} - \int_0^\pi (\log 2)\cos n\theta\,d\theta \\ &= (-1)^{n+1}\frac{\pi}{2n} - \log 2\,\frac{\sin n\theta}{n}\Big|_0^\pi \\ &= (-1)^{n+1}\frac{\pi}{2n}.\end{aligned}$$

[4.149] Putting $x = 1/e^2$ in [4.123], we get

$$\begin{aligned}&\int_0^\pi \log\left(1+\frac{1}{e^4}+\frac{2}{e^2}\cos\theta\right)d\theta = 0 \\ \iff &\int_0^\pi \log\left(\frac{1}{e^2}\left(e^2+\frac{1}{e^2}+2\cos\theta\right)\right)d\theta = 0 \\ \iff &\int_0^\pi \log\left(e^2+\frac{1}{e^2}+2\cos\theta\right)d\theta = \int_0^\pi \log\left(e^2\right)d\theta = 2\pi.\end{aligned}$$

[4.150] Put $y = -1/4x$ in [4.121] for $x \in \left[-1, -\frac{1}{4}\right] \cup \left[\frac{1}{4}, 1\right]$.

[4.151] Put $y = -1/4x$ in [4.122] for $(x, n) \in \left(\left[-1, -\frac{1}{4}\right] \cup \left[\frac{1}{4}, 1\right]\right) \times \mathbb{N}$.

[4.152] Put $x = 1/2$ in [4.150].

[4.153] Put $x = 1/2$ in [4.151] for $n \in 2\mathbb{N}$.

[4.154] Put $x = 1/2$ in [4.151] for $n \in 2\mathbb{N} - 1$.

[4.155] Put $y = 1/x$ in [4.117].

[4.156] Put $y = -1/x$ in [4.117].

[4.157] Put $x = y = 1$ in [4.117].

[4.158] Put $x = y = 1$ in [4.118].

[4.159] Put $x = 1$ and $y = -1$ in [4.117].

[4.160] Put $x = 1$ and $y = -1$ in [4.118] for $(m, n) \in (\mathbb{N} - 1) \times (2\mathbb{N} - 2)$.

[4.161] Put $x = 1$ and $y = -1$ in [4.118] for $(m, n) \in (\mathbb{N} - 1) \times (2\mathbb{N} - 1)$.

[4.162] Put $x = y = 1$ in [4.119].

[4.163] Put $x = y = 1$ in [4.120].

# Chapter 11
# Solutions to Generalized Bessel Functions

[5.1] Putting $\alpha = 2$ in (5.6), and using (5.5) and [3.6], we get

$$\begin{aligned} E_2(z^2) &= \sum_{j=0}^{\infty} \frac{z^{2j}}{\Gamma(2j+1)} \\ &= \sum_{j=0}^{\infty} \frac{z^{2j}}{(2j)!} \\ &= \cosh z. \end{aligned}$$

[5.2] From (5.7), we have

$$\begin{aligned} E_{\alpha,\beta}(z) &= \sum_{j=0}^{\infty} \frac{z^j}{\Gamma(\alpha j+\beta)} \\ &= \frac{1}{\Gamma(\beta)} + z \sum_{j=1}^{\infty} \frac{z^{j-1}}{\Gamma(\alpha(j-1)+\alpha+\beta)} \\ &= \frac{1}{\Gamma(\beta)} + z \sum_{j=0}^{\infty} \frac{z^j}{\Gamma(\alpha j+\alpha+\beta)} \\ &= \frac{1}{\Gamma(\beta)} + z E_{\alpha,\alpha+\beta}. \end{aligned}$$

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[5.3] From (5.8), we have

$$\begin{aligned}
E^{\gamma}_{\alpha,\beta-\alpha}(z) - E^{\gamma-1}_{\alpha,\beta-\alpha}(z) &= \sum_{j=0}^{\infty} \frac{\prod_{k=0}^{j-1}(\gamma+k)}{j!\Gamma(\alpha j+\beta-\alpha)} z^j - \sum_{j=0}^{\infty} \frac{\prod_{k=0}^{j-1}(\gamma+k-1)}{j!\Gamma(\alpha j+\beta-\alpha)} z^j \\
&= \sum_{j=0}^{\infty} \frac{(\gamma+j-1-(\gamma-1))\prod_{k=0}^{j-2}(\gamma+k)}{j!\Gamma(\alpha j+\beta-\alpha)} z^j \\
&= z\sum_{j=1}^{\infty} \frac{\prod_{k=0}^{j-2}(\gamma+k)}{(j-1)!\Gamma(\alpha(j-1)+\beta)} z^{j-1} \\
&= z\sum_{j=0}^{\infty} \frac{\prod_{k=0}^{j-1}(\gamma+k)}{j!\Gamma(\alpha j+\beta)} z^j \\
&= zE^{\gamma}_{\alpha,\beta}(z).
\end{aligned}$$

[5.4] From (5.9) and (3.6), we have

$$\begin{aligned}
&\beta E^{\gamma,q}_{\alpha,\beta+1}(z) + \alpha z\frac{d}{dz}\left(E^{\gamma,q}_{\alpha,\beta+1}(z)\right) \\
&= \beta\sum_{j=0}^{\infty} \frac{\prod_{k=0}^{qj-1}(\gamma+k)}{j!\Gamma(\alpha j+\beta+1)} z^j + \alpha z\frac{d}{dz}\left(\sum_{j=0}^{\infty} \frac{\prod_{k=0}^{qj-1}(\gamma+k)}{j!\Gamma(\alpha j+\beta+1)} z^j\right) \\
&= \beta\sum_{j=0}^{\infty} \frac{\prod_{k=0}^{qj-1}(\gamma+k)}{j!(\alpha j+\beta)\Gamma(\alpha j+\beta)} z^j + \alpha z\sum_{j=1}^{\infty} \frac{j\prod_{k=0}^{qj-1}(\gamma+k)}{j!(\alpha j+\beta)\Gamma(\alpha j+\beta)} z^{j-1} \\
&= \sum_{j=0}^{\infty} \frac{(\alpha j+\beta)\prod_{k=0}^{qj-1}(\gamma+k)}{j!(\alpha j+\beta)\Gamma(\alpha j+\beta)} z^j
\end{aligned}$$

$$= \sum_{j=0}^{\infty} \frac{\prod_{k=0}^{qj-1} (\gamma + k)}{j!\Gamma(\alpha j + \beta)} z^j$$
$$= E_{\alpha,\beta}^{\gamma,q}(z).$$

[5.5] From (5.10) and (3.6), we have

$$\begin{aligned}
&\phi(\alpha, \beta - 1; z) + (1 - \beta)\phi(\alpha, \beta; z) \\
&= \sum_{j=0}^{\infty} \frac{z^j}{j!\Gamma(\alpha j + \beta - 1)} + (1 - \beta) \sum_{j=0}^{\infty} \frac{z^j}{j!\Gamma(\alpha j + \beta)} \\
&= \sum_{j=-1}^{\infty} \frac{z^{j+1}}{(j+1)!\Gamma(\alpha(j+1) + \beta - 1)} + \sum_{j=-1}^{\infty} \frac{(1-\beta)z^{j+1}}{(j+1)!\Gamma(\alpha(j+1) + \beta)} \\
&= \frac{1}{\Gamma(\beta - 1)} + \frac{1 - \beta}{\Gamma(\beta)} + \sum_{j=0}^{\infty} \frac{z^{j+1}}{(j+1)!\Gamma(\alpha j + \alpha + \beta - 1)} \\
&\quad + \sum_{j=0}^{\infty} \frac{(1-\beta)z^{j+1}}{(j+1)!\Gamma(\alpha j + \alpha + \beta)} \\
&= \frac{\beta - 1}{(\beta - 1)\Gamma(\beta - 1)} + \frac{1 - \beta}{\Gamma(\beta)} + z \sum_{j=0}^{\infty} \left( \frac{1}{\Gamma(\alpha j + \alpha + \beta - 1)} \right. \\
&\quad \left. + \frac{1 - \beta}{\Gamma(\alpha j + \alpha + \beta)} \right) \frac{z^j}{(j+1)!} \\
&= \frac{\beta - 1}{\Gamma(\beta)} + \frac{1 - \beta}{\Gamma(\beta)} + z \sum_{j=0}^{\infty} \left( \frac{\alpha j + \alpha + \beta - 1}{(\alpha j + \alpha + \beta - 1)\Gamma(\alpha j + \alpha + \beta - 1)} \right. \\
&\quad \left. + \frac{1 - \beta}{\Gamma(\alpha j + \alpha + \beta)} \right) \frac{z^j}{(j+1)!} \\
&= z \sum_{j=0}^{\infty} \left( \frac{\alpha j + \alpha + \beta - 1}{\Gamma(\alpha j + \alpha + \beta)} + \frac{1 - \beta}{\Gamma(\alpha j + \alpha + \beta)} \right) \frac{z^j}{(j+1)!} \\
&= z \sum_{j=0}^{\infty} \frac{\alpha(j+1)z^j}{(j+1)!\Gamma(\alpha j + \alpha + \beta)} \\
&= \alpha z \sum_{j=0}^{\infty} \frac{z^j}{j!\Gamma(\alpha j + \alpha + \beta)} \\
&= \alpha z \phi(\alpha, \alpha + \beta; z).
\end{aligned}$$

[5.6] From (5.10), we have

$$\begin{aligned}\frac{d\phi(\alpha,\beta;z)}{dz} &= \frac{d}{dz}\left(\sum_{j=0}^{\infty}\frac{z^j}{j!\Gamma(\alpha j+\beta)}\right)\\ &= \sum_{j=1}^{\infty}\frac{jz^{j-1}}{j!\Gamma(\alpha j+\beta)}\\ &= \sum_{j=1}^{\infty}\frac{z^{j-1}}{(j-1)!\Gamma(\alpha j+\beta)}\\ &= \sum_{j=0}^{\infty}\frac{z^j}{j!\Gamma(\alpha(j+1)+\beta)}\\ &= \sum_{j=0}^{\infty}\frac{z^j}{j!\Gamma(\alpha j+\alpha+\beta)}\\ &= \phi(\alpha,\alpha+\beta;z).\end{aligned}$$

[5.7] Multiply [5.6] by $az$ and use [5.5].

[5.8] Use (5.11) and [4.22].

[5.9] Use (5.11) and [4.23]. Proved in [4.23] for $n \in \mathbb{N}-1$. For $n \in -\mathbb{N}$, using (5.12) and [4.23], we have

$$\begin{aligned}e(1,n) &= e(1,-n)\\ &= \frac{1}{\pi}\int_0^{\pi} e^{2\cos\theta}\cos(-n\theta)\,d\theta\\ &= \frac{1}{\pi}\int_0^{\pi} e^{2\cos\theta}\cos n\theta\,d\theta.\end{aligned}$$

[5.10] Use (5.11) and [4.24].

[5.11] Use (5.11) and [4.25]. Proved in [4.25] for $n \in \mathbb{N}-1$. For $n \in -\mathbb{N}$, using (5.12) and [4.25], we have

$$\begin{aligned}e\left(x^2,n\right) &= x^{-2n}e\left(x^2,-n\right)\\ &= \frac{x^{-2n}}{\pi x^{-n}}\int_0^{\pi} e^{2x\cos\theta}\cos(-n\theta)\,d\theta\\ &= \frac{1}{\pi x^n}\int_0^{\pi} e^{2x\cos\theta}\cos n\theta\,d\theta.\end{aligned}$$

[5.12] Use (5.11) and [4.77]. Proved in [4.77] for $n \in 2\mathbb{N} - 1$. For $n \in -(2\mathbb{N} - 1)$, using (5.12), [3.9], and [4.77], we have

$$\begin{aligned} e\left(-x^2, n\right) &= (-1)^{-n} x^{-2n} e\left(-x^2, -n\right) \\ &= \frac{(-1)^{(-n-1)/2}(-1)^{-n} x^{-2n}}{\pi x^{-n}} \int_0^{\pi} \sin(2x\cos\theta)\cos(-n\theta)\, d\theta \\ &= \frac{(-1)^{(n-1)/2}}{\pi x^n} \int_0^{\pi} \sin(2x\cos\theta)\cos n\theta\, d\theta \\ &= -\frac{J_{-n}(2x)}{x^n} \\ &= \frac{J_n(2x)}{x^n}. \end{aligned}$$

[5.13] Use (5.11) and [4.85]. Proved in [4.85] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.12), [3.9], and [4.85], we have

$$\begin{aligned} e\left(-x^2, n\right) &= (-1)^{-n} x^{-2n} e\left(-x^2, -n\right) \\ &= \frac{(-1)^{-n/2}(-1)^{-n} x^{-2n}}{\pi x^{-n}} \int_0^{\pi} \cos(2x\cos\theta)\cos(-n\theta)\, d\theta \\ &= \frac{(-1)^{n/2}}{\pi x^n} \int_0^{\pi} \cos(2x\cos\theta)\cos n\theta\, d\theta \\ &= \frac{J_{-n}(2x)}{x^n} \\ &= \frac{J_n(2x)}{x^n}. \end{aligned}$$

[5.14] Use (5.11) and [4.79]. Proved in [4.79] for $n \in 2\mathbb{N} - 1$. For $n \in -(2\mathbb{N} - 1)$, using (5.12), [3.19], and [4.79], we have

$$\begin{aligned} e\left(x^2, n\right) &= x^{-2n} e\left(x^2, -n\right) \\ &= \frac{(-1)^{(-n-1)/2} x^{-2n}}{\pi x^{-n}} \int_0^{\pi} \sinh(2x\sin\theta)\sin(-n\theta)\, d\theta \\ &= \frac{(-1)^{(n-1)/2}}{\pi x^n} \int_0^{\pi} \sinh(2x\sin\theta)\sin n\theta\, d\theta \\ &= \frac{I_{-n}(2x)}{x^n} \\ &= \frac{I_n(2x)}{x^n}. \end{aligned}$$

[5.15] Use (5.11) and [4.87]. Proved in [4.87] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.12), [3.19], and [4.87], we have

$$\begin{aligned} e\left(x^2, n\right) &= x^{-2n} e\left(x^2, -n\right) \\ &= \frac{(-1)^{-n/2} x^{-2n}}{\pi x^{-n}} \int_0^{\pi} \cosh\left(2x \sin\theta\right) \cos(-n\theta)\, d\theta \\ &= \frac{(-1)^{n/2}}{\pi x^n} \int_0^{\pi} \cosh\left(2x \sin\theta\right) \cos n\theta\, d\theta \\ &= \frac{I_{-n}(2x)}{x^n} \\ &= \frac{I_n(2x)}{x^n}. \end{aligned}$$

[5.16] Use (5.11) and [4.81]. Proved in [4.81] for $n \in 2\mathbb{N} - 1$. For $n \in -(2\mathbb{N} - 1)$, using (5.12), [3.19], and [4.81], we have

$$\begin{aligned} e\left(x^2, n\right) &= x^{-2n} e\left(x^2, -n\right) \\ &= \frac{x^{-2n}}{\pi x^{-n}} \int_0^{\pi} \sinh\left(2x \cos\theta\right) \cos(-n\theta)\, d\theta \\ &= \frac{1}{\pi x^n} \int_0^{\pi} \sinh\left(2x \cos\theta\right) \cos n\theta\, d\theta \\ &= \frac{I_{-n}(2x)}{x^n} \\ &= \frac{I_n(2x)}{x^n}. \end{aligned}$$

[5.17] Use (5.11) and [4.89]. Proved in [4.89] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.12), [3.19], and [4.89], we have

$$\begin{aligned} e\left(x^2, n\right) &= x^{-2n} e\left(x^2, -n\right) \\ &= \frac{x^{-2n}}{\pi x^{-n}} \int_0^{\pi} \cosh\left(2x \cos\theta\right) \cos(-n\theta)\, d\theta \\ &= \frac{1}{\pi x^n} \int_0^{\pi} \cosh\left(2x \cos\theta\right) \cos n\theta\, d\theta \\ &= \frac{I_{-n}(2x)}{x^n} \\ &= \frac{I_n(2x)}{x^n}. \end{aligned}$$

[5.18] Use (5.11) and [4.83]. Proved in [4.83] for $n \in 2\mathbb{N} - 1$. For $n \in -(2\mathbb{N} - 1)$, using (5.12), [3.9], and [4.83], we have

$$\begin{aligned} e\left(-x^2, n\right) &= (-1)^{-n} x^{-2n} e\left(-x^2, -n\right) \\ &= \frac{(-1)^n x^{-2n}}{\pi x^{-n}} \int_0^\pi \sin(2x \sin\theta) \sin(-n\theta)\, d\theta \\ &= \frac{1}{\pi x^n} \int_0^\pi \sin(2x \sin\theta) \sin n\theta\, d\theta \\ &= -\frac{J_{-n}(2x)}{x^n} \\ &= \frac{J_n(2x)}{x^n}. \end{aligned}$$

[5.19] Use (5.11) and [4.91]. Proved in [4.91] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.12), [3.9], and [4.91], we have

$$\begin{aligned} e\left(-x^2, n\right) &= (-1)^{-n} x^{-2n} e\left(-x^2, -n\right) \\ &= \frac{(-1)^n x^{-2n}}{\pi x^{-n}} \int_0^\pi \cos(2x \sin\theta) \cos(-n\theta)\, d\theta \\ &= \frac{1}{\pi x^n} \int_0^\pi \cos(2x \sin\theta) \cos n\theta\, d\theta \\ &= \frac{J_{-n}(2x)}{x^n} \\ &= \frac{J_n(2x)}{x^n}. \end{aligned}$$

[5.20] Put $n = 0$ in [5.21].

[5.21] Put $y = 1$ in [5.47].

[5.22] Put $y = 1$ in [5.54].

[5.23] Put $y = 1$ in [5.52].

[5.24] From [5.26], we have

$$\begin{aligned}
\pi &= \int_0^{\pi} e^{(x+1)\cos\theta} \cos\left((x+1)\sin\theta\right)\, d\theta \\
\iff 1 &= \frac{1}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \cos\left(\sin\theta\right)\cos\left(x\sin\theta\right)\, d\theta \\
&\quad - \frac{1}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \sin\left(\sin\theta\right)\sin\left(x\sin\theta\right)\, d\theta.
\end{aligned} \tag{11.1}$$

From [5.20], we have

$$\begin{aligned}
e(x,0) &= \frac{1}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \cos\left((x-1)\sin\theta\right)\, d\theta \\
&= \frac{1}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \cos\left(\sin\theta\right)\cos\left(x\sin\theta\right)\, d\theta \\
&\quad + \frac{1}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \sin\left(\sin\theta\right)\sin\left(x\sin\theta\right)\, d\theta.
\end{aligned} \tag{11.2}$$

Adding (11.1) and (11.2), we get

$$\begin{aligned}
e(x,0)+1 &= \frac{2}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \cos\left(\sin\theta\right)\cos\left(x\sin\theta\right)\, d\theta \\
\iff e(x,0) &= -1 + \frac{2}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \cos\left(\sin\theta\right)\cos\left(x\sin\theta\right)\, d\theta.
\end{aligned}$$

[5.25] Subtracting (11.1) from (11.2), we get

$$\begin{aligned}
e(x,0)-1 &= \frac{2}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \sin\left(\sin\theta\right)\sin\left(x\sin\theta\right)\, d\theta \\
\iff e(x,0) &= 1 + \frac{2}{\pi}\int_0^{\pi} e^{(x+1)\cos\theta} \sin\left(\sin\theta\right)\sin\left(x\sin\theta\right)\, d\theta.
\end{aligned}$$

[5.26] Replace $x$ with $x+1$ in [4.96].

[5.27] Use (5.19) and [4.32].

[5.28] Use (5.19) and [4.31].

[5.29] Use (5.19) and [4.33]. Proved in [4.33] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.19) and [4.33], we have

$$\begin{aligned}\sinh\left(x^2, n\right) &= x^{-2n} \sinh\left(x^2, -n\right) \\ &= \frac{(-1)^{-n/2} x^{-2n}}{\pi x^{-n}} \int_0^\pi \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta)\right. \\ &\qquad \left. + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos(-n\theta)\,d\theta \\ &= \frac{(-1)^{n/2}}{\pi x^n} \int_0^\pi \left(\sin^2(x\cos\theta)\cosh^2(x\sin\theta)\right. \\ &\qquad \left. + \cos^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos n\theta\,d\theta.\end{aligned}$$

[5.30] Use (5.19) and [4.35]. Proved in [4.35] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.19) and [4.35], we have

$$\begin{aligned}\sinh\left(x^2, n\right) &= x^{-2n} \sinh\left(x^2, -n\right) \\ &= \frac{x^{-2n}}{\pi x^{-n}} \int_0^\pi \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta)\right. \\ &\qquad \left. + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos(-n\theta)\,d\theta \\ &= \frac{1}{\pi x^n} \int_0^\pi \left(\sinh^2(x\cos\theta)\cos^2(x\sin\theta)\right. \\ &\qquad \left. + \cosh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\,d\theta.\end{aligned}$$

[5.31] Use (5.20) and [4.37].

[5.32] Use (5.20) and [4.38].

[5.33] Use (5.20) and [4.39]. Proved in [4.39] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.20) and [4.39], we have

$$\begin{aligned}\cosh\left(x^2, n\right) &= x^{-2n} \cosh\left(x^2, -n\right) \\ &= \frac{(-1)^{-n/2} x^{-2n}}{\pi x^{-n}} \int_0^\pi \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta)\right. \\ &\qquad \left. + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos(-n\theta)\,d\theta \\ &= \frac{(-1)^{n/2}}{\pi x^n} \int_0^\pi \left(\cos^2(x\cos\theta)\cosh^2(x\sin\theta)\right. \\ &\qquad \left. + \sin^2(x\cos\theta)\sinh^2(x\sin\theta)\right)\cos n\theta\,d\theta.\end{aligned}$$

[5.34] Use (5.20) and [4.41]. Proved in [4.41] for $n \in 2\mathbb{N} - 2$. For $n \in -2\mathbb{N}$, using (5.20) and [4.41], we have

$$\begin{aligned}
\cosh\left(x^2, n\right) &= x^{-2n} \cosh\left(x^2, -n\right) \\
&= \frac{x^{-2n}}{\pi x^{-n}} \int_0^\pi \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta)\right. \\
&\qquad \left. + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos(-n\theta)\, d\theta \\
&= \frac{1}{\pi x^n} \int_0^\pi \left(\cosh^2(x\cos\theta)\cos^2(x\sin\theta)\right. \\
&\qquad \left. + \sinh^2(x\cos\theta)\sin^2(x\sin\theta)\right)\cos n\theta\, d\theta.
\end{aligned}$$

[5.35] Replacing $x$ with $ix$ in [5.21] and using (5.14), (5.26), and (5.17), we get

$$\begin{aligned}
& e(ix, n) = \frac{2}{\pi(1 + i^n x^n)} \int_0^\pi e^{(ix+1)\cos\theta} \cos\left((ix - 1)\sin\theta\right)\cos n\theta d\theta \\
\iff\ & \cos(x, n) + i\sin(x, n) \\
&= \frac{2}{\pi(1 + i^n x^n)} \int_0^\pi e^{ix\cos\theta} \cos\left(ix\sin\theta - \sin\theta\right) e^{\cos\theta} \cos n\theta\, d\theta \\
&= \frac{2}{\pi(1 + i^n x^n)} \int_0^\pi (\cos(x\cos\theta) + i\sin(x\cos\theta)) \\
&\quad (\cos(ix\sin\theta)\cos(\sin\theta) + \sin(ix\sin\theta)\sin(\sin\theta)) e^{\cos\theta} \cos n\theta\, d\theta \\
&= \frac{2}{\pi(1 + i^n x^n)} \int_0^\pi (\cos(x\cos\theta) + i\sin(x\cos\theta)) \\
&\quad (\cosh(x\sin\theta)\cos(\sin\theta) + i\sinh(x\sin\theta)\sin(\sin\theta)) e^{\cos\theta} \cos n\theta\, d\theta \\
&= \frac{2}{\pi(1 + i^n x^n)} \int_0^\pi ((\cos(x\cos\theta)\cosh(x\sin\theta)\cos(\sin\theta) \\
&\quad - \sin(x\cos\theta)\sinh(x\sin\theta)\sin(\sin\theta)) \\
&\quad + i\,(\sin(x\cos\theta)\cosh(x\sin\theta)\cos(\sin\theta) \\
&\quad + \cos(x\cos\theta)\sinh(x\sin\theta)\sin(\sin\theta))\, e^{\cos\theta} \cos n\theta\, d\theta \\
&= \frac{2}{\pi(1 + i^n x^n)} \int_0^\pi (C_1(x, \theta) + iS_1(x, \theta)) e^{\cos\theta} \cos n\theta\, d\theta.
\end{aligned} \tag{11.3}$$

For $(x, n) \in \mathbb{R} \times 2\mathbb{Z} - \{-1, 1\} \times 2(2\mathbb{Z} + 1)$, (11.3) becomes

$$\begin{aligned}
& \cos(x, n) + i\sin(x, n) \\
&= \frac{2}{\pi(1 + (-1)^{n/2} x^n)} \int_0^\pi C_1(x, \theta) e^{\cos\theta} \cos n\theta\, d\theta \\
&\quad + \frac{2i}{\pi(1 + (-1)^{n/2} x^n)} \int_0^\pi S_1(x, \theta) e^{\cos\theta} \cos n\theta\, d\theta.
\end{aligned} \tag{11.4}$$

Equating the imaginary part of (11.4) for $(x, n) \in \mathbb{R} \times 2\mathbb{Z} - \{-1, 1\} \times 2(2\mathbb{Z}+1)$, we get

$$\sin(x, n) = \frac{2}{\pi\left(1 + (-1)^{n/2}x^n\right)} \int_0^\pi S_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta. \tag{11.5}$$

[5.36] From (11.5), for $(x, n) \in \{-1, 1\} \times 2(2\mathbb{Z}+1)$, we get

$$\int_0^\pi S_1(x, \theta) e^{\cos\theta} \cos n\theta d\theta = \frac{\pi}{2}\left(1 + (-1)^{n/2}x^n\right) \sin(x, n) = 0.$$

[5.37] For $(x, n) \in \mathbb{R} \times (2\mathbb{Z}+1)$, (11.3) becomes

$$\begin{aligned}
&\cos(x, n) + i \sin(x, n) \\
&= \frac{2\,(1 - i^n x^n)}{\pi(1 + i^n x^n)(1 - i^n x^n)} \int_0^\pi (C_1(x, \theta) + i S_1(x, \theta)) e^{\cos\theta} \cos n\theta \, d\theta \\
&= \frac{2\left(1 + (-1)^{(n+1)/2} i x^n\right)}{\pi(1 + x^{2n})} \int_0^\pi (C_1(x, \theta) + i S_1(x, \theta)) e^{\cos\theta} \cos n\theta \, d\theta \\
&= \left( \frac{2}{\pi(1 + x^{2n})} \int_0^\pi C_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta \right. \\
&\quad \left. + \frac{2(-1)^{(n-1)/2} x^n}{\pi(1 + x^{2n})} \int_0^\pi S_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta \right) \\
&+ i \left( \frac{2}{\pi(1 + x^{2n})} \int_0^\pi S_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta \right. \\
&\quad \left. + \frac{2(-1)^{(n+1)/2} x^n}{\pi(1 + x^{2n})} \int_0^\pi C_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta \right).
\end{aligned} \tag{11.6}$$

Equating the imaginary part of (11.6) for $(x, n) \in \mathbb{R} \times (2\mathbb{Z}+1)$, we get

$$\begin{aligned}
\sin(x, n) = &\frac{2}{\pi\left(1 + x^{2n}\right)} \int_0^\pi S_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta \\
&+ (-1)^{(n+1)/2} \frac{2x^n}{\pi\left(1 + x^{2n}\right)} \int_0^\pi C_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta.
\end{aligned}$$

[5.38] Equating the real part of (11.4) for $(x, n) \in \mathbb{R} \times 2\mathbb{Z} - \{-1, 1\} \times 2(2\mathbb{Z} + 1)$, we get

$$\cos(x, n) = \frac{2}{\pi\left(1 + (-1)^{n/2}x^n\right)} \int_0^\pi C_1(x, \theta) e^{\cos\theta} \cos n\theta \, d\theta. \tag{11.7}$$

[5.39] From (11.7) for $(x, n) \in \{-1, 1\} \times 2(2\mathbb{Z}+1)$, we get

$$\int_0^\pi C_1(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta = \frac{\pi}{2}\left(1+(-1)^{n/2}x^n\right)\cos(x,n) = 0.$$

[5.40] Equating the real part of (11.6) for $(x, n) \in \mathbb{R} \times (2\mathbb{Z}+1)$, we get

$$\begin{aligned}\cos(x,n) = {} & \frac{2}{\pi\left(1+x^{2n}\right)}\int_0^\pi C_1(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta \\ & +(-1)^{(n-1)/2}\frac{2x^n}{\pi\left(1+x^{2n}\right)}\int_0^\pi S_1(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta.\end{aligned}$$

[5.41] Replace $x$ with $ix$ in [5.35].

[5.42] Replace $x$ with $ix$ in [5.37].

[5.43] From [5.42] for $(x, n) \in \{-1, 1\} \times (2\mathbb{Z}+1)$, we get

$$\begin{aligned}& \int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta - x^n\int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta \\ & = \frac{\pi}{2}\left(1-x^{2n}\right)\sinh(x,n) = 0 \\ \Longleftrightarrow\ & \int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta d\theta = x^n\int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta.\end{aligned}$$

[5.44] Replace $x$ with $ix$ in [5.38].

[5.45] Replace $x$ with $ix$ in [5.40].

[5.46] From [5.45] for $(x, n) \in \{-1, 1\} \times (2\mathbb{Z}+1)$, we get

$$\begin{aligned}& \int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta d\theta - x^n\int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta \\ & = \frac{\pi}{2}\left(1-x^{2n}\right)\cosh(x,n) = 0 \\ \Longleftrightarrow\ & \int_0^\pi C_2(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta = x^n\int_0^\pi S_2(x,\theta)e^{\cos\theta}\cos n\theta\, d\theta.\end{aligned}$$

[5.47] Applying Theorem 4.17 to $f(z) = e^z$ for $n \in \mathbb{N} - 1$, and using (5.1) and (5.11), we get

$$\begin{aligned}
\frac{\pi}{2} T_n(x, y) &= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\
&= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{(xy)^j}{(n+2j)!} \\
&= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \frac{(xy)^j}{j!(n+j)!} \\
&= \frac{\pi}{2} \left(x^n + y^n\right) e(xy, n) \\
&= \int_0^{\pi} \frac{e^{xe^{i\theta} + ye^{-i\theta}} + e^{xe^{-i\theta} + ye^{i\theta}}}{2} \cos n\theta \, d\theta \\
&= \int_0^{\pi} \frac{e^{(x+y)\cos\theta + i(x-y)\sin\theta} + e^{(x+y)\cos\theta - i(x-y)\sin\theta}}{2} \cos n\theta \, d\theta \\
&= \int_0^{\pi} e^{(x+y)\cos\theta} \frac{e^{i(x-y)\sin\theta} + e^{-i(x-y)\sin\theta}}{2} \cos n\theta \, d\theta \\
&= \int_0^{\pi} e^{(x+y)\cos\theta} \cos((x-y)\sin\theta) \cos n\theta \, d\theta. \qquad (11.8)
\end{aligned}$$

For $n \in -\mathbb{N}$, using (5.12) and (11.8), we get

$$\begin{aligned}
&\int_0^{\pi} e^{(x+y)\cos\theta} \cos((x-y)\sin\theta) \cos n\theta \, d\theta \\
&= \int_0^{\pi} e^{(x+y)\cos\theta} \cos((x-y)\sin\theta) \cos(-n\theta) \, d\theta \\
&= \frac{\pi}{2} \left(x^{-n} + y^{-n}\right) e(xy, -n) \\
&= \frac{\pi}{2} \left(x^{-n} + y^{-n}\right) (xy)^n e(xy, n) \\
&= \frac{\pi}{2} \left(x^n + y^n\right) e(xy, n).
\end{aligned}$$

[5.48] Applying Theorem 4.17 to $f(z) = \sin z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$, and using (5.2) and (5.11), we get

$$\frac{\pi}{2} T_n(x, y) = \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j$$

$$
\begin{aligned}
&= \frac{\pi}{2}\left(x^n + y^n\right) \sum_{j=0}^{\infty} (-1)^{(n+2j-1)/2} \binom{n+2j}{j} \frac{(xy)^j}{(n+2j)!} \\
&= (-1)^{(n-1)/2} \frac{\pi}{2}\left(x^n + y^n\right) \sum_{j=0}^{\infty} \frac{(-xy)^j}{j!(n+j)!} \\
&= (-1)^{(n-1)/2} \frac{\pi}{2}\left(x^n + y^n\right) e(-xy, n) \\
&= \int_0^{\pi} \frac{\sin\left(xe^{i\theta} + ye^{-i\theta}\right) + \sin\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^{\pi} \sin\left((x+y)\frac{e^{i\theta} + e^{-i\theta}}{2}\right) \cos\left((x-y)\frac{e^{i\theta} - e^{-i\theta}}{2}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \sin\left((x+y)\cos\theta\right) \cos\left(i(x-y)\sin\theta\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \sin\left((x+y)\cos\theta\right) \cosh\left((x-y)\sin\theta\right) \cos n\theta \, d\theta. \qquad (11.9)
\end{aligned}
$$

For $n \in -(2\mathbb{N}-1)$, using (5.12) and (11.9), we get

$$
\begin{aligned}
&\int_0^{\pi} \sin\left((x+y)\cos\theta\right) \cosh\left((x-y)\sin\theta\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \sin\left((x+y)\cos\theta\right) \cosh\left((x-y)\sin\theta\right) \cos(-n\theta) \, d\theta \\
&= (-1)^{(-n-1)/2} \frac{\pi}{2}\left(x^{-n} + y^{-n}\right) e(-xy, -n) \\
&= (-1)^{(-n-1)/2} \frac{\pi}{2}\left(x^{-n} + y^{-n}\right) (-1)^n (xy)^n e(-xy, n) \\
&= (-1)^{(n-1)/2} \frac{\pi}{2}\left(x^n + y^n\right) e(-xy, n).
\end{aligned}
$$

[5.49] Applying Theorem 4.17 to $f(z) = \sin z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N}-2)$, and using (5.2) and (11.9), we get

$$
\begin{aligned}
&\frac{\pi}{2} T_n(x, y) = \frac{\pi}{2}\left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\
&= 0 \\
&= \int_0^{\pi} \frac{\sin\left(xe^{i\theta} + ye^{-i\theta}\right) + \sin\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^{\pi} \sin\left((x+y)\cos\theta\right) \cosh\left((x-y)\sin\theta\right) \cos n\theta \, d\theta.
\end{aligned}
$$

The above identity is also true for $n \in -2\mathbb{N}$ because $\cos n\theta$ is an even function.

[5.50] Applying Theorem 4.17 to $f(z) = \cos z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 2)$, and using (5.3) and (5.11), we get

$$\begin{aligned}
\frac{\pi}{2} T_n(x, y) &= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\
&= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} (-1)^{(n+2j)/2} \binom{n+2j}{j} \frac{(xy)^j}{(n+2j)!} \\
&= (-1)^{n/2} \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \frac{(-xy)^j}{j!(n+j)!} \\
&= (-1)^{n/2} \frac{\pi}{2} \left(x^n + y^n\right) e(-xy, n) \\
&= \int_0^{\pi} \frac{\cos\left(xe^{i\theta} + ye^{-i\theta}\right) + \cos\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^{\pi} \cos\left((x+y)\frac{e^{i\theta} + e^{-i\theta}}{2}\right) \cos\left((x-y)\frac{e^{i\theta} - e^{-i\theta}}{2}\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \cos\left((x+y)\cos\theta\right) \cos\left(i(x-y)\sin\theta\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \cos\left((x+y)\cos\theta\right) \cosh\left((x-y)\sin\theta\right) \cos n\theta \, d\theta. \qquad (11.10)
\end{aligned}$$

For $n \in -2\mathbb{N}$, using (5.12) and (11.10), we get

$$\begin{aligned}
&\int_0^{\pi} \cos\left((x+y)\cos\theta\right) \cosh\left((x-y)\sin\theta\right) \cos n\theta \, d\theta \\
&= \int_0^{\pi} \cos\left((x+y)\cos\theta\right) \cosh\left((x-y)\sin\theta\right) \cos(-n\theta) \, d\theta \\
&= (-1)^{-n/2} \frac{\pi}{2} \left(x^{-n} + y^{-n}\right) e(-xy, -n) \\
&= (-1)^{-n/2} \frac{\pi}{2} \left(x^{-n} + y^{-n}\right) (-1)^n (xy)^n e(-xy, n) \\
&= (-1)^{n/2} \frac{\pi}{2} \left(x^n + y^n\right) e(-xy, n).
\end{aligned}$$

[5.51] Applying Theorem 4.17 to $f(z) = \cos z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$, and using (5.3) and (11.10), we get

$$\begin{aligned}\frac{\pi}{2} T_n(x, y) &= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\ &= 0 \\ &= \int_0^{\pi} \frac{\cos\left(xe^{i\theta} + ye^{-i\theta}\right) + \cos\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\ &= \int_0^{\pi} \cos\left((x + y)\cos\theta\right) \cosh\left((x - y)\sin\theta\right) \cos n\theta \, d\theta.\end{aligned}$$

The above identity is also true for $n \in -(2\mathbb{N} - 1)$ because $\cos n\theta$ is an even function.

[5.52] Applying Theorem 4.17 to $f(z) = \sinh z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$, and using (5.4) and (5.11), we get

$$\begin{aligned}&\frac{\pi}{2} T_n(x, y) = \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\ &= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} \frac{(xy)^j}{(n+2j)!} \\ &= \frac{\pi}{2} \left(x^n + y^n\right) \sum_{j=0}^{\infty} \frac{(xy)^j}{j!(n+j)!} \\ &= \frac{\pi}{2} \left(x^n + y^n\right) e(xy, n) \\ &= \int_0^{\pi} \frac{\sinh\left(xe^{i\theta} + ye^{-i\theta}\right) + \sinh\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\ &= \int_0^{\pi} \sinh\left((x + y)\frac{e^{i\theta} + e^{-i\theta}}{2}\right) \cosh\left((x - y)\frac{e^{i\theta} - e^{-i\theta}}{2}\right) \cos n\theta \, d\theta \\ &= \int_0^{\pi} \sinh\left((x + y)\cos\theta\right) \cosh\left(i(x - y)\sin\theta\right) \cos n\theta \, d\theta \\ &= \int_0^{\pi} \sinh\left((x + y)\cos\theta\right) \cos\left((x - y)\sin\theta\right) \cos n\theta \, d\theta.\end{aligned} \tag{11.11}$$

For $n \in -(2\mathbb{N} - 1)$, using (5.12) and (11.11), we get

$$
\begin{aligned}
&\int_0^\pi \sinh\left((x+y)\cos\theta\right)\cos\left((x-y)\sin\theta\right)\cos n\theta\, d\theta \\
&= \int_0^\pi \sinh\left((x+y)\cos\theta\right)\cos\left((x-y)\sin\theta\right)\cos(-n\theta)\, d\theta \\
&= \frac{\pi}{2}\left(x^{-n}+y^{-n}\right)e(xy,-n) \\
&= \frac{\pi}{2}\left(x^{-n}+y^{-n}\right)(xy)^n e(xy,n) \\
&= \frac{\pi}{2}\left(x^n+y^n\right)e(xy,n).
\end{aligned}
$$

[5.53] Applying Theorem 4.17 to $f(z)=\sinh z$ for $(x,y,n)\in\mathbb{R}^2\times(2\mathbb{N}-2)$, and using (5.4) and (11.11), we get

$$
\begin{aligned}
\frac{\pi}{2}T_n(x,y) &= \frac{\pi}{2}\left(x^n+y^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}(xy)^j \\
&= 0 \\
&= \int_0^\pi \frac{\sinh\left(xe^{i\theta}+ye^{-i\theta}\right)+\sinh\left(xe^{-i\theta}+ye^{i\theta}\right)}{2}\cos n\theta\, d\theta \\
&= \int_0^\pi \sinh\left((x+y)\cos\theta\right)\cos\left((x-y)\sin\theta\right)\cos n\theta\, d\theta.
\end{aligned}
$$

The above identity is also true for $n\in-2\mathbb{N}$ because $\cos n\theta$ is an even function.

[5.54] Applying Theorem 4.17 to $f(z)=\cosh z$ for $(x,y,n)\in\mathbb{R}^2\times(2\mathbb{N}-2)$, and using (5.5) and (5.11), we get

$$
\begin{aligned}
\frac{\pi}{2}T_n(x,y) &= \frac{\pi}{2}\left(x^n+y^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}a_{n+2j}(xy)^j \\
&= \frac{\pi}{2}\left(x^n+y^n\right)\sum_{j=0}^{\infty}\binom{n+2j}{j}\frac{(xy)^j}{(n+2j)!} \\
&= \frac{\pi}{2}\left(x^n+y^n\right)\sum_{j=0}^{\infty}\frac{(xy)^j}{j!(n+j)!} \\
&= \frac{\pi}{2}\left(x^n+y^n\right)e(xy,n)
\end{aligned}
\tag{11.12}
$$

$$
\begin{aligned}
&= \int_0^\pi \frac{\cosh\left(xe^{i\theta} + ye^{-i\theta}\right) + \cosh\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^\pi \cosh\left((x+y)\frac{e^{i\theta} + e^{-i\theta}}{2}\right) \cosh\left((x-y)\frac{e^{i\theta} - e^{-i\theta}}{2}\right) \cos n\theta \, d\theta \\
&= \int_0^\pi \cosh\left((x+y)\cos\theta\right) \cosh\left(i(x-y)\sin\theta\right) \cos n\theta \, d\theta \\
&= \int_0^\pi \cosh\left((x+y)\cos\theta\right) \cos\left((x-y)\sin\theta\right) \cos n\theta \, d\theta.
\end{aligned}
$$

For $n \in -2\mathbb{N}$, using (5.12) and (11.12), we get

$$
\begin{aligned}
&\int_0^\pi \cosh\left((x+y)\cos\theta\right) \cos\left((x-y)\sin\theta\right) \cos n\theta \, d\theta \\
&= \int_0^\pi \cosh\left((x+y)\cos\theta\right) \cos\left((x-y)\sin\theta\right) \cos(-n\theta) \, d\theta \\
&= \frac{\pi}{2}\left(x^{-n} + y^{-n}\right) e(xy, -n) \\
&= \frac{\pi}{2}\left(x^{-n} + y^{-n}\right) (xy)^n e(xy, n) \\
&= \frac{\pi}{2}\left(x^n + y^n\right) e(xy, n).
\end{aligned}
$$

[5.55] Applying Theorem 4.1 to $f(z) = \cosh z$ for $(x, y, n) \in \mathbb{R}^2 \times (2\mathbb{N} - 1)$, and using (5.5) and (11.12), we get

$$
\begin{aligned}
\frac{\pi}{2} T_n(x, y) &= \frac{\pi}{2}\left(x^n + y^n\right) \sum_{j=0}^{\infty} \binom{n+2j}{j} a_{n+2j} (xy)^j \\
&= 0 \\
&= \int_0^\pi \frac{\cosh\left(xe^{i\theta} + ye^{-i\theta}\right) + \cosh\left(xe^{-i\theta} + ye^{i\theta}\right)}{2} \cos n\theta \, d\theta \\
&= \int_0^\pi \cosh\left((x+y)\cos\theta\right) \cos\left((x-y)\sin\theta\right) \cos n\theta \, d\theta.
\end{aligned}
$$

The above identity is also true for $n \in -(2\mathbb{N} - 1)$ because $\cos n\theta$ is an even function.

[5.56] Put $y = 1/x$ in [5.47].

[5.57] Put $y = 1/x$ in [5.48].

[5.58] Put $y = 1/x$ in [5.49].

[5.59] Put $y = 1/x$ in [5.50].

[5.60] Put $y = 1/x$ in [5.51].

[5.61] Put $y = 1/x$ in [5.52].

[5.62] Put $y = 1/x$ in [5.53].

[5.63] Put $y = 1/x$ in [5.54].

[5.64] Put $y = 1/x$ in [5.55].

[5.65] Put $n = 0$ in [5.56].

[5.66] Put $y = -1/x$ and $n = 0$ in [5.50].

[5.67] Put $n = 0$ in [5.63].

[5.68] Put $x = 1$ in [5.56].

[5.69] Put $x = 1$ in [5.57].

[5.70] Put $x = 1$ in [5.58].

[5.71] Put $x = 1$ in [5.59].

[5.72] Put $x = 1$ in [5.60].

[5.73] Put $x = 1$ in [5.61].

[5.74] Put $x = 1$ in [5.62].

[5.75] Put $x = 1$ in [5.63].

[5.76] Put $x = 1$ in [5.64].

[5.77] Put $x = 1$ in [5.29].

[5.78] Put $x = 1$ in [4.34].

[5.79] Put $x = 1$ in [5.30].

[5.80] Put $x = 1$ in [4.36].

[5.81] Put $x = 1$ in [5.33].

[5.82] Put $x = 1$ in [4.40].

[5.83] Put $x = 1$ in [5.34].

[5.84] Put $x = 1$ in [4.42].

[5.85] Put $x = 1$ in [4.96].

[5.86] Put $x = 1$ in [4.97].

[5.87] Putting $x = 1$ in [4.97], we get

$$\int_0^{\pi} e^{\cos\theta} \cos(\sin\theta) \cos n\theta \, d\theta = \frac{\pi}{2n!}. \tag{11.13}$$

Putting $x = -1$ in [4.97], we get

$$\int_0^{\pi} e^{-\cos\theta} \cos(\sin\theta) \cos n\theta d\theta = (-1)^n \frac{\pi}{2n!}. \tag{11.14}$$

Taking the average of (11.13) and (11.14), we get

$$\begin{aligned} \int_0^{\pi} \frac{e^{\cos\theta} + e^{-\cos\theta}}{2} \cos(\sin\theta) \cos n\theta \, d\theta &= \int_0^{\pi} \cosh(\cos\theta) \cos(\sin\theta) \cos n\theta \, d\theta \\ &= \left( \frac{1 + (-1)^n}{2} \right) \frac{\pi}{2n!}. \end{aligned} \tag{11.15}$$

For $n \in 2\mathbb{N} - 2$, (11.15) becomes

$$\int_0^{\pi} \cosh(\cos\theta) \cos(\sin\theta) \cos n\theta d\theta = \frac{\pi}{2n!}.$$

[5.88] For $n \in 2\mathbb{N} - 1$, (11.15) becomes

$$\int_0^{\pi} \cosh(\cos\theta) \cos(\sin\theta) \cos n\theta \, d\theta = 0.$$

The above identity is also true for $n \in -(2\mathbb{N} - 1)$ because $\cos n\theta$ is an even function.

[5.89] Subtracting (11.14) from (11.13) and dividing the result by 2, we get

$$\int_0^{\pi} \frac{e^{\cos\theta} - e^{-\cos\theta}}{2} \cos(\sin\theta)\cos n\theta \, d\theta = \int_0^{\pi} \sinh(\cos\theta)\cos(\sin\theta)\cos n\theta \, d\theta = \left(\frac{1-(-1)^n}{2}\right)\frac{\pi}{2n!}. \tag{11.16}$$

For $n \in 2\mathbb{N} - 1$, (11.16) becomes

$$\int_0^{\pi} \sinh(\cos\theta)\cos(\sin\theta)\cos n\theta \, d\theta = \frac{\pi}{2n!}.$$

[5.90] For $n \in 2\mathbb{N} - 2$, (11.16) becomes

$$\int_0^{\pi} \sinh(\cos\theta)\cos(\sin\theta)\cos n\theta \, d\theta = 0.$$

The above identity is also true for $n \in -2\mathbb{N}$ because $\cos n\theta$ is an even function.

[5.91] Put $x = 1$ and $y = -1$ in [5.54].

[5.92] Put $x = 1$ and $y = -1$ in [5.55].

[5.93] From (5.14), (5.15) and (5.16), we have

$$\begin{aligned}
&\frac{e(iz,n) - e(-iz,n)}{2i} \\
&= \frac{(\cos(z,n) + i\sin(z,n)) - (\cos(-z,n) + i\sin(-z,n))}{2i} \\
&= \frac{(\cos(z,n) + i\sin(z,n)) - (\cos(z,n) - i\sin(z,n))}{2i} \\
&= \frac{2i\sin(z,n)}{2i} \\
&= \sin(z,n).
\end{aligned}$$

[5.94] From (5.14), (5.15), and (5.16), we have

$$\begin{aligned}
&\frac{e(iz,n)+e(-iz,n)}{2}\\
&=\frac{(\cos(z,n)+i\sin(z,n))+(\cos(-z,n)+i\sin(-z,n))}{2}\\
&=\frac{(\cos(z,n)+i\sin(z,n))+(\cos(z,n)-i\sin(z,n))}{2}\\
&=\frac{2\cos(z,n)}{2}\\
&=\cos(z,n).
\end{aligned}$$

[5.95] From (5.14), (5.17), (5.18), (5.19), and (5.20), we have

$$\begin{aligned}
&\frac{e(z,n)-e(-z,n)}{2}\\
&=\frac{(\cos(-iz,n)+i\sin(-iz,n))-(\cos(iz,n)+i\sin(iz,n))}{2}\\
&=\frac{(\cosh(-z,n)-\sinh(-z,n))-(\cosh(z,n)-\sinh(z,n))}{2}\\
&=\frac{(\cosh(z,n)+\sinh(z,n))-(\cosh(z,n)-\sinh(z,n))}{2}\\
&=\frac{2\sinh(z,n)}{2}\\
&=\sinh(z,n).
\end{aligned}$$

[5.96] From (5.14), (5.17), (5.18), (5.19), and (5.20), we have

$$\begin{aligned}
&\frac{e(z,n)+e(-z,n)}{2}\\
&=\frac{(\cos(-iz,n)+i\sin(-iz,n))+(\cos(iz,n)+i\sin(iz,n))}{2}\\
&=\frac{(\cosh(-z,n)-\sinh(-z,n))+(\cosh(z,n)-\sinh(z,n))}{2}\\
&=\frac{(\cosh(z,n)+\sinh(z,n))+(\cosh(z,n)-\sinh(z,n))}{2}\\
&=\frac{2\cosh(z,n)}{2}\\
&=\cosh(z,n).
\end{aligned}$$

# Chapter 12
# Solutions to Series Sums Using Special Integrals




[6.1] Replacing $x$ with $-1+ix$ in [5.21], using (5.14), and $\phi(x)=\tan^{-1}x$, we get the LHS of [5.21] as

$$\begin{aligned}
&e(-1+ix,n)=e\left(-\sqrt{1+x^2}e^{-i\phi(x)},n\right)\\
&=\sum_{j=0}^{\infty}(-1)^j\frac{\left(1+x^2\right)^{j/2}e^{-ij\phi(x)}}{j!(n+j)!}\\
&=\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}e^{-2ij\phi(x)}}{(2j)!(n+2j)!}-\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}e^{-i(2j+1)\phi(x)}}{(2j+1)!(n+2j+1)!}\\
&=\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}(\cos 2j\phi(x)-i\sin 2j\phi(x))}{(2j)!(n+2j)!}\\
&\quad-\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}(\cos(2j+1)\phi(x)-i\sin(2j+1)\phi(x))}{(2j+1)!(n+2j+1)!}\\
&=\left(\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos 2j\phi(x)}{(2j)!(n+2j)!}-\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}\right)\\
&+i\left(\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\sin(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}-\sum_{j=1}^{\infty}\frac{(1+x^2)^j\sin 2j\phi(x)}{(2j)!(n+2j)!}\right).
\end{aligned}\tag{12.1}$$

Replacing $x$ with $-1+ix$ in [5.21] and using $\phi(x)=\tan^{-1}x$, we simplify the factor of the integral on RHS of [5.21] as

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$$\begin{aligned}
&\frac{2}{\pi}\cdot\frac{1}{1+(-1+ix)^n}=\frac{2}{\pi}\cdot\frac{1}{1+\left(1+x^2\right)^{n/2}e^{in(\pi-\phi(x))}}\\
&=\frac{2}{\pi}\cdot\frac{1}{1+\left(1+x^2\right)^{n/2}\cos n(\pi-\phi(x))+i\left(1+x^2\right)^{n/2}\sin n(\pi-\phi(x))}\\
&=\frac{2}{\pi}\cdot\frac{1+\left(1+x^2\right)^{n/2}\cos n(\pi-\phi(x))-i\left(1+x^2\right)^{n/2}\sin n(\pi-\phi(x))}{1+\left(1+x^2\right)^{n}+2\left(1+x^2\right)^{n/2}\cos n(\pi-\phi(x))}.
\end{aligned}\tag{12.2}$$

Replacing $x$ with $-1+ix$ in [5.21] and using (6.28) and (6.29), we simplify the integral on RHS of [5.21] as

$$\begin{aligned}
&\int_0^\pi e^{ix\cos\theta}\cos\left((ix-2)\sin\theta\right)\cos n\theta d\theta\\
&=\int_0^\pi(\cos(x\cos\theta)+i\sin(x\cos\theta))\\
&\quad(\cos(ix\sin\theta)\cos(2\sin\theta)+\sin(ix\sin\theta)\sin(2\sin\theta))\cos n\theta\,d\theta\\
&=\int_0^\pi(\cos(x\cos\theta)+i\sin(x\cos\theta))\\
&\quad(\cosh(x\sin\theta)\cos(2\sin\theta)+i\sinh(x\sin\theta)\sin(2\sin\theta))\cos n\theta\,d\theta\\
&=\int_0^\pi(\cos(x\cos\theta)\cosh(x\sin\theta)\cos(2\sin\theta)\\
&\quad-\sin(x\cos\theta)\sinh(x\sin\theta)\sin(2\sin\theta))\cos n\,\theta\,d\theta\\
&\quad+i\int_0^\pi(\cos(x\cos\theta)\sinh(x\sin\theta)\sin(2\sin\theta)\\
&\quad+\sin(x\cos\theta)\cosh(x\sin\theta)\cos(2\sin\theta))\cos n\theta\,d\theta\\
&=\int_0^\pi C_3(x,\theta)\cos n\theta d\theta+i\int_0^\pi S_3(x,\theta)\cos n\theta\,d\theta.
\end{aligned}\tag{12.3}$$

Combining (12.1), (12.2), and (12.3), using [5.21], and $\phi(x)=\tan^{-1}x$, we get

$$\begin{aligned}
&\left(\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos 2j\phi(x)}{(2j)!(n+2j)!}-\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}\right)\\
&+i\left(\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\sin(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}-\sum_{j=1}^{\infty}\frac{(1+x^2)^j\sin 2j\phi(x)}{(2j)!(n+2j)!}\right)\\
=&\frac{2}{\pi}\cdot\frac{1+\left(1+x^2\right)^{n/2}\cos n(\pi-\phi(x))-i\left(1+x^2\right)^{n/2}\sin n(\pi-\phi(x))}{1+\left(1+x^2\right)^{n}+2\left(1+x^2\right)^{n/2}\cos n(\pi-\phi(x))}\\
&\left(\int_0^\pi C_3(x,\theta)\cos n\theta d\theta+i\int_0^\pi S_3(x,\theta)\cos n\theta d\theta\right).
\end{aligned}\tag{12.4}$$

We get the result by equating the real part of (12.4).

[6.2] Put $n = 0$ in [6.1].

[6.3] Equate the imaginary part of (12.4).

[6.4] Put $n = 0$ in [6.3].

[6.5] Put $x = 1$ in [6.4].

[6.6] Put $x = 1$ in [6.2].

[6.7] Replace $x$ with $ix$ in [6.1].

[6.8] Put $n = 0$ in [6.7].

[6.9] Replace $x$ with $ix$ in [6.3].

[6.10] Put $n = 0$ in [6.9].

[6.11] Replacing $x$ with $1 + ix$ in [5.21], and using (5.14) and $\phi(x) = \tan^{-1} x$, we get the LHS of [5.21] as

$$
\begin{aligned}
&e(1+ix,n) = e\left(\sqrt{1+x^2}e^{i\phi(x)},n\right)\\
&=\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j/2}e^{ij\phi(x)}}{j!(n+j)!}\\
&=\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}e^{2ij\phi(x)}}{(2j)!(n+2j)!}+\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}e^{i(2j+1)\phi(x)}}{(2j+1)!(n+2j+1)!}\\
&=\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}(\cos 2j\phi(x)+i\sin 2j\phi(x))}{(2j)!(n+2j)!}\\
&\quad+\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{\left(1+x^2\right)^{j}(\cos(2j+1)\phi(x)+i\sin(2j+1)\phi(x))}{(2j+1)!(n+2j+1)!}\\
&=\left(\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos 2j\phi(x)}{(2j)!(n+2j)!}+\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}\right)\\
&+i\left(\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\sin(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}+\sum_{j=1}^{\infty}\frac{(1+x^2)^j\sin 2j\phi(x)}{(2j)!(n+2j)!}\right).
\end{aligned}
\tag{12.5}
$$

Replacing $x$ with $1+ix$ in [5.21] and using $\phi(x)=\tan^{-1}x$, we simplify the factor of the integral on RHS of [5.21] as

$$\begin{aligned}
&\frac{2}{\pi}\cdot\frac{1}{1+(1+ix)^n}=\frac{2}{\pi}\cdot\frac{1}{1+\left(1+x^2\right)^{n/2}e^{in\phi(x)}}\\
&=\frac{2}{\pi}\cdot\frac{1}{1+\left(1+x^2\right)^{n/2}\cos n\phi(x)+i\left(1+x^2\right)^{n/2}\sin n\phi(x)}\\
&=\frac{2}{\pi}\cdot\frac{1+\left(1+x^2\right)^{n/2}\cos n\phi(x)-i\left(1+x^2\right)^{n/2}\sin n\phi(x)}{1+\left(1+x^2\right)^{n}+2\left(1+x^2\right)^{n/2}\cos n\phi(x)}.
\end{aligned}\tag{12.6}$$

Replacing $x$ with $1+ix$ in [5.21], we simplify the integral on RHS of [5.21] as

$$\begin{aligned}
&\int_0^\pi e^{(ix+2)\cos\theta}\cos(ix\sin\theta)\cos n\theta\,d\theta\\
&\quad=\int_0^\pi e^{2\cos\theta}e^{ix\cos\theta}\cosh(x\sin\theta)\cos n\theta\,d\theta\\
&\quad=\int_0^\pi e^{2\cos\theta}(\cos(x\cos\theta)+i\sin(x\cos\theta))\cosh(x\sin\theta)\cos n\theta\,d\theta\\
&\quad=\int_0^\pi e^{2\cos\theta}\cos(x\cos\theta)\cosh(x\sin\theta)\cos n\theta\,d\theta\\
&\qquad+i\int_0^\pi e^{2\cos\theta}\sin(x\cos\theta)\cosh(x\sin\theta)\cos n\theta\,d\theta.
\end{aligned}\tag{12.7}$$

Combining (12.5), (12.6), and (12.7), using [5.21], and $\phi(x)=\tan^{-1}x$, we get

$$\begin{aligned}
&\left(\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos 2j\phi(x)}{(2j)!(n+2j)!}+\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\cos(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}\right)\\
&+i\left(\sqrt{1+x^2}\sum_{j=0}^{\infty}\frac{(1+x^2)^j\sin(2j+1)\phi(x)}{(2j+1)!(n+2j+1)!}+\sum_{j=1}^{\infty}\frac{(1+x^2)^j\sin 2j\phi(x)}{(2j)!(n+2j)!}\right)\\
&=\frac{2}{\pi}\cdot\frac{1+\left(1+x^2\right)^{n/2}\cos n\phi(x)-i\left(1+x^2\right)^{n/2}\sin n\phi(x)}{1+\left(1+x^2\right)^{n}+2\left(1+x^2\right)^{n/2}\cos n\phi(x)}\\
&\left(\int_0^\pi e^{2\cos\theta}\cos(x\cos\theta)\cosh(x\sin\theta)\cos n\theta\,d\theta\right.\\
&\left.+i\int_0^\pi e^{2\cos\theta}\sin(x\cos\theta)\cosh(x\sin\theta)\cos n\theta\,d\theta\right).
\end{aligned}\tag{12.8}$$

We get the result by equating the real part of (12.8).

[6.12] Put $n = 0$ in [6.11].

[6.13] Equate the imaginary part of (12.8).

[6.14] Put $n = 0$ in [6.13].

[6.15] Put $x = 1$ in [6.12].

[6.16] Put $x = 1$ in [6.14].

[6.17] Replace $x$ with $ix$ in [6.11].

[6.18] Put $n = 0$ in [6.17].

[6.19] Replace $x$ with $ix$ in [6.13].

[6.20] Put $n = 0$ in [6.19].

[6.21] Replacing $x$ with $xe^{i\phi}$ in [5.21] and using (5.14), we get the LHS of [5.21] as

$$\begin{aligned} e\left(xe^{i\phi}, n\right) &= \sum_{j=0}^{\infty} \frac{x^j e^{ij\phi}}{j!(n+j)!} \\ &= \sum_{j=0}^{\infty} \frac{x^j (\cos j\phi + i \sin j\phi)}{j!(n+j)!} \\ &= \sum_{j=0}^{\infty} \frac{x^j \cos j\phi}{j!(n+j)!} + i \sum_{j=1}^{\infty} \frac{x^j \sin j\phi}{j!(n+j)!}. \end{aligned} \tag{12.9}$$

Replacing $x$ with $xe^{i\phi}$ in [5.21], we simplify the factor of the integral on RHS of [5.21] as

$$\begin{aligned} \frac{2}{\pi} \cdot \frac{1}{1 + x^n e^{in\phi}} &= \frac{2}{\pi} \cdot \frac{1}{1 + x^n \cos n\phi + ix^n \sin n\phi} \\ &= \frac{2}{\pi} \cdot \frac{1 + x^n \cos n\phi - ix^n \sin n\phi}{1 + x^{2n} + 2x^n \cos n\phi}. \end{aligned} \tag{12.10}$$

Replacing $x$ with $xe^{i\phi}$ in [5.21], and using (6.32) and (6.33), we simplify the integral on RHS of [5.21] as

$$
\begin{aligned}
&\int_0^{\pi} e^{\left(xe^{i\phi}+1\right)\cos\theta} \cos\left(\left(xe^{i\phi}-1\right)\sin\theta\right)\cos n\theta\, d\theta \\
&= \int_0^{\pi} e^{(1+x\cos\phi+ix\sin\phi)\cos\theta} \cos\left((x\cos\phi-1)\sin\theta + ix\sin\phi\sin\theta\right)\cos n\theta\, d\theta \\
&= \int_0^{\pi} e^{(1+x\cos\phi)\cos\theta}(\cos(x\sin\phi\cos\theta) + i\sin(x\sin\phi\cos\theta)) \\
&\quad (\cos((x\cos\phi-1)\sin\theta)\cosh(x\sin\phi\sin\theta) \\
&\quad - i\sin((x\cos\phi-1)\sin\theta)\sinh(x\sin\phi\sin\theta))\cos n\theta\, d\theta \\
&= \int_0^{\pi} (\cos(x\sin\phi\cos\theta)\cos((x\cos\phi-1)\sin\theta)\cosh(x\sin\phi\sin\theta) \\
&\quad + \sin(x\sin\phi\cos\theta)\sin((x\cos\phi-1)\sin\theta)\sinh(x\sin\phi\sin\theta)) \\
&\quad e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta \\
&\quad + i\int_0^{\pi} (\sin(x\sin\phi\cos\theta)\cos((x\cos\phi-1)\sin\theta)\cosh(x\sin\phi\sin\theta) \\
&\quad - \cos(x\sin\phi\cos\theta)\sin((x\cos\phi-1)\sin\theta)\sinh(x\sin\phi\sin\theta)) \\
&\quad e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta \\
&= \int_0^{\pi} C_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta + i\int_0^{\pi} S_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta} \\
&\qquad\qquad \cos n\theta\, d\theta.
\end{aligned}
\tag{12.11}
$$

Combining (12.9), (12.10), and (12.11), and using [5.21], we get

$$
\begin{aligned}
&\sum_{j=0}^{\infty}\frac{x^j\cos j\phi}{j!(n+j)!} + i\sum_{j=1}^{\infty}\frac{x^j\sin j\phi}{j!(n+j)!} \\
&= \frac{2}{\pi}\cdot\frac{1+x^n\cos n\phi - ix^n\sin n\phi}{1+x^{2n}+2x^n\cos n\phi} \\
&\quad \left(\int_0^{\pi} C_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta\right. \\
&\qquad \left. + i\int_0^{\pi} S_5(x,\phi,\theta)e^{(1+x\cos\phi)\cos\theta}\cos n\theta\, d\theta\right).
\end{aligned}
\tag{12.12}
$$

We get the result by equating the real part of (12.12).

[6.22] Put $n = 0$ in [6.21].

[6.23] Equate the imaginary part of (12.12).

[6.24] Put $n = 0$ in [6.23].

[6.25] Replace $\phi$ with $i\phi$ in [6.21].

[6.26] Put $n = 0$ in [6.25].

[6.27] Replace $\phi$ with $i\phi$ in [6.23].

[6.28] Put $n = 0$ in [6.27].

[6.29] Put $x = \sec\phi$ in [6.22].

[6.30] Put $x = -\sec\phi$ in [6.22].

[6.31] Put $x = \sec\phi$ in [6.24].

[6.32] Put $x = -\sec\phi$ in [6.24].

[6.33] Put $x = \operatorname{sech}\phi$ in [6.26].

[6.34] Put $x = -\operatorname{sech}\phi$ in [6.26].

[6.35] Put $x = \operatorname{sech}\phi$ in [6.28].

[6.36] Put $x = -\operatorname{sech}\phi$ in [6.28].

[6.37] Applying Theorem 6.13 to $g(z) = e^z$, we get

$$
\begin{aligned}
f(1) \star e^x &= \sum_{j=0}^{\infty} \frac{a_j}{j!} x^j \\
&= -a_0 + \frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \left( e^{xe^{i\theta}} + e^{xe^{-i\theta}} \right) d\theta \\
&= -a_0 + \frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \left( e^{x(\cos\theta + i\sin\theta)} + e^{x(\cos\theta - i\sin\theta)} \right) d\theta \\
&= -a_0 + \frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \left( e^{ix\sin\theta} + e^{-ix\sin\theta} \right) e^{x\cos\theta} \, d\theta \\
&= -f(0) + \frac{1}{\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \cos(x\sin\theta) e^{x\cos\theta} \, d\theta.
\end{aligned}
$$

[6.38] Applying Theorem 6.13 to $g(z) = 1/(1-z) = \sum_{j=0}^{\infty} z^j$, we get

$$
\begin{aligned}
&f(1) \star \frac{1}{1-x} = \sum_{j=0}^{\infty} a_j x^j \\
&= -a_0 + \frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \left( \frac{1}{1 - x e^{i\theta}} + \frac{1}{1 - x e^{-i\theta}} \right) d\theta \\
&= -a_0 + \frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \frac{2 - x\left(e^{i\theta} + e^{-i\theta}\right)}{1 + x^2 - x\left(e^{i\theta} + e^{-i\theta}\right)} d\theta \\
&= -f(0) + \frac{1}{\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta} d\theta.
\end{aligned}
$$

[6.39] Applying Theorem 6.13 to $g(z) = -\log(1-z) = \sum_{j=0}^{\infty} z^j / j$, we get

$$
\begin{aligned}
&f(1) \star \frac{1}{1-x} = \sum_{j=1}^{\infty} \frac{a_j}{j} x^j \\
&= -\frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \left( \log\left(1 - x e^{i\theta}\right) + \log\left(1 - x e^{-i\theta}\right) \right) d\theta \\
&= -\frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \log\left(1 + x^2 - x\left(e^{i\theta} + e^{-i\theta}\right)\right) d\theta \\
&= -\frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \log\left(1 + x^2 - 2x\cos\theta\right) d\theta.
\end{aligned}
$$

[6.40] Applying Theorem 6.13 to $g(z) = z/(1-z)^2 = \sum_{j=1}^{\infty} j z^j$, we get

$$
\begin{aligned}
&f(1) \star \frac{x}{(1-x)^2} = \sum_{j=1}^{\infty} j a_j x^j \\
&= \frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \left( \frac{x e^{i\theta}}{\left(1 - x e^{i\theta}\right)^2} + \frac{x e^{-i\theta}}{\left(1 - x e^{-i\theta}\right)^2} \right) d\theta \\
&= \frac{1}{2\pi} \int_0^{\pi} x \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \\
&\quad \frac{e^{i\theta}\left(1 + x^2 e^{-2i\theta} - 2x e^{-i\theta}\right) + e^{-i\theta}\left(1 + x^2 e^{2i\theta} - 2x e^{i\theta}\right)}{\left(1 + x^2 - x\left(e^{i\theta} + e^{-i\theta}\right)\right)^2} d\theta \\
&= \frac{1}{2\pi} \int_0^{\pi} x \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \frac{e^{i\theta} + e^{-i\theta} + x^2\left(e^{i\theta} + e^{-i\theta}\right) - 4x}{\left(1 + x^2 - 2x\cos\theta\right)^2} d\theta \\
&= \frac{1}{\pi} \int_0^{\pi} x \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \frac{(1 + x^2)\cos\theta - 2x}{\left(1 + x^2 - 2x\cos\theta\right)^2} d\theta.
\end{aligned}
$$

[6.41] *First Method:* Proved in [5.24].
*Second Method:* Applying Theorem 6.13 to $f(z) = e^z$ and $g(z) = e^z$, we get

$$\begin{aligned}
e^1 \star e^x &= \sum_{j=0}^{\infty} \frac{x^j}{(j!)^2} = e(x,0) \\
&= -1 + \frac{1}{2\pi} \int_0^{\pi} \left(e^{e^{i\theta}} + e^{e^{-i\theta}}\right) \left(e^{xe^{i\theta}} + e^{xe^{-i\theta}}\right) d\theta \\
&= -1 + \frac{1}{2\pi} \int_0^{\pi} \left(e^{\cos\theta + i\sin\theta} + e^{\cos\theta - i\sin\theta}\right) \left(e^{x\cos\theta + ix\sin\theta} + e^{x\cos\theta - ix\sin\theta}\right) d\theta \\
&= -1 + \frac{1}{2\pi} \int_0^{\pi} e^{(x+1)\cos\theta} \left(e^{i\sin\theta} + e^{-i\sin\theta}\right) \left(e^{ix\sin\theta} + e^{-ix\sin\theta}\right) d\theta \\
&= -1 + \frac{2}{\pi} \int_0^{\pi} e^{(x+1)\cos\theta} \cos(\sin\theta)\cos(x\sin\theta)\, d\theta.
\end{aligned}$$

[6.42] *First Method:* Proved in [5.25].
*Second Method:* Applying Theorem 6.14 to $f(z) = e^z$ and $g(z) = e^z$, we get

$$\begin{aligned}
e^1 \star e^x &= \sum_{j=0}^{\infty} \frac{x^j}{(j!)^2} = e(x,0) \\
&= 1 - \frac{1}{2\pi} \int_0^{\pi} \left(e^{e^{i\theta}} - e^{e^{-i\theta}}\right) \left(e^{xe^{i\theta}} - e^{xe^{-i\theta}}\right) d\theta \\
&= 1 - \frac{1}{2\pi} \int_0^{\pi} \left(e^{\cos\theta + i\sin\theta} - e^{\cos\theta - i\sin\theta}\right) \left(e^{x\cos\theta + ix\sin\theta} - e^{x\cos\theta - ix\sin\theta}\right) d\theta \\
&= 1 - \frac{1}{2\pi} \int_0^{\pi} e^{(x+1)\cos\theta} \left(e^{i\sin\theta} - e^{-i\sin\theta}\right) \left(e^{ix\sin\theta} - e^{-ix\sin\theta}\right) d\theta \\
&= 1 + \frac{2}{\pi} \int_0^{\pi} e^{(x+1)\cos\theta} \sin(\sin\theta)\sin(x\sin\theta)\, d\theta.
\end{aligned}$$

[6.43] Subtract [6.42] from [6.41].

[6.44] Applying Theorem 6.13 to $g(z) = f(z)$, we get

$$f(1) \star f(1) = \sum_{j=0}^{\infty} a_j^2$$
$$= -f(0)^2 + \frac{1}{2\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right)^2 d\theta.$$

[6.45] Applying Theorem 6.13 to $f(z) = e^z = \sum_{j=0}^{\infty} z^j/j!$ and $g(z) = 1/(1-z) = \sum_{j=0}^{\infty} z^j$, we get

$$e^x \star \frac{1}{1-y} = \sum_{j=0}^{\infty} \frac{(xy)^j}{j!} = e^{xy}$$
$$= -1 + \frac{1}{2\pi} \int_0^{\pi} \left( e^{xe^{i\theta}} + e^{xe^{-i\theta}} \right) \left( \frac{1}{1-ye^{i\theta}} + \frac{1}{1-ye^{-i\theta}} \right) d\theta$$
$$= -1 + \frac{1}{2\pi} \int_0^{\pi} \left( e^{x\cos\theta + ix\sin\theta} + e^{x\cos\theta - ix\sin\theta} \right) \frac{2 - y\left(e^{i\theta} + e^{-i\theta}\right)}{1 + y^2 - y\left(e^{i\theta} + e^{-i\theta}\right)} d\theta$$
$$= -1 + \frac{1}{2\pi} \int_0^{\pi} e^{x\cos\theta} \left( e^{ix\sin\theta} + e^{-ix\sin\theta} \right) \frac{2 - 2y\cos\theta}{1 + y^2 - 2y\cos\theta} d\theta$$
$$= -1 + \frac{2}{\pi} \int_0^{\pi} e^{x\cos\theta} \cos(x\sin\theta) \frac{1 - y\cos\theta}{1 + y^2 - 2y\cos\theta} d\theta$$
$$\iff \int_0^{\pi} \frac{1 - y\cos\theta}{1 + y^2 - 2y\cos\theta} e^{x\cos\theta} \cos(x\sin\theta)\, d\theta = \frac{\pi}{2}\left(e^{xy} + 1\right).$$

[6.46] Use [6.39].

[6.47] Putting $f(x) = \log(1+x)$ in [6.46], we get

$$\int_0^x \frac{\log(1+x)}{x} dx$$
$$= -\frac{1}{2\pi} \int_0^{\pi} \left( \log\left(1 + e^{i\theta}\right) + \log\left(1 + e^{-i\theta}\right) \right) \log\left(1 + x^2 - 2x\cos\theta\right) d\theta$$
$$= -\frac{1}{2\pi} \int_0^{\pi} \log\left(2 + e^{i\theta} + e^{-i\theta}\right) \log\left(1 + x^2 - 2x\cos\theta\right) d\theta$$
$$= -\frac{1}{2\pi} \int_0^{\pi} \log\left(2 + 2\cos\theta\right) \log\left(1 + x^2 - 2x\cos\theta\right) d\theta$$
$$= -\frac{1}{2\pi} \int_0^{\pi} \log\left(4\cos^2\frac{\theta}{2}\right) \log\left(1 + x^2 - 2x\cos\theta\right) d\theta$$
$$= -\frac{1}{\pi} \int_0^{\pi} \log\left(2\cos\frac{\theta}{2}\right) \log\left(1 + x^2 - 2x\cos\theta\right) d\theta.$$

[6.48] Putting $f(x) = \sin x$ in [6.46], we get

$$\begin{aligned}
&\int_0^x \frac{\sin x}{x}\,dx = -\frac{1}{2\pi}\int_0^\pi \left(\sin\left(e^{i\theta}\right) + \sin\left(e^{-i\theta}\right)\right)\log\left(1+x^2-2x\cos\theta\right)\,d\theta \\
&= -\frac{1}{\pi}\int_0^\pi \sin\left(\frac{e^{i\theta}+e^{-i\theta}}{2}\right)\cos\left(\frac{e^{i\theta}-e^{-i\theta}}{2}\right)\log\left(1+x^2-2x\cos\theta\right)\,d\theta \\
&= -\frac{1}{\pi}\int_0^\pi \sin(\cos\theta)\cos(i\sin\theta)\log\left(1+x^2-2x\cos\theta\right)\,d\theta \\
&= -\frac{1}{\pi}\int_0^\pi \sin(\cos\theta)\cosh(\sin\theta)\log\left(1+x^2-2x\cos\theta\right)\,d\theta.
\end{aligned}$$

[6.49] Putting $f(x) = \sinh x$ in [6.46], we get

$$\begin{aligned}
&\int_0^x \frac{\sinh x}{x}\,dx = -\frac{1}{2\pi}\int_0^\pi \left(\sinh\left(e^{i\theta}\right) + \sinh\left(e^{-i\theta}\right)\right)\log\left(1+x^2-2x\cos\theta\right)\,d\theta \\
&= -\frac{1}{\pi}\int_0^\pi \sinh\left(\frac{e^{i\theta}+e^{-i\theta}}{2}\right)\cosh\left(\frac{e^{i\theta}-e^{-i\theta}}{2}\right)\log\left(1+x^2-2x\cos\theta\right)\,d\theta \\
&= -\frac{1}{\pi}\int_0^\pi \sinh(\cos\theta)\cosh(i\sin\theta)\log\left(1+x^2-2x\cos\theta\right)\,d\theta \\
&= -\frac{1}{\pi}\int_0^\pi \sinh(\cos\theta)\cos(\sin\theta)\log\left(1+x^2-2x\cos\theta\right)\,d\theta.
\end{aligned}$$

[6.50] Applying Theorem 6.13 to $f(z) = \log(1-z) = -\sum_{j=1}^\infty z^j/j$ and $g(z) = 1/(1-z) = \sum_{j=0}^\infty z^j$, we get

$$\begin{aligned}
&\log(1-x) \star \frac{1}{1-y} = -\sum_{j=1}^\infty \frac{(xy)^j}{j} = \log(1-xy) \\
&= \frac{1}{2\pi}\int_0^\pi \left(\frac{1}{1-ye^{i\theta}} + \frac{1}{1-ye^{-i\theta}}\right)\left(\log\left(1-xe^{i\theta}\right) + \log\left(1-xe^{-i\theta}\right)\right)\,d\theta \\
&= \frac{1}{2\pi}\int_0^\pi \frac{2-y\left(e^{i\theta}+e^{-i\theta}\right)}{1+y^2-y\left(e^{i\theta}+e^{-i\theta}\right)}\log\left(1+x^2-x\left(e^{i\theta}+e^{-i\theta}\right)\right)\,d\theta \\
&= \frac{1}{\pi}\int_0^\pi \frac{1-y\cos\theta}{1+y^2-2y\cos\theta}\log\left(1+x^2-2x\cos\theta\right)\,d\theta.
\end{aligned}$$

[6.51] Putting $f(z) = \log(1+z)$ in [6.38], we get

$$\begin{aligned}
\log(1+x) &= \frac{1}{\pi}\int_0^\pi \left(\log\left(1+e^{i\theta}\right) + \log\left(1+e^{-i\theta}\right)\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{1}{\pi}\int_0^\pi \log\left(2+e^{i\theta}+e^{-i\theta}\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{1}{\pi}\int_0^\pi \log\left(2+2\cos\theta\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{1}{\pi}\int_0^\pi \log\left(4\cos^2\frac{\theta}{2}\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{2}{\pi}\int_0^\pi \log\left(2\cos\frac{\theta}{2}\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta.
\end{aligned}$$

[6.52] Putting $f(z) = \sin z$ in [6.38], we get

$$\begin{aligned}
\sin x &= \frac{1}{\pi}\int_0^\pi \left(\sin\left(e^{i\theta}\right) + \sin\left(e^{-i\theta}\right)\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{2}{\pi}\int_0^\pi \sin\left(\frac{e^{i\theta}+e^{-i\theta}}{2}\right)\cos\left(\frac{e^{i\theta}-e^{-i\theta}}{2}\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{2}{\pi}\int_0^\pi \sin\left(\cos\theta\right)\cos\left(i\sin\theta\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{2}{\pi}\int_0^\pi \sin\left(\cos\theta\right)\cosh\left(\sin\theta\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta.
\end{aligned}$$

[6.53] Putting $f(z) = \sinh z$ in [6.38], we get

$$\begin{aligned}
\sinh x &= \frac{1}{\pi}\int_0^\pi \left(\sinh\left(e^{i\theta}\right) + \sinh\left(e^{-i\theta}\right)\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{2}{\pi}\int_0^\pi \sinh\left(\frac{e^{i\theta}+e^{-i\theta}}{2}\right)\cosh\left(\frac{e^{i\theta}-e^{-i\theta}}{2}\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{2}{\pi}\int_0^\pi \sinh\left(\cos\theta\right)\cosh\left(i\sin\theta\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta \\
&= \frac{2}{\pi}\int_0^\pi \sinh\left(\cos\theta\right)\cos\left(\sin\theta\right)\frac{1-x\cos\theta}{1+x^2-2x\cos\theta}\,d\theta.
\end{aligned}$$

[6.54] Applying Theorem 6.13 to $f(z) = g(z) = 1/(1-z) = \sum_{j=0}^{\infty} z^j$, we get

$$\begin{aligned}
&\frac{1}{1-x} \star \frac{1}{1-y} = \sum_{j=0}^{\infty} (xy)^j = \frac{1}{1-xy} \\
&= -1 + \frac{1}{2\pi} \int_0^{\pi} \left( \frac{1}{1-xe^{i\theta}} + \frac{1}{1-xe^{-i\theta}} \right) \left( \frac{1}{1-ye^{i\theta}} + \frac{1}{1-ye^{-i\theta}} \right) d\theta \\
&= -1 + \frac{1}{2\pi} \int_0^{\pi} \frac{2 - x\left(e^{i\theta} + e^{-i\theta}\right)}{1 + x^2 - x\left(e^{i\theta} + e^{-i\theta}\right)} \cdot \frac{2 - y\left(e^{i\theta} + e^{-i\theta}\right)}{1 + y^2 - y\left(e^{i\theta} + e^{-i\theta}\right)} \, d\theta \\
&= -1 + \frac{1}{2\pi} \int_0^{\pi} \frac{2 - 2x\cos\theta}{1 + x^2 - 2x\cos\theta} \cdot \frac{2 - 2y\cos\theta}{1 + y^2 - 2y\cos\theta} \, d\theta \\
&= -1 + \frac{2}{\pi} \int_0^{\pi} \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta} \cdot \frac{1 - y\cos\theta}{1 + y^2 - 2y\cos\theta} \, d\theta \\
\iff &\int_0^{\pi} \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta} \cdot \frac{1 - y\cos\theta}{1 + y^2 - 2y\cos\theta} \, d\theta = \frac{\pi}{2} \cdot \frac{2 - xy}{1 - xy}.
\end{aligned}$$

[6.55] Applying Theorem 6.13 to $f(z) = z/(1-z)^2 = \sum_{j=1}^{\infty} j z^j$ and $g(z) = \log(1-z) = -\sum_{j=1}^{\infty} z^j/j$, we get

$$\begin{aligned}
&\frac{x}{(1-x)^2} \star \log(1-y) = -\sum_{j=1}^{\infty} (xy)^j = -\frac{xy}{1-xy} \\
&= \frac{1}{2\pi} \int_0^{\pi} \left( \frac{xe^{i\theta}}{\left(1 - xe^{i\theta}\right)^2} + \frac{xe^{-i\theta}}{\left(1 - xe^{-i\theta}\right)^2} \right) \left( \log\left(1 - ye^{i\theta}\right) + \log\left(1 - ye^{-i\theta}\right) \right) \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad d\theta \\
&= \frac{1}{2\pi} \int_0^{\pi} x \frac{e^{i\theta}\left(1 + x^2 e^{-2i\theta} - 2xe^{-i\theta}\right) + e^{-i\theta}\left(1 + x^2 e^{2i\theta} - 2xe^{i\theta}\right)}{\left(1 + x^2 - x\left(e^{i\theta} + e^{-i\theta}\right)\right)^2} \\
&\qquad \log\left(1 + y^2 - y\left(e^{i\theta} + e^{-i\theta}\right)\right) \, d\theta \\
&= \frac{1}{2\pi} \int_0^{\pi} x \frac{e^{i\theta} + e^{-i\theta} + x^2\left(e^{i\theta} + e^{-i\theta}\right) - 4x}{\left(1 + x^2 - 2x\cos\theta\right)^2} \log\left(1 + y^2 - 2y\cos\theta\right) \, d\theta \\
&= \frac{1}{\pi} \int_0^{\pi} x \log\left(1 + y^2 - 2y\cos\theta\right) \frac{(1 + x^2)\cos\theta - 2x}{\left(1 + x^2 - 2x\cos\theta\right)^2} \, d\theta \\
\iff &\int_0^{\pi} x \log\left(1 + y^2 - 2y\cos\theta\right) \frac{(1 + x^2)\cos\theta - 2x}{\left(1 + x^2 - 2x\cos\theta\right)^2} \, d\theta = -\frac{\pi xy}{1 - xy}.
\end{aligned}$$

[6.56] Putting $f(x) = \sum_{j=0}^{n} a_j x^j$ in [6.38], we get

$$\begin{aligned} 2a_0 + \sum_{j=1}^{n} a_j x^j &= a_0 + f(x) \\ &= \frac{1}{\pi} \int_0^{\pi} \left( f\left(e^{i\theta}\right) + f\left(e^{-i\theta}\right) \right) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta}\, d\theta \\ &= \frac{1}{\pi} \int_0^{\pi} \left( \sum_{j=0}^{n} a_j \left(e^{ji\theta} + e^{-ji\theta}\right) \right) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta}\, d\theta \\ &= \frac{2}{\pi} \int_0^{\pi} \left( \sum_{j=0}^{n} a_j \cos j\theta \right) \frac{1 - x\cos\theta}{1 + x^2 - 2x\cos\theta}\, d\theta. \end{aligned}$$

[6.57] Applying Theorem 6.13 to $f(z) = g(z) = 1/\sqrt{1-z} = \sum_{j=0}^{\infty} \binom{2j}{j} z^j / 2^{2j}$, we get

$$\begin{aligned} \frac{1}{\sqrt{1-x}} \star \frac{1}{\sqrt{1-x}} &= \sum_{j=0}^{\infty} \binom{2j}{j}^2 \frac{x^{2j}}{2^{4j}} \\ &= -1 + \frac{1}{2\pi} \int_0^{\pi} \left( \frac{1}{\sqrt{1 - xe^{i\theta}}} + \frac{1}{\sqrt{1 - xe^{-i\theta}}} \right)^2 d\theta. \end{aligned} \tag{12.13}$$

Putting $x = 1/4$ in (12.13), we get

$$\begin{aligned} \sum_{j=0}^{\infty} \frac{1}{2^{8j}} \binom{2j}{j}^2 &= -1 + \frac{1}{2\pi} \int_0^{\pi} \left( \frac{1}{\sqrt{1 - \frac{1}{4}e^{i\theta}}} + \frac{1}{\sqrt{1 - \frac{1}{4}e^{-i\theta}}} \right)^2 d\theta \\ &= -1 + \frac{1}{2\pi} \int_0^{\pi} \frac{2 - \frac{1}{4}\left(e^{i\theta} + e^{i\theta}\right) + 2\sqrt{\frac{17}{16} - \frac{1}{4}\left(e^{i\theta} + e^{i\theta}\right)}}{\frac{17}{16} - \frac{1}{4}\left(e^{i\theta} + e^{i\theta}\right)}\, d\theta \\ &= -1 + \frac{1}{2\pi} \int_0^{\pi} \frac{2 - \frac{1}{2}\cos\theta + 2\sqrt{\frac{17}{16} - \frac{1}{2}\cos\theta}}{\frac{17}{16} - \frac{1}{2}\cos\theta}\, d\theta \end{aligned}$$

$$= -1 + \frac{1}{2\pi}\int_0^{\pi} \frac{32 - 8\cos\theta + 8\sqrt{17 - 8\cos\theta}}{17 - 8\cos\theta}\, d\theta$$

$$= -1 + \frac{1}{2\pi}\int_0^{\pi} \left(1 + \frac{15 + 8\sqrt{9 + 16\sin^2\dfrac{\theta}{2}}}{9 + 16\sin^2\dfrac{\theta}{2}}\right) d\theta$$

$$= -\frac{1}{2} + \frac{1}{2\pi}\int_0^{\pi} \frac{15 + 8\sqrt{9 + 16\sin^2\dfrac{\theta}{2}}}{9 + 16\sin^2\dfrac{\theta}{2}}\, d\theta.$$

[6.58] Put $x = 1$ in [6.41].

[6.59] Put $x = 1$ in [6.42].

# Appendix A
# Mapping from the Manuscript to the Book

| Manuscript page# | Manuscript entry# | Book entry# |
|---|---|---|
| 1 | (1) | [1.1] |
| 1 | (2) | [1.2] |
| 1 | (3) | [1.3] |
| 1 | (4) *Incorrect Identity* | [1.4] *Corrected Identity* |
| 1 | (5) | [1.5] |
| 1 | (6) *Incorrect Identity* | [1.6] *Corrected Identity* |
| 1 | (7) | [1.7] |
| 1 | (8) | [1.8] |
| 2 | (9) | [1.9] |
| 2 | (10) | [1.10] |
| 2 | (11) | [1.11] |
| 2 | (12) | [1.12] |
| 2 | (13) | [1.13] |
| 3 | (14) | [1.14] |
| 3 | (15) | [1.15] |
| 3 | (16) | [1.16] |
| 3 | (17) | [1.17] |
| 3 | (18) | [1.18] |
| 4 | (19) | [1.20] |
| 4 | (20) | [1.19] |
| 4 | (21) | [1.21] |
| 4 | (22) | [1.22] |
| 4 | (23) | [1.23] |
| 4 | (24) | [1.24] |
| 4 | (25) | [1.25] |
| 5 | (26) | [1.26] |
| 5 | (27) | [1.27] |
| 5 | (28) | [1.28] |
| 5 | (29) | [1.29] |
| 5 | (30) | [1.30] |
| 5 | (31) | [1.31] |
| 5 | (32) | [1.32] |

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| Manuscript page# | Manuscript entry# | Book entry# |
|---|---|---|
| 6 | (33) | [1.33] |
| 6 | (34) | [4.22] |
| 6 | (35) | [1.34] |
| 6 | (36) | [1.35] |
| 6 | (37) | [1.36] |
| 6 | (38) | [1.37] |
| 6 | (39) | [1.38] |
| 7 | (40) | [1.39] |
| 7 | (41) | [1.40] |
| 7 | (42) | [1.41] |
| 7 | (43) | [1.42] |
| 7 | (44) | [1.43] |
| 7 | (45) | [1.44] |
| 7 | (46) | [1.45] |
| 7 | (47) | [1.46] |
| 8 | (48) | [1.47] |
| 8 | (49) | [1.48] |
| 8 | (50) | [1.49] |
| 8 | (51) | [1.50] |
| 8 | (52) | [1.51] |
| 8 | (53) | [1.52] |
| 8 | (54) | [1.53] |
| 9 | (55) | [1.54] |
| 9 | (56) | [1.55] |
| 9 | (57) | [1.56] |
| 9 | (58) | [1.57] |
| 9 | (59) | [1.58] |
| 9 | (60) | [1.59] |
| 9 | (61) | [1.60] |
| 9 | (62) | [1.61] |
| 10 | (63) | [1.62] |
| 10 | (64) | [1.63] |
| 10 | (65) | [1.64] |
| 10 | (66) | [1.65] |
| 10 | (67) | [1.66] |
| 10 | (68) | [1.67] |
| 11 | (69) | [1.68] |
| 11 | (70) | [1.69] |
| 11 | (71) | [1.73] |
| 11 | (72) | [1.83] |
| 12 | (1) *Incorrect Identity* | [2.1] *Corrected Identity* |
| 12 | (2) | [2.2] |
| 12 | (3) | [2.3] |
| 13 | (4) | [2.4] |
| 13 | (5) | [2.5] |
| 13 | (6) | [2.6] |
| 13 | (7) | [2.7] |
| 13 | (8) | [2.8] |
| 13 | (9) | [2.9] |

| Manuscript page# | Manuscript entry# | Book entry# |
|---|---|---|
| 14 | (10) | [2.10] |
| 14 | (11) | [2.11] |
| 14 | (12) | [2.12] |
| 14 | (13) | [2.13] |
| 14 | (14) | [2.14] |
| 15 | (15) | [2.15] |
| 15 | (16) | [2.16] |
| 15 | (17) | [2.17] |
| 15 | (18) | [2.18] |
| 15 | (19) | [2.19] |
| 16 | (20) | [2.20] |
| 16 | (21) | [2.21] |
| 16 | (22) | [2.22] |
| 16 | (23) | [2.23] |
| 16 | (24) | [2.24] |
| 16 | (25) | [2.25] |
| 16 | (26) | [2.26] |
| 17 | (1) | [2.27] |
| 17 | (2) | [2.28] |
| 17 | (3) | [2.29] |
| 17 | (4) | [2.30] |
| 18 | (5) | [2.31] |
| 18 | (6) | [2.32] |
| 18 | (7) | [2.33] |
| 18 | (8) | [2.34] |
| 18 | (9) | [2.35] |
| 18 | (10) | [2.36] |
| 19 | (11) | [2.37] |
| 19 | (12) | [2.38] |
| 19 | (13) | [2.39] |
| 19 | (14) | [2.40] |
| 20 | (15) | [2.41] |
| 20 | (16) | [2.42] |
| 20 | (17) | [2.43] |
| 20 | (18) | [2.44] |
| 21 | (19) | [2.45] |
| 21 | (20) | [2.46] |
| 21 | (21) | [2.47] |
| 21 | (22) | [2.48] |
| 21 | (23) | [2.49] |
| 21 | (24) | [2.50] |
| 21 | (25) | [2.51] |
| 22 | (26) | [2.52] |
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| 39 | (5) | [4.8–4.9] |
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| 40 | (7) | [4.12–4.13] |
| 40 | (8) | [4.14–4.15] |
| 40 | (9) | [4.16–4.17] |
| 41 | (10) | [4.18–4.19] |
| 41 | (11) | [4.20–4.21] |
| 41 | (12) | [4.22] |
| 41 | (13) | [4.23] |
| 41 | (14) | [4.24] |
| 42 | (15) | [4.25] |
| 42 | (16) | [4.26–4.27] |
| 43 | (17) | [4.28] |
| 43 | (18) *Incorrect Identity* | [4.29–4.30] *Corrected Identities* |
| 43 | (19) | [4.31–4.32] |
| 43 | (20) | [4.33–4.34] |
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| 44 | (23) | [4.39–4.40] |
| 44 | (24) | [4.41–4.42] |
| 45 | (25) | [4.43–4.44] |
| 45 | (26) | [4.45–4.46] |
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| 47 | (28) | [4.49–4.50] |
| 47 | (29) | [4.51] |
| 48 | (30) | [4.52–4.53] |
| 49 | (31) | [4.54] |
| 49 | Theorem 5 *Incorrect Theorem* | Theorem 4.16 *Corrected Theorem* |
| 50 | (32) | [4.55–4.56] |
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| 53 | (38) | [4.65–4.66] |
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| 54 | (41) | [4.71–4.72] |
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| 54 | (43) | [4.75–4.76] |
| 55 | (44) | [4.77–4.78] |
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| 62 | Generalization of Theorem 1 | Theorem 4.2 |
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| 69 | (4) | Corollary 4.21 |
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| 79 | (11) (Second Part) | Corollary 4.53 |
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| 81 | (3) *Incorrect Identity* | [4.110–4.111] *Corrected Identities* |
| 81 | (4) *Incorrect Identity* | [4.112–4.113] *Corrected Identities* |
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| 90 | (37) | [4.155] |
| 91 | (38) | [4.156] |
| 91 | (39) | [4.157] |
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| 96 | (6) | [5.26] |
| 97 | (1) | [5.8] |
| 97 | (2) | [5.9] |
| 97 | (3) | [5.10] |
| 97 | (4) | [5.11] |
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| 98 | (8) | [5.18–5.19] |
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| 100 | Circulo$-Z$ Functions | Definition 5.11 |
| 100 | Hypo$-Z$ Functions | (5.17) and (5.18) |
| 101 | Hypo$-Z$ Functions | Definition 5.12 |
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| 110 | (5) *Incorrect Identities* | [5.54–5.55] *Corrected Identities* |
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| 110 | (7) | [5.57–5.58] |
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| 112 | (13) | [5.69–5.70] |
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| 113 | (20) | [5.83–5.84] |
| 114 | (21) | [5.75] |
| 114 | (22) | [5.75–5.76] |
| 114 | (23) | [5.73–5.74] |
| 114 | (24) | [5.85] |
| 114 | (25) *Incorrect Identity* | [4.1] *Corrected Identity* |
| 114 | (26) | [5.86] |
| 115 | (27) | [5.87–5.88] |
| 115 | (28) | [5.89–5.90] |
| 115 | (29) | [5.91] |
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| 115 | (31) *Incorrect Identity* | [5.8] *Corrected Identity* |
| 115 | (32) *Incorrect Identity* | [5.9] *Corrected Identity* |
| 115 | (33) *Incorrect Identity* | |
| 116 | (34) *Incorrect Identity* | |
| 116 | (35) *Incorrect Identity* | |
| 116 | (36) *Incorrect Identity* | |
| 117 | $C_1(\theta)$ | (6.28) |
| 117 | $C_2(\theta)$ | (6.30) |
| 117 | $S_1(\theta)$ | (6.29) |
| 117 | $S_2(\theta)$ | (6.31) |
| 118 | (1) | [6.1] |
| 118 | (2) | [6.2] |
| 119 | (3) | [6.3] |
| 119 | (4) | [6.4] |
| 120 | (5) *Incorrect Identity* | [6.5] *Corrected Identity* |
| 120 | (6) *Incorrect Identity* | [6.6] *Corrected Identity* |
| 121 | (7) *Incorrect Identity* | [6.7] *Corrected Identity* |
| 121 | (8) | [6.8] |
| 122 | (9) *Incorrect Identity* | [6.9] *Corrected Identity* |
| 122 | (10) | [6.10] |
| 123 | (11) *Incorrect Identity* | [6.11] *Corrected Identity* |
| 123 | (12) *Incorrect Identity* | [6.12] *Corrected Identity* |
| 124 | (13) *Incorrect Identity* | [6.13] *Corrected Identity* |
| 124 | (14) *Incorrect Identity* | [6.14] *Corrected Identity* |
| 125 | (15) *Incorrect Identity* | [6.15] *Corrected Identity* |
| 125 | (16) *Incorrect Identity* | [6.16] *Corrected Identity* |
| 126 | (17) *Incorrect Identity* | [6.17] *Corrected Identity* |
| 126 | (18) *Incorrect Identity* | [6.18] *Corrected Identity* |
| 127 | (19) *Incorrect Identity* | [6.19] *Corrected Identity* |
| 127 | (20) *Incorrect Identity* | [6.20] *Corrected Identity* |
| 128 | $C_1(\theta)$ | (6.32) |
| 128 | $S_1(\theta)$ | (6.33) |
| 128 | $C_2(\theta)$ | (6.34) |
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| 129 | (21) | [6.21] |
| 129 | (22) | [6.22] |

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| 129 | (23) | [6.23] |
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| 130 | (25) | [6.25] |
| 130 | (26) | [6.26] |
| 131 | (27) | [6.27] |
| 131 | (28) | [6.28] |
| 132 | (29) | [6.29] |
| 132 | (30) | [6.30] |
| 132 | (31) | [6.31] |
| 132 | (32) | [6.32] |
| 133 | (33) | [6.33] |
| 133 | (34) | [6.34] |
| 133 | (35) | [6.35] |
| 133 | (36) | [6.36] |
| 134 | Note *Incorrect Identities* | |
| 135 | Definition | Definition 6.12 |
| 136 | General Formula for the Functional Transformation | Theorem 6.13 |
| 137 | (1) | [6.37] |
| 137 | (2) | [6.38] |
| 137 | (3) | [6.39] |
| 138 | (4) *Incorrect Identity* | [6.40] *Corrected Identity* |
| 138 | (5) | [6.41] |
| 138 | (6) | [6.42] |
| 138 | (7) | [6.43] |
| 138 | (8) | [6.44] |
| 139 | (9) | [6.45] *Generalized Identity* |
| 139 | (10) | [6.46] |
| 139 | (11) | [6.47] |
| 139 | (12) | [6.48] |
| 140 | (13) | [6.49] |
| 140 | (14) | [6.50] *Generalized Identity* |
| 140 | (15) | [6.51] |
| 140 | (16) | [6.52] |
| 141 | (17) | [6.53] |
| 141 | (18) | [6.54] *Generalized Identity* |
| 141 | (19) *Incorrect Identity* | [6.55] *Corrected Identity* |
| 141 | (20) | [6.56] |
| 142 | (21), Line 1 *Incorrect Identity* | [1.19]*Corrected Identity* |
| 142 | (21), Lines 2–4 *Incorrect Identity* | [6.57] *Corrected Identity* |
| 142 | (22) | [6.58] |
| 142 | (23) | [6.59] |
| 143 | (1) | [1.73] |
| 143 | (2) | [1.74] |
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| 144 | (4) | [1.76] |
| 144 | (5) | [1.77] |
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| 145 | (8) | [1.80] |
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| 146 | (10) *Incorrect Identity* | [1.82] *Corrected Identity* |
| 146 | (20) | [1.92] |
| 147 | (11) | [1.83] |
| 147 | (12) | [1.84] |
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| 148 | (14) | [1.86] |
| 148 | (15) | [1.87] |
| 149 | (16) | [1.88] |
| 149 | (17) | [1.89] |
| 150 | (18) | [1.90] |
| 150 | (19) | [1.91] |

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